



WRITING BOOKS IN THE TERMINAL

# The Inkhaven Manual

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*The Complete Guide to Writing a Book in the Terminal*

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2026 · examples assume Inkhaven 3.0.0 or newer

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# About This Manual

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Most writing tools ask you to leave the page. You open a browser to research, a second app to track your characters, a third to check continuity, a spreadsheet to count words, a design program to typeset. Inkhaven's premise is the opposite: **the whole of writing a book lives in one terminal window** — the prose, the world it is set in, the facts it must not contradict, the assistant you steer, and the finished PDF — and you never leave the keyboard to reach any of it.

This manual is the complete tour of that window. It assumes **no prior knowledge** — not of Inkhaven, not of the terminal beyond opening one — and it runs in reading order from installing the binary to publishing a book. Read it front to back once for the shape of the thing; return to any chapter when you reach the workflow it covers.

## What Inkhaven is

Inkhaven is a single command-line program for writing books and long-form documentation. It pairs a full-screen editor for Typst — a modern typesetting language — with a local semantic index, an AI writing assistant you fully control, versioned snapshots, a backup pipeline, and an unusually deep set of **reading** intelligences that watch a manuscript for the things a long book gets wrong. Your prose lives as plain `.typ` files on disk; a local database tracks the hierarchy and the search index, but the words are always text you can read, diff, and version-control yourself.

### TERM The manuscript is plain files

A paragraph is a `.typ` file on disk, inside a tree of folders for books, chapters, and subchapters. Inkhaven manages that tree and indexes it, but it never locks your words inside a proprietary format. You can open any paragraph in any editor, at any time, and Inkhaven will notice.

## How this manual is organised

The book is in nine parts and a set of reference appendices:

### ● ● ● *The shape of the manual*

|      |                        |                                                                             |
|------|------------------------|-----------------------------------------------------------------------------|
| I    | Getting Started .....  | install, first project, the panes                                           |
| II   | Writing .....          | the editor, the tree, snapshots, search                                     |
| III  | The AI Assistant ..... | scopes, chat-with-your-book, prompts, cost                                  |
| IV   | The World & Facts .... | places, characters, world, graph, timeline                                  |
| V    | The Intelligences .... | continuity, knowledge, read-through, voices,<br>revision, the inner readers |
| VI   | Language & Verse ..... | conlang, poetry, research, scholarship                                      |
| VII  | Producing the Book ... | PDF, EPUB, web, technical docs                                              |
| VIII | Scripting .....        | the embedded Bund language                                                  |
| IX   | Keeping It Healthy ... | backup, doctor, configuration                                               |
| A–D  | Reference .....        | keys, commands, config, the feature index                                   |

Parts I through III are the ground floor — everything you need to write and get help. Parts IV and V are Inkhaven’s distinctive depth: the machinery for a book with a world, a cast, and secrets that must stay kept. Parts VI through IX are the specialist tracks and the plumbing.

## This manual, and the companions

Inkhaven ships a small library of **companion books**, each a deep dive on one domain — **Know Your Book** on the reading intelligences, **Building the World** on worldbuilding, **Poetry with Inkhaven**, **Grounding Your Book in Fact** on research, and others. This manual is deliberately **breadth-first**: it tells you what each feature is, how to reach it, and how to run it, then points you to the companion when you want the full treatment.



## FICTION

If you write **fiction**, the heart of the book is Parts IV and V — a world and cast Inkhaven can hold in mind, and the readers that watch continuity, knowledge, voice, and pacing across a whole manuscript.

## NON-FICTION

If you write **non-fiction**, your centre of gravity is the facts and the graph (Part IV), research and sources (Part VI), and the technical-doc and index tools (Part VII) — the machinery of claims that must be right.

## A word on the pictures

Inkhaven is a terminal application, so this manual illustrates it with **terminal frames** rather than photographs — a faithful rendering of what you will see on screen, set in a monospace type. A frame is truer than a screenshot for a text interface, and it keeps the book self-contained. When you see one, read it as “this is what the screen shows here.”

## WHAT YOU LEARNED

- Inkhaven is one terminal program for the **whole** of writing a book — prose, world, facts, an AI you steer, and the finished document.
- Your manuscript is always **plain .typ files** on disk; nothing is locked in a proprietary format.
- This manual is the **breadth-first tour**, install to publish; the companion books hold the deep dives.

PART I

---

# Getting Started

## CHAPTER 1

## 1

# Installing Inkhaven

---

Inkhaven is a single program. When it is installed you have one command, `inkhaven`, and everything else in this book happens inside it or through it — the editor, the world, the facts, the assistant, the finished PDF. There is no server to run, no account to create, no companion app to keep open. This chapter gets that one command onto your machine, explains what happens the first time you run it, and tells you how to check that all is well before you write a word.

It assumes you can open a terminal and type into it, and nothing more. If you have never compiled a Rust program, that is fine — you will not have to understand Rust, only to install its toolchain and let it work for a few minutes. Read this chapter

once, top to bottom; you will not need it again until you set Inkhaven up on a second machine.

## What you are installing

Inkhaven ships as one self-contained binary. It carries its own database engine, its own semantic-search index, and — after the first run — its own local embedding model. It does **not** depend on any external program being present on your system: no Node, no Python, no Docker, no database server, no paid AI account. The one genuinely optional companion is the Typst compiler, and only if you want to render your manuscript to PDF from inside the editor; everything else, including the plain-text and single-`.typ` exports, works with Inkhaven alone.

### TERM Binary

A single executable file — here, one called `inkhaven`. “Installing” it means getting that file onto your machine and onto your `PATH` (the list of directories your shell searches for commands), so that typing `inkhaven` anywhere runs it. Everything Inkhaven does lives in that one file plus the per-user model cache it downloads once.

There are two ways to get the binary: let Cargo **compile** it for you (from crates.io or from source), or download a **prebuilt** one. Compiling is the path this chapter treats as canonical, because it always produces a binary matched to your exact machine and needs nothing but a Rust toolchain. The prebuilt paths are faster and are covered too.

## Prerequisites

You need three things: a Rust toolchain, a terminal, and a little disk space. Only the first requires any action.

### A Rust toolchain

Inkhaven is written in Rust, edition 2024, and requires a compiler of at least **version 1.85**. That minimum is not arbitrary — the 2024 edition and several language features Inkhaven relies on landed in that release. Anything newer is fine; Rust is strongly backward-compatible.

The official way to install Rust is

#### TERM **rustup**

the standard Rust toolchain installer — a small script that places **rustc** (the compiler) and **cargo** (Rust’s build tool and package manager) under your home directory and adds them to your shell’s **PATH**. It needs no administrator rights and installs entirely inside your own account.

. On macOS and Linux one line does it:

#### ● ● ● *Installing the Rust toolchain via rustup*

```
$ curl --proto '=https' --tlsv1.2 -sSf \  
https://sh.rustup.rs | sh
```

Accept the default when it prompts (option 1 is correct). When it finishes, either open a fresh terminal or reload your shell so that **cargo** is on the **PATH**:

#### ● ● ● *Making cargo available in the current shell*

```
$ source "$HOME/.cargo/env"
```

Then confirm both tools are present and new enough:

#### ● ● ● *Verifying the toolchain*

```
$ rustc --version  
rustc 1.85.0 (a028ae42f 2026-01-09)  
$ cargo --version  
cargo 1.85.0 (d73d2caf9 2026-01-06)
```

If either prints “command not found”, your shell has not yet picked up **\$HOME/.cargo/bin** — close and reopen the terminal, or add that directory to your **PATH** by hand.

**NOTE**

The Rust toolchain is a one-time, whole-system install of roughly 600 MB. It is not part of Inkhaven and is reused by every Rust program you ever build. You install it once and forget it exists.

**A terminal**

Inkhaven is a terminal application. It runs in whatever terminal emulator you already have — the built-in Terminal on macOS, or any of the common Linux ones (GNOME Terminal, Konsole, Alacritty, kitty, WezTerm, xterm). It also runs perfectly over SSH, inside `tmux`, and on a tiling window manager, because it is text the whole way down and never reaches for a browser or a graphical window. Any terminal from the last decade with 256-colour support will do; the richer ones (kitty, iTerm2, WezTerm) additionally give you inline image previews, but those are a luxury, not a requirement.

**Disk space**

A fresh, complete install occupies a few hundred megabytes, most of it the one-time Rust toolchain. The parts that are actually Inkhaven are modest:

● ● ● *Disk footprint of a fresh install*

|                     |               |                                 |
|---------------------|---------------|---------------------------------|
| Rust toolchain      | ..... ~600 MB | one-time, shared by all Rust    |
| Inkhaven binary     | ..... ~90 MB  | the compiled `inkhaven`         |
| Embedding model     | ..... ~120 MB | downloaded once, per-user cache |
| A new empty project | . ~5 MB       | database + index scaffolding    |

The embedding model is a single shared download, not a per-project cost: a hundred projects reuse the same cached model. A project's own footprint grows only with your prose and its snapshots.

**Supported platforms**

Inkhaven is released and supported on two platforms:

### Supported release targets

|       |                          |                 |
|-------|--------------------------|-----------------|
| Linux | x86_64-unknown-linux-gnu |                 |
| macOS | aarch64-apple-darwin     | (Apple Silicon) |

These are the targets the release pipeline builds, tests, and ships prebuilt binaries for. Compiling from source with `cargo` works on any host Rust and the dependencies support — an Intel Mac, for instance, compiles cleanly even though there is no prebuilt asset for it — but the two above are the tested, first-class combination.

### Why not Windows

Windows is **deferred**: there is no released Windows binary, and you should not expect one yet. The reason is specific and worth stating plainly, because it is not a matter of effort or interest.

Inkhaven computes its embeddings locally through `fastembed`, which in turn runs models through the ONNX Runtime by way of the `ort` crate. `ort` ships prebuilt runtime binaries for the Windows **MSVC** toolchain but not for the Windows **GNU** toolchain, and Inkhaven's build has historically targeted the GNU toolchain (alongside the DuckDB and font-handling C++ that MSYS2 builds cleanly). The build therefore fails at the ONNX Runtime bindings — upstream, in a dependency, not in Inkhaven's own code.

#### WHY IT IS DEFERRED, NOT ABANDONED

The fix is real and known; it is simply not on the critical path. Any one of three routes unblocks Windows: `ort` shipping a Windows-GNU prebuilt (the CI job is kept ready and would simply go green), adopting the Windows-MSVC target so that `ort`'s existing prebuilds apply, or bundling a build-time-verified runtime library in the release archive. The one route deliberately **rejected** is fetching and loading a runtime library at startup — that would run unsandboxed native code pulled over the network, a far larger attack surface than the model data Inkhaven does download, and against the project's no-external-binary principle. Until one of the sound routes is taken, Windows users should build under WSL (a Linux environment) and treat it as the Linux target.

## Installing with Cargo

If you have the Rust toolchain, the shortest path to a working `inkhaven` is to let Cargo compile the published crate.

### **cargo install inkhaven — compile from crates.io**

Inkhaven is published on crates.io; every release tag pushes a new version. This one command downloads the crate and its dependencies and compiles the lot:

● ● ● *Installing the published crate*

```
$ cargo install inkhaven
```

Be patient the first time. The build pulls roughly a hundred crates and then compiles the heavy ones — DuckDB, `fastembed`, and the ONNX Runtime bindings are large C and C++ projects — so a first build takes on the order of **ten minutes** on a modern laptop. This is a one-off: Cargo caches the compiled dependencies, and you will not pay that cost again.

When it finishes, the binary lands in Cargo’s binary directory, which is already on your `PATH` if rustup set your shell up:

● ● ● *Where cargo install puts the binary*

```
~/ .cargo/bin/inkhaven
```

That is the whole install. Skip ahead to “The first run” once `inkhaven --version` prints a version.

### **cargo binstall — the prebuilt fast path**

If you would rather not wait for a compile and you have



**TERM cargo-binstall**

a small Cargo extension that downloads a **prebuilt** binary from a project's GitHub releases instead of compiling it. Install it once with `cargo install cargo-binstall`; thereafter `cargo binstall <crate>` fetches the right prebuilt asset for your platform.

installed, one command fetches the prebuilt binary and drops it into `~/.cargo/bin`:

● ● ● *Installing a prebuilt binary*

```
$ cargo binstall inkhaven
```

`cargo-binstall` reads the packaging metadata from Inkhaven's `Cargo.toml`, picks the asset matching your platform off the GitHub releases, and installs it without compiling anything. This is the quickest route on the two supported platforms.

**GitHub Releases — a direct download**

You can also bypass Cargo entirely. Each release publishes a tarball per platform on the project's Releases page. Download the one for your platform, unpack it, and place the `inkhaven` binary anywhere on your `PATH`:

● ● ● *Installing from a release tarball*

```
$ tar xzf inkhaven-<version>-<platform>.tar.gz
$ sudo install inkhaven /usr/local/bin/inkhaven
```

The releases live at [github.com/vulgov/blackInkhaven/releases](https://github.com/vulgov/blackInkhaven/releases) the asset names encode the version and the target triple so you can pick the right one at a glance.

**cargo install --git — a specific tag or branch**

When you want a particular tagged version, a pre-release branch, or your own fork, install straight from the git repository and name the tag:

### ● ● ● Installing a specific tagged version

```
$ cargo install \  
  --git https://github.com/vulogov/blackInkhaven \  
  --tag v3.0.0
```

This compiles exactly the named revision. Drop `--tag` to build the default branch's current tip. Check the Releases page for the tag you actually want; the examples here name `v3.0.0`, the stable edition this manual documents.

## Building from source

Compiling from a clone is the path to choose if you want to read or modify the code, track a development branch, or simply keep the whole thing under your own eye. It needs only the Rust toolchain and `git`.

First, clone the repository and enter it:

### ● ● ● Cloning the source

```
$ git clone \  
  https://github.com/vulogov/blackInkhaven.git  
$ cd blackInkhaven
```

If you have no `git`, the GitHub page offers the same source as a zip; unpack it and `cd` into the folder. Then build an optimised binary:

### ● ● ● Building the release binary

```
$ cargo build --release
```

The first build downloads about a hundred crates and compiles them; expect several minutes — commonly three to eight on a modern laptop, longer on older hardware because of the same heavy C/C++ dependencies noted above. Later builds are incremental and reuse the cache, so they finish in seconds. When it is done, the binary is here:

### ● ● ● *Where the source build puts the binary*

```
./target/release/inkhaven
```

You can run it straight from that path. To type just `inkhaven` from anywhere, copy it onto your `PATH`:

### ● ● ● *Putting inkhaven on your PATH*

```
# system-wide (asks for your password)
$ sudo install target/release/inkhaven \
  /usr/local/bin/inkhaven

# or, no sudo, if ~/.local/bin is on PATH
$ cp target/release/inkhaven ~/.local/bin/
```

## WHY THE `--RELEASE` FLAG MATTERS

Without `--release`, Cargo builds an unoptimised debug binary. It runs, but it is several times slower and a few hundred megabytes larger — embedding and search in particular feel sluggish. Always build `--release` for real writing; reserve the debug build for hacking on Inkhaven’s own code.

## Optional: the Typst compiler

Inkhaven writes and manages Typst, but it does not bundle the Typst compiler. If you want `inkhaven export pdf` (and the in-editor “build to PDF” chords) to actually produce a PDF, install Typst separately — it, too, is a single static binary on every platform, available from its own project page. Everything else, including the `export typst` path that emits one combined `.typ` file, works without it.

## The first run

The install is inert until the first time Inkhaven opens a project. That first open does two one-time things worth understanding, so that a slightly longer startup does not alarm you.

## The embedding model download

Inkhaven’s semantic search — finding “the moment the lighthouse fails” even in a paragraph that never says **lighthouse** — is powered by a local embedding model. That model is not shipped inside the binary; it is downloaded the first time you initialise or open a project. The default is

### TERM **MultilingualE5Small**

the default embedding model — a compact, multilingual model (English, Russian, German, French, Spanish, and more) about 120 MB in size. It is chosen as a sensible balance of quality against footprint; the configuration lets you switch to the Base or Large variant, at the cost of a larger download and slower embedding.

, roughly 120 MB, fetched once into a per-user cache and reused by every project thereafter.

The cache location depends on your platform:

● ● ● *Where the embedding model is cached*

```
macOS  ~/Library/Caches/
        dev.inkhaven.inkhaven/embeddings/
Linux  $XDG_CACHE_HOME/inkhaven/embeddings/
        (defaults to ~/.cache/inkhaven/)
```

On a normal connection the download completes in well under two minutes, behind a splash screen that shows the elapsed time. It happens once per model: switching the configured model later downloads the new one on the next open and leaves the old one on disk until you remove the cache directory yourself. If you ever change `embeddings.model` in a project’s configuration, Inkhaven re-embeds every paragraph against the new model the next time it opens.

## Model init and the project lock

After the model is present, Inkhaven initialises it into memory and takes an **advisory** lock on the project so that two sessions do not write the same database at once. The lock is a small file, `.inkhaven.lock`, in the project root, held for as long as the session runs and released automatically by the operating system when

the process exits — even on a crash or a `kill -9`, so there is never a stale lock to clean up by hand.

#### THE LOCK INFORMS, IT NEVER BLOCKS

In keeping with Inkhaven’s permissive character, opening a project that another session already holds does not fail. The launcher tells you who holds it — a process id, host, and time — and lets you open it anyway. Only genuine data-safety is at stake, and the choice stays yours. This is the same principle you will meet throughout Inkhaven: it warns, it does not forbid.

Once the model is cached and warm, subsequent launches are quick — there is no download and the lock is instantaneous.

## Verifying the install

Two commands confirm a healthy install without touching any project. First, the version:

#### ● ● ● *Confirming the binary runs*

```
$ inkhaven --version
inkhaven 3.0.0
```

A version number means the binary is on your `PATH` and runs. If instead you see “command not found”, the binary is not on your `PATH` — invoke it by its full path (`./target/release/inkhaven` from a source build, or `~/.cargo/bin/inkhaven` from a Cargo install), or copy it to a directory that is on your `PATH`.

Second, the help surface, which lists every top-level subcommand and the global flags:

● ● ● *Listing the command surface*

```

$ inkhaven --help
TUI literary work editor for Typst books

Usage: inkhaven [OPTIONS] [COMMAND]

Commands:
  init      Initialize a new project
  add       Add a node to the hierarchy
  list      Print the hierarchy as a tree
  search    Run a semantic search
  export    Export the book(s)
  backup    Zip the whole project
  ...

Options:
  -p, --project <PROJECT>  Path to a project root
  -h, --help                Print help
  -V, --version             Print version

```

Any subcommand takes `--help` of its own — `inkhaven export --help`, `inkhaven init --help` — which prints that command’s full set of flags. This is the authoritative reference for the exact surface of the version you have installed; when this book and the binary ever disagree, the binary’s `--help` is right.

## Troubleshooting the install

Most installs are uneventful. The handful of things that can go wrong have short answers.

### command not found: inkhaven

Your shell cannot find the binary on its `PATH`. From a source build it lives at `./target/release/inkhaven`; from `cargo install` at `~/.cargo/bin/inkhaven`. Either call it by that full path, copy it into a directory that is on your `PATH` (`/usr/local/bin` or `~/.local/bin`), or alias it. Confirm your `PATH` contains `$HOME/.cargo/bin` if you expected Cargo’s install to be found automatically.

## The first-run download is slow or stalls

The model download is the one part of startup that reaches the network. If the splash screen's elapsed counter climbs past a couple of minutes on a connection you know is working, something upstream is slow rather than broken. You can:

### ● ● ● *Recovering from a stalled first-run download*

- Check the machine actually has connectivity.
- Watch the splash's elapsed timer – under ~2 min is normal for MultilingualE5Small.
- Press Ctrl+Q to abort startup, then retry later.

Aborting is safe: nothing is half-written into the cache in a way that a later retry cannot replace.

## Offline, air-gapped, or behind a proxy

Inkhaven never needs the network **except** for that one first-run model download (and, separately, for any external AI provider you choose to configure – which is entirely optional). Two consequences follow.

On a **proxy**, make sure the standard `HTTPS_PROXY` environment variable is set before the first run so the download can reach out. On a genuinely **air-gapped** machine, you cannot download at all – so seed the cache from a networked machine instead: install and run Inkhaven once on a connected computer, then copy the populated cache directory (the platform paths listed under “The first run”) onto the air-gapped machine before its first launch. With the model already present, Inkhaven starts fully offline and stays offline for the whole of writing, worldbuilding, search, and snapshots – every subsystem but the external providers is on-device.

## A configuration error on open

If opening a project reports a missing configuration field, you are opening a project whose `inkhaven.hjson` was written by an older release than the binary you just installed. Add the missing field by hand – the error names it, and Appendix C lists every key with its default. Do **not** reach for `inkhaven init --force` to “refresh” the config: `--force` deletes and recreates the **entire project directory** – every book, database, and word of prose – not just `inkhaven.hjson`. New projects created by the version you installed never hit this.

## The CLI and the TUI

One binary, two shapes. Understanding the split now will save confusion in every later chapter, because this manual moves freely between them.

Run `inkhaven` with **no subcommand** — inside a project directory, or with a `--project` path — and it launches the full-screen editor: the

### TERM TUI

the Text User Interface — Inkhaven’s full-screen, keyboard-driven editor, with its tree, editor, search, AI, and output panes. It takes over the terminal, and you leave it with `Ctrl+Q`. This is where you actually write.

, the interactive program that fills the terminal and is the subject of most of this book:

#### ● ● ● Opening the editor

```
$ cd ~/Books/my-novel
$ inkhaven                # opens the TUI here
# ...or from anywhere:
$ inkhaven --project ~/Books/my-novel
```

Run `inkhaven` **with** a subcommand and it does one job and exits, printing to the terminal without ever entering full-screen mode. These headless commands are what you script, pipe, and run in continuous integration:

#### ● ● ● Headless, one-shot commands

```
$ inkhaven init ~/Books/my-novel
$ inkhaven --project ~/Books/my-novel list
$ inkhaven --project ~/Books/my-novel \
  search "the lighthouse fails"
$ inkhaven --project ~/Books/my-novel \
  export pdf --status ready
```



The global `--project` flag (also `-p`, or the long alias `--project-directory`) tells any command which project to act on, and defaults to the current directory — so `cd` into a project first and you can drop it entirely. The top-level command families you will meet across the book group roughly like this:

|                          |                                   |
|--------------------------|-----------------------------------|
| <b>init · add · mv</b>   | Create and shape the hierarchy    |
| <b>list · outline</b>    | Inspect the tree from the shell   |
| <b>search · book-rag</b> | Semantic search and retrieval     |
| <b>export · index</b>    | Produce PDF, Typst, EPUB, indexes |
| <b>backup · restore</b>  | Project-level disaster recovery   |
| <b>reindex · doctor</b>  | Reconcile store with disk; health |
| <b>ai</b>                | One-shot inference from the shell |

A few subcommands — the standalone configuration editor, the research and worldbuilder and linguistic workspaces — are themselves full-screen TUIs rather than one-shot commands; the book flags each where it introduces it. Everything else you type after `inkhaven` runs, prints, and returns you to your prompt.

#### FICTION

If you write **fiction**, your usual entry is the bare `inkhaven` — you live in the editor, with the tree, the world, and the readers a keystroke away. The headless commands matter mainly at the edges: `init` to begin, `backup` to be safe, `export` to ship.

#### NON-FICTION

If you write **non-fiction** or documentation, you will lean on the headless side far more — `search`, `export`, `index`, and the check commands slot naturally into build scripts and continuous integration, where an editor cannot go.

With `inkhaven` installed, verified, and its two shapes clear, you are ready to create a project — which is where the next chapter begins.

## WHAT YOU LEARNED

- Inkhaven is **one self-contained binary**; the only prerequisite you must install yourself is a **Rust toolchain of version 1.85 or newer** via rustup.
- Install it by compiling — `cargo install inkhaven` (a 10-minute first build) or `cargo build --release` from a clone — or grab a prebuilt binary with `cargo binstall` or from GitHub Releases.
- The supported targets are **Linux x86\_64** and **macOS Apple Silicon**; Windows is deferred because the ONNX Runtime bindings `fastembed` needs have no Windows-GNU prebuilt.
- The **first run** downloads a 120 MB embedding model once into a per-user cache, then takes an advisory project lock that **informs but never blocks**.
- Verify with `inkhaven --version` and `inkhaven --help`; running `inkhaven` bare opens the **TUI**, while `inkhaven <subcommand>` runs **headless** and exits.

## CHAPTER 2

## 2

## Your First Project

---

A book, to Inkhaven, is a **directory**. Not a file with an opaque extension, not a row in some cloud database you cannot see — a plain folder on your own disk, with your prose in readable text files and a small local database beside them that remembers the shape of the thing. This chapter creates one, opens it up, and names every part, so that when a later chapter says “the Facts book” or “the `.inkhaven/` sidecars” or “a structural leaf” you already know exactly where on disk it lives and what it is for.

We will run one command, read what it laid down, meet the twenty **system books** Inkhaven seeds into every project, and then make a book, a chapter, and a

paragraph of your own. By the end you will have a project you can open, close, back up, and put under version control — and a mental map of every file in it.

## The one command that starts everything

Everything begins with `inkhaven init`. It takes a single positional argument — the directory the project should live in — and creates it if it does not yet exist. The convention is one project per book (or per tightly-related set), kept somewhere like `~/Books/`.

### ● ● ● *Creating a project*

```
$ inkhaven init ~/Books/my-first-project
Initialized inkhaven project at /home/you/Books/my-first-project
config:    .../my-first-project/inkhaven.hjson
prompts:   .../my-first-project/prompts.hjson
store db:  .../my-first-project/metadata.db
vecstore:  .../my-first-project/vectors
books:     .../my-first-project/books
```

The directory need not exist beforehand — Inkhaven makes it. If it **does** exist, Inkhaven refuses to touch it silently, because `init` starts from a clean database and re-initialising means **wiping** what is there. It asks first:

### ● ● ● *The overwrite guard*

```
$ inkhaven init ~/Books/my-first-project
Directory `/home/you/Books/my-first-project` already exists.
Remove it and re-initialise? [y/N]
```

Only `y` or `yes` proceeds; a bare Enter, an `n`, or end-of-input all abort with your files left untouched. There is one further safety interlock: Inkhaven will **refuse** to wipe a directory that your current shell is sitting inside, because deleting the ground beneath your own feet leaves the terminal in a broken state. `cd` out first.

## The two flags

`init` has exactly two options beyond the path.

- force** Skip the [y/N] prompt and overwrite an existing directory unattended. For scripts and fresh-checkout automation — never needed interactively.
- template** Scaffold a starting tree instead of an empty one. Defaults to empty.
- <name>**

The templates are pre-built manuscript skeletons applied **after** the standard `init`, so a template never changes what `init` itself lays down — it only adds books and chapters on top. Run `inkhaven template list` to see them all; the most common are:

```
● ● ● inkhaven init --template <name>
```

|                |                                                     |
|----------------|-----------------------------------------------------|
| empty          | the default — system books only, no manuscript      |
| novel          | three-act manuscript + Characters stubs             |
| nonfiction     | intro / parts / conclusion + Research method        |
| rpg-sourcebook | Setting/Rules/Adventures + Places/Artefacts/Threads |
| technical      | Overview / Reference / Tutorials / Index            |
| nanowrimo      | like novel, with a 50 000-word goal + pacing        |

If you are unsure, take `empty`. You can build the same tree by hand in the TUI in a minute, and this chapter shows you how. A template is a convenience, never a commitment.

#### FIRST RUN DOWNLOADS A MODEL

The very first `init` on a machine fetches a multilingual embedding model (about 120 MB) into a **per-user** cache — not into the project. On macOS that is `~/Library/Caches/dev.inkhaven.inkhaven/embeddings/`; on Linux `~/.cache/inkhaven/embeddings/`; on Windows under `%LOCALAPPDATA%`. Every later `init` reuses it, so only the first project pays the download. This is the model that powers semantic search and the reading intelligences — all of it runs on your own machine, offline.

## What init writes to disk

`cd` into the new project and list it. Ignoring the model cache (which lives elsewhere), a fresh project is small and entirely legible:

● ● ● *A fresh project, top level*

```
my-first-project/
├─ inkhaven.hjson      the configuration file
├─ prompts.hjson      your prompt library
├─ metadata.db         hierarchy + node metadata (DuckDB)
├─ blobs.db           paragraph bodies, snapshots, blobs
├─ frequency.db        the full-text search index
├─ vectors/           the HNSW semantic-search index
└─ books/             your prose – one folder per book
```

Two of these you will edit by hand; the rest Inkhaven owns. The two yours are `inkhaven.hjson` (theme, language, AI provider, autosave cadence – Chapter and appendix on configuration cover every field) and `prompts.hjson` (your library of reusable AI prompts). The four database entries – `metadata.db`, `blobs.db`, `frequency.db`, and the `vectors/` directory – are the local store, created and maintained by the embedded database engine. You never open them with a text editor.

### TERM The store is three databases and an index

Inkhaven’s metadata lives in DuckDB files beside your prose. `metadata.db` holds the **hierarchy** – every book, chapter, and paragraph as a node with a stable UUID, its title, order, and file path. `blobs.db` holds paragraph bodies and versioned snapshots. `frequency.db` is the full-text index behind keyword search. `vectors/` is the HNSW index that makes **semantic** search and the reading intelligences possible. Your words themselves, though, are never trapped in there – they are the `.typ` files under `books/`, and the database merely points at them.

This split is the heart of Inkhaven’s data model, and worth dwelling on for a moment. The **prose is the filesystem**. The **database is an index over the filesystem**. If the two ever drift apart – you edited a paragraph in another editor,

or a crash left them out of step — `inkhaven reindex` re-reads the `.typ` tree and rebuilds the database from it. The files are the source of truth; the database is derived, and therefore always rebuildable. That is why your manuscript is safe even if you never trust the database at all: it is ordinary text, in ordinary folders, that you can read, `grep`, diff, and commit to Git without Inkhaven’s help.

### Where session, backups, and caches live

Beyond the store, Inkhaven keeps a handful of **dotfiles** and one **dot-directory** in the project root. None of these exist right after `init` — they appear as you use the project — but knowing them now saves confusion later.

● ● ● *The runtime sidecars (appear with use)*

```
my-first-project/
├─ .session.json          cursor + which paragraph was open
├─ .inkhaven-backup.json  timestamp of the last backup
├─ .inkhaven.log          the runtime log (append-only)
└─ .inkhaven/            advisory sidecars + local caches
    ├─ drift.json         continuity-drift findings
    ├─ facts_scan.json    fact-check results
    ├─ tensions.json      pacing / tension curve
    ├─ continuity.json     the continuity bible cache
    ├─ submissions.json   submission-package tracker
    ├─ ai_usage.json       rolling AI cost ledger
    ├─ prose.duckdb        per-chapter prose metrics
    ├─ voices/             character-voice profiles
    └─ digest-<slug>.json  per-book cached digests
```

The three top-level dotfiles are small and transient. `.session.json` remembers which paragraph you had open and where the cursor sat, so re-opening the project puts you back exactly where you left off. `.inkhaven-backup.json` is a single timestamp the auto-backup logic checks against your configured `backup.max_age`. `.inkhaven.log` is the append-only log — the first place to look when something misbehaves.

The `.inkhaven/` directory is a growing collection of **advisory sidecars**: the cached output of scans and the reading intelligences. Every file in it is disposable. Each records what some analysis last found — drift, fact-checks, tension curves, voice profiles — so that opening a view is instant instead of re-computing from scratch. Delete any of them and Inkhaven simply recomputes on next use. Crucially, **nothing here is your prose**: these are derived opinions about your prose, kept to one side. Chapters throughout Parts IV and V describe each sidecar where its

feature is covered; for now, treat `.inkhaven/` as “the scratchpad the intelligences share.”

#### FOR VERSION CONTROL

Commit `inkhaven.hjson`, `prompts.hjson`, and `books/` — the prose and its configuration. You may commit the `.db` files too (they are your index and snapshots), but they are large and rebuildable, so many authors add `*.db`, `vectors/`, `.inkhaven/`, `.session.json`, and `.inkhaven.log` to `.gitignore` and let `inkhaven` `reindex` rebuild the store on a fresh clone.

## The anatomy of the manuscript tree

Open `books/` and you find the real structure. Inkhaven’s hierarchy — **Book** → **Chapter** → **Subchapter** → **Paragraph** — is mirrored one-to-one onto the filesystem: branches are folders, leaves are files, and every entry carries a zero-padded numeric prefix so a plain directory listing sorts in reading order.

● ● ● *One book on disk*

```
books/
├─ my-first-book/
│   ├─ 01-chapter-one/
│   │   ├─ 01-the-storm.typ
│   │   ├─ 02-landfall.typ
│   │   └─ 02-a-quiet-town/      ← a subchapter
│   │       ├─ 01-the-inn.typ
│   │       └─ 02-the-innkeeper.typ
│   └─ 02-chapter-two/
│       └─ 01-morning.typ
```

Read that tree carefully, because its rules are the whole model:

- A **book** is a folder named by its bare slug (`my-first-book`) — no number, because books are ordered among themselves in the database, and their folders sit at the top level of `books/`.
- A **chapter** and a **subchapter** are numbered folders (`01-chapter-one`, `02-a-quiet-town`). A subchapter is simply a chapter-level folder that lives inside a chapter — the same kind of container, one level down.



- A **paragraph** is a numbered **.typ file** (01-the-storm.typ). This is the leaf: the actual unit of prose you edit. Despite the name, a “paragraph” can hold as much text as you like — think of it as a titled **section** of the manuscript, the smallest thing Inkhaven versions, embeds, and re-orders.

The numeric prefixes are Inkhaven’s bookkeeping, not yours to hand-edit. When you reorder rows in the tree, Inkhaven rennumbers the files on disk to match; the slug (the human-readable part after the number) comes from the title, and a paragraph with an empty title gets its slug — and title — from the first sentence you type on first save.

#### TERM The manuscript is a Typst project in disguise

Each paragraph file is a fragment of Typst, the typesetting language. A new paragraph opens with a single = heading line — Typst’s level-one heading marker — so the paragraph renders as its own section in the final book. When you export, Inkhaven walks the tree in order, assembles the fragments into one Typst document, and hands it to the typesetter. Nothing about the on-disk format is proprietary: it is Typst, all the way down.

## The node model at a glance

Every row in the tree is a **node**, and every node has a **kind**. Four kinds are the structural spine you have already met — Book, Chapter, Subchapter, Paragraph. Alongside the prose paragraph, though, Inkhaven allows several **non-prose leaves**: siblings in the same tree that carry something other than Typst prose. Each has its own on-disk extension and its own chapter later in this manual; here they are only named, so you recognise them in a tree.

#### ● ● ● Leaf kinds at a glance

| KIND                                                                                      | ON DISK       | WHAT IT HOLDS                          |
|-------------------------------------------------------------------------------------------|---------------|----------------------------------------|
| paragraph                                                                                 | NN-slug.typ   | prose (Typst) — the default leaf       |
| image                                                                                     | NN-slug.png   | a picture: png / jpg / webp / svg      |
| hjson {                                                                                   | NN-slug.hjson | structured data (a place, a source...) |
| jinja  | NN-slug.jinja | a template rendered to Typst on export |
| bund λ                                                                                    | NN-slug.bund  | an embedded Bund script                |

A few notes to fix the picture:

- An **image** is a first-class node, not a paragraph. Drop a `.png` / `.jpg` / `.webp` / `.svg` into the tree and Inkhaven ships the bytes into the assembled book with the right image call; a caption and alt-text ride along in the metadata.
- An **hjson** leaf is a paragraph whose content type is data rather than prose. This is how the world books store structure — a place, a character, a bibliography entry — as an HJSON record instead of paragraphs of text.
- A **jinja** leaf is a template: it holds Jinja markup that Inkhaven renders down to Typst at assembly time, for repeated or generated structure.
- A **bund** leaf is a script in Inkhaven’s embedded Bund language, evaluated into the scripting engine when the project opens — the home of custom hooks and rules. Its default home is the Scripts system book, but it can live anywhere.

There is one more flavour that is **not** a separate file type but a **subtype** of an ordinary Typst paragraph — the **structural paragraph**. In the editor, the structural-subtype picker turns a paragraph into a piece of recognised scaffolding — code block, admonition, mathematics, procedure, or table — each with its own glyph in the tree (`▢` `△` `∫` `≡` `▣`). It is still a `.typ` file; the subtype only tells Inkhaven how to treat and render it. A single leaf can be cycled through its types — `typst` → `hjson` → `jinja` → `bund` — as your needs change. Every one of these gets a full treatment in a later chapter; for now, the point is simply that **the tree holds more than prose**, and each non-prose leaf is still a plain, readable file.

## The system books

Here is the part that surprises people opening a fresh project: it is not empty. Before you have written a word, `inkhaven list` shows a stack of books already there. These are the **system books** — reference and machinery books Inkhaven seeds into **every** project, on every open, whether you use them or not. They carry stable internal tags, they are marked **protected** (you cannot delete or rename them), and anything you create lands **above** them, at the top of the tree.

## HOW MANY, REALLY

Older documentation counted “six” and then “nine” system books; the project has grown since. The authoritative list is the `SYSTEM_BOOKS` table in the source, and as of this edition it seeds **twenty**. If a future release adds one, it appears automatically the next time you open any existing project — the seeder is idempotent and back-fills.

They divide cleanly into three groups: reference books you write into, world books that hold your fiction’s ground truth, and machinery books Inkhaven and its tools own. First, the reference and prose books:

● ● ● *System books · reference & prose*

|          |                                                                |
|----------|----------------------------------------------------------------|
| Notes    | free-form scratch prose, kept out of the manuscript            |
| Research | research notes and gathered source material                    |
| Sources  | bibliography entries (HJSON) → compiled to a <code>.bib</code> |
| Glossary | canonical terms + banned synonyms (terminology)                |
| Snippets | reusable Typst blocks, <code>#include'd</code> elsewhere       |
| Prompts  | your AI prompt library (seeded with examples)                  |
| Help     | Inkhaven's own manual — F1 answers by RAG here                 |

Then the **world** books — the ground truth a work of fiction must stay consistent with. These are where Parts IV and V of this manual do their work:

● ● ● *System books · the world*

|            |                                                                     |
|------------|---------------------------------------------------------------------|
| Facts      | the world's invariants — the fact-check reference                   |
| Places     | your gazetteer — locations, named and highlighted                   |
| Characters | the cast — names light up in the editor                             |
| Artefacts  | objects of significance — items, relics, devices                    |
| World      | simulation output (the <code>realworld</code> compiler writes here) |
| Threads    | narrative plot threads and their arc shapes                         |
| Planning   | story structure — beats and acts of a framework                     |
| Mythology  | declared symbols, motifs, and archetypes                            |
| Intent     | your declared authorial choices (silences findings)                 |

And finally the **machinery** books — homes for the tooling that surrounds the prose:

● ● ● *System books · machinery*

|             |                                                      |
|-------------|------------------------------------------------------|
| Typst       | reusable Typst templates and skeletons               |
| Scripts     | .bund scripts auto-loaded into the engine at open    |
| Language    | invented-language books (one child book per conlang) |
| Submissions | the submission package – query letter, synopsis...   |

A handful of these deserve a sentence of **why** now, because you will meet them again and again:

- **Notes** and **Research** are the free-form catch-alls — prose that supports the book without being in it. The AI can be scoped to read them.
- **Facts** is the reference the continuity and fact-checking tools ground against: the climate, the geography, the chronology your chapters must not contradict. **Grounding Your Book in Fact** is the companion book for it.
- **Places**, **Characters**, and **Artefacts** are **lexicon** books — Inkhaven highlights their names where they appear in your prose, and you can ask the AI about any of them by name.
- **Prompts** ships pre-seeded with `.example` prompts for the built-in AI actions; rename one to drop the `.example` suffix and it overrides the default. **Help** is Inkhaven’s own documentation, which is why pressing **F1** can answer questions about the program itself.
- **Intent** is where you record “I meant to do that” — a declared choice that tells the reading intelligences to stop flagging something they would otherwise report.

You do not need to touch most of these on day one. They are there so that when a later feature needs a home — a bibliography, a cast list, a plot thread — it already has one, in a known place, tagged the same in every project you will ever open.

## Opening the project and making your first book

You have a project on disk. Now open it. Running `inkhaven` with no arguments opens the project in the **current** directory; `--project <path>` opens one elsewhere.

● ● ● *Opening the editor*

```
$ cd ~/Books/my-first-project
$ inkhaven
# - or, from anywhere -
$ inkhaven --project ~/Books/my-first-project
```

The full-screen editor comes up in a multi-pane layout — a search bar across the top, the **Tree** on the left, the **Editor** in the middle, an **AI** pane on the right, and a status line along the bottom. Chapter three walks the panes in detail; here we only need the Tree, which has focus on startup, and where you see the system books stacked and waiting.

The Tree is driven by single letters. To build the spine of a book you need just three of them — one for each level — plus Enter to open a leaf for editing.

|              |                                                 |
|--------------|-------------------------------------------------|
| <b>B</b>     | add a book at the root (above the system books) |
| <b>C</b>     | append a chapter under the selected book        |
| <b>A</b>     | append a subchapter under the selected chapter  |
| <b>+</b>     | append a paragraph under the selected branch    |
| <b>Enter</b> | open the selected paragraph in the Editor       |
| <b>F2</b>    | rename the selected node                        |

Each of **C**, **A**, and **+** has an **insert-after** twin — **V**, **S**, and **P** respectively — that drops the new node immediately below the current row rather than at the end of its parent. You will reach for those when a manuscript grows and you need to slot something in the middle; for a first pass, append is all you want.

Press **B**. A small dialog asks for the title; type one and press Enter, and your book appears at the very top of the tree, above **Notes**:

## ● ● ● Adding a book

```

┌ Add book ────────────┐
│ Parent: <books root>  │
│   Where: above the system block │
│ Title : My First Book│
│                               │
│ Enter to confirm · Esc to cancel │
└────────────────────────┘

```

With the cursor on your new book, press **C** to add a chapter beneath it; with the cursor on the chapter, press **+** to add a paragraph. You may leave the paragraph's title blank — Inkhaven will name it from your first sentence when you save. The tree now shows your work sitting above the seeded books:

## ● ● ● Your first structure

```

▼ My First Book
  ▼ Chapter One
    ¶ Untitled paragraph
  ▶ Notes           [book · notes]
  ▶ Research        [book · research]
  ▶ Sources         [book · sources]
  ...the other system books...

```

Put the cursor on the paragraph row and press Enter. Focus moves to the Editor, which shows the paragraph's starting template — a single Typst heading line:

## ● ● ● A new paragraph in the Editor

```

= Untitled paragraph
|

```

Press **End**, then Enter twice to leave the heading behind, and write. The border around the Editor turns **yellow** the moment the buffer is dirty. Press **Ctrl+S** to save: the **.typ** file is written under **books/**, the metadata is updated, and a fresh embedding is computed for search — and the border returns to **green**. Because you

left the title empty, the Tree row now reads back your opening sentence, and the file on disk has been renamed to match its new slug.

That is a complete round trip: a book, a chapter, a paragraph, saved to disk as plain Typst, indexed for search, and ready to reopen exactly where you left it. To leave, press **Ctrl+Q** — Inkhaven autosaves anything dirty and records the session so your next launch reopens this very paragraph.

## Two ways to shape the tree first

Before you pour in words, it is worth a minute's thought about **shape**, because the tree you build is the outline you will write against — and a novelist and a non-fiction author reach for it differently.

### FICTION

Start with the **spine of the story**, not the prose. Make one book, then a chapter per act or major movement, and let paragraphs be **scenes**. Lean on the world books early: drop your cast into **Characters** and your locations into **Places** as you invent them, so their names highlight in the prose and the continuity readers have something to watch. The novel template lays exactly this down if you would rather not start empty.

### NON-FICTION

Start from the **table of contents**. Make a book, a chapter per part or major section, and subchapters for the sections beneath — the tree **is** your outline, and it will become the document's structure verbatim. Put your gathered material and citations into **Research** and **Sources** from the first day, so claims have a home and a bibliography assembles itself. The nonfiction and technical templates scaffold this shape.

Neither choice is binding. The tree reorders freely — rows move up and down, paragraphs move between chapters, whole branches relocate — and every move rennumbers the files on disk for you. Build the shape you can see today; reshape it the moment the book tells you to.

## WHAT YOU LEARNED

- `inkhaven init <dir>` creates a project directory; `--force` skips the overwrite prompt and `--template` scaffolds a starting tree, defaulting to **empty**. The first ever init downloads a shared embedding model to a per-user cache, not into the project.
- A project is **plain files**: `inkhaven.hjson` and `prompts.hjson` are yours to edit; `metadata.db`, `blobs.db`, `frequency.db`, and `vectors/` are the local store; `books/` is your prose. The files are the source of truth and the store is a rebuildable index — `inkhaven reindex` restores it from the tree.
- The hierarchy **Book** → **Chapter** → **Subchapter** → **Paragraph** is mirrored onto disk as folders and numbered `.typ` files; alongside prose paragraphs the tree can hold **image**, **hjson**, **jinja**, and **bund** leaves, plus structural-paragraph subtypes.
- Every project is seeded with **twenty protected system books** — reference (Notes, Research, Sources, Glossary, Snippets, Prompts, Help), world (Facts, Places, Characters, Artefacts, World, Threads, Planning, Mythology, Intent), and machinery (Typst, Scripts, Language, Submissions) — that you cannot delete and that your own books sit above.
- In the TUI the Tree is driven by single letters — B book, C chapter, A subchapter, + paragraph, Enter to edit, Ctrl+S to save — and the `.inkhaven/` directory plus the `.session.json` / `.inkhaven-backup.json` / `.inkhaven.log` dotfiles hold disposable session, backup, and cache state.



CHAPTER 3

# 3

## The Four Panes

---

Everything in the last two chapters happened somewhere on one screen. Now we name the places. Inkhaven fills your terminal edge to edge with a single full-screen window, and the whole of writing a book takes place inside it — you never open a second app, never reach for a mouse you do not need, never lose your place. This chapter is the map of that window: the four regions it is divided into, how you move your attention between them, and the two-key **chord** language that reaches every command the tool has. Learn this chapter and the rest of the manual becomes a matter of looking things up; skip it and every later chapter will feel like a room you entered through the wrong door.

## One window, four regions

Open a project and the terminal divides into four working areas. On the left, a narrow **Tree** holds the shape of your book — its parts, chapters, and paragraphs, folded and unfolded like an outline. Filling the centre is the **Editor**, where the open paragraph's words live. To its right sits a region that shows one of three things at a time — most often the **AI** pane, a conversation with your assistant. Across the bottom run two thin input bars — a Search bar and the AI prompt — and beneath them a single **status bar** summing up the moment. That is the whole instrument.

### TERM Pane

A **pane** is one of the window's working regions — the Tree, the Editor, the AI pane, and the others. Exactly one pane has **focus** at any moment: the pane your keystrokes are talking to. Everything you type is read as input to the focused pane (or, when a chord is in flight, as a command).

The book you are reading calls this “the four panes” for the shape you meet first — Tree, Editor, AI, and the region beside the Editor — but the window is really a small family of surfaces, and the right-hand region alone can show three different panes. Hold the four in mind as the frame; the rest hang off it.

● ● ● *The whole window — Tree, Editor, and the right region*

Tree

▼ Rain

► 1 Arrival

▼ 2 The City

► the-quay

¶ the-inn 88w

► 3 Departure

► Facts

► Notes

Editor · the-quay [modified]

1 The rain came sideways off the harbour,

2 and Mara counted the lamps as they went

3 dark, one by one, the length of the

4 quay. Six days' ride still lay ahead.

AI · llama · done · scope=Paragraph

you Is the pacing of this opening too

slow for a first page?

ai The image is strong, but four beats

land before any question is raised...

search: |

AI prompt · scope ...

2 The City ► the-quay

● 214w

scope=None

infer=Local

Read it top to bottom. The Tree names the book; the green ► marks the paragraph currently loaded in the Editor, wherever your tree cursor happens to be. The Editor holds that paragraph's lines with a number gutter down the left. The right region — here showing the AI pane — carries the conversation. The two bottom bars wait for a search query and an AI prompt. The status bar, last of all, tells you where you stand: the breadcrumb to the open paragraph, its word count, the red ● that means unsaved edits, and the current AI scope and inference mode.

#### TERMINAL-FIRST

Because it is only text, this window runs anywhere a terminal does — over SSH, inside `tmux`, on a bare tiling window manager, on a laptop on a train with the lid half-closed. There is no separate GUI, no browser tab, nothing to install beyond the one binary. The frames in this book are faithful to what you will actually see.

## The right-hand region — three panes in one place

The region to the right of the Editor is not a single pane but a slot that holds **one of three** at a time. This is the part of the layout newcomers find surprising, so it is worth naming plainly.

#### TERM The right region

The right-hand slot cycles between three panes: the **AI pane** (a conversation you drive), the **Output pane** (a notice board subsystems post to), and the **Thoughts pane** (a quiet home for long reflective text). Only one is visible at a time; switching between them is a single chord.

The three are three different **kinds** of surface, which is why they are three panes and not one. The AI pane is a **dialogue** — you ask, the model answers, the history scrolls. The Output pane is a **notice board** — structured, one-way messages that the fact-checker, the continuity watch, the translation engine, a finished background job, or a Bund script post for you to read, act on, and dismiss; each carries a severity glyph (● info, ▲ warning, ⊗ contradiction, ♻ still-running) and expires on its own so the board never becomes a junk drawer. The Thoughts pane is a **reading** surface

— a scrollable, read-only place where the longer, essayistic output lands, such as the Inner Theologian’s questions or a developmental brief.

● ● ● *The right region showing the Output pane — a notice board*

```

┌ Output · 3/4 · fact-check ────────────┐
│ ◎ fact_conflict                        │
│   │ "they reached Rillmark by nightfall" — the bible │
│   │ puts Rillmark six days' ride away.              │
│ ● review_complete                      │
│   3 checks clean · 1 contradiction · 0 warnings      │
│ △ uncovered_words                      │
│   2 word(s) uncovered: sword, sun                  │
└────────────────────────────────────────┘
└─ select  o expand  Enter jump  a ask-AI  d dismiss ─┘

```

Which pane the slot shows when you launch is remembered: on a project’s very first start it opens on the Output pane, and after that Inkhaven restores whichever of the three you last left active, exactly as it restores which paragraph was open. When fresh content arrives for a pane it will quietly surface that pane for you — unless you are actively working in a right pane at the time, in which case it holds still rather than steal your place mid-read.

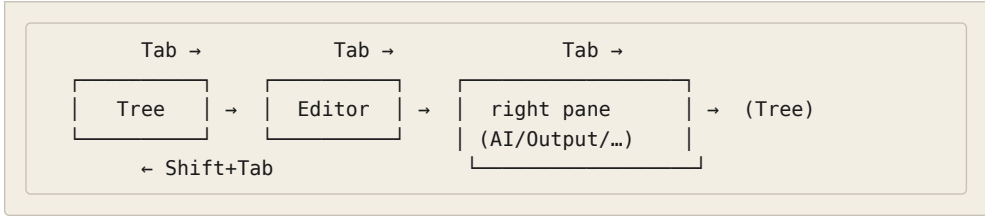
## Moving your attention — focus and navigation

Only one pane listens to your keys at a time. Moving that attention around is the most common thing you will do, so Inkhaven gives you both a cycle and a set of direct jumps.

### The plain cycle — Tab

The everyday key is **Tab**. It walks focus through a short ring — the Tree, the Editor, and **whichever right pane is currently shown** — then back to the Tree. It is a three-stop loop, not a tour of every surface: the right region counts as one stop, so if the slot is on the Output pane, **Tab** visits the Output pane; if it is on the AI pane, **Tab** visits the AI pane. **Shift+Tab** walks the same ring in reverse. You will reach for **Tab** far more than any other navigation key, and because Inkhaven intercepts it before the text editor sees it, pressing **Tab** never inserts a literal tab into your prose.

● ● ● *Tab cycles a three-stop ring*



### Cycling the right region — **Ctrl+B Tab**

To change **which** pane the right slot shows, you cycle the region itself with **Ctrl+B Tab**. Each press advances the slot through the three panes in order — Output, then AI, then Thoughts — and moves focus onto it; **Ctrl+B Shift+Tab** steps backward. This is a different motion from plain **Tab**: **Tab** moves your focus **to** the region, **Ctrl+B Tab** changes **what the region is**. The chord works from anywhere — even mid-edit — because the **Ctrl+B** prefix (which you will meet properly in a moment) swallows the keystroke before the editor could read **Tab** as a tab.

### Jumping straight to a pane

When you know exactly where you want to be, skip the cycle and jump. Each of the main surfaces has a direct key, so no pane is ever more than one chord away.

|                          |                                                              |
|--------------------------|--------------------------------------------------------------|
| <b>Tab</b>               | Cycle focus: Tree → Editor → current right pane → Tree.      |
| <b>Shift+Tab</b>         | Cycle focus in reverse.                                      |
| <b>Ctrl+B Tab</b>        | Cycle the right region Output → AI → Thoughts, and focus it. |
| <b>Ctrl+B Shift+Tab</b>  | Cycle the right region backward.                             |
| <b>Ctrl+T</b>            | Focus the Tree pane.                                         |
| <b>Ctrl+1</b>            | Focus the Editor pane.                                       |
| <b>Ctrl+3</b>            | Focus the AI pane.                                           |
| <b>Ctrl+/<br/>Ctrl+I</b> | Focus the Search bar (top). Ctrl+4 does the same.            |
| <b>Ctrl+I</b>            | Focus the AI prompt bar (bottom). Ctrl+5 does the same.      |

### How you know which pane has focus

Focus is never a guess — every pane shows it. The Editor's border carries the signal most vividly: **green** when the pane is focused and saved, **yellow** when focused

with unsaved edits, and plain **white** when it is not focused at all. In the Tree, the open paragraph keeps its green ► marker whether or not the Tree has focus, so you can always find your place. And when you have started a chord but not finished it, a yellow **META** chip lights up in the status bar to tell you the tool is waiting for your next key. Between the border, the marker, and the chip, the window always tells you what it is listening to.

#### FOCUS-LOSS AUTOSAVE

Whenever focus leaves the Editor — by **Tab**, by a direct jump, by **Esc** from another input — the open paragraph is saved for you if it was dirty. You can shift your attention mid-sentence without a thought for losing work; the words are on disk before the next pane has your keystrokes.

## The chord system — the heart of the interface

Inkhaven has more commands than a keyboard has keys, and it refuses to bury the useful ones behind menus. Its answer is the **chord**: a short, deliberate two-key sequence, led by a prefix that tells the tool “a command is coming.” Once the idea clicks, the whole interface opens up, because every serious action in the program is one chord away from wherever you are.

#### TERM Chord

A **chord** is a two-key command: a **prefix** key, then an **action** key. You press the prefix, release it, then press the second key — they are struck in sequence, not held down together. Between the two presses the tool is in a brief **pending** state, waiting for the action key and showing you it is waiting.

There are two prefixes, and the difference between them is a genuine distinction of **kind**, not an accident of which keys were free.

### Ctrl+B — the meta prefix

**Ctrl+B** is the **meta** prefix: it introduces commands that **act on your book** — add a chapter, take a snapshot, run a check, tag a paragraph, translate a line. Press **Ctrl+B** and release it, and the tool enters **meta mode**: the status bar lights its yellow **META** chip and prints the actions available for the pane you are in. The next

key you press selects one of them. **Esc** backs out of a pending chord without doing anything, and any key the table does not recognise cancels with a hint naming which pane's table it consulted.

### TERM **Meta prefix**

**Ctrl+B** — the key that opens **meta mode**. The single most important thing to learn about it is that the action table is **pane-specific**: **Ctrl+B S** means **read the paragraph aloud** in the Editor, **add a subchapter** in the Tree, and **clear the inference** is one of the AI pane's few actions. The same second key, read against a different pane, is a different command.

● ● ● *Meta mode pending — the status bar prompts for the second key*

```

┌ Editor · the-quay [modified] ────────────┐
│ 1 The rain came sideways off the harbour, and... │
├───────────────────────────────────────────┤
│ META S read-aloud · N snapshot · R cycle-status · │
│ T retitle · P place-RAG · C char-RAG · H help │
└───────────────────────────────────────────┘

```

That the table is pane-specific is not a complication to memorise but a simplification to lean on. You do not learn one enormous list of chords; you learn the handful that belong to the pane you are working in, and the same letters mean the natural thing in each place. In the Tree, **Ctrl+B C**, **S**, **P** add a **chapter**, a **subchapter**, and a **paragraph** — the second key is the word's first letter — while adding a **book** is the bare Tree key **B** (it has no meta-chord, since **Ctrl+B B** builds). In the Editor, **Ctrl+B N** takes a **new** snapshot and **F6** opens its **history** (**Ctrl+B R** cycles the paragraph's status). A last tier of meta chords — the ones that run whole-book intelligences, like **Ctrl+B Shift+C** to run every fast checker at once — work from **any** pane, because they belong to the book rather than to a place in it.

### **Ctrl+V** — the view prefix

**Ctrl+V** is the **view** prefix, and it introduces a different kind of command: things that **show you a view** or **reach a tool** rather than change the manuscript's structure — export this paragraph to markdown, open the fuzzy paragraph picker, float a rendered preview, jump a bookmark, run the Editorial Pass, open the command palette. Where **Ctrl+B** is about **acting on the book**, **Ctrl+V** is about **looking**

**through it** and reaching the pickers and panels that help you look. It reads the same way — press and release `Ctrl+V`, then the action key — and like the meta layer it is fully rebindable.

The mnemonic is worth stating out loud, because it is what lets you **guess** a chord correctly instead of looking it up: `Ctrl+B` for the **book** (things you do to it), `Ctrl+V` for the **view** (things you do to see it). Nearly every chord you meet in this manual sits under one of these two prefixes, and the prefix tells you which half of your mind the command lives in.

### Overlays — the modal stack on top of it all

Some chords do not run and return; they **open something** — a picker, a confirmation, a full-screen editor for your config, a floating preview. These are **overlays**, and while one is up it sits on top of the panes and takes your keys for itself. The pane beneath keeps its visual focus but does not see input until the overlay closes, and `ESC` is the near-universal way to close the top overlay and drop back to what was underneath. Overlays stack: a picker can open a confirm on top of itself, and closing the confirm returns you to the picker. This is why `ESC` is the key you can always press when you feel lost — it peels one layer off the top and never does anything destructive.

## The status bar and the title bars

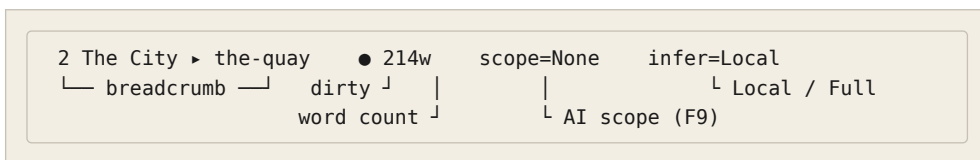
The bottom line and the little labels along the tops of panes are not decoration; they are a continuous, quiet readout of state. Learning to glance at them is most of what “knowing where you are” amounts to.

### The status bar

The status bar is the window’s single most information-dense line. In calm moments it shows the breadcrumb path to your cursor, the open paragraph’s word count, a red ● when there are unsaved edits, and the current AI scope and inference mode. In busy moments it is also where transient messages land — a search reporting `match 1 / N`, a diagnostic jump reading `diag 2/5 line 40:12`, a marked N count while you multi-select in the Tree, the yellow **META** prompt while a chord is pending, or a notice that a file changed on disk and was reloaded. It is the first place to look when you wonder what just happened.



● ● ● *The status bar — a running readout of the moment*



Optional chips ride the same line when you turn them on — a POV chip naming the paragraph’s viewpoint character, a **lang=** chip for the prompt language, a live syllable readout while a verse paragraph is open. They are quiet by default and each has its own toggle; the point is that the status bar is the one place Inkhaven speaks to you without being asked.

## The title bars

Every pane wears a thin title along its top, and each title earns its space. The Editor’s names the open paragraph and appends `[modified]` when it is dirty. The Tree’s is simply the pane’s name. The right region’s title tells you which of the three panes you are looking at and its live state — the AI pane’s reads like `AI · llama · done · infer=Full · scope=Paragraph`, folding in the provider, the stream state, the inference mode, and the scope; the Output pane’s shows a `shown/total · filter` count; the AI prompt bar’s title echoes the scope as `AI prompt · scope: Paragraph`. You rarely read them word for word, but they are the reason you are never guessing which mode a pane is in.

## Finding a chord – the palette and the quick reference

Two prefixes and a pane-specific table is a system you can **reason** about, but nobody memorises every chord, and Inkhaven does not ask you to. Three surfaces exist precisely so you can find a command without remembering its keys.

## The command palette – Ctrl+V Space

**Ctrl+V Space** opens the **command palette**: a fuzzy search over the entire keybind registry. Start typing any part of a command's name, its chord, or its description, and the list narrows to match; **↑↓** moves the selection, **Enter** runs it, **Esc** closes. This is the fast way to reach **any** command without knowing its chord — type “snapshot”, or “fact”, or “outline”, and run what comes up. It is also how you **learn** the chords: the palette shows each command's key beside its name, so the thing you searched for by word today you will strike by chord tomorrow.



|                     |                                                                                  |
|---------------------|----------------------------------------------------------------------------------|
| <b>Ctrl+V Space</b> | Command palette — fuzzy-find and run any command by name, chord, or description. |
|---------------------|----------------------------------------------------------------------------------|

? Quick reference, but only when the Tree pane has focus.

Between them the two surfaces cover both ways a person forgets a chord. When you know **what you want** but not the keys, reach for the palette. When you want to **see what is possible** from where you are, open the quick reference. Neither asks you to leave the window or read this book.

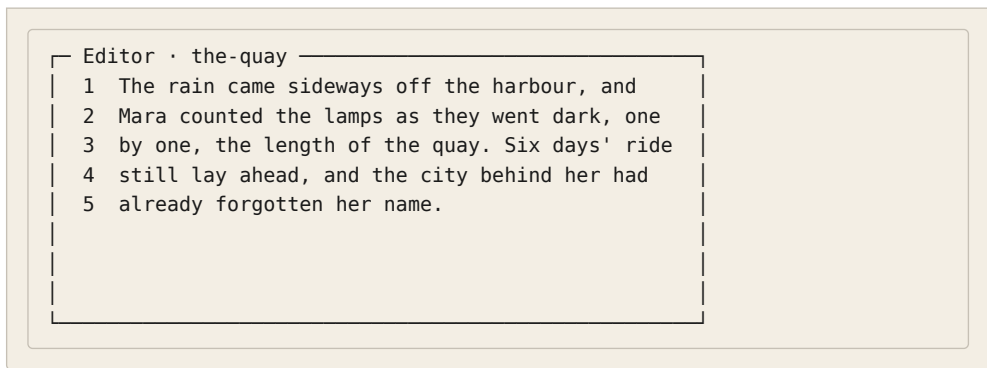
The four-pane layout is the working default, but sometimes you want less on the screen — the prose alone while you draft, or one right pane blown up while you

read a long result. Inkhaven has a mode for each, and each is a toggle: press to enter, press again to return.

### Focus mode — **Ctrl+B Shift+W**

**Ctrl+B Shift+W** enters **focus mode** — the distraction-free layout. It hides the Tree, the AI pane, the Search bar, and the AI prompt, and gives the whole window to the Editor, forcing focus there as it opens. It is the mode for the moment you have decided to stop tending the book and simply write it: nothing on screen but your paragraph. Press **Ctrl+B Shift+W** again to restore the four panes. (The setting is called “typewriter mode” in a few older strings and in the config file, for backward compatibility — same mode, the friendlier name won.)

● ● ● *Focus mode — the Editor alone, everything else hidden*



### Fullscreen panes — **Ctrl+Z f** and **Ctrl+B K**

When it is a **result** you want to sink into rather than the prose, fullscreen the right pane instead. **Ctrl+Z f** blows up the currently shown right pane — the Output pane or the Thoughts pane — to fill the window, for reading a long notice board or a developmental brief without the Editor crowding it. The AI pane has its own dedicated fullscreen, **Ctrl+B K**, which lays out the AI pane, its scrollable chat history, and the prompt across the whole window for a long conversation. Focus mode and AI-fullscreen are mutually exclusive — you are either writing alone or talking at length, not both — and each toggle returns you to the four-pane layout.

**Ctrl+B Shift+W**      Focus mode — hide everything but the Editor. Press again to restore.

**Ctrl+Z f**              Fullscreen the current right pane (Output / Thoughts).

**Ctrl+B K**              Fullscreen the AI pane with its chat history and prompt.

## A note on the mouse

Inkhaven is a keyboard instrument, but it does not ignore the mouse. It captures mouse input on start: a left-click moves focus to the pane you click, and inside a pane the click lands where you expect — the row cursor in the Tree, the character cursor in the Editor (clicks in the number gutter are ignored). The scroll wheel scrolls whatever it is over — three rows per tick in the Tree, three lines in the Editor, the chat history in the AI pane, the message list in the Output pane. It is enough to point and scroll when a hand is already on the mouse, and it is entirely optional — nothing in this book requires it.

### SELECTING TEXT THROUGH THE TERMINAL

Because Inkhaven captures the mouse, dragging to select does not reach your terminal's own copy by default. Hold `Shift` (or `Option`, depending on the terminal) while you drag to bypass the capture and select through the terminal, or toggle capture off entirely with `Ctrl+Shift+M` when you want the terminal's native clipboard for a while.

The window you have just mapped does not change from here. Every later chapter — the editor's depths, the AI's scopes, the world and its facts, the reading intelligences — plays out on these same four panes, reached by these same chords, read off this same status bar. You will spend the rest of the book learning what to **do** in this window; you now know what the window **is**.

## WHAT YOU LEARNED

- The window is **four regions**: the **Tree** (left), the **Editor** (centre), and a **right slot** that shows one of the **AI**, **Output**, or **Thoughts** panes, over a Search bar, an AI prompt, and a status bar.
- Plain **Tab** cycles focus **Tree** → **Editor** → **current right pane**; **Ctrl+B Tab** cycles **which** pane the right slot shows (**Output** → **AI** → **Thoughts**); **Ctrl+T**, **Ctrl+1**, and **Ctrl+3** jump straight to a pane.
- A **chord** is a two-key command: **Ctrl+B** (the **meta** prefix — **act on the book**) or **Ctrl+V** (the **view** prefix — **reach a view or tool**), then a **pane-specific** action key. **Esc** cancels a pending chord or closes the top overlay.
- The **status bar** and **title bars** are a running readout — breadcrumb, word count, the red ● dirty chip, AI scope and inference mode, and the yellow **META** prompt while a chord waits.
- You never have to memorise a chord: **Ctrl+V Space** fuzzy-finds and runs any command, **Ctrl+B H** (or **?** in the Tree) opens the pane-aware quick reference.
- **Ctrl+B Shift+W** gives the window to the Editor alone; **Ctrl+Z f** fullscreens the current right pane and **Ctrl+B K** the AI pane — each a toggle back to the four panes.

PART II

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# Writing

CHAPTER 4

# 4

## The Editor

---

The Editor is the pane you will live in. The last chapter mapped the whole window; this one goes to the centre of it and stays there, because the great majority of a writing session is one paragraph open in front of you and your hands on the keys. Everything the Editor does — move, select, cut, find, replace, save, split, check — is reachable without the mouse and without a menu, and almost all of it follows the plain conventions your fingers already know. The handful of places where Inkhaven departs from those conventions it departs for a reason, and this chapter names each one so nothing surprises you mid-sentence. Read it once for the shape, then let the muscle memory take over.

## ONE PARAGRAPH AT A TIME

The Editor holds a **single paragraph** — one `.typ` file — not a whole chapter scrolled end to end. This is deliberate: a paragraph is Inkhaven’s unit of prose, of snapshot, of search, and of every reading intelligence in the later chapters. You move between paragraphs in the Tree (Chapter 5); you write **within** one here. Opening another paragraph saves this one first, so the boundary costs you nothing.

## Opening a paragraph, and reading its state

You open a paragraph from the Tree: move the row cursor to it and press `Enter`. Focus jumps to the Editor, the file’s text loads into the buffer with a line-number gutter down the left, and from that moment the pane is a running readout of where the words stand. You never have to wonder whether your work is safe — the Editor tells you, in three places at once.

The most vivid signal is the **border colour**, and it speaks only while the Editor has focus. A **green** border means the buffer matches what is on disk — you are saved, nothing is pending. A **yellow** border means you have unsaved edits. When the pane is **unfocused** the border goes plain white and hands the dirty signal off to two always-on indicators: the `[modified]` suffix in the title, and the red ● chip in the status bar. Between them you can tell the state of your paragraph from anywhere in the window.

### TERM Dirty

A buffer is **dirty** when it holds edits not yet written to disk. Inkhaven shows dirtiness three ways — a yellow border (focused), a `[modified]` title suffix, and the red ● in the status bar — and it clears all three the instant a save completes. “Clean” is the opposite: buffer and file agree.

The Editor’s title carries the paragraph’s name, the `[modified]` chip when dirty, and a live cursor read-out in the form `L<row> C<col>` — the line and column your cursor sits on, updated as you move. It is the quiet coordinate you glance at when a message says “jump to line 40” or when you want to know how far down a long paragraph you have wandered.



● ● ● The Editor title — name, dirty chip, live cursor

```

— Editor · Opening Scene [modified] · L4 C18 —
1 The rain came sideways off the harbour, and
2 Mara counted the lamps as they went dark, one
3 by one, the length of the quay. Six days'
4 ride still lay ahead of her, and the city|
5 behind had already begun to forget her name.

```

One kind of paragraph opens in a state you cannot type into. If the paragraph lives inside the **Help** system book, the Editor goes read-only: the border turns teal, the title gains a `[read-only]` mark, and every key that would change the text is intercepted with a `Help is read-only` status message. Movement, copy, search, and scrolling all still work — you can read and lift from Help freely — but `Ctrl+S` is a no-op and nothing you press alters the file. Help is reserved for the material that answers `F1` lookups; your own prose simply never lives under it.

## Moving the cursor

Movement is entirely conventional, which is the point — your existing habits transfer whole. The arrow keys move a cell or a line at a time. `Ctrl+←` and `Ctrl+→` jump by **word**, landing on each word boundary, for crossing a line faster than a character at a time. `Home` and `End` snap to the start and end of the current line; `Ctrl+Home` and `Ctrl+End` leap to the very top and bottom of the paragraph. `PageUp` and `PageDown` move by a viewport for the longer paragraphs. There is nothing here to unlearn.

|                         |                                     |
|-------------------------|-------------------------------------|
| <code>← → ↑ ↓</code>    | Move one character or one line.     |
| <code>Ctrl+←</code>     | Jump to the previous word boundary. |
| <code>Ctrl+→</code>     | Jump to the next word boundary.     |
| <code>Home / End</code> | Start / end of the current line.    |
| <code>Ctrl+Home</code>  | Top of the paragraph.               |
| <code>Ctrl+End</code>   | Bottom of the paragraph.            |

**PageUp** / **Page-** Move one viewport up / down.  
**Down**

## Jumping between scene breaks

A long paragraph of fiction is often a run of scenes separated by a break line — `* * *`, `***`, `---`, `___`, `###`, `~~~`, or a lone `§`. Inkhaven recognises all of these, and gives you a pair of chords to leap between them without hunting: **Ctrl+B >** jumps the cursor forward to the next scene break, **Ctrl+B <** back to the previous one. They are the fast way to walk the beats of a scene-heavy passage. (The mnemonic borrows vim's angle brackets; the `<` avoids a collision with the tag-search chord that wanted the same key.)

**Ctrl+B >** Jump to the next scene break in the paragraph.

**Ctrl+B <** Jump to the previous scene break.

## Selecting text

Inkhaven has two kinds of selection: the ordinary **linear** run you know from every editor, and a **rectangular** block for column work. They are separate models, and only one is ever active at a time.

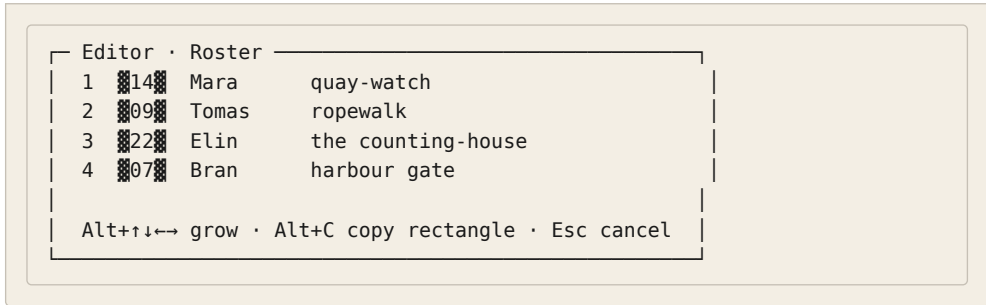
### Linear selection

Hold **Shift** and move: **Shift+←** and **Shift+→** extend the selection a character at a time, **Shift+↑** and **Shift+↓** a line at a time. Combine the two — a few **Shift+→** to reach across a phrase, **Shift+↓** to take whole lines — and you build up exactly the run you want, drawn in reversed video so it stands out against the syntax colours. **Ctrl+A** selects the entire paragraph at a stroke. The cut, copy, and paste chords in the next section all operate on whatever linear selection is live; with none, they act on the cursor position instead.

### Vertical block selection

Sometimes what you want is not a run of text but a **rectangle** of it — a column of leading numbers, a stack of names, the aligned left edge of a verse stanza. For that, hold **Alt** and move. The first **Alt+arrow** drops an anchor at the cursor and enters block-select mode; every further **Alt+arrow** grows a rectangle from that anchor to the cursor, redrawn each frame in reversed video. When the rectangle covers what you want, **Alt+C** copies it to the system clipboard as a multi-line string — one line per row of the block. **Esc** cancels the block without copying, and any ordinary key ends block mode and falls back to normal editing.

● ● ● *A block selection — a rectangle, not a run*



#### BLOCK MODE IS COPY-ONLY

Rectangular **copy** is supported; rectangular **cut** and **paste** are not, in this release — the multi-line character surgery they need is more than the underlying text widget exposes cleanly. Copy-only still covers the everyday cases: pulling out a column of figures, a list of names, or the aligned start of a stanza to reuse elsewhere.

## The clipboard — cut, copy, and paste

Here is the one place the Editor departs from the keys your fingers expect, and it departs for a concrete reason: the conventional **Ctrl+V** collides with the terminal's own paste on many setups, and **Ctrl+X** and **Ctrl+Z** are already spoken for elsewhere in Inkhaven. So the clipboard chords are chosen to sit on distinct, unambiguous keys. **Ctrl+C** copies, **Ctrl+K** cuts, and **Ctrl+P** pastes; **Ctrl+A** selects everything first. Learn this one triad and the rest of the editor is convention.

- Ctrl+C** Copy the selection to the system clipboard.
- Ctrl+K** Cut the selection to the clipboard (marks dirty).
- Ctrl+P** Paste at the cursor, replacing any selection.
- Ctrl+A** Select the entire paragraph.

The clipboard is the **real, system** clipboard — Inkhaven talks to it through the **arboard** library, so what you copy here you can paste into another application and vice versa. On a headless or restricted session where no system clipboard is

reachable — a bare SSH login, some Wayland setups — the chords do not fail. They fall back to an internal yank buffer, so cut, copy, and paste keep working **within** the editor session; the only thing you lose is crossing the process boundary to other apps.

#### WHY CTRL+P FOR PASTE

If your muscle memory reaches for `Ctrl+V`, retrain it here to `Ctrl+P`. `Ctrl+V` is Inkhaven’s **view** prefix (the pickers and panels of Chapter 3), and on many terminals it is also the native paste — leaving it as an editor chord would mean two pastes fighting over one key. `Ctrl+P` is the explicit, unambiguous Inkhaven paste, and it behaves identically everywhere.

## Deleting by the chunk

Beyond `Backspace` and `Delete`, four chords remove whole **pieces** of a line at once — the fast erasers for revision. `Ctrl+Backspace` deletes the word before the cursor. `Ctrl+D` removes the entire current line and closes the gap. `Ctrl+E` clears from the cursor to the end of the line; `Ctrl+W` clears from the cursor back to the start. None of them touch your clipboard — each saves and restores the yank buffer around itself — so a big delete never costs you the last thing you copied.

|                       |                                                   |
|-----------------------|---------------------------------------------------|
| <b>Ctrl+Backspace</b> | Delete the word before the cursor.                |
| <b>Ctrl+D</b>         | Delete the whole current line.                    |
| <b>Ctrl+E</b>         | Delete from the cursor to the end of the line.    |
| <b>Ctrl+W</b>         | Delete from the cursor back to the start of line. |

#### IF YOUR TERMINAL EATS CTRL+W

Some shells and multiplexers (bash, tmux) grab `Ctrl+W` for their own “delete previous word” before Inkhaven ever sees it. If `Ctrl+W` seems inert, that is your terminal layer intercepting it, not a bug — the fix is to rebind it in `inkhaven.hjson` or reach the same delete another way. Chapter 27 on configuration covers the rebinding.

## Undo and redo

Undo is `Ctrl+U`; redo is `Ctrl+Y`. The history is **per paragraph** — each open buffer keeps its own stack — so undoing in one paragraph never reaches back into another. If you wonder why undo is not the usual `Ctrl+Z`, the answer is the same distinctness principle as the clipboard: `Ctrl+Z` is Inkhaven’s **Bund** scripting prefix (Part VIII), and on many terminals the bare `Ctrl+Z` also suspends the process to the shell’s job control. Putting undo on `Ctrl+U` sidesteps both. Reach for `Ctrl+U` to step back, `Ctrl+Y` to step forward again.

**Ctrl+U** Undo the last edit (per-paragraph history).

**Ctrl+Y** Redo.

## Every change since the last save

While you write, Inkhaven quietly renders **bold** every character you have added since the paragraph was last saved. It is a running visual diff against the on-disk version: at a glance you can see exactly what this editing pass has touched, which is invaluable when you have been picking at a paragraph and want to know what actually changed. The bolding is computed with a fast per-line diff, so it keeps up with typing at literary scale, and it **clears the moment you save** — bold means “new since disk”, and a save makes the buffer and the disk agree, so there is nothing left to mark. Watching the bold appear and then dissolve on `Ctrl+S` is the simplest confirmation that your work has landed.

## Finding and replacing

Search inside a paragraph is regex-powered and lives on two chords: `Ctrl+F` to find, `Ctrl+R` to find-and-replace. Both take the full Rust regular-expression syntax, so a plain word matches literally while `\bword\b` pins a whole-word match, `(?i)` makes it case-insensitive, and `(?s)` lets `.` cross a line.

Press `Ctrl+F` and a magenta-bordered Find modal opens. Type a pattern, press `Enter`, and every match in the buffer lights up in red while the cursor jumps to the first; the status bar reports `match 1 / N` so you know how many you have. From there, `Ctrl+X` is “repeat” — it advances to the next match and wraps at the

end — and the match your cursor currently sits on is drawn a brighter red-and-bold so it stands out among its siblings. **Esc** clears the search and drops the highlights, returning you to plain editing.

● ● ● *Ctrl+F — the regex Find modal*

```
Find (regex) _____
Search: the\s+thunder|
Enter find · Ctrl+X next · Esc cancel
```

For replacement, press **Ctrl+R** instead. The modal grows a second field; **Tab** switches between Search and Replace. The **first** press of **Enter** applies one replacement — the current match — and keeps you in replace mode so you can walk the rest one at a time with **Ctrl+X**. A **second** press of **Ctrl+R** while replace mode is live does the wholesale thing: replace every remaining match and exit. Re-opening either modal after a search pre-fills your last pattern and replacement, so refining a search is a matter of editing what is already there.

● ● ● *Ctrl+R — Find & Replace, two fields*

```
Find & Replace (regex) _____
Search: the\s+thunder
Replace: the storm|
Enter one · Ctrl+R all · Tab field · Esc cancel
```

There is one more reach in the replace modal. **Ctrl+B** toggles the **scope** of the replacement between this paragraph and the **whole book**. In book scope, **Enter** scans every paragraph in your user books and opens a review modal: matches shown in context, **↑↓** to move, **Space** to skip the one under the cursor, **a** to apply all, **n** to skip all, **Enter** to apply, **Esc** to cancel. Book-wide replace starts in the safe whole-word literal mode, and **w**, **i**, **x** toggle whole-word, ignore-case, and full-regex in place. Every paragraph it changes is snapshotted first — annotated

with the substitution — so **F6** is always your undo. The same operation is available

headless as `inkhaven replace`, covered in the reference.

#### SEARCH MATCHES A LINE AT A TIME

In-buffer find works line by line, so a pattern that must span a line break will not match. In practice the literary tasks — swapping a word, changing a character’s name, fixing a repeated phrase — all sit within a line, so this rarely bites; when you truly need a cross-line change, the whole-book replace and the CLI are the wider tools.

## Saving, and the round trip

Saving is **Ctrl+S**. It writes the paragraph’s `.typ` file to disk, updates the metadata, re-embeds the text into the semantic index so search stays current, clears the dirty flag, and reloads the Tree so word counts refresh. The status bar confirms it with the path and a word count. Because your prose is **plain files**, this is a genuine round trip: the words Inkhaven holds and the words on disk are the same bytes, and you are free to open, diff, or version-control that file with any tool you like — Inkhaven will notice a change made from outside and reload it.

#### TERM The round trip

Inkhaven never locks your prose inside a private format. A paragraph is a `.typ` file; a save is an ordinary write of that file. Edit it in another program while Inkhaven is open and, if your buffer here is clean, Inkhaven silently reloads the newer version; if your buffer is **dirty**, it warns rather than clobber your edits and leaves the choice — overwrite with **Ctrl+S**, or reconcile — to you.

You will reach for **Ctrl+S** less than you might expect, because Inkhaven saves for you at three natural moments. It saves on **idle** — after a few seconds of not typing (the interval is `editor.autosave_seconds`, default five). It saves on **paragraph switch** — opening another paragraph writes this one first. And it saves on **focus loss** — the moment your attention leaves the Editor by **Tab**, a direct jump, or **Esc** from another input, a dirty paragraph is written before the next pane sees a

keystroke. Between the three, moving around the window costs you no work; the deliberate `Ctrl+S` is there for the moments you want the confirmation.

### THE CRASH MIRROR

Against the rare case of a crash between saves, Inkhaven keeps a **mirror** of your unsaved buffer. Every couple of seconds while a paragraph is dirty (the window is `editor.crash_mirror_seconds`, default two) it pushes the current text and cursor position into the crash-report context, so if the program ever panics its handler can flush that buffer to a rescue file on the way down. The worst you can lose is one debounce window of typing — and a split or side-by-side second buffer is mirrored just the same.

## Split-edit — two versions at once

Some revisions are easier when you can see where you came from. `F4` toggles **split-edit**: the Editor area divides in half horizontally. The **upper** pane is your live, read-write editor, exactly as full-featured as ever. The **lower** pane is a frozen, read-only snapshot of the buffer as it stood the instant you pressed `F4`, drawn in dim grey. You rewrite above while the earlier version stays visible below.

Because a long earlier passage may not line up with where you are editing, the lower pane scrolls on its own: `Ctrl+H` nudges it up a line, `Ctrl+J` down. These two chords are routed to the snapshot pane **only while split is active**, so they shadow nothing in ordinary use. When you are done, `F4` again closes the split and drops the snapshot — or `Ctrl+F4` **accepts** it, replacing your live buffer with the frozen copy. Acceptance is the clean way to roll back a change you decided against: split, rewrite, and if the rewrite is worse, take the old version wholesale.

● ● ● *F4 split-edit — live above, frozen snapshot below*

```

┌ Editor · Opening Scene [modified] ───────────┐
│ 1 The rain came hard off the harbour, and Mara │
│ 2 counted the lamps as they failed, one by one. │
├──────────┐
│ snapshot · line 1/4 · Ctrl+H/J scroll ─────────┐
│ 1 The rain came sideways off the harbour, and │
│ 2 Mara counted the lamps as they went dark, one │
└──────────┘

```



The lower half is one particular **snapshot**, and snapshots are Inkhaven’s point-in-time bookmarks of a paragraph: **F5** takes a fresh one, **F6** opens the picker of every snapshot for this paragraph to load or compare. They are the subject of their own chapter; here it is enough to know that split-edit builds on the same idea — a captured earlier version you can hold beside the present one and, if you wish, restore.

|                |                                                   |
|----------------|---------------------------------------------------|
| <b>F4</b>      | Toggle split-edit; capture the snapshot on enter. |
| <b>Ctrl+F4</b> | Accept the snapshot — replace the live buffer.    |
| <b>Ctrl+H</b>  | Scroll the lower snapshot pane up (split only).   |
| <b>Ctrl+J</b>  | Scroll the lower snapshot pane down (split only). |
| <b>F5</b>      | Take a fresh snapshot of the buffer.              |
| <b>F6</b>      | Open the snapshot picker (load / diff / delete).  |

## Grammar check — F7 and the g-apply

Grammar checking is the Editor’s most-used piece of AI, and it is built to be safe: it never touches your prose until you say so. Press **F7** and Inkhaven sends the open paragraph to your configured model with a grammar-and-punctuation prompt keyed to the project’s language, explicitly told to preserve every bit of Typst markup. The review streams into the AI pane and focus moves there, so you watch it arrive in real time. Nothing has changed in your buffer yet — the check is a proposal, not an edit.

To **take** the correction, press **g** in the AI pane. This is the “grammar-check apply”: it lifts only the corrected paragraph out of the response — the model returns it between markers so Inkhaven knows exactly which part is the clean text — and overwrites the editor buffer with it. Because the grammar prompt preserves Typst markup verbatim, the apply skips the usual markdown conversion and drops the corrected prose in as-is. Every character that changed is then rendered in red, so you can read the correction as a diff against what you wrote.

● ● ● F7 review in the AI pane, ready for the g-apply

```
AI · llama · done · scope=Paragraph
ai <<<CORRECTED>>>
  The rain came sideways off the harbour, and
  Mara counted the lamps as they went dark,
  one by one, the length of the quay.
  <<<END>>>

g apply · c copy · r replace · Esc back to prompt
```

Those red change-highlights stay up while you read back over what the grammar pass altered, then clear on your **next save** — saving is the gesture that says **I accept the corrections**. They also clear when you move to another paragraph or the editor loses focus. If the response has no correction Inkhaven can lift, **g** refuses with a status message rather than guessing.

#### THE AI NEVER WRITES WITHOUT YOUR KEY

F7 only **shows** you a correction; **g** is the separate, deliberate keystroke that accepts it. This two-step — propose in the AI pane, apply on demand — is the same contract every AI-to-prose path in Inkhaven honours: the assistant advises, and no word of your manuscript changes until you press the key that says so. The change-highlights then let you audit exactly what it did.

## The reading-pace preview

One editor-scoped tool is worth meeting here because it changes how a paragraph **reads** rather than how it is edited. **Ctrl+B Shift+E** opens the **reading-pace preview** — a teleprompter that walks a highlight through the open paragraph one word at a time, at a reader's speed rather than an editor's glance (the rate is `editor.reading_wpm`, default two hundred words a minute). Words already passed dim, the current word is reverse-highlighted, the words ahead sit normal, and a footer tracks your position and the time left. Experiencing your own prose at the pace a reader will meet it surfaces problems the editing eye skims straight over — a sentence that drags, a beat that lands too abruptly. **Space** pauses and resumes, **←** and **→** step the highlight a word at a time, **r** restarts from the top, and

**Esc** closes it. It reads the same clean prose the audiobook export uses, with the structural markup stripped away.

|                       |                                                            |
|-----------------------|------------------------------------------------------------|
| <b>F7</b>             | Grammar-check the open paragraph (streams to the AI pane). |
| <b>g</b>              | In the AI pane: apply the correction to the buffer.        |
| <b>Ctrl+S</b>         | Save the paragraph — clears the grammar change-highlights. |
| <b>Ctrl+B Shift+E</b> | Reading-pace preview (teleprompter over the paragraph).    |

### WHAT YOU LEARNED

- The Editor holds **one paragraph** — a plain `.typ` file — and shows its save state three ways: a **green** (saved) or **yellow** (dirty) focused border, the `[modified]` title chip, and the red ● in the status bar.
- Movement is conventional — arrows, `Ctrl+←/→` by word, `Home/End`, `Ctrl+Home/Ctrl+End`, `PageUp/PageDown` — with `Ctrl+B > / Ctrl+B <` to hop between scene breaks.
- Selection is either **linear** (`Shift+arrows`, `Ctrl+A` for all) or a **rectangular block** (`Alt+arrows`, `Alt+C` to copy — copy-only this release).
- The clipboard breaks convention on purpose: **copy** `Ctrl+C`, **cut** `Ctrl+K`, **paste** `Ctrl+P` — the real system clipboard, with an in-editor fallback when none is reachable.
- Chunk-deletes (`Ctrl+Backspace` word, `Ctrl+D` line, `Ctrl+E` to end, `Ctrl+W` to start) never touch the clipboard; undo is `Ctrl+U`, redo `Ctrl+Y`, per paragraph.
- `Ctrl+F` finds by regex (`Ctrl+X` next), `Ctrl+R` replaces (again for replace-all, `Ctrl+B` to widen to whole-book with a review modal).
- `Ctrl+S` saves, but Inkhaven also autosaves on idle, on paragraph switch, and on focus loss, and mirrors your dirty buffer against a crash.
- `F4` splits the pane to edit against a frozen snapshot (`Ctrl+F4` accepts it, `Ctrl+H/Ctrl+J` scroll the lower half); `F7` grammar-checks and `g` applies the correction, with red change-highlights you dismiss via `Ctrl+B C`.

## CHAPTER 5

## 5

## The Tree and Its Files

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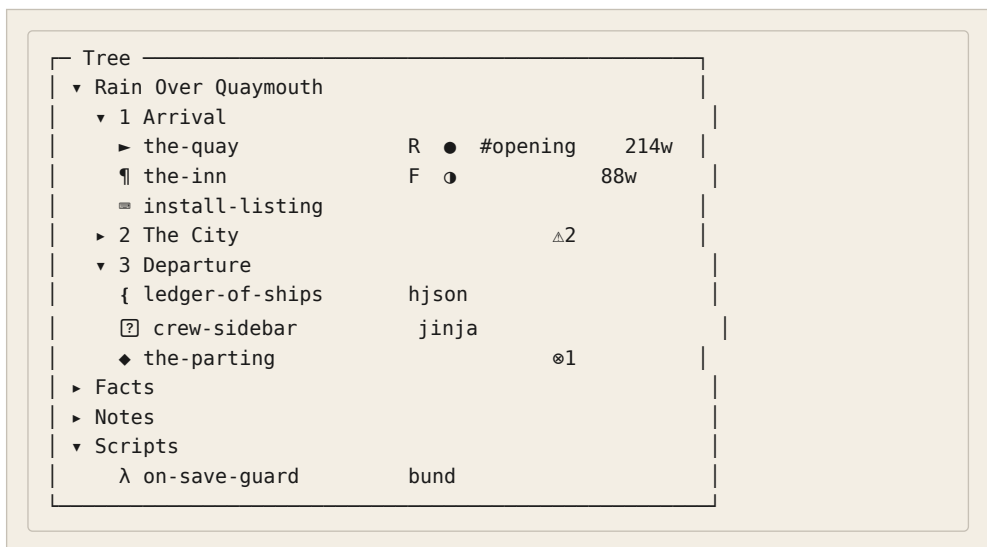
The last chapter mapped the window; this one goes into the pane on its left and never really leaves it. The Tree is where a book **has a shape** — where its parts and chapters and paragraphs stand in order, where you add a scene and move it three chapters earlier, where you fold a finished act out of the way and open the one you are still fighting. It is also, quietly, the pane where Inkhaven stops being only a prose editor: the same outline that holds your chapters holds data paragraphs, templates, scripts, images, and a family of **structural** paragraphs for the code, tables, and admonitions a technical book needs. This chapter teaches the Tree as a builder's tool first — how to grow and reshape the outline, how to delete without fear — and then walks every kind of leaf that can hang from it, one at a time, in real depth.

Everything here happens with the Tree pane focused. Reach it with `Ctrl+T`, or `Tab` around to it; it is the pane that has focus when a project first opens. The plain-letter shortcuts in this chapter — `B`, `C`, `+`, `D`, `t`, `i`, `e`, and the rest — fire **only** while the Tree has focus, and they reject `Ctrl` and `Alt`, so `Ctrl+A` will never quietly add a subchapter. Some carry a `Ctrl+B` chord form too, but those structural adds are **Tree-scoped**: `Ctrl+B C`, `S`, `P` add a chapter, subchapter, and paragraph only while the Tree has focus — from the Editor the same chords are character-lookup, read-aloud, and place-lookup — and `Ctrl+B B` builds, so a book is the bare `B` alone. The bare letters exist because terminals and multiplexers so often eat the meta prefix.

## What a tree row says

The Tree draws your project as an indented outline, one node per row, each row carrying more information than its width suggests. Read a row left to right and it tells you, in order: how deep it sits (by indentation), whether it is marked for a batch operation, what **kind** of node it is (a glyph), how far along its draft is (a status letter), its title, and then a train of small **pips** — a progress gauge, tag chips, and any findings a reader has posted against it. Paragraphs also carry a dim `Nw` word count at the end.

● ● ● A tree, read top to bottom — glyphs, badges, and pips



**TERM Node**

A **node** is one entry in the tree — a book, a chapter, a subchapter, a paragraph, an image, or a script. Branches (book, chapter, subchapter) can hold children; leaves (paragraph, image, script) cannot. The whole book is a tree of nodes, and almost everything you do in this pane is adding a node, moving a node, changing a node's kind, or deleting one.

**The kind glyph**

The glyph in front of a title is the fastest read on the row — it tells you what the node **is** without opening it. Branches show a fold arrow: ▾ when expanded, ▸ when collapsed. Leaves show their nature. Prose is the plain ¶; the other leaves each earn a glyph of their own, and this chapter's second half is a tour of exactly what stands behind each one.

● ● ● *Every glyph you will meet in the Tree*

```

▾ / ▸  branch, expanded / collapsed (book·chapter·subchapter)
▸       the paragraph currently open in the Editor (green, bold)
¶       prose paragraph — the default leaf (.typ)
{       HJSON data paragraph (.hjson)
[?]     Jinja template paragraph (.jinja)
λ       Bund script (.bund)
■       image (.png · .jpg · .webp · .svg)
◆       timeline-event paragraph
≡ △ ∫ ≡ ▣ structural: code·admonition·math·procedure·table
|| ∩ = ∴ ∴ ≠ verse: line·stanza·couplet·tercet·quatrain·translation

```

The open paragraph is special: whichever leaf is loaded in the Editor is drawn with a green, bold ▸ in place of its usual glyph, and it keeps that marker whether or not the Tree has focus. This is your anchor — however far your tree cursor wanders, the ▸ always shows you which paragraph the Editor is holding. If the cursor happens to land on that same row, the reversed cursor highlight wins the foreground, but the green underneath still marks it.

**The badges and pips**

After the glyph comes a run of optional marks. None of them is noise; each is a readout you can act on.

- A **status letter** rides just before the title on paragraphs — the draft-status ladder as a single initial (N apkin, F irst, S econd, T hird, F inal, R eady). Cycle it up a rung with **o** in the Tree, or from the Editor. A paragraph with no status shows nothing here.
- A **target gauge** appears when a paragraph carries a word-count goal — a single filling pip (○ ◐ ◑ ◒ ◓) that greens as the draft approaches its target and turns bold green at 100%. Set or clear the target with **Ctrl+V t**.
- **Tag chips** show up to two of a paragraph's tags as dim **#tag** labels, with a **+N** when there are more. Manage tags with **g** in the Tree or **Ctrl+B ]** in the Editor.
- A **report-card badge** appears when a reader has posted findings against a node — a worst-severity glyph and a count (⊗2 a contradiction, △3 a warning, ●1 an informational note). On a branch the badge is **aggregated** up from the paragraphs beneath it, so a collapsed chapter still tells you something inside it needs attention. These come from the review pass (**Ctrl+B Shift+C**) and the intelligences in Part V.
- A dim **Nw** word count closes every paragraph row.

## Moving through the tree

Navigation is arrow-driven and quiet. **↑** and **↓** walk the cursor one visible row at a time; **Home** and **End** jump to the first and last rows; **PageUp** and **PageDown** move ten rows at once. **Enter** **opens** the cursor's node — a paragraph loads into the Editor and pulls focus there (autosaving whatever was open if it was dirty), an image pops a preview, and a branch simply prints a status hint and stays put, since a chapter is a container, not a document.

## Folding — the four keys

A long book is unreadable in the Tree unless you can fold the parts you are not working in. Two motions expand and collapse a single branch; two more fold in bulk.

- **→** **expands** the branch under the cursor, revealing its children. It is a no-op on a paragraph or an already-open branch.
- **←** **collapses** an expanded branch. If the branch is already collapsed — or the cursor is on a leaf — **←** instead walks the cursor **up to the parent**, so pressing it repeatedly climbs you out of a deep subtree toward the root.
- **Z** folds the cursor's **enclosing subchapter** (or the cursor's node itself if it already is one) and lands the cursor on the folded row — the fast way to tuck away the section you just finished.

- **X collapses everything** — folds every expanded branch in the whole tree, leaving you the top-level outline to navigate from. Paragraphs and empty branches are left untouched.

#### THE CURSOR IS NOT THE OPEN PARAGRAPH

Two different things move in the Tree: the **cursor** (the reversed highlight you drive with the arrows) and the **open paragraph** (the green ►). Moving the cursor changes nothing on disk and opens nothing — it only chooses where the next action lands. You have to press **Enter** to actually open a node. This separation is what lets you reorder, tag, or delete a paragraph without first loading it into the Editor.

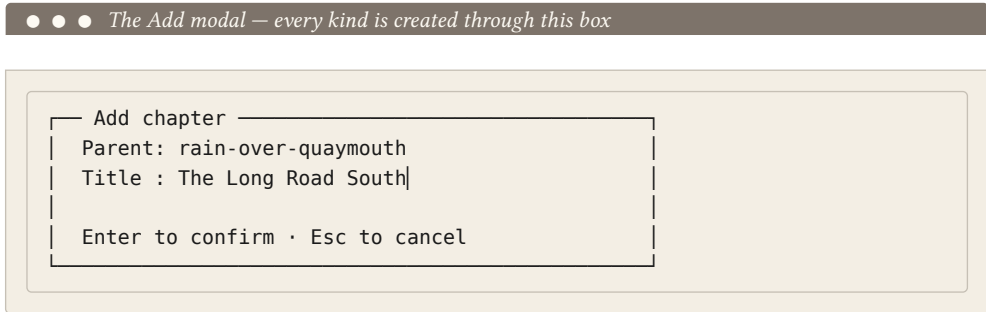
|                           |                                                     |
|---------------------------|-----------------------------------------------------|
| ↑ / ↓                     | Move the cursor one visible row up / down.          |
| →                         | Expand the cursor's branch.                         |
| ←                         | Collapse the branch, or step up to the parent.      |
| <b>Home / End</b>         | Jump to the first / last row.                       |
| <b>PageUp / Page-Down</b> | Move the cursor ten rows.                           |
| <b>Enter</b>              | Open the node — load a paragraph, preview an image. |
| <b>Z</b>                  | Collapse the cursor's enclosing subchapter.         |
| <b>X</b>                  | Collapse every expanded branch in the tree.         |
| <b>Space</b>              | Mark / unmark the row for a batch operation.        |
| <b>Esc</b>                | Cycle focus onward to the Search bar.               |

## Building the tree

A new project is a nearly empty tree — a book and the system books Inkhaven maintains for you. You grow it by adding nodes, and the Tree gives you a small, learnable vocabulary for doing so. Every add opens the same green **Add modal**: a floating box that names the parent it will slot into and takes a title. Type the title, press **Enter**, and Inkhaven derives a slug, writes the file to disk, inserts the record,



reloads the tree, and moves the cursor onto the new node. **Esc** cancels; an empty title keeps the modal open with a hint (except for paragraphs, where an empty title is allowed — the first sentence of the body becomes the title on the next save).



### The four building blocks — and where they land

The everyday structure keys come in two flavours, and the difference is **where the new node lands among its siblings**. This is the single most useful thing to understand about building a tree.

- **Append at end.** **B** (book), **C** (chapter), **A** (subchapter), and **+** (paragraph) add the new node at the **end** of its parent's children. Inkhaven chooses the parent by walking up from your cursor to the nearest node that can legally host the kind you asked for — press **+** deep inside a subchapter and the paragraph appends to that subchapter; press **C** anywhere in a book and the chapter appends to the book.
- **Insert after current.** **V** (chapter), **S** (subchapter), and **P** (paragraph) add the new node **immediately after** the cursor's same-kind ancestor, shoving every later sibling down by one — the way you add a scene **between** two scenes rather than at the end of the act. If there is no same-kind ancestor to insert after, these fall back to append-at-end so the key still does something.

The mnemonic is the shape of the keyboard row: the append keys sit under your fingers as whole words (**B**ook, **C**hapter, **A** for subchapter, **+** for one more), and the insert-after keys are the ones you reach for when order matters. These also carry **Ctrl+B** chord forms — **Ctrl+B C**, **Ctrl+B S**, **Ctrl+B P** for chapter, subchapter, and paragraph — but only while the Tree has focus, not from any pane; adding a book stays the bare **B**, since **Ctrl+B B** builds instead.

● ● ● *Append versus insert-after, on the same chapter*

cursor on 'the-inn', press +

```

┌ 1 Arrival ────┐
│  ¶ the-quay   │
│  ¶ the-inn   ←here │
│  ¶ the-market │
│  ¶ NEW       ←end  │
└───────────┘

```

cursor on 'the-inn', press P

```

┌ 1 Arrival ────┐
│  ¶ the-quay   │
│  ¶ the-inn   ←here │
│  ¶ NEW       ←after │
│  ¶ the-market │
└───────────┘

```

### BOOKS SLOT IN ABOVE THE SYSTEM BLOCK

A new user book (B) is inserted **above** the built-in system books — Notes, Research, Prompts, Places, Characters, Help, and the rest — which shift down to make room. Those system books are **protected**: Inkhaven will not let you rename or delete them, because features across the tool find them by tag. One context-sensitive exception is worth knowing: pressing **b** on the Language system book scaffolds a whole conlang sub-book under it, five chapters deep — a special case covered in the language chapters.

### Renaming and the file picker

**F2** opens the **Rename** modal, pre-filled with the current node's title. It changes only the displayed title and re-embeds the node for search; the slug and the on-disk filename stay put, so a rename never breaks a cross-reference or a snippet include. **F3** opens a **file picker** rooted at your shell's working directory: **Enter** on a file creates a paragraph after the cursor with that file's contents, and **Enter** on a **directory** recursively imports the whole tree — subfolders become branches, files become paragraphs, flattened to fit the hierarchy's depth. (An image file picked this way is routed to the image import path described later, not turned into prose.)

### Reshaping the tree

Structure is never right the first time. The Tree gives you three ways to move a node that already exists: reorder it among its siblings, change its nesting level, and relocate it to a different parent entirely. All three reuse the same filesystem-aware store primitives, so the **.typ** files on disk are renamed and renumbered to match whatever you do — the tree and the folder never drift apart.

## Reordering among siblings

**U** moves the cursor's node **up** — swapping it with its previous sibling — and **J** moves it **down**. These are the plain-letter forms of **Ctrl+B ↑** and **Ctrl+B ↓**; use the letters when your terminal swallows Ctrl with the arrow keys. Each swap rennumbers the two nodes' filesystem entries, so the order you see is the order the book assembles in. The CLI mirror is `inkhaven mv ... up`.

## Promote and demote — changing nesting level

Reordering keeps a node at its level; **promote** and **demote** change the level. These live in the full-screen Outline pane (**Ctrl+2**), the structural sibling of the side Tree, where **<** **promotes** a childless node one level out (appending it under its grand-parent) and **>** **demotes** it one level in (nesting it into the preceding sibling). Any move that would break a placement rule — a paragraph where a chapter belongs — leaves the manuscript untouched and tells you why. The Outline pane is also where you get a wide, foldable view of the whole book with a detail panel; it shares the Tree's clipboard, so the two stay in lock-step.

## Copy and move across parents

To relocate a paragraph to an entirely different chapter, use the cross-pane clipboard. **y** **copies** the cursor paragraph onto it; **m** **moves** (cuts) it; then navigate to the destination and press **f** to **affix** the clipboard paragraph as the last child of the cursor's effective parent — into a branch when the cursor is a branch, alongside it when the cursor is a paragraph. A copy duplicates with a fresh UUID and keeps the clipboard loaded; a move relocates the original and clears the clipboard. The same **y** / **m** / **f** work in the Outline pane, and the CLI equivalent is `inkhaven paragraph copy|move <src> <dest>`.

|                              |                                                            |
|------------------------------|------------------------------------------------------------|
| <b>U · Ctrl+B ↑</b>          | Move the node up one place among its siblings.             |
| <b>J · Ctrl+B ↓</b>          | Move the node down one place among its siblings.           |
| <b>&lt; / &gt; (Outline)</b> | Promote / demote a childless node one nesting level.       |
| <b>y</b>                     | Copy the cursor paragraph onto the clipboard.              |
| <b>m</b>                     | Move (cut) the cursor paragraph onto the clipboard.        |
| <b>f</b>                     | Affix the clipboard paragraph under the cursor's parent.   |
| <b>F2</b>                    | Rename the node's displayed title (slug + file unchanged). |

## Deleting, and getting it back

Deletion is the one action that can lose work, so Inkhaven wraps it in three layers of safety — a confirmation that tells you the cost, an immediate undo, and a durable snapshot for the large deletes. Understanding the layers is what lets you delete freely.

### The two delete keys

Delete is kind-specific on purpose. `-` deletes the cursor's node **only if it is a paragraph**; press it on a branch and you get a hint to use `D` instead. `D` deletes **only a branch** (book, chapter, subchapter); on a paragraph it points you back to `-`. The split is a guard rail: `-` will not nuke a whole chapter because your cursor drifted onto it, and `D` will not kill a paragraph you meant to keep. If you genuinely want a kind-blind delete, the global `Ctrl+B D` does it. Either way a red **Confirm delete** modal opens first.

### The confirmation tells you the cost

The confirm names the kind, the title, the descendant count, and — the part that matters — **the word count you are about to lose**:

● ● ● *The delete confirmation counts the words at risk*

```

Confirm delete
Delete chapter `Act II` and 12 descendants
(15,342 words)?

Removes files from disk AND records from the
store. y / Enter confirm · n / Esc cancel

```

A single paragraph reads `Delete paragraph the-quay (342 words)?`. Zero-word deletes — an empty paragraph, an HJSON or Jinja leaf — omit the count, since there is no prose to lose. `y`, `Y`, or `Enter` confirms; `n`, `N`, or `Esc` backs out. If the paragraph you had open was inside the deleted subtree, the Editor closes with it.

### The kill-ring — immediate undo

Every deleted paragraph is stashed on a **kill-ring**, and a branch delete stashes **every paragraph leaf** under it, not just the one you were pointing at. `Ctrl+B U` restores the front of the ring — the most recent deletion — instantly; the status line

reports the word count that came back. For a branch delete, repeated **Ctrl+B U** restores the paragraphs one at a time, in their original order.

#### TERM **Kill-ring**

A bounded stack (up to ten entries, capped by `editor.deleted_paragraph_history`) of recently deleted paragraphs. **Ctrl+B U** pops the front; **Ctrl+V Shift+U** opens a **picker** over the whole ring so you can choose which deletion to bring back. Restored paragraphs return at their original tree position but get a **fresh UUID**, so any cross-reference that pointed at the old node stays broken — Inkhaven flags this in the status rather than silently mis-linking.

● ● ● *Ctrl+V Shift+U — the kill-ring restore picker*

```
Restore deleted paragraph
┌ the-parting      Act III · 512w · 2m ago
├ the-market      Act I  · 88w  · 14m ago
└ ledger-of-ships  Act III · hjson · 20m ago

↑↓ select · Enter restore in place · Esc cancel
```

### Pre-delete snapshots — the long-term net

A very large branch can overflow a ten-slot ring, so before a **branch** delete Inkhaven takes an annotated snapshot of every paragraph leaf, labelled `pre-delete: <title> · <date>`. These live in the ordinary snapshot system, so you can find and restore any of them from the **F6** snapshot picker long after the kill-ring has cycled — even in a later session. The snapshots are taken **before** the delete fires, so they survive even a delete that partially fails. Together the three layers mean a branch delete is fully recoverable: the confirmation tells you what is at stake, the kill-ring is instant undo, and the **F6** snapshots are the durable safety net. (Single-paragraph deletes skip the snapshot — it would only clutter the list, and the kill-ring already covers them.)

- Delete the cursor's node — paragraphs only.

**D · Ctrl+B D** Delete the cursor's node — branches (Ctrl+B D is kind-blind).

|                       |                                                               |
|-----------------------|---------------------------------------------------------------|
| <b>Ctrl+B U</b>       | Restore the most recently deleted paragraph.                  |
| <b>Ctrl+V Shift+U</b> | Open the kill-ring picker — choose which deletion to restore. |
| <b>F6</b>             | Snapshot picker — reach the pre-delete snapshots of a branch. |

## The leaf types

So far a leaf has mostly meant a prose paragraph. It need not. A leaf is any childless node, and Inkhaven ships six kinds of them, each stored as its own file on disk and each with a job. The rest of this chapter takes them one at a time. The through-line: **the manuscript is always plain files** — a prose paragraph is a `.typ`, a data paragraph a `.hjson`, a template a `.jinja`, a script a `.bund`, an image its own bytes — and the `content_type` field (or the node kind) is all that distinguishes them.

### Prose — the `.typ` paragraph

The default leaf, and the one you will make thousands of, is a **prose paragraph**: a Typst `.typ` file holding the words of your book. It shows the ¶ glyph, counts toward your word totals, and is the surface every prose intelligence reads — the Inner Editor, the continuity watch, the fact-checker, the read-through. There is nothing to configure; `+` or `P` makes one, and you write. Everything the next chapter says about the Editor is about editing these. The three other **text** leaves below are all, on disk, close cousins — the same kind of node with a different `content_type` — and you convert between them with a single morph key covered at the end of the chapter.

### HJSON — the data paragraph

Some content is **data**, not prose: a character's dossier, a ship's manifest, a glossary entry, a table of API fields. For these Inkhaven offers the **HJSON data paragraph** — a paragraph whose `content_type` is `"hjson"`, stored as a `.hjson` file and shown with the { glyph. HJSON is JSON for humans: unquoted keys, comments, trailing commas forgiven. The Editor switches to an HJSON highlighter and header badge, and the prose intelligences stand down — a data blob is not prose, so it draws no style notes and does not inflate the word count.

● ● ● *An HJSON data paragraph — structured, not prose*

```

Editor · 01-aria [hjson]
1 {
2   name:    "Aria"
3   species: "fox"           // unquoted keys, //
4   role:    "scout"        // comments, no fuss
5   age:     19
6 }

```

Data paragraphs are useful on their own — the Characters, Places, and Language system books store their entries as HJSON — but they come into their own as the **input to a template**. That is the next leaf.

### Jinja — the template paragraph

A **Jinja template paragraph** is a paragraph whose `content_type` is `"jinja"`, stored as `.jinja`, shown with the `[?]` glyph, and marked `[jinja]` in the Editor header. It is a Jinja2-style template — rendered by `minijinja` — that the assembler **compiles to Typst** before Typst compiles anything to PDF. The two layers are strictly sequential, never nested:

● ● ● *Two layers, one direction — Jinja out, then Typst*

```

your .jinja      assembly      typst
paragraph  → (minijinja) → .typ → PDF
              renders              compiles

```

Templates exist so that **structured** content stays in one place instead of being copy-pasted and left to drift: a character sidebar that reads a linked dossier, an endpoint table that loops over fields, a shared admonition written once and pulled into a dozen chapters. The feature is **self-gating** — a project with no Jinja paragraphs assembles exactly as before, and there is no flag to turn on.

**The render context.** Every template renders with a fixed set of variables: its own `title` and `slug`; the enclosing `book.title` / `book.slug` / `book.genre` and `chapter.title` / `chapter.slug`; the project `language` and `genre`; and, most importantly, `linked[...]`.

**TERM linked**

The data-injection mechanism. Link an **HJSON** paragraph to a template with the ordinary add-link chord `Ctrl+V a`; at assembly its parsed data is exposed under `linked["<that paragraph's slug>"]`. Only HJSON-bodied links land here — a linked prose paragraph is skipped, because its raw Typst is not meaningful template data. This is how a template reads facts instead of repeating them.

● ● ● *A template reading a linked HJSON dossier*

```
= {{ linked["01-aria"].name }}

#block[
  A {{ linked["01-aria"].species }} who
  serves as {{ linked["01-aria"].role }}.
]

{% if language == "ru" %}Глава{% else %}Chapter{% endif %}
```

**Reusable fragments — the Snippets book.** A `.jinja` paragraph placed in the built-in **Snippets** system book is registered as a named template before any rendering, so manuscript templates can pull it in with `{% include "snippets/<path>.jinja" %}`. The include name is the snippet's tree path, lowercased — chapter and subchapter titles become path segments, the paragraph slug is the filename — so you read the name straight off the tree.

**Assembly order.** At `Ctrl+B A` (or `inkhaven build`) the assembler first registers every Snippets template, then renders the snippets to `.typ`, then renders each manuscript template to a `.typ` in the generated tree, then runs `typst compile`. Because registration precedes rendering, every `{% include %}` resolves in one pass — no nesting, no two-pass surprises. By default a render failure — bad syntax, a missing include, a typo'd variable — **aborts the whole assembly** with the offending paragraph named, so a broken template can never silently drop content from the PDF; set `jinja.continue_on_error: true` to write a visible error block into that paragraph's place and keep going while you fix templates one at a time.



## TWO KINDS OF "SNIPPET" — DO NOT CONFUSE THEM

A Jinja template paragraph generates structured Typst **at assembly**. The edit-time text-expansion snippets (the `bund:` expansions that fire **while you type**) are a different system entirely. Both coexist; a Jinja paragraph is not prose and is skipped by the Inner Editor, Inner Socrates, and the idle fact-checker.

**Bund — the script paragraph**

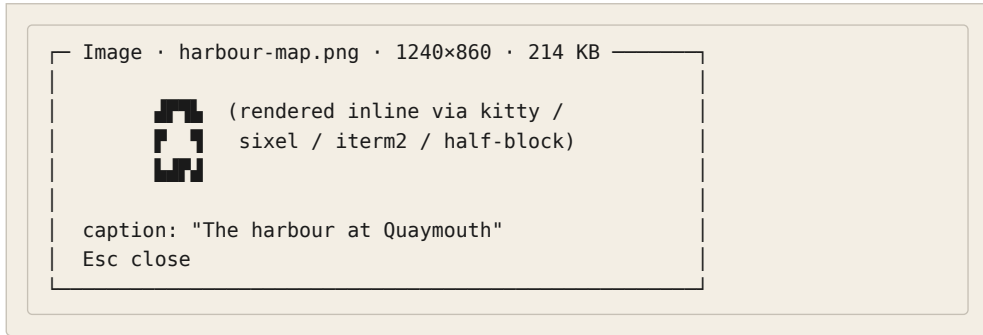
A **Bund script** is a first-class leaf — a distinct node kind, not a paragraph flavour — stored on disk as a `.bund` file and shown with the `λ` glyph. Its natural home is the built-in **Scripts** system book, whose `.bund` files are `eval`'d into Inkhaven's embedded Bund virtual machine when the project opens. That is where user-authored hook lambdas live — `hook.on_save` and its kin — so a script under Scripts can watch your saves, enforce a house rule, or extend the tool. Bund is Inkhaven's embedded scripting language; Part VIII is its full treatment. For the Tree's purposes what matters is that a script is a real node you can add, move, and delete like any other, and that you **reach** the bund rung by morphing a paragraph — the last stop on the type cycle described below.

**Images — the picture leaf**

Images are first-class nodes too — `NodeKind::Image`, shown with the `■` glyph, holding their own bytes on disk (`.png`, `.jpg` / `.jpeg`, `.webp`, `.svg`, `.gif`). You add one by importing it: the `F3` file picker routes any image file to the image-import path instead of treating it as prose, dropping a new Image node into the tree near your cursor. An Image node carries an optional caption and alt-text; at book assembly Inkhaven emits the right `wrap_image_*` calls and ships the bytes into the generated Typst tree, so the picture lands in the PDF with its caption and accessible alt-text intact.

Two conveniences ride on images. `Enter` on an Image row pops a **preview** — a real in-terminal rendering through `ratatui-image`, using your terminal's best protocol (kitty, sixel, iTerm2, or a half-block fallback). And from **inside** the Editor, with the cursor in a Typst `#image("...")` call, `Ctrl+B P` opens a sibling-image picker so you can fill the path from the images already in the tree rather than typing it.

● ● ● Enter on an image row — an in-terminal preview



### Structural paragraphs — the `para:*` family

A technical or nonfiction chapter is not all prose: it has code listings, warning boxes, display math, numbered procedures, tables. Left as ordinary paragraphs these would all read as ¶, draw prose-style craft notes that make no sense on a code block, and inflate the word count with text that is not narrative. **Structural paragraphs** fix this. A structural paragraph is an **ordinary .typ paragraph carrying a `para:*` tag** — the tag, not a new content type, is the whole mechanism.

● ● ● The structural subtypes and their glyphs

|                                      |   |                |
|--------------------------------------|---|----------------|
| <code>para:code</code>               | ≡ | code listing   |
| <code>para:admonition-note</code>    | △ | note box       |
| <code>para:admonition-warning</code> | △ | warning box    |
| <code>para:admonition-tip</code>     | △ | tip box        |
| <code>para:admonition-caution</code> | △ | caution box    |
| <code>para:math</code>               | ∫ | display math   |
| <code>para:procedure</code>          | ≡ | numbered steps |
| <code>para:table</code>              | ≡ | table          |

Marking a paragraph structural does three things: it swaps the tree glyph for the subtype's icon, it **excuses the paragraph from the prose companions** (the Inner Editor and Inner Socrates no longer fire on it — it is not prose), and it **removes it from the prose word count**, counting it separately (Book Info, `Ctrl+B I`, shows a `structural: N` line). One subtype breaks the pattern: `para:procedure` is prose the author writes — real steps — so the companions **do** still run on it; it is only excluded from the word count.

**Create one — i in the Tree.** Press **i** and a picker of the subtypes opens. Choose one, **Enter**, name it, and you get a `.typ` paragraph tagged `para:*`, its glyph set, and **matching Typst boilerplate seeded** — a `#figure` code block, a coloured `#block` admonition, a `#table`, a `$ ... $` math block, or a `+`-stepped procedure — so you start from a working skeleton rather than a blank buffer.

● ● ● *The i picker — pick a structural subtype*

```
┌ Add structural paragraph · i ────────────┐
│ ▣ code listing                          │
│ ▴ admonition: note / warning / tip / caution │
│ ∫ math                                  │
│ ≡ procedure                             │
│ ▤ table                                 │
│ ↑↓ select · Enter create · Esc cancel    │
└──────────────────────────────────────────┘
```

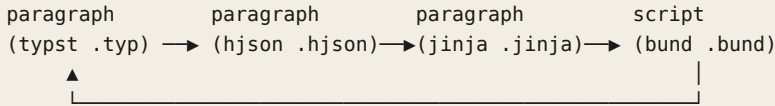
#### TAG, NOT TYPE — THE ESCAPE HATCH

Because a structural subtype is only a tag, you manage it like any tag. Add `para:code` to an existing prose paragraph with `Ctrl+B ]` to make it structural, or remove the tag to turn it back into prose — the body is untouched either way. There is no morph cycle to walk; the tag **is** the mechanism. (Poetry rides the same machinery: a stanza is a paragraph tagged `para:verse-*`, a sixth family with its own glyphs `|| ↓ = ∴ ⌘` and its own reader, the Inner Poet — the Poetry companion has the full treatment.)

## The morph cycle — turning one leaf into another

The four **text** leaves — prose, HJSON, Jinja, and Bund — are close relatives on disk, so Inkhaven lets you cycle a leaf from one to the next in place. **t** (or **T**) **morphs the node type** through a fixed loop:

● ● ● *t / T cycles a leaf through four types*



With no rows marked, **t** cycles just the cursor's leaf (branches are skipped); with a set of paragraphs marked ( **Space** ), it cycles every marked leaf at once. The morph **only flips the type and renames the file's extension** — it does not seed a starter body. That is the practical difference from the two Tree **pickers**, which create a **fresh, seeded** leaf:

- **i** creates a new **structural** paragraph from the subtype picker, with boilerplate seeded (covered above).
- **e** creates a new **Jinja** template — seeded with a documented starter that lists the variables available to you — under a user book, or a reusable `{% include %}` fragment under the Snippets book. It is rejected elsewhere (Notes, Characters, and the other system books do not take templates).

So the rule of thumb is simple: reach for **e** (or **i**) when you want a **new leaf with a working skeleton**; reach for **t** when you have an **existing** leaf whose type you want to change and you will write the body yourself. To land on the Bund rung there is no seeded creator — you morph a paragraph around the cycle to **script**, or add a **.bund** under the Scripts book and let the tree hold it.

|                 |                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>t / T</b>    | Morph the leaf's type: <code>typst</code> → <code>hjson</code> → <code>jinja</code> → <code>bund</code> (marked set, or the cursor). |
| <b>i</b>        | Create a new structural paragraph ( <code>para:*</code> ) from the subtype picker, boilerplate seeded.                               |
| <b>e</b>        | Create a new Jinja template ( <code>[?]</code> ), seeded — a manuscript template, or a Snippets fragment.                            |
| <b>Ctrl+B ]</b> | Add / remove a tag — including a <code>para:*</code> structural tag — on the open paragraph.                                         |
| <b>g</b>        | Tag the marked set (or the cursor row) from the Tree.                                                                                |

## WHAT YOU LEARNED

- The **Tree** is where the book has a shape. A row reads left to right — depth, mark, **kind glyph**, status letter, title, then **pips**: a target gauge, tag chips, a finding badge, and a dim word count. The open paragraph always wears a green ►.
- Navigate with the arrows; fold with → / ← (single branch), Z (enclosing subchapter), and X (everything). The **cursor** and the **open paragraph** are two different things — moving the cursor opens nothing until you press Enter.
- Build with B / C / A / + to **append at end** and V / S / P to **insert after** the cursor's same-kind ancestor; each has a Ctrl+B twin. New books slot above the protected system block.
- Reshape with U / J (reorder siblings), < / > in the Outline (promote / demote a level), and y / m / f (copy / move / affix across parents). The files on disk follow every move.
- Delete kind-specifically — - a paragraph, D a branch — behind a confirm that **counts the words at risk**. Recover with the **kill-ring** (Ctrl+B U, or the Ctrl+V Shift+U picker) and, for branches, the **pre-delete snapshots** in F6.
- Six leaf kinds: prose .typ (¶), HJSON data .hjson ({}), Jinja templates .jinja (?), rendered to Typst at assembly, fed by linked HJSON), Bund scripts .bund (λ), images (■, previewed with Enter), and the para:\* **structural** subtypes (▣ ▴ ∫ ≡ ▢) — the last a tag, not a type.
- Turn one text leaf into another with the morph key t (typst → hjson → jinja → bund); use the **pickers** i and e when you want a fresh, **seeded** leaf instead.

CHAPTER 6

# 6

## Snapshots and History

---

The last chapter taught you to write into a paragraph; this one teaches you to write **without fear**. Every long book is a graveyard of sentences the author was glad to lose and a handful the author would give a finger to have back, and the difference is never obvious in the moment you cut them. Inkhaven's answer is to make going back cheap and reliable at three different grains: a versioned history **inside** each paragraph, a project-wide index over **all** of it, and a recovery layer beneath both that survives the tool crashing under you. Learn this chapter and the delete key stops being a decision — it becomes a gesture you can always take back.

## What a snapshot is

A snapshot is the smallest unit of Inkhaven's memory: a frozen, dated copy of one paragraph's prose, kept beside the paragraph but never confused with it. When you take a snapshot the current bytes of the open buffer are written into the project database as a document of their own, tagged with the paragraph they came from, the moment they were taken, a word count, a one-line preview, and an optional note. The paragraph goes on changing; the snapshot does not. It is a photograph, not a mirror.

### TERM Snapshot

A **snapshot** is an independent, dated copy of a single paragraph's body, stored in the project database as its own document ( `kind: "snapshot"` ) with a `parent_id` back-reference to the paragraph it was cut from. It is not the live file, not an autosave, and not part of the search index — it is a point-in-time bookmark you can read, compare, and restore.

Three properties follow from that independence, and each one matters when you need a snapshot most.

First, **snapshots outlive saves**. Autosave and `Ctrl+S` overwrite the paragraph in place — they commit the **latest** version and keep no history. A snapshot is a separate document, so saving the paragraph a thousand times never touches it.

Second, **snapshots outlive the paragraph**. Because a snapshot is not stored inside the paragraph, deleting the paragraph does not delete its snapshots; they remain in the database, findable, as a recovery hatch when a whole branch goes. This is the seam that connects the two halves of this chapter — the versioning layer and the deletion layer meet here.

Third, **snapshots are never collected**. Inkhaven does not garbage-collect them, expire them, or cap them. A paragraph you have revised for a year carries its whole revision history, oldest to newest, until you delete a row by hand. They cost only disk, and disk at literary scale is free.

## NOT IN SEARCH

Snapshots are deliberately excluded from the vector index — they carry no embedding and never surface in semantic search or the AI’s grounding. A history of forty drafts of one paragraph would otherwise drown the one live version in every “find similar” result. Snapshots are a versioning surface, not a corpus.

## When snapshots are taken

You take snapshots two ways: deliberately, with a keystroke, and automatically, whenever Inkhaven is about to do something to your prose that you might regret. The second kind is the one that saves manuscripts, because it fires exactly at the moments you are least likely to think of it yourself.

The **manual** trigger is a single key, covered in the next section. The **automatic** triggers all share one rule — **nothing rewrites your prose without a restorable copy landing first** — and there are five of them:

- **Before an AI rewrite is applied.** When you accept a diff from any AI rewrite — the sentence-rhythm rewrite, an Editorial Pass fix, a reconciliation — the pre-rewrite paragraph is snapshotted before the new text replaces it, under an annotation naming the rewrite. Reject the diff and nothing is written at all.
- **Before a project-wide find-and-replace.** Each paragraph a book-scope replace touches is snapshotted first, annotated `replace: X → Y`, so **F6** is the undo for a substitution that went wider than you meant.
- **Before a branch is deleted.** Deleting a chapter, subchapter, or book snapshots **every paragraph leaf** inside it first, annotated `pre-delete`, so any single paragraph from a vanished branch can be found and restored even though the branch itself is gone for good.
- **Before a snapshot restore.** Loading an old snapshot over the live buffer first snapshotts the live buffer — the **pre-restore safety net**, described below. Restoring is itself a restorable act.
- **On demand from a Bund script.** The `hook.on_snapshot` hook fires on every snapshot however it was taken, so a script can mirror snapshots to git or stamp them with the paragraph’s workflow status.



## THE THROUGH-LINE

Read the list once more and notice what it guarantees: there is **no path** in Inkhaven that overwrites or destroys prose without a dated, restorable copy existing first. Manual **F5** is the belt; the five automatic triggers are the braces. You never have to remember to protect yourself before a risky edit — the tool has already done it.

## Taking a snapshot — F5 and the annotation prompt

Press **F5** (or its chord form, **Ctrl+B N**, for **new** snapshot) with a paragraph open. Rather than snapshot silently, Inkhaven floats a one-line **annotation prompt** — a chance to label this version with a note you will thank yourself for later.

● ● ● *F5 — the annotation prompt over the editor*

```
Snapshot annotation · F5 —
|
| Snapshot `the-quay` — annotation:
| > before the lighthouse rewrite|
|
| Enter commits (empty = no note) · Esc cancels
```

The prompt is forgiving in both directions. Type a note and press **Enter** to commit a **labelled** snapshot; press **Enter** on an empty line to commit an **unlabelled** one — the old reflex of “F5, Enter” still fires as fast as it ever did. Press **Esc** to cancel, and **no snapshot is written at all**.

One subtlety earns its keep: the buffer to be snapshotted is captured the **instant the prompt opens**, not when you press **Enter**. You can keep typing in the editor behind the prompt — or take your time composing the note — without changing what gets saved. The annotation lives in the snapshot’s metadata, never in its body, so it never contaminates the prose you are versioning.

## When to annotate

You do not need a note on every snapshot, and most of yours will not have one. A label earns its place on the **decision-point** snapshots — the versions you might actually want to walk back to:

- Before a structural rewrite — before the lighthouse rewrite.
- After a milestone draft — first complete draft, before pass 1.
- A version that landed — the version Maria loved.
- A risky experiment — trying second-person POV.

Annotations are what turn a long history from a wall of identical timestamps into a labelled list of states you can scan. They are how you find “the good one” three weeks later without reading every row.

## The per-paragraph picker — F6

Press **F6** with a paragraph open and Inkhaven lists every snapshot of **that** paragraph, newest first. This is the workbench for one paragraph's past — where you read it, diff it, restore it, and prune it.

- ● ● *F6 — the snapshot picker for the open paragraph*

```

└─ Snapshots · the-quay
  › 2026-06-19 14:23:11 421w The rain came sideways...
    ⚡ after the lighthouse rewrite
  2026-06-19 11:02:48 388w The rain came sideways...
    ⚡ first complete draft
  2026-06-18 16:40:09 205w She counted the lamps...

↑↓ move · Enter load · V diff · D delete · / filter
Shift+Enter pin to split · Esc close

```

Every row shows a timestamp, the snapshot's word count, and a preview of its first prose line. Annotated snapshots carry the note on a second indented line under a  glyph, so labelled rows stand out from the unlabelled ones without your reading a single preview. The keys are:

↑↓ Move the cursor row through the history.

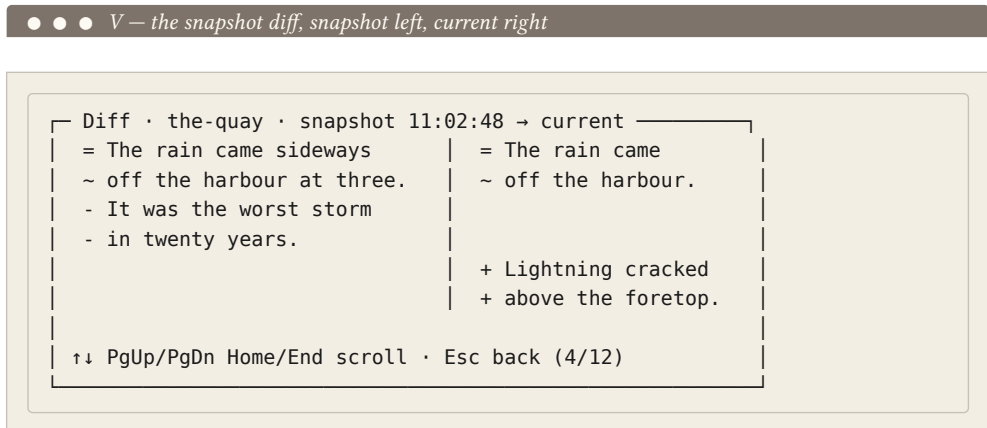
**Home / End**      Jump to the newest / oldest snapshot.

|                    |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>Enter</b>       | Load the selected snapshot into the buffer (safety-snapshotting the live buffer first). |
| <b>V</b>           | Side-by-side diff of the snapshot against the live buffer.                              |
| <b>D / Del</b>     | Delete the selected snapshot.                                                           |
| <b>/</b>           | Filter the list by annotation text; Esc leaves the filter, keeping the query.           |
| <b>Shift+Enter</b> | Pin the snapshot into the split-view secondary pane, read-only.                         |
| <b>Esc</b>         | Close the picker.                                                                       |

The **/** filter narrows the visible rows to those whose annotation contains what you type — a case-insensitive substring match. It is filter-focus: typed characters build the query, **Backspace** edits it, and **Esc** exits the filter back to row navigation while keeping the narrowed list. The filter resets each time you open **F6**, so a previous session’s query never haunts the next picker.

### Viewing a diff — V

Reading a raw old version tells you little; seeing **what changed** tells you everything. Press **V** on a row and Inkhaven opens a side-by-side diff — the snapshot on the left, the paragraph’s current text on the right.



The diff is computed with the Myers algorithm and sorted into four colour buckets: unchanged lines are dim, lines **removed** since the snapshot are red, lines **added** since are green, and a one-line rewrite — a delete and an insert that sit adjacent — is **fused** into a single yellow “changed” row rather than shown as red-then-green. The result reads the way you think about an edit.

**Esc** in the diff returns you to the picker, **not** all the way out — the picker is stashed underneath so you can scan a row, diff it, scan the next, diff that, without ever losing your place in the list. One more **Esc** closes the picker.

### Restoring an old version — Enter

Press **Enter** on a row to load that snapshot's bytes into the editor. The buffer is replaced and marked **dirty** — nothing is committed to disk until you save, so a restore is a proposal you can still walk away from. The status bar reports both the load and the safety snapshot it took first.

Because every restore leaves the live buffer dirty rather than saved, and because every restore first snapshots what **was** there, you can flip back and forth between “what the snapshot has” and “what was in the editor” indefinitely, comparing them in place, until one of them is the one you save.

### The pre-restore safety net

The single most important thing **F6** does happens where you cannot see it. When you press **Enter** to restore, Inkhaven first takes a fresh snapshot of whatever is currently in the editor — **then** replaces it. Without this, restoring an old version would silently discard any unsaved typing. With it, the state you were about to lose becomes one more row at the top of the history.

#### SAFETY BEFORE RESTORE, ALWAYS

If that pre-restore snapshot cannot be written — the disk is full, the store is offline — the restore **aborts entirely** and the buffer is left untouched. The whole point of the net is data safety, so Inkhaven would rather do nothing than do the replace without the copy. Fix the underlying cause and try again.

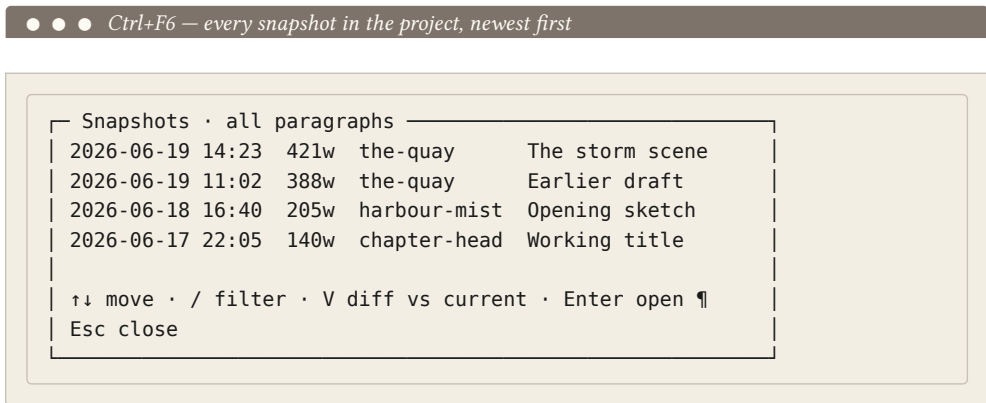
This is also the undo for a restore you did not mean. Pressed **Enter** on the wrong row? Press **F6** again: the safety snapshot of your real work is now the top row, newest timestamp. **Enter** on it and you are back where you started — and **that** restore made its own safety snapshot, so the chain never breaks.

### Deleting a snapshot — D

**D** (or **Del**) removes the selected snapshot from the history. There is no confirmation dialog: snapshots are explicit creations, the list regenerates fresh after a delete, and the safety-snapshot chain from any recent restore has you covered if you prune the wrong one. Use it to keep a decision-heavy paragraph's history readable — but you rarely need to, since nothing forces you to look at the old rows.

## The project-wide snapshot browser — Ctrl+F6

The **F6** picker answers “what has this paragraph been?” The browser answers a different question — “what was I touching last Tuesday?” — because it shows every snapshot in the **whole project** at once, one flat list, newest first, regardless of which paragraph each came from. Press **Ctrl+F6** from **any** pane; you need no paragraph open and no particular place in the tree.



Each row is **timestamp · words · paragraph · annotation**. Because the list is flat and sorted purely by time, one paragraph’s history interleaves with every other’s — which is exactly what you want when you are hunting by **when** you did something rather than **where**. The browser is the same engine as **F6**, widened; it is how you find the row, and it hands the acting-on-it back to **F6**:

- ↑↓ Move through the flat, project-wide list.
- / Filter by paragraph title OR annotation; Enter or Esc leaves the filter, a second Esc closes.
- V Diff the selected snapshot against its OWN paragraph’s current text; Esc returns to the browser.
- Enter Open that snapshot’s paragraph in the editor and drop straight into its F6 picker.
- Esc Close the browser.

Two of these repay a second look. The **V** diff resolves the snapshot’s own paragraph and diffs against **that** paragraph’s live text — even though you opened the browser

from somewhere else entirely, the comparison is always apples-to-apples. And **Enter** is a hand-off, not an action: the browser deliberately **does not restore or delete** from the project-wide view, because those are destructive, paragraph-local acts. Instead it opens the owning paragraph and drops you into its **F6** picker, where the pre-restore safety net is already wired up and you can act with full local context.

The browser lists itself in the command palette (**Ctrl+V Space**) under “snapshot”, so you can reach it by name without remembering **Ctrl+F6**.

## Deleted-paragraph history — the kill-ring

Snapshots version a paragraph that still exists. A different problem is the paragraph you **deleted** — cut from the Tree in a fit of restructuring, then wanted back. For this Inkhaven keeps a **kill-ring**: a short, ordered stash of the most recently deleted paragraphs, whole and restorable.

### TERM Kill-ring

The **kill-ring** is a session-local stash of your most recently deleted paragraphs — up to `editor.deleted_paragraph_history` of them (default 10), oldest rolling off as new deletes arrive. Each entry captures the paragraph whole, enough to re-create it in place. It lives in memory only and is not written to `.session.json`, so restarting Inkhaven clears it.

When you delete a single paragraph from the Tree (**-**, then confirm), it is pushed onto the front of the ring, and the status bar tells you both that it is gone and that **Ctrl+B U** will bring it back. What comes back is not just the prose but the whole paragraph: its title and slug (so the file lands at the same path), its body, its tags, its workflow status, its per-paragraph word target, its content type, its outgoing paragraph links, and any timeline event data.

### A NEW IDENTITY

A restored paragraph gets a **fresh** uuid. It returns to the same place with the same words, but paragraph links elsewhere that pointed at the **old** uuid stay broken — the status line is upfront about this. If preserving identity matters (an incoming link you need to keep live), reach instead for the branch-level snapshot restore, which keeps the uuid.

## Restore the most recent — Ctrl+B U

**Ctrl+B U** restores the paragraph at the **front** of the ring — the one you just deleted — without opening anything. It re-creates the paragraph at its original position: back in its old slot if a sibling anchor survives, or at the end of the (now shorter) child list if it had been the first child. This is the one-key “undo that delete” you reach for the instant you realise you cut the wrong row. Press it with an empty ring and it is a harmless no-op — the status reads **nothing to restore**.

## The restore picker — Ctrl+V Shift+U

**Ctrl+B U** only ever gives you back the **latest** delete. When the one you want is further down the ring — you deleted three paragraphs, then wanted the first of them — open the **restore picker** with **Ctrl+V Shift+U**. It lists the ring’s entries, most recent first, each with its word count and where it came from, and **Enter** restores the highlighted one at its original position.

● ● ● *Ctrl+V Shift+U — pick which deleted paragraph to restore*

```

┌ Recently deleted paragraphs ────────────┐
│ > morning          128w  Rillmark ▶ 2 The City │
│   the-ledger       64w  Rillmark ▶ 3 Departure │
│   a-false-start    212w  Rillmark ▶ 1 Arrival  │
│ ────────────────────────────────────────────┘
│ ↑↓ move · Enter restore at original position · Esc

```

The word count on each row is the quiet detail that makes the picker usable — when three deleted paragraphs share a forgettable slug, the size is how you recognise the two hundred words you actually want among the throwaways.

## What the ring does and does not cover

The kill-ring is for **single-paragraph** deletes only. Deleting a whole branch — a chapter, a subchapter, a book — is a larger and more consequential act that the ring does not try to undo, and a branch delete no longer clears the ring either, so older single-paragraph entries stay valid recoveries after one. What protects a deleted branch is the **other** half of this chapter: the automatic **pre-delete** snapshot of every paragraph in it, which you recover through the **Ctrl+F6** browser and the **F6** picker. The two systems divide the work — the kill-ring for the everyday single cut you want back in one keystroke, snapshots for the durable, branch-scale safety net.

**THREE GRAINS OF UNDO**

Keep the layers straight and you will always reach for the right one. A **typo** is `Ctrl+U` in the editor. A **deleted paragraph** is the kill-ring (`Ctrl+B U` / `Ctrl+V Shift+U`). A **deleted chapter** — or any older version of living prose — is a snapshot (`F6` / `Ctrl+F6`). The zip backup, from the health chapter, is the grain beneath all three.

## The crash mirror and recovery

Everything so far assumes Inkhaven exits cleanly. The last layer assumes it does not — that the process panics, the host loses power, or something sends a `kill -9` between two keystrokes. The promise here is concrete: your dirty buffers survive it, and the next run picks them back up. You should never need this section; when you do, it should feel like a non-event.

### The crash mirror

While you edit, a background task quietly mirrors every **dirty** buffer to disk on a short cadence — `editor.crash_mirror_seconds`, two seconds by default. This is not autosave and it is not a snapshot; it is a side copy that exists so a sudden death loses as little typing as possible. A lower value loses fewer keystrokes at the cost of more disk churn; `0` mirrors on every tick. The worst case is narrow: if the power fails between two mirror ticks, only the last couple of seconds of typing can be lost, and everything older is already on disk.

### What a panic writes

If Inkhaven panics, its handler writes two kinds of artefact before it goes down, both atomically (temp file, `fsync`, rename, directory `fsync`), and both silent on failure — the tool would rather get partial state out than panic inside its own panic handler:

- A **crash report**, `inkhaven-crash-<UTC>.hjson`, in the launch directory: the panic message and location, the project root and the paragraph open at the time, a manifest of the rescued buffers, the last fifty actions you took, and a small environment fingerprint (`$TERM`, `$LANG`, OS, version).
- A **rescue file** per dirty paragraph, `<paragraph>.typ.inkhaven-rescue`, beside the real file — the in-memory buffer flushed to disk. These are the actual prose; the report is only the index over them.



The report is deliberately conservative about what it records. It does **not** include your LLM prompts or responses, your search queries, snapshot bodies, the project database, your full config, or any environment variable beyond `$TERM` and `$LANG`. You can read it yourself before attaching it to a bug report.

## Running `inkhaven recover`

After a crash, restart Inkhaven on the project, then run `inkhaven recover` from a shell, pointing it at the crash report. It walks each rescued buffer, showing its size and the delta from what is currently on disk, and asks what to do.

● ● ● *inkhaven recover — the per-buffer walk*

```
$ inkhaven recover ./inkhaven-crash-20260805T0321.hjson

inkhaven version : 3.0.0
panic at         : src/foo.rs:42:7
message          : called Option::unwrap() on None

Found 3 rescued buffer(s).

[1/3] manuscript/01-arrival/the-quay.typ
  rescue : ../the-quay.typ.inkhaven-rescue (4823 b)
  on-disk: ../the-quay.typ (4801 b, delta +22)
  apply? [y/N/diff]:
```

At each prompt, `y` applies the rescue — atomically replacing the on-disk file, but first copying the current on-disk contents to `<original>.pre-recover-<UTC>` so you can roll the decision back later with a `mv`. `N` (or a bare Enter) skips the buffer. `d` or `diff` shows a side-by-side line diff of rescue against on-disk, then re-prompts. When the two are byte-identical — you saved just before the crash and the rescue is redundant — the walk skips the prompt entirely and reports “no action needed”, making no backup because there is nothing to back up. After the walk, the report and its rescues are moved into `<project>/.inkhaven/recovered/` so a second run does not re-apply the same set; pass `--keep` to leave them in place. For scripted cleanup, `--yes` applies every rescue without prompting, and a path-traversal guard rejects any rescue entry whose path would escape the project root — a crafted crash report cannot be used to write arbitrary files.

## THE BACKSTOPS BENEATH RECOVERY

If a `kill -9` lands inside the mirror window, the last couple of seconds may not be in the rescue. The established backstops still hold under it: `Ctrl+S` saves the open paragraph immediately and atomically; `editor.autosave_seconds` flushes dirty buffers after an idle window; and `inkhaven backup` (or `Ctrl+B Shift+B`) writes the whole project to a time-stamped `.zip`, surfacing an overdue warning in the health chip. Recovery is the floor, not the only net.

## The configuration knobs

Four settings in the `editor` block of `inkhaven.hjson` govern the machinery in this chapter. The defaults are chosen so you never have to touch them, but they are here when your workflow wants otherwise.

● ● ● *The history and recovery knobs, with their defaults*

```
editor: {
  crash_mirror_seconds:      2    // dirty-buffer mirror cadence
  deleted_paragraph_history: 10    // kill-ring depth
  external_change_auto_reload: true // silent reload of clean files
  autosave_seconds:         5    // idle autosave window
}
```

`crash_mirror_seconds` sets how often the crash mirror flushes dirty buffers — lower narrows the worst-case loss window at the cost of more writes; `0` mirrors every tick. `deleted_paragraph_history` sets how many deleted paragraphs the kill-ring holds before the oldest rolls off. `external_change_auto_reload` decides what happens when a **clean** open file changes on disk under you — `true` reloads it silently, `false` warns and leaves your view untouched, which is the setting you want when an external `git pull` or a script rewrites files while you work. `autosave_seconds` is the idle window before an unsaved buffer is written for you; `0` disables idle autosave, leaving the quit-time and paragraph-switch autosaves in force. Together they tune how eagerly Inkhaven protects your words against how much it writes to your disk to do it.

## WHAT YOU LEARNED

- A **snapshot** is a dated, independent copy of one paragraph's prose — separate from the live file, excluded from search, and never garbage-collected. It **outlives saves and even the paragraph's own deletion**.
- Snapshots are taken manually with **F5** (the annotation prompt; **Ctrl+B N**), and **automatically** before every risky op — AI rewrites, book-scope replace, branch deletes, and each restore — so **no path overwrites prose without a restorable copy landing first**.
- **F6** is the per-paragraph picker: **Enter** restores (behind a **pre-restore safety snapshot**), **V** diffs against the live buffer, **D** deletes, **/** filters by annotation. **Ctrl+F6** is the project-wide browser over every snapshot, which hands acting-on-a-row back to **F6**.
- The **kill-ring** restores deleted paragraphs — **Ctrl+B U** brings back the most recent, **Ctrl+V Shift+U** opens a word-counted picker over the last `deleted_paragraph_history` (default 10). It is single-paragraph and session-local; deleted **branches** are recovered through their **pre-delete** snapshots instead.
- The **crash mirror** copies dirty buffers to disk every `crash_mirror_seconds` (default 2); a panic writes a crash report plus per-paragraph `.inkhaven-rescue` files, and **inkhaven recover** walks them back with per-buffer `y / N / diff`, `.pre-recover` backups, and a path-traversal guard.

## CHAPTER 7

## 7

Search

---

There are two questions a writer asks a long manuscript, and they are not the same question. The first is “**where did I write that exact phrase?**” — you remember the words, you just cannot find the paragraph. The second is harder and more interesting: “**where did I write the scene where the light goes out and she loses her way?**” — you remember the **feeling** of the passage, the shape of what happened, but not one word of how you actually put it. A tool that can only answer the first question makes you the index. A tool that can answer the second reads with you.

Inkhaven answers both, but it leans hard on the second, because that is the one worth building. This chapter is the whole of finding things: the two kinds of search

and what each is for; the Search bar you type a query into and the ranked results it drops down; the machinery underneath that lets a search for words you never wrote land on the paragraph you meant; searching inside a single part of the book; keeping the index honest; and reaching the same search from a shell or a script. By the end you will treat “where is that passage?” as a keystroke, not a hunt.

## Two kinds of search

The word **search** covers two genuinely different operations, and knowing which one you want is most of using them well.

### TERM Full-text (literal) search

**Literal** search looks for a run of characters exactly as you typed them — the string `lantern`, the phrase `she counted the lamps`. It is precise and unforgiving: it finds every occurrence of those characters and nothing else, and it misses a paragraph that means the same thing in different words. This is the `Ctrl+F` find inside the open paragraph, and `rg` (ripgrep) or `grep` over the `.typ` files on disk from a shell.

### TERM Semantic (meaning-based) search

**Semantic** search looks for passages whose **meaning** is near your query, whether or not they share any words with it. It represents both the query and every paragraph as points in a “meaning space” and returns the paragraphs that sit closest to the query. This is the project-wide Search bar, and it is the one this chapter is mostly about.

Hold the division in mind, because it draws a clean line down the middle of the tool. Literal search is **inside** the text you are looking at (the open buffer) or **outside** it entirely (the files on disk, via your shell). Inkhaven’s **project-wide** search — the one that reaches every paragraph, note, fact, place, and character at once — is **semantic only**. There is no inverted full-text index over the whole project, by design: the meaning-based index is the one that earns its keep for a book, and a second literal index would be a large thing to maintain for a job `Ctrl+F` and `rg` already do well.

## WHEN TO REACH FOR WHICH

Want a **specific word or spelling** — a character's name, a coined term, a phrase you are hunting to replace? Use `Ctrl+F` in the editor (it does regex on the open paragraph) or `rg <pattern> books/` from a shell. Want **the passage about a thing**, remembered by sense rather than wording? Use the Search bar. The rest of this chapter is the Search bar.

## The classic case — the lantern that never appears

Here is the example that shows why semantic search is worth the machinery. Suppose, three hundred pages into a draft, you want the scene where the last light fails and your protagonist is left to find her way in the dark. You type into the Search bar:

● ● ● *A query for a scene you remember by feeling, not by words*

Search \_\_\_\_\_  
the moment the lantern fails

Press `Enter`, and the top hit is a paragraph that reads:

● ● ● *The paragraph the query finds — which never says any of it*

The last lamp guttered and went out, and Mara counted the dark the way she had once counted the lamps: one, then the next, then nothing at all, the length of the quay gone to soot and the harbour with it.

Read the two side by side. Your query said **lantern**; the paragraph says **lamp**. Your query said **fails**; the paragraph says **guttered and went out**. Your query said **the moment**; the paragraph names no moment at all. A literal search for `lantern fails` would have returned **nothing** — not one of those characters is in the passage. Yet it is unmistakably the paragraph you meant, and semantic search puts it first, because **the meanings are neighbours** even though the words are

strangers. That is the whole promise in one example: you search by what a passage is **about**, and the tool bridges the gap between how you remember it and how you wrote it.

## The Search bar — asking the whole book a question

The Search bar is the thin input that runs along the bottom of the window, and it is the front door to project-wide search. You reach it, type, run, read, and jump — five small moves you will wear a groove into.

### Opening the bar — Ctrl+/

Ctrl+/ focuses the Search bar from anywhere; Ctrl+4 does the same, for hands that prefer the number row. The bar takes your keystrokes and a cursor appears in it. (You can also click it, if a hand is already on the mouse.) Nothing else about the window changes — the panes stay where they were; you have simply pointed your typing at the search input instead of the prose.

### Typing a query

Anything goes in the bar — there is no syntax to learn and no operators to remember. A single keyword (**mariner**), an exact-ish phrase you half-recall (**stood at the rail at dawn**), or, best of all, a loose paraphrase of the thing you are after (**the moment the storm broke and someone left the deck**). The query is read as a natural-language description of the passage you want, not as a pattern to match, so write it the way you would describe the scene to a friend.

#### LONGER QUERIES SEARCH BETTER

It is tempting to type three terse words. Resist it. A fuller query — a sentence, even two — gives the model more to work with and usually retrieves a sharper result than a key-  
a quiet argument between them just before  
word does. the funeral will beat  
argument funeral almost every time. Semantic search rewards description, not keyword-thrift.

### Running it and reading the results

Press Enter. A ranked results overlay drops down over the body — the nearest paragraphs, best match first.

● ● ● The results overlay — ranked by meaning, best first

```

— Results · "the moment the lantern fails" (5) —
0.907 [paragraph] Rain > The City > the-quay
      The quay at dusk
      The last lamp guttered and went out, and Mara...

0.861 [paragraph] Rain > Arrival > the-crossing
      Losing the road
      With the beacon dark she had only the sound...

0.842 [note]      Notes > light-and-dark
      On darkness as a threat
      I keep returning to the image of a guide...

↑↓ select · Enter open · Esc close

```

Every row is three lines and tells you three things. The first line carries the **score**, the **kind** of node (a **paragraph**, a **note**, a fact, a place), and a human-readable

**breadcrumb** built from the node titles — **Rain > The City > the-quay**, the trail you would walk down the Tree to reach it. The second line is the node's **title**. The third is a one-line **snippet** from the body, enough to recognise the passage on sight.

#### TERM The score

The number at the head of each row is a **cosine similarity** — how close the paragraph's meaning sits to the query's, from roughly **0.0** (unrelated) toward **1.0** (nearly identical in meaning). It is a **relative** ranking signal, not an absolute grade: read it to compare hits against each other in one result set, not to set a universal pass mark. A **0.72** top hit in a sparse book can be exactly right; a **0.91** in a dense one can still be the wrong scene.

The list reaches across the **whole** project in one ranked column — manuscript prose, your Notes, the Research book, the Places and Characters entries, the Facts. A search for a mood can surface a research note you took weeks ago alongside the scene it informed, because both were indexed the same way. That breadth is a feature: the thing you are looking for is not always in the chapter you think it is in.



## Jumping to a hit

↑ and ↓ move the selection through the results. Enter on the highlighted row opens that paragraph in the Editor — focus moves there, the Editor loads the body, and the Tree cursor repositions onto the very same row, so you land in context and can move to neighbours immediately. One keystroke took you from “the scene where the light fails” to the cursor blinking inside it.

## Dismissing the overlay

Esc once closes the overlay while leaving focus in the Search bar, so you can refine the query and run again. Esc a second time steps focus on — to the Editor if a paragraph is open, otherwise onward through the focus ring. Nothing Esc does here is destructive; it only peels away the overlay.

|                                           |                                                                                                                                                                                                                                                                                               |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ctrl+/<br>Enter<br>↑ / ↓<br>Esc<br>Ctrl+F | Focus the Search bar (top). Ctrl+4 does the same.<br>Run the query; on an open result, open that paragraph.<br>Move the selection through the ranked results.<br>Close the results overlay; press again to move focus on.<br>Literal regex find inside the open paragraph (not project-wide). |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## What makes a good query — and what does not

Semantic search is powerful and a little unlike the search boxes you are used to, so it pays to learn where it shines and where it is the wrong tool.

### The cases it is best at

**Paraphrase** is its home ground — the lantern example above. Describe the passage in your own words and it bridges to however you actually wrote it.

**Mood and situation** work beautifully, because a passage’s emotional register is part of its meaning. Try a quiet pause in the action, someone delivers bad news, the protagonist doubts herself — you will find the passages that fit even when you have forgotten every literal word.

**Character viewpoint** is findable by name plus a verb phrase — Mara realises, the captain's hand — because the embedding learns that a passage is about a person from far more than the bare occurrence of their name.

The **long query** beats the short one, as the callout above warned: a sentence of description retrieves better than a keyword, every time you are patient enough to type it.

### Multilingual, by construction

The search is **multilingual** down to its bones, not as a bolt-on. Set the project language to Russian and a query like `утренний рассвет на палубе` finds Russian prose written in different inflections; the model that powers the search understands roughly a hundred languages, and it maps meaning across the morphology that trips up a literal search. A book written in Russian, French, German, or Spanish searches exactly as well as one in English — the same model, the same index, the same bar. (You will meet the details of **why** in the next section.)

#### SEARCH ACROSS THE LANGUAGES OF A TRANSLATION

Because meaning is language-agnostic in the index, a query in one language can surface near-neighbours written in another — useful when a manuscript and its translation live in the same project. It is not a translation tool, but it will often land you on the corresponding passage in the sibling book.

### What it will not do

It is as important to know the edges. **Literal single-word** search is **not** its strength: a query for one specific word may skip a paragraph that contains that exact word, if the paragraph’s overall meaning lands far from the query. When you truly need “every paragraph containing this string,” that is a job for `Ctrl+F` in the editor or `rg` over `books/` — not the Search bar. **Boolean operators** (`AND`, `OR`, `NOT`) do nothing; the interface is natural language, and the way to combine ideas is to describe them together in one query or to run two queries. **Field-restricted** search (`title:foo`) is not supported, and **regex** belongs to the in-buffer `Ctrl+F`, not to the project-wide bar. None of these are oversights; they are the deliberate shape of a meaning-first search.

## Finding the paragraph like this one — similar-paragraph mode

The Search bar starts from a query you type. There is a second, quieter way into the same vector index that starts from **a paragraph you are already reading**: similar-paragraph mode, on `Ctrl+V S`.

**TERM Similar-paragraph mode**

**Ctrl+V S** (from the editor) treats the **open paragraph itself** as the query. It saves the buffer, runs a vector-similarity search for the paragraphs nearest it in meaning, and opens the chosen hit in a second editor **side by side** with the first. Press **Ctrl+V S** again to save both and leave. It answers “what else in the book reads like this?” without your having to phrase the question.

● ● ● *Similar-paragraph mode — nearest neighbours of the open buffer*

```
Similar to · the-quay · 4 near _____
0.884 Rain > Departure > the-harbour-gate
      She had watched other ships leave at dusk...

0.850 Rain > The City > the-inn
      The common room was loud with the same rain...

↑↓ select · Enter open beside · Esc close
```

This is the tool for a different kind of question than the Search bar answers. The Search bar is for **retrieval** — you know roughly what you want and you go get it. Similar-mode is for **discovery** — you are standing in one passage and want to know what else in the book rhymes with it: the other rain scene, the echo you did not plan, the note that anticipates this beat. It is also how you catch **unintended** repetition, the paragraph you have written twice in different chapters without noticing. Both surfaces read the same underlying index; they differ only in where the query comes from — your fingers, or the paragraph under your cursor.

## Under the hood — how meaning becomes a search

The magic in the lantern example is worth demystifying, because it is not magic and understanding it tells you when to trust it. Two ideas do all the work: an **embedding** that turns text into numbers, and a **vector index** that finds nearby numbers fast. Both run **entirely on your machine**.

## Embeddings — text as a point in meaning space

### TERM **Embedding**

An **embedding** is a passage of text rendered as a list of numbers — a **vector** — such that passages with similar meanings produce vectors that sit close together. Inkhaven’s default model outputs a vector of roughly 384 numbers per passage. “The last lamp guttered” and “the lantern fails” land near each other in that 384-dimensional space; “the wedding was at noon” lands far away. Nearness of vectors **is** nearness of meaning — that is the entire trick.

The model that computes these vectors is **MultilingualE5Small**, run locally through fastembed. It is a small, fast, multilingual model — the source of the roughly-a-hundred-languages reach — and it is what makes a Russian query find Russian prose and an English paraphrase find English prose without a word of either being sent anywhere. The ONNX weights load once per session (a one-time cost of about half a second, paid lazily the first time you actually save or search, never at launch), then stay resident and cheap.

### ON-DEVICE AND PRIVATE

Every embedding — of your prose and of your query — is computed **on your own machine**. No paragraph, no search term, no scrap of the book leaves the computer to be searched. There is no cloud call, no API key, and no network round-trip in the search path; it works on a train with the wifi off. The model weights live in a per-user cache after their first download, and an air-gapped install can be pre-seeded with them. Search is a local capability, full stop.

## The HNSW vector index — finding neighbours fast

Comparing a query against every paragraph’s vector one by one would be slow in a big book. Instead the vectors live in a structure built for exactly this.

**TERM HNSW index**

**HNSW** (Hierarchical Navigable Small World) is a graph that links each vector to its near neighbours, so a search can **navigate** toward the closest points instead of checking every one. Inkhaven stores its vectors in vecstore, an HNSW index kept beside the metadata database. A query embeds once, then walks the graph to the nearest paragraphs in time that barely grows as the book does.

The index returns **cosine distance** (lower means closer), which Inkhaven flips into the **cosine similarity** score you see in the results (higher means closer), so the number reads the natural way — bigger is better. Each node actually carries **two** vectors in the index: one for its metadata fingerprint and one for its content, so both the substance of a paragraph and its identifying details are findable.

**How a paragraph gets embedded — on save**

The index stays current because embedding is wired into saving. Every time you save a paragraph — **Ctrl+S**, or the autosave that fires when focus leaves the Editor — Inkhaven does two things beyond writing the **.typ** file to disk: it feeds the prose to the embedding model and updates that paragraph’s vector in the HNSW index, and it refreshes the paragraph’s metadata (word count, modified time, a derived title if the title was still a placeholder). By the time your focus has reached the next pane, the words are on disk **and** the search index already knows about them. This is why a passage you wrote a moment ago is findable a moment later, and it is why the tutorial’s advice to “just keep writing” costs the search nothing — the index maintains itself, one save at a time.

**WHY RE-EMBED ON EVERY SAVE**

Re-embedding on each save (rather than once, long ago) means searches always run against the **live** model and the **current** text. Change the embedding model in config and every future save re-embeds with it; edit a paragraph and its vector moves to match the new words. The index is never a stale snapshot of last week’s draft — it is a running mirror of what is on disk now.

## Searching within a scope — Facts search

Project-wide search is the default, but sometimes the question belongs to **one part** of the book. The clearest case is the Facts book — the store of things your world holds to be true — and it has its own scoped search on **Ctrl+B Shift+S**.

### The Facts search modal — Ctrl+B Shift+S

**Ctrl+B Shift+S** opens a two-phase modal that runs a semantic search **restricted to the Facts book**. It uses the very same vector index as the project-wide search and similar-mode, then post-filters the results to the Facts subtree — so you are searching by meaning, but only among your facts.

● ● ● *Facts search — semantic search scoped to the Facts book*

```

— Facts search —
| query: siege supplies at the north gate|
|                                     |
| [x] 0.891  Facts > Logistics > grain-stores |
|       Three weeks of grain remained in the keep... |
| [ ] 0.864  Facts > Defences > the-north-gate |
|       The north gate had not been rehung since... |
|                                     |
| ↑↓ select · Space mark · Enter → chat · Esc close |

```

The first phase is the query: type a description (multi-word is fine) and press **Enter** to run the scoped search. The second phase is the ranked matches: **↑↓** navigate, **Space** marks several facts for multi-select, and **Enter** sends the marked facts — or, if you marked none, the cursor’s row — into a **targeted Facts chat** grounded in exactly those facts. Any printable key or **Backspace** in the results drops you back to refine the query; **Esc** closes.

### How it differs from the Search bar

Three differences matter. It is **scoped** — it never returns prose or notes, only facts, so a big Facts book does not drown your answer in manuscript hits. It is **two-phase** — search then act, with multi-select in between, where the Search bar is search-then-jump. And its payoff is not “open the paragraph” but “**ground a conversation** in these facts”: the marked handful becomes the context for a focused chat, which is the scalable way to reason over a large fact base — you pull in the relevant few rather

than loading the whole book into the model. Where the Search bar takes you **to** a passage, Facts search hands a passage **to the assistant**. Same index underneath; a different destination on top.

## Keeping the index healthy — reindex

The index maintains itself for everything you do **inside** Inkhaven. It cannot know about changes made **outside** it, and it does not automatically follow a change of model. For those two situations there is one command.

### What re-embeds automatically

Saving a paragraph re-embeds it. Renaming a node re-embeds it (the title is part of what is indexed). Creating, editing, and deleting through the TUI all keep the index in step. In normal writing you never think about the index at all — it is already current.

### When to reindex

#### TERM **inkhaven reindex**

`inkhaven reindex` walks every `.typ` file under `books/` and reconciles the store with what is on disk — re-reading each file’s content and re-embedding it. It is the command that re-aligns the index with reality after something changed the files without going through Inkhaven, or after you changed the embedding model. It is idempotent: safe to run as often as you like.

Reach for it after any of these:

- You **switched `embeddings.model`** in `inkhaven.hjson`. The new model must re-embed every paragraph, since old vectors were made by the old model and are not comparable to new ones. Search quality looks wrong until you do.
- You **edited a `.typ` file in another editor**, or a `git checkout` brought back paragraphs the database had forgotten.
- You **moved or deleted files** under `books/` from a shell, outside the TUI.
- You simply want a “**are my files and the index aligned?**” sanity check.

Two flags refine the walk. `--prune` removes store records whose `.typ` file has gone missing from disk (use it after deleting files by hand). `--adopt` finds `.typ` files on disk the store does not yet know about and registers them under the

matching hierarchy branch (use it after dropping new files into a chapter directory). They combine: `reindex --prune --adopt` does both passes.

● ● ● *Re-embedding the whole project after a model switch*

```
$ inkhaven --project ~/Books/my-novel reindex
re-read 412 paragraphs · re-embedded 412 · pruned 0 · adopted 0
index aligned with disk.
```

#### SWITCH THE MODEL, THEN REINDEX

The one sequence to remember: change `embeddings.model` in the config, then run `inkhaven reindex`. The bigger multilingual models (`MultilingualE5Base`, `MultilingualE5Large`, or `BGEM3`) buy better recall at the cost of disk and inference time; whichever you choose, the reindex is what makes existing paragraphs searchable under it. Skip the reindex and old and new vectors mix, and the scores stop meaning anything.

## Search from the shell — the CLI

Everything the Search bar does is available from a terminal, which is what you want when you are scripting around a manuscript or piping results into other tools. The subcommand is `search`.



● ● ● *The same semantic search, from a shell*

```
$ inkhaven --project ~/Books/my-novel \
  search "the moment the lantern fails"

0.907 [paragraph ] rain/the-city/the-quay
      The quay at dusk
      The last lamp guttered and went out, and Mara counted...
      id: 7b3e9a04-...

0.861 [paragraph ] rain/arrival/the-crossing
      Losing the road
      With the beacon dark she had only the sound of the...
      id: alf6c2d8-...
```

Each hit prints four lines: the score with the node kind and the slug path; the title; a one-line snippet; and the node's `id`. That last line is the point of the CLI form — the `id` is a stable handle you can feed to other `inkhaven` commands or grep out of the stream. The default is the top ten hits; `--limit N` changes the count.

● ● ● *Capping the result count*

```
$ inkhaven --project ~/Books/my-novel \
  search "someone delivers bad news" --limit 3
```

An empty result set prints `No results.` to standard error and exits cleanly, so a script can tell “nothing matched” from “the command failed.” Typical uses: listing the top matches to a file before a manuscript-wide edit, piping to `grep` to pull the `id` column, or feeding a shortlist of paragraphs into a batch job.

## Search from a script — the Bund word

Inkhaven's embedded Bund language exposes the same search to your scripts, so a macro can find passages and act on them. The word is `ink.search.text`.

**TERM** `ink.search.text`

Stack effect ( `query limit -- list` ). Takes a query string and a positive integer limit, runs the same semantic `search_text` the Search bar and CLI use, and pushes a list of hit dictionaries — each with `id`, `title`, `score`, `document` (the body), and `kind`. It is the scripting front door to the vector index.

● ● ● *Searching from Bund, then reading the top hit*

```
"the moment the lantern fails" 5 ink.search.text
; -> a list of 5 hit dicts: { id title score document kind }
```

A companion word, `ink.search.load`, goes one step further: given a query and an index, it runs the search and **opens the Nth hit in the editor**, returning false when the search came back empty. It is the scripted equivalent of running a search and pressing `Enter` on a result — useful for a macro that jumps you to “the passage about X” as one bound key. Between the two, a Bund script can find passages by meaning and either process them or navigate to them, with the same index the whole rest of the chapter has been about.

## WHAT YOU LEARNED

- There are **two kinds of search**: **literal** (exact characters — `Ctrl+F` in the buffer, or `rg` over the files) and **semantic** (meaning-based). Inkhaven’s **project-wide** search is semantic only.
- Semantic search finds a passage by **meaning**, not words: a query for the moment the lantern fails lands on a paragraph that says “the last lamp guttered and went out” and never uses one of your words.
- The **Search bar** opens with `Ctrl+/ (or Ctrl+4): type a query — longer and more descriptive is better — Enter runs it, ↑↓ picks a ranked hit, and Enter opens that paragraph in the Editor. Ctrl+V S searches from the open paragraph, for what reads like it.`
- Under the hood: an **embedding** (fastembed’s `MultilingualE5Small`, 384 numbers per passage) turns text into a point in meaning space; an **HNSW** vector index finds the nearest points fast. It is **multilingual** and runs **entirely on-device** — nothing leaves your machine. Every save re-embeds the paragraph.
- **Facts search** (`Ctrl+B Shift+S`) is the same index **scoped to the Facts book**, two-phase: search, multi-select, then ground a targeted chat in the chosen facts.
- Run `inkhaven reindex` after switching `embeddings.model` or after any change to the files made outside Inkhaven; `--prune` and `--adopt` reconcile deletions and additions. The CLI `inkhaven search "<query>" [--limit N]` and the Bund word `ink.search.text` reach the same search from a shell or a script.

## CHAPTER 8

## 8

## The Style Overlays

---

The last chapter left you writing prose into the Editor. This one is about the faint marks that appear **underneath** it — amber underlines beneath a hedge word, a purple one beneath a word you have used three times in as many paragraphs, a teal one beneath “was angry,” a chip in the status bar naming whose eyes you are seeing through. Inkhaven calls these the **style overlays**, and they are the most easily misunderstood thing in the whole instrument, so let us be plain at the outset about what they are for.

They are not a grammar checker, and they are certainly not a style **authority**. Every one of them is a question the tool poses and then leaves entirely to you: **you used “just” here — did you mean to?** The overlay has no opinion about

the answer. A filter word is sometimes exactly the right word; a repeated image is sometimes a deliberate refrain; a line of pure telling is sometimes the fastest way across a dull stretch of plot. The overlays exist to make those choices **visible** so they become choices at all, rather than habits you never see. The whole family is built on one principle, and it is worth stating in its own box before we open a single one.

#### TERM An overlay is advisory

A **style overlay** marks a pattern in your prose — a word, a phrase, a construction — that **often** rewards a second look. It never changes a character of your text, never blocks a save, never scores you. It is a coaching mark you toggle on when you want the second pair of eyes and off when you want the page clean. You question it; you do not obey it.

Two more things are true of the whole family, and both matter enough to say up front. First, every overlay is **multilingual** — its word lists and its stemming are keyed to the project's top-level `language` field, so a Russian manuscript is measured against Russian hedges and a French one against French, never against English words bleeding into foreign prose. Second, every overlay has a **master switch** in `inkhaven.hjson` and a **session-local chord** that flips it for the current run without touching the file. The chords are how you audition an overlay; the config is how you decide, once, that you want it on for good. We will meet each overlay in turn and close on the switchboard that governs them all.

## The filter-word overlay — Ctrl+B Shift+F

The oldest and busiest of the overlays flags **filter words**: the intensifiers and hedges that leak into a first draft and dilute it — `just`, `really`, `very`, `quite`, `actually`, `simply`, `seem`, `perhaps`, and their kin. Press `Ctrl+B Shift+F` in the Editor and every one of them in the open paragraph gains a faint amber underline. Press it again and they vanish. That is the whole gesture; the depth is in what gets underlined, and in which language.

**TERM Filter word**

A word that **filters** the reader’s experience through a layer of the author’s hedging or emphasis instead of letting the image land directly. “The room was very cold” asks the reader to trust an intensifier; “the room was cold” and a shiver ask them to feel it. Not wrong — but almost always worth a glance.

**The amber underline**

The mark is a coloured underline drawn beneath the word in the Editor, in the amber `theme.style_warning_filter_word_fg` (default `#f9c44e`). It is deliberately quiet — a hint at the edge of vision, not a red-pen slash — and its weight is yours to set: `theme.style_warning_filter_word_modifier` accepts `underline` (the default), `bold`, `dim`, `reversed`, `italic`, `none`, or `+`-combined forms like `underline+bold` for terminals where the plain underline reads faint.

● ● ● *The amber filter-word underline in the Editor*

```
Editor · the-quay [modified]
1 She just wanted to see the harbour, and it
2 really did seem very far, almost too far.
(amber — = a filter word — question, don't obey)
```

Four words are marked here — `just`, `really`, `seem`, `very` — and two words that look like candidates are **not**: “almost” and “too” are not on the English list, and “harbour” is plainly innocent. The underline is a prompt to reread the sentence, nothing more. Strike `Ctrl+B Shift+F` again and the page goes clean for the next stretch of drafting.

**The per-language word lists**

Inkhaven ships a curated built-in list for each of its five first-class languages — English, Russian, French, German, Spanish — and picks the one that matches your project’s `language`. The English list runs to about thirty entries and gathers the usual suspects into two families: the **intensifier crutches and hedges** (`just`, `really`, `very`, `pretty`, `quite`, `rather`, `fairly`, `somewhat`, `slightly`, `actually`, `basically`, `literally`, `simply`, `definitely`, `absolutely`,

totally, completely, and the notoriously load-bearing `that`) and the **sensory or hedging verbs** listed in base form (`seem`, `feel`, `look`, `appear`, `sound`, `notice`, `begin`, `start`), plus `suddenly`, `perhaps`, and `maybe`. The Russian list is its own vocabulary, not a translation of the English one — `очень`, `просто`, `именно`, `довольно`, `слишком`, `весьма`, `крайне`, `вполне`, `достаточно`, and so on — because the hedges of one language are not the hedges of another.

#### DOES IT WORK IN RUSSIAN?

Yes, and by design rather than accident. Tokenisation is UAX-29-aware (Cyrillic, Latin, Greek, and Devanagari word boundaries all resolve), the columns are counted in **characters** not bytes so a multi-byte word never shifts the underline, and a `ë → e` fold is applied so a list entry spelled `ещё` still catches the `еще` you actually typed. A language Inkhaven ships no list for is left **silent** — never measured against English words — until you give it a list of its own.

### Replacing the list versus adding to it

Two config fields shape the active list, and the difference between them is the one thing to get right. The per-language array — `editor.style_warnings.filter_words.english` (or `.russian`, and so on) — is a **replacement**: leave it empty (the default) and Inkhaven uses its built-in list; put anything in it and your list **replaces** the built-in one wholesale. The `extra_words` field is an **addition**: whatever you list there is unioned on top of whichever list is active, in every language. So the everyday move — “keep the defaults, but also flag my own two crutch words” — is `extra_words`, not the language array. Reach for the language array only when you want to curate the whole list yourself.

#### FICTION

A novelist who over-uses “somehow” and “a little” adds just those two to `extra_words` and keeps the thirty built-in English defaults underneath — the fastest way to hunt a personal tic without losing the general net.

#### NON-FICTION

A technical writer who finds the literary defaults noisy in reference prose can **replace** the English list with a short array of the three or four hedges that actually creep into documentation, and leave the rest clean.

There is a third door for a language outside the curated five. The `filter_words.languages` map takes any language by its lowercased name — an Italian project can be enabled with a `languages: { italian: ["molto", "solo"] }` entry, typically written for you by `inkhaven lang bootstrap`. A mapped entry takes precedence over everything; without one, a non-curated language simply stays quiet.

### Stemming — one entry catches every inflection

You will notice the built-in lists carry **lemmas**, not every surface form — `seem`, never `seemed` / `seems` / `seeming`. That is because the detector stems both the list and your prose through the project’s Snowball algorithm before matching, so a single entry catches the whole inflectional family. `seem` on the list flags `seemed`; Russian `казаться` flags `казался`, `казалась`, `казалось`, and `казались` alike. The stemmer is also what keeps the overlay honest at the edges: English Snowball reduces `justice` to `justic`, not `just`, so “justice” is never flagged for containing “just.” If you would rather match exact forms — listing every inflection yourself — set `filter_words.use_stemming = false`.

### The siblings under the same switch

`Ctrl+B Shift+F` is labelled “toggle style warnings,” in the plural, because it governs more than filter words. The `style_warnings` master switch (and the one chord that flips it) covers a small family of always-on inline detectors that share the same rendering machinery and the same multilingual foundation:

- The **repeated-phrase** detector slides an `n`-word window (default `n = 4`) across the open paragraph and marks, in magenta (`style_warning_repeated_phrase_fg`, default `#eb6f92`), any phrase that recurs `threshold` times or more (default `3`) — with stop-words excluded so “the dog and” does not inflate a count, and stemming applied so “lifted her shoulders” and “lifting her shoulders” are recognised as the same gesture.
- The **show-don’t-tell** inline detector marks telling constructions in teal; it has its own section below.
- The **anachronism** detector marks, in an era-caution orange (`style_warning_anachronism_fg`, default `#eba672`), any word that post-dates your manuscript’s setting — a `wristwatch` in an 1840 novel. It is off until you set `editor.style_warnings.anachronism.year`, and it ships a built-in lexicon (each term carrying its earliest plausible year) that you extend with a project `terms` list.



Each has its own per-detector **enabled** flag beneath the master switch, so a writer who wants filter-word marks without repeated-phrase magenta can silence just that one. The master switch turns the whole family off at once.

**Ctrl+B Shift+F** Toggle the style-warning overlays (filter words, repeated phrase, show-don't-tell, anachronism) for the session.

## The echo overlay — Ctrl+B Shift+K

The filter-word family looks **within** the open paragraph. The **echo** overlay looks **between** paragraphs — it is the live, in-editor companion to the **echo-repetition** doctor scan, and it catches the revision-stage tic where a distinctive word gets reused a few paragraphs apart: “she **walked** to the window... he **walked** across the room... they **walked** out.” Each sentence reads fine on its own; the cluster clangs. Press **Ctrl+B Shift+K** in the Editor and every word in the **open** paragraph that echoes across nearby paragraphs of the chapter gains an underline in its own muted purple (`style_warning_echo_fg`, default `#b48ead`) — deliberately distinct from the repeated-phrase magenta, so a within-paragraph repeat and a cross-paragraph echo read as two different findings.

● ● ● *The echo overlay — a distinctive word reused nearby*

```
Editor · the-market —
12 The lantern swung. A second lantern hung by
13 the door, and a third lantern by the dry well.

echo: "lantern" ×3 within a 5-paragraph window
```

### Cross-paragraph, and live

The overlay reads the chapter’s paragraphs around the one you are editing — using the **open** paragraph live, unsaved edits and all, so a fresh repeat you type this instant is caught without a save. Its sensitivity is governed by three tunables under **editor**, shared with the doctor scan so the live overlay and the batch report agree:

- **echo\_window** (default **5**) — how many consecutive paragraphs count as “nearby.” A word must cluster within this span to flag.

- `echo_min_repeats` (default 3) — how many occurrences within the window it takes to raise the echo. Lower is more sensitive.
- `echo_max_global` (default 40) — the distinctiveness ceiling. A word used more than this many times across the chapter is treated as ordinary vocabulary an author legitimately reuses, and skipped even when it clusters.

That last knob is the clever part, and it is why the overlay does not need a frequency dictionary for every language. Distinctiveness is measured **relative to this chapter's own distribution**: a word must repeat but not be common. A word used two hundred times across the book is vocabulary, not an echo, however it clusters; a word used three times total, all within four paragraphs, is a glaring one. The measure is corpus-relative and language-agnostic, which is what lets the same detector serve every language honestly.

#### MULTILINGUAL, WITH A DOCUMENTED EDGE

Echo detection stems through the project's Snowball algorithm (with the Russian `ë`→`e` fold applied first), so `walked/walking` collapse to one echo and Russian `корабль / корабля / корабле` likewise. Languages **outside** the Snowball set — Japanese, Chinese — degrade gracefully to exact-surface matching: inflected variants will not collapse, but identical repeats still flag. The behaviour is documented and consistent with the other overlays' fallback, never a silent failure.

The master switch is `editor.echo_overlay` (default `false` — it is an opt-in); `Ctrl+B Shift+K` is the session-local override that flips it without rewriting your config.

**Ctrl+B Shift+K**      Toggle the echo overlay — words in the open paragraph echoing across nearby paragraphs of the chapter.

## Show, don't tell — the overlay and the AI scan

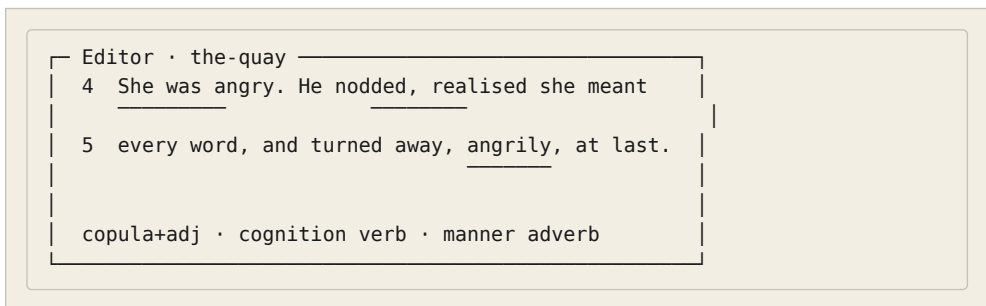
The most quoted rule of prose craft gets two tools, and the split between them is the model for how Inkhaven pairs a cheap deterministic detector with an optional LLM pass throughout the program. One is the always-on inline overlay under the `Ctrl+B Shift+F` family; the other is a deliberate AI scan on `Ctrl+B Shift+T`.

## The inline overlay — three telling patterns

The `show-don't-tell` detector marks, in teal (`style_warning_show_dont_tell_fg`, default `#94e2d5`), three specific constructions that **tell** the reader a feeling instead of **showing** the behaviour that would let them infer it:

- A **copula and an emotion adjective** — a linking verb (`be`, `seem`, `feel`, `appear`, `look`, `become`, `remain`, `grow`, `sound`) followed by an emotion word (`angry`, `sad`, `afraid`, ...). “She was angry” is flagged across both words; “she was running” is not, because “running” carries no emotion label.
- A **manner-of-emotion adverb** — `angrily`, `sadly`, `nervously` — flagged on its own, because such adverbs almost always name the feeling outright.
- A **cognition verb** — `realised`, `knew`, `understood`, `wondered`, `decided` — flagged on its own, because it reports a character’s interior state directly to the reader.

● ● ● *Show-don't-tell — teal marks on three telling patterns*



Curated built-in lists ship for all five first-class languages; a language Inkhaven does not ship lists for produces an empty (silent) detector rather than noise. Matching is Snowball-stemmed by default, so “seemed nervous” is caught through the `seem` lemma; set `show_dont_tell.use_stemming = false` for exact forms. The detector is **enabled** by default **under** the master switch — so it rides `Ctrl+B Shift+F` with its siblings.

## The AI scan — Ctrl+B Shift+T

The inline overlay catches the obvious two-grams; it cannot catch the paragraph that tells you a character is grief-stricken through a page of narrated summary with no flagged word in it. That is what `Ctrl+B Shift+T` is for. It sends the open paragraph to your configured model with a prompt asking for telling passages **and suggested rewrites**, and streams the answer into the AI pane. The two tools

are complementary by intent: the regex overlay is free, instant, and always on, catching the mechanical cases; the AI scan costs a call and a moment, and earns it by catching the subtle instances and proposing alternatives. Neither touches your prose — the scan is advisory, its rewrites are suggestions in the AI pane, and any change you make from them is yours to type or to route through the Editorial Pass. The mnemonic is **T for tell**.

**Ctrl+B Shift+T** AI show-don't-tell scan — send the open paragraph to the model for telling passages plus suggested rewrites, streamed into the AI pane.

## The terminology overlay — Ctrl+V z

Where the filter-word overlay is about literary reflex, the **terminology** overlay is about discipline: keeping one name for one thing. It reads the **Glossary** system book — a book of canonical terms, each with a list of **banned synonyms** — and red-underlines any synonym that appears in your prose, so the canonical form reads clean while its rivals are flagged. Define “access token” as canonical with “auth token” among its banned synonyms, and every “auth token” you type earns a red underline (`style_warning_banned_synonym_fg`, default `#e05a5a`) while “access token” stays untouched.

● ● ● *The terminology overlay — a banned synonym flagged*

```

Editor · setup-guide —
3 Paste the auth token into the header field,
4 then reload. The access token never expires.
   (canonical — clean, no mark)
terms: "auth token" → use "access token"

```

The overlay is **self-gating**: with an empty Glossary it flags nothing and costs nothing, so a project that does not govern terminology never sees it. Synonyms may be one to three words, matched longest-first over UAX-29 word tokens (so “auth-token” tokenises and matches the two-word form), and it works in Cyrillic as readily as Latin. When your cursor sits on a flagged word the Editor footer names the fix — `terms: "auth token" → use "access token"` — so the correction

is one glance away. It is on by default within the master style toggle; `Ctrl+V z` is its own session-local override, distinct from `Ctrl+B Shift+F`.

### The deliberate-variant escape hatch

Sometimes the variant is intentional — a character who says “auth token” in dialogue because that is how they speak. With the cursor on a red-underlined synonym, `Ctrl+V Shift+Z` records that canonical term as a **deliberate variant**

in the intent ledger, and both the overlay and the CI-ready `check` stop flagging it. It is the “I meant to write it this way” door, and it is a first-class part of the design rather than a way to silence the tool wholesale.

**Ctrl+V z** Toggle the terminology overlay — red-underline Glossary banned synonyms of canonical terms.

**Ctrl+V Shift+Z** Declare the banned synonym under the cursor a deliberate variant, so it stops being flagged.

### The POV chip — Ctrl+B Shift+P

Not every overlay draws under your words; some speak from the status bar. The **POV chip** answers a question a long manuscript makes surprisingly easy to lose track of: whose eyes is this paragraph seen through? Press `Ctrl+B Shift+P` and the status bar shows, for the open paragraph, the most-mentioned character as its heuristic point-of-view figure, followed by up to three other named characters present in the scene.

● ● ● *The POV chip riding the status bar*

```
the-quay ● 214w POV Mara ·Corin ·Tyr scope=None
└ breadcrumb ┘ dirty+words ┘ └ heuristic POV chip ┘
```

The chip is a **heuristic**, and honest about it. It counts mentions against the project’s existing `characters` lexicon — no separate per-paragraph tagging needed — takes the most-mentioned name as POV, and breaks ties by order of first mention. It will be wrong in a scene where the viewpoint character is silent while another is named repeatedly, which is precisely the kind of place worth a second look: a mismatch between who the chip names and who you **intend** is a prompt, not a verdict. The

master switch is `editor.pov_chip_enabled` (default `true`); `Ctrl+B Shift+P` is the session-local flip. Its colours are `theme.pov_chip_bg` / `pov_chip_fg` (defaults `#8b1d88` on `#ffffff`).

**Ctrl+B Shift+P** Toggle the status-bar POV chip — the most-mentioned character in the open paragraph, plus up to three others present.

## The sentence-rhythm gauge — Ctrl+B Shift+H

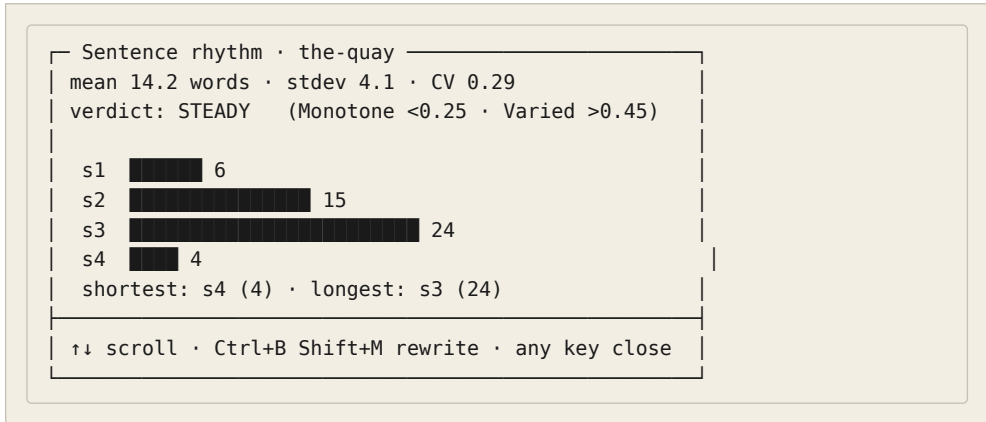
Prose has a heartbeat, and monotony in it — every sentence the same length — drones however clean the words are. `Ctrl+B Shift+H` opens the **sentence-rhythm gauge**, a modal that splits the open paragraph into sentences (with a hand-rolled walker that suppresses abbreviations — `Mr.`, `Mrs.`, `Dr.`, `e.g.`, `i.e.`, `Ph.D.` — so they do not cut a sentence short), measures each one's word count, and reports the mean, the standard deviation, and above all the **coefficient of variation** (CV) — the standard deviation divided by the mean, a single number for how much your sentence lengths vary.

### TERM Coefficient of variation

The CV is the spread of your sentence lengths **relative to their average**, so it compares fairly across dense and sparse prose alike. A low CV means every sentence is about the same length — a drone. A high CV means you mix the short and the long — the rhythm most strong prose has.

The gauge maps the CV to one of four verdicts: **Monotone** (`CV < 0.25` — the prose drones), **Steady** (`0.25 – 0.45`), **Varied** (`0.45 – 0.80` — the sweet spot of strong rhythm), and **Choppy** (`CV ≥ 0.80` — so jagged it may jar). It lists each sentence as a bar and calls out the three shortest and three longest so you can see the outliers at a glance.

● ● ● *The sentence-rhythm gauge — CV to verdict*



The gauge diagnoses; it does not fix. But because the natural next step from a **MONOTONE** verdict is to break the rhythm up, **Ctrl+B Shift+M** — the AI sentence-rhythm rewrite — fires from **inside** the gauge as well as from the Editor, so the diagnose-then-rewrite path needs no extra keystroke: open the gauge, read the verdict, and if you want it, ask for a rewrite that mixes short and long while preserving your voice. That rewrite, like every AI prose change in Inkhaven, arrives as a reviewable diff you accept or reject — snapshot first, never an unconfirmed write. The mnemonic is **H for heartbeat**.

**Ctrl+B Shift+H** Open the sentence-rhythm gauge — per-sentence bars, mean/stdev/CV, and a Monotone/Steady/Varied/Choppy verdict.

**Ctrl+B Shift+M** AI sentence-rhythm rewrite — mix short and long sentences while preserving voice; arrives as a reviewable diff. Also fires from inside the gauge.

## The reader-pace preview — Ctrl+B Shift+E

The last overlay is the one that changes not how your prose **looks** but how you **read** it. You read your own draft at editing-glance speed — skimming, because you already know what it says — and at that speed a run-on that drags or a beat that lands too abruptly is invisible. **Ctrl+B Shift+E** opens the **reader-pace preview**: a teleprompter that advances a highlight word by word through the open paragraph at a reader's speed, so you experience the prose the way a first reader will.

● ● ● *The reader-pace preview — prose at reading speed*

```

— Reader pace · 200 wpm —
| The rain came sideways off the harbour, and Mara |
| counted the ||lamps|| as they went dark, one by one. |
|
| dim = already read · ||reverse|| = now · plain = ahead |
|
| word 9/48 · 00:12 left · Space pause · ↔ step · r |

```

The words behind the highlight `dim`, the current word is reverse-highlighted, and the words ahead read normally; the footer shows your position and the time left. The pace is `editor.reading_wpm` (default `200`, the common silent-reading average — drop it toward `150` to feel an audiobook narration). `Space` pauses and resumes (elapsed time carries across cycles), `←` and `→` step the highlight by a word, `r` restarts from the top, and `Esc` closes. It reads **clean** prose — the same markup-stripping pass the audiobook export uses — so your Typst markers never break the flow. The mnemonic is **E for experience**.

**Ctrl+B Shift+E**      Reader-pace preview — a word-by-word teleprompter through the open paragraph at `editor.reading_wpm`, to feel your prose at a reader's speed.

## Master switches and session overrides

Now the switchboard. Every overlay in this chapter has the same two-level control, and understanding the pattern once saves you configuring each one from scratch. The **master switch** lives in `inkhaven.hjson` and is the persistent default — on or off every time you open the project. The **session-local chord** is the audition: it flips the overlay for the current run only and never rewrites your config, so you can turn something on to check a passage and let it lapse when you quit. When you decide, once, that you want an overlay on for good, you set its master switch; day to day, you reach for the chord.

The overlays cluster under a handful of config keys, most of them inside the `editor` block:



● ● ● *The overlay config, gathered in `inkhaven.hjson`*

```
editor: {
  style_warnings: {
    enabled: false           // master – Ctrl+B Shift+F
  }
  filter_words: {
    enabled: true
    use_stemming: true
    extra_words: [ "somehow", "a bit" ]
    english: []             // [] = use the built-in list
  }
  show_dont_tell: { enabled: true }
  repeated_phrases: { enabled: true, n: 4, threshold: 3 }
  anachronism: { year: 1840 } // off until set
}
echo_overlay: false        // master – Ctrl+B Shift+K
echo_window: 5
echo_min_repeats: 3
echo_max_global: 40
pov_chip_enabled: true     // status chip – Ctrl+B Shift+P
reading_wpm: 200          // reader pace – Ctrl+B Shift+E
}
```

A few defaults are worth committing to memory because they surprise people. The `style_warnings.enabled` master is `false` out of the box — the whole filter-word family is **opt-in**, so a fresh project shows no underlines until you either flip the master or press `Ctrl+B Shift+F`. The per-detector flags beneath it (`filter_words.enabled`, `show_dont_tell.enabled`, `repeated_phrases.enabled`) default `true`, so once the master is on the family comes on together; you silence an individual detector by setting **its** flag `false`. The echo overlay's master (`echo_overlay`) is likewise `false` by default. The POV chip's master (`pov_chip_enabled`) is `true`. And the terminology overlay is on within the master style toggle but self-gates to silence on an empty Glossary.

#### EDIT THE CONFIG WITHOUT LEAVING THE EDITOR

You do not have to leave Inkhaven to change any of this. `Ctrl+B 0` opens `inkhaven.hjson` in a full-screen editor with syntax highlighting; `Ctrl+S` saves, and because these keys are read at launch a **Restart required** notice appears when a change needs the next run to take effect. The session chords are there precisely so you rarely need the round-trip mid-draft.

Here is the whole family in one table — the chord, and the master key it overrides — so you can find any overlay from any pane.

|                       |                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+B Shift+F</b> | Style warnings (filter word / repeated phrase / show-don't-tell / anachronism) — master editor.style_warnings.enabled. |
| <b>Ctrl+B Shift+K</b> | Echo overlay — master editor.echo_overlay.                                                                             |
| <b>Ctrl+B Shift+T</b> | AI show-don't-tell scan (per-invocation; complements the inline overlay).                                              |
| <b>Ctrl+V z</b>       | Terminology overlay — on within the style master; self-gates on an empty Glossary.                                     |
| <b>Ctrl+V Shift+Z</b> | Declare a banned synonym a deliberate variant.                                                                         |
| <b>Ctrl+B Shift+P</b> | POV chip — master editor.pov_chip_enabled.                                                                             |
| <b>Ctrl+B Shift+H</b> | Sentence-rhythm gauge (modal; Ctrl+B Shift+M rewrites from inside it).                                                 |
| <b>Ctrl+B Shift+E</b> | Reader-pace preview (modal; paced by editor.reading_wpm).                                                              |

The overlays are the quietest intelligence in the tool — no reports, no letters, no scans you wait on — just a set of faint marks you can raise and lower over the page as you draft. Turn them on when you want the second pair of eyes on a finished paragraph; turn them off when you want to write without one watching. They only ever ask; the answer, in every case, is yours.

## WHAT YOU LEARNED

- The style overlays are **advisory** — they mark patterns worth a second look (filter words, echoes, telling, banned synonyms, POV, rhythm, pace) and never edit your prose, block a save, or score you. You **question** them.
- `Ctrl+B Shift+F` toggles the **filter-word** family — amber underlines on intensifiers and hedges, keyed to the project language, Snowball-stemmed; `extra_words` **adds** to the list, the per-language array **replaces** it. The same switch governs repeated-phrase, inline show-don't-tell, and anachronism.
- `Ctrl+B Shift+K` toggles the **echo** overlay — a distinctive word reused across nearby paragraphs — tuned by `echo_window` (5), `echo_min_repeats` (3), and the `echo_max_global` (40) distinctiveness ceiling.
- **Show-don't-tell** is two tools: the always-on inline overlay (copula+emotion, manner adverb, cognition verb) under `Ctrl+B Shift+F`, and the AI scan on `Ctrl+B Shift+T` for the subtler cases with rewrites.
- The **terminology** overlay (`Ctrl+V z`) red-underlines Glossary banned synonyms and self-gates to silence when the Glossary is empty; `Ctrl+V Shift+Z` marks a variant deliberate. The **POV chip** (`Ctrl+B Shift+P`), **rhythm gauge** (`Ctrl+B Shift+H`), and **reader-pace preview** (`Ctrl+B Shift+E`) round out the family.
- Every overlay has a **master switch** in `editor.*` (the persistent default, most of them off) and a **session-local chord** (the audition that never rewrites config). `Ctrl+B 0` edits the config in place.

PART III

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# The AI Assistant

## CHAPTER 9

## 9

## The AI Assistant

---

Every other chapter so far has been about you and the manuscript. This one adds a third party to the desk: a language model you can consult without leaving the window. Inkhaven’s stance on that model is stated in a single line of its own README — *AI is a co-author you steer* — and the word that matters is *steer*. The assistant never acts on its own, never edits a word you did not ask it to edit, and never sees more of your book than you decide to show it. Everything in this chapter is machinery for that steering: how you point the model at a slice of the manuscript (the **scope**), how much of its own training you let it draw on (the **mode**), where its answer is allowed to land (the **destination**), and which of six providers is doing

the thinking. Learn these four dials and the AI pane stops being a chat box and becomes an instrument.

The pane itself you already met in Chapter 3, as one of the three surfaces the right-hand region can show. Here we open it up.

## Opening and focusing the pane

The AI feature is really two surfaces working as a pair: the **AI pane**, where answers stream in and the conversation scrolls, and the **AI prompt bar**, the thin input line along the bottom where you type. They are bound together so tightly that Inkhaven treats moving between them as a bounce rather than a full focus change.

To reach the prompt from anywhere, press **Ctrl+I** (or **Ctrl+5**) — focus drops to the AI prompt bar and you can start typing immediately. To read and scroll the answer, **Ctrl+3** focuses the AI pane itself. Once you are on one of the two, **Esc** *bounces* you to the other: **Esc** from the prompt bar lifts focus up into the pane so you can scroll the history; **Esc** from the pane drops focus back to the prompt so you can type a follow-up. This AI ↔ AI-prompt pairing is its own little loop, independent of the Tree / Editor / Search rotation that **Tab** walks.

### TERM The bounce

From the AI prompt bar, **Esc** moves focus to the AI pane (to read); from the AI pane, **Esc** moves focus back to the prompt bar (to type). You ping-pong between asking and reading with one key, and never fall out of the AI surfaces by accident.

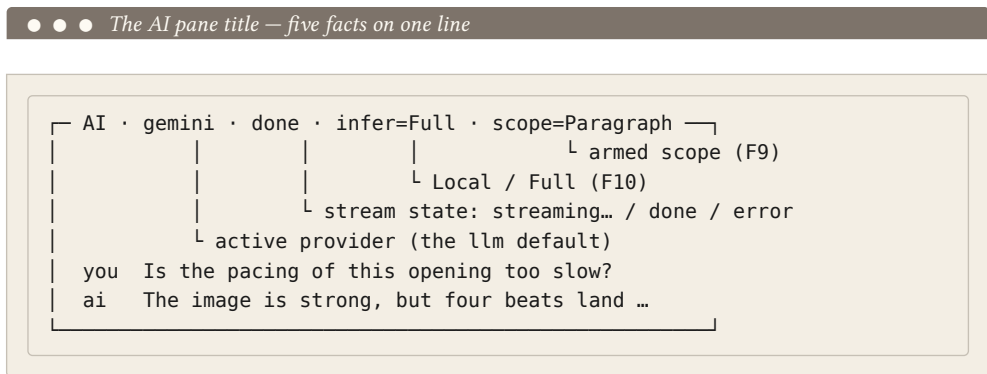
If the right region is currently showing the Output or Thoughts pane instead, **Ctrl+B Tab** cycles it around to the AI pane (Output → AI → Thoughts), or **Ctrl+3** jumps straight there and surfaces it. And when a conversation grows long enough that the four-pane layout feels cramped, **Ctrl+B K** fullscreens the AI pane — the chat history and the prompt bar spread across the whole window for an extended session. Press it again to return to four panes.

|                        |                                                         |
|------------------------|---------------------------------------------------------|
| <b>Ctrl+I / Ctrl+5</b> | Focus the AI prompt bar (bottom). Start typing at once. |
| <b>Ctrl+3</b>          | Focus the AI pane and surface it in the right region.   |
| <b>Esc</b>             | Bounce between the AI pane and the AI prompt bar.       |

|                   |                                                              |
|-------------------|--------------------------------------------------------------|
| <b>Ctrl+B Tab</b> | Cycle the right region Output → AI → Thoughts.               |
| <b>Ctrl+B K</b>   | Fullscreen the AI pane with its history and prompt (toggle). |
| <b>Enter</b>      | Send the prompt (from the prompt bar).                       |

## The title bar — the pane's state at a glance

The AI pane wears a dense title strip, and it repays a glance. Read left to right, it folds in everything about the pane's current state: the provider doing the work, whether a stream is in flight, the inference mode, the armed scope, and how deep the conversation has run.



While tokens are arriving the stream field reads `streaming...`; when the model finishes it flips to `done`; if the call fails it shows `error` and the failure text lands in the pane body. Once a conversation is under way the strip also carries the turn depth — `3 turn(s)` — so you know how much history the next prompt will replay. Two of the fields are colour-coded by default so an accidental setting is obvious at a glance: the scope chip is peach (`theme.ai_scope_fg`) and the inference-mode chip is teal (`theme.ai_infer_fg`). When prompt-language resolution is active a `lang=` chip joins them, reading `ru (book)` or `ru (paragraph)` depending on how the language was decided (see the last section of this chapter).

The prompt bar has a title of its own, and it echoes the armed scope so you see it while you type: `AI prompt · scope: Paragraph`. You rarely read either strip word for word, but between them you are never guessing what the next `Enter` will do.

## Streaming inference

Press `Ctrl+I`, type a question, press `Enter`, and the answer begins to appear immediately — not a spinner, then a wall of text, but characters arriving left to right as the model produces them. Inkhaven streams the tokens into the pane as they land. This is worth internalising: the model is *literally* writing the answer one piece at a time, and you are watching it think, so you can start reading — or decide the answer is going the wrong way and clear it — before it finishes.

● ● ● *A streaming answer, tokens arriving live*

```
AI · gemini · streaming... · infer=Full
Several darker alternatives:
• stunned sailor — keeps the alliteration
• shaken helmsman — slightly nautical
• dazed deckhand — leans on disorientation

"Thunderstruck" reads pre-20th-century; if the |
```

The pane renders the response as *markdown* as it streams: `*bold*`, `_italic_`, `inline code`, `# headings`, bullet and numbered lists, code fences, and block-quotes all display as formatted text rather than raw asterisks and hashes. That rendering is purely a display convenience — the raw markdown is what Inkhaven actually stores and what flows through the apply pipeline described below.

Underneath, the status bar narrates the call. While the tokens arrive it reports something like `streaming from gemini (chat turn #3 · scope=Paragraph)...`; when the model finishes it prints the elapsed time, `gemini responded in 1.8s`. If the call fails — a missing API key, a mistyped model name, a network hiccup — the status flips to an error, the title carries `error`, and the error text sits in the pane body. Fix the cause and re-send; nothing is lost.



## READING WHILE IT WRITES

You do not have to wait for `done`. Scroll the pane (`Ctrl+3`, then the arrows or the wheel) while a stream is still running, or type your next question into the prompt bar — Inkhaven queues nothing behind your back, so a follow-up sends a fresh call once the current one settles.

## Scope — telling the AI what to look at

By default the model sees only the prompt you typed plus the running chat history. It knows nothing about your manuscript unless you attach a slice of it. The dial that attaches that slice is the *scope*, and you cycle it with a single key: `F9`, from any pane.

### TERM Scope

The *scope* decides what chunk of your book is prepended to the next prompt as context. `F9` cycles it. A non-`None` scope prepends its matching content to the query you send, then — for the manuscript scopes — *auto- resets to* `None` after that one submission, so a follow-up prompt is never surprised by stale context you forgot was armed.

`F9` walks a ten-stop ring and wraps:

● ● ● The `F9` scope ring

None → Selection → Paragraph → Subchapter → Chapter →  
Book → Facts → Socrates → Editor → Graph → None

The armed scope shows in three places while it is set: the AI pane title (`scope=Paragraph`), the prompt-bar title (AI prompt · `scope: Paragraph`), and the status bar, which spells out AI scope: Paragraph (will prepend matching context to next prompt). After you send, a manuscript scope's chip disappears as it resets to `None`.

### The manuscript scopes

The first six steps draw text from the tree around your cursor — progressively wider slices of the book you are actually writing.

• • • *What each manuscript scope sends*

| Scope      | Sends to the model                                                              |
|------------|---------------------------------------------------------------------------------|
| None       | Nothing extra — the prompt + prior chat.                                        |
| Selection  | The current editor selection. Errors if no selection is active.                 |
| Paragraph  | The whole open paragraph. In split-edit, both the snapshot and the live buffer. |
| Subchapter | Every paragraph under the cursor's enclosing subchapter.                        |
| Chapter    | Same, for the enclosing chapter.                                                |
| Book       | Same, for the enclosing book — but retrieval-grounded (see below).              |

Two of these have a wrinkle worth stating plainly. *Selection* errors out with a status message if there is no active selection — it has nothing to attach, so it refuses rather than send an empty scope. And *Book* does not blindly stuff an entire novel into one prompt; that would blow past every model's context window and cost a fortune. Instead it is *retrieval-grounded*: Inkhaven retrieves the passages most relevant to your question from the book's semantic index and cites those, rather than sending the whole text. A book-wide question therefore stays affordable and pointed, at the price of the model seeing the relevant passages rather than literally every word.

FICTION

Match the scope to the question's reach. Brainstorming one sentence? *Selection* or *None*. Tightening a paragraph? *Paragraph*. Checking that a scene stays consistent with itself? *Subchapter*. Chasing a plot thread across a chapter? *Chapter*. A whole-book sweep? *Book* — long, and best on a large-context provider.

NON-FICTION

For reference and documentation the same ladder holds: *Selection* to reword a clause, *Paragraph* to fact-check a claim in place, *Chapter* to ask whether a section contradicts an earlier one, *Book* to ask a question that ranges across the whole document and let retrieval find the relevant passages.

## The structured scopes — Facts, Socrates, Editor, Graph

The last four stops on the ring are a different animal. Where the manuscript scopes attach *prose from around your cursor*, these attach a *curated body of knowledge* and pre-seed the conversation with a matching system prompt. And unlike the manuscript scopes, they are *sticky*: they persist across follow-up prompts until you cycle **F9** away from them, because they open a standing conversation rather than a one-shot attachment.

### ● ● ● The four structured scopes

| Scope    | Opens a conversation with...                                                                                                    |
|----------|---------------------------------------------------------------------------------------------------------------------------------|
| Facts    | every paragraph of the Facts system book – the world's invariants – as ground truth, seeded with a fact-analysis system prompt. |
| Socrates | the open paragraph read by the active Reader Persona (the Inner Socrates).                                                      |
| Editor   | the Inner Editor's craft observations on the open paragraph.                                                                    |
| Graph    | your knowledge graph – relevant passages plus the edges that connect them (contradicts / sourced_from / cites / ...).           |

These are the doorways into Inkhaven's reading intelligences from the AI pane, and each has a whole world behind it. *Facts* turns the AI pane into an interrogation of your worldbuilding — ask whether a claim holds against the established canon and the model answers from your Facts book, not from the internet. *Graph*, like **Book**, is retrieval-grounded, but it additionally folds in the graph relations touching each retrieved passage, so the answer is grounded in how the book's parts *connect*, not just in the prose. *Socrates* and *Editor* are the conversational faces of the Inner Socrates and Inner Editor respectively. All four are covered in full in Chapter 10 and the companion *Know Your Book* here it is enough to know that **F9** reaches them and that they stay armed until you cycle past **Graph** back to **None**.

### STICKY VS. RESETTING

The six manuscript scopes reset to `None` the instant you send — one prompt, one attachment. The four structured scopes (`Facts`, `Socrates`, `Editor`, `Graph`) stay armed across every follow-up until you press `F9` to leave them. The pane title's `scope= chip` is your reminder of which world you are still talking to.

## Mode — local-only RAG versus full knowledge

Scope decides *what the model sees*. Mode decides *how much of its own training it may lean on*. There are two settings, and `F10` toggles between them from any pane.

### ● ● ● The two inference modes

| Mode  | The model is told...                                                                              | Reach for it when...                                                                                     |
|-------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Local | use ONLY the supplied context and prior chat — refuse rather than fall back on general knowledge. | summarising your own canon, fact-checking against worldbuilding, working strictly inside the manuscript. |
| Full  | treat the context as ground truth where present, but general knowledge is fair game.              | brainstorming, craft questions, anything that benefits from the model's wider reading.                   |

The default is *Full*, because most fresh chats are exploratory and you *want* the model to suggest a trope, name a craft book, or reach for a comparison. Flip to *Local* when you are asking the model to work only from what you have shown it — “summarise what the `Facts` book says about the northern climate” should never be answered from the model’s guess about real-world northern climates. The active mode is always shown in the title (`infer=Local` / `infer=Full`, teal by default), precisely so an accidentally- armed `Local` mode is never a silent surprise.

**TERM Local mode is not privacy**

*Local* mode constrains what the model may *draw on* — its own training versus your context. It does *not* mean the call stays on your machine. With an external provider, a **Local** -mode prompt still travels to that provider's servers. On-device privacy is a *provider* choice (Ollama), not a mode choice — see the providers section below.

Two flows override the toggle. Help inferences (**F1**, or a **Help!** prefix in the prompt bar) and grammar checks (**F7**) are *pinned to Local* no matter where **F10** sits — each has its own strict system prompt, and neither should ever invent a feature or paraphrase a rule from general training data.

**Destination — where an answer lands**

An answer in the pane is inert until you decide what to do with it. When an inference is *done* and has non-empty content, a row of action chips appears in the pane's footer, and each key sends the response somewhere different. The chip keys fire only when the *AI pane itself is focused*. After you press **Enter** to send a prompt, focus stays on the prompt bar, so pressing **r** there just types the letter — press **Esc** (or **Ctrl+3**) first to focus the pane, then press the chip key.

● ● ● *The apply chips, shown when an inference is done*

r replace   i insert   t top   b bottom   c copy   g grammar

• • • What each destination does

| Key | Destination                                                                                                |
|-----|------------------------------------------------------------------------------------------------------------|
| r/R | Replace – overwrite the editor selection with the AI text. With no selection, replace the whole paragraph. |
| i/I | Insert – drop the text in at the cursor.                                                                   |
| t/T | Top – prepend to the paragraph (blank-line gap).                                                           |
| b/B | Bottom – append to the paragraph.                                                                          |
| c/C | Copy – to the clipboard only; the buffer is not touched.                                                   |
| g/G | Grammar – lift the corrected text from an F7 grammar-check result (see Chapter 6 / grammar).               |

Three things hold for every apply. First, *markdown becomes Typst* on the way in: when the answer lands in the editor (`r`, `i`, `t`, `b`), Inkhaven converts its markdown to Typst syntax — `#` headings become `=`, `*bold*` stays `*bold*` (Typst’s own bold), `1.` lists become `+`, and so on — so the pasted text is valid manuscript markup, not raw markdown. Only `c` keeps the raw markdown, precisely so you can paste it somewhere outside Inkhaven. Second, an apply sets the buffer *dirty* press `Ctrl+S` to commit it (or let idle autosave do so). Third, after `r` / `i` / `t` / `b` the *focus jumps to the editor* so you land where the change did and can review it in context.

#### THE GRAMMAR CHIP IS SPECIAL

`g` is not a general “apply”. It only works on the output of the F7 grammar check, whose response is wrapped in markers; `g` lifts the corrected text from between those markers and applies just that. On any other inference `g` has nothing to lift. The full grammar-review flow, with its change highlighting, lives in Chapter 6.

## Chat history — the conversation is continuous

Every ordinary inference (everything except a one-shot Help or grammar call) appends a `(you, ai)` turn to an in-memory chat history, and the next prompt replays the whole history to the model. The conversation is therefore *continuous*: when you say “now do another pass, especially on the dialogue tags,” the model

knows what “this” was because the previous turn is replayed along with it. The title’s **N** **turn(s)** chip tracks the depth.

A typical multi-turn pass reads like a dialogue with an editor who remembers:

● ● ● *A continuous revision loop*

1. F9 → Paragraph. Prompt: "Tighten this." Enter.
2. Esc – focus the pane (chips only fire there).
3. r – replace the paragraph with the tightened text.
4. Prompt: "Now another pass, harder on the dialogue tags." Enter. (History knows what "this" was.)
5. Esc, then r – focus the pane and apply again.
6. Ctrl+B C – clear the thread and start fresh.

**Ctrl+B C** clears the chat history along with the currently displayed inference — the way to start a genuinely fresh conversation when the old context would only confuse the next question. From the AI pane it doubles as the cancel key for a stream you no longer want. (Help answers never enter the history at all — they are deliberately one-shot, so a question about Inkhaven itself never pollutes the thread you are having about your prose.)

## Prompt templates and the Help! prefix

Two shortcuts live in the prompt bar itself. Typing **/** opens the *prompt picker*, a list of reusable templates drawn from two places: the **prompts.hjson** file in your project root (shown as *system* prompts) and the paragraphs under the *Prompts* system book (shown as *book* prompts). Arrow keys move, **Enter** or **Tab** expands the chosen template into the bar — with any **{{selection}}** / **{{context}}** substitutions already applied — and you can edit the expanded text before sending. Direct invocation works too: type **/tighten** and **Enter** with no picker at all. **Esc** closes it without expanding. The full template schema is its own subject; the point here is that a prompt you type often need never be typed twice.

The second shortcut is the **Help!** prefix. Typing **Help!** followed by a question routes the rest of the line through the **F1** help-manual flow — a retrieval-grounded answer over Inkhaven’s own Help book, pinned to **Local** so it never invents a feature — without your leaving the prompt bar. The capitalisation is exact: **Help!**,

not `help!` or `HELP!`, so an actual sentence that happens to start with the word “help” is never hijacked.

## Providers — five pre-configured, and any the router reaches

Behind the pane sits a *provider*: the company or runtime that actually runs the model. Inkhaven speaks to all of them through the `genai` crate, which picks the right adapter from the *model name* alone — so adding a new model is a matter of naming it, not writing code. Five providers ship pre-configured in the default `llm.providers`; Ollama is supported (keyless, local) — add it yourself.

### ● ● ● The five pre-configured providers

| Provider | Default model     | API-key env var   |
|----------|-------------------|-------------------|
| gemini   | gemini-2.5-pro    | GEMINI_API_KEY    |
| claude   | claude-sonnet-4-5 | ANTHROPIC_API_KEY |
| openai   | gpt-4o            | OPENAI_API_KEY    |
| deepseek | deepseek-chat     | DEEPSEEK_API_KEY  |
| grok     | grok-2-latest     | XAI_API_KEY       |

You configure them in the `llm` block of `inkhaven.hjson`. Each entry names a `model` and, for the cloud providers, the environment variable that holds your key; `default` names which one the TUI uses. Local runtimes like Ollama omit the key entirely — the auth check is simply skipped when `api_key_env` is absent.

### ● ● ● The llm config block

```
llm: {
  default: gemini          # which provider the TUI streams from
  auto_fallback: true      # use any working key if default's unset
  providers: {
    gemini: { model: gemini-2.5-pro, api_key_env: GEMINI_API_KEY }
    claude: { model: claude-sonnet-4-5, api_key_env: ANTHROPIC_API_KEY }
    deepseek: { model: deepseek-chat, api_key_env: DEEPSEEK_API_KEY }
    ollama: { model: llama3.2 }      # no key – runs locally
  }
}
```



Set a key by exporting its variable before you launch — `export GEMINI_API_KEY='...'` — and Ollama needs nothing but the daemon running and a model pulled (`ollama pull llama3.2`). The `default` provider is the one the TUI streams from; to change it, edit `default` and relaunch. There is no per-inference provider switch inside the TUI today — the model is a session-level choice — but the CLI takes a one-shot override: `inkhaven ai "summarise this" --provider deepseek`.

#### TERM `auto_fallback`

When `auto_fallback` is `true` (the default) and your *default* provider's key is unset, Inkhaven quietly uses any other configured provider whose key is available — including a keyless local one like Ollama — rather than failing. It is the “use whatever works” setting. Set it to `false` if you want the configured provider or a clear error, and nothing else.

#### PRIVACY — EXTERNAL PROVIDERS SEE YOUR PROMPTS

This is the one privacy fact to internalise. When you use an external provider (Gemini, Claude, OpenAI, DeepSeek, Grok), your prompts — including whatever manuscript context the scope attaches — *travel to that provider's servers* under their terms. Inkhaven adds no inherent privacy over that channel, and Local mode does not change it. For genuine on-device privacy, set the default to `ollama` and run a local model: the prompt never leaves your machine. Every *other* Inkhaven subsystem — the RAG embeddings, semantic search, snapshot diffs — is already fully local, so switching to Ollama makes the whole tool on-device.

## Prompt-language resolution

Inkhaven is multilingual to its core, and the AI pane is no exception: a prompt about a Russian paragraph should be answered in Russian, keyed off Russian word lists and a Russian-aware system prompt. The dial that decides which language a prompt resolves against is *prompt-language resolution*, set

by `editor.prompt_language_mode` in HJSON and toggled at runtime with `Ctrl+B Shift+N`.

There are two strategies. *book\_defined* (the default) resolves *every* prompt against the project's top-level `language` field — simplest, and right for a manuscript written in one language. *paragraph\_detected* instead runs the `whatlang` detector over the *live* paragraph body and resolves against whatever language it finds — the mode for a genuinely bilingual project where one chapter is English and the next is Russian. Because `whatlang` is unreliable on very short text, `paragraph_detected` only attempts detection once the paragraph has at least `prompt_language_detection_min_chars` of non-whitespace text (default 50); below that floor it silently falls back to the book language.

`Ctrl+B Shift+N` cycles the mode *for the session* without rewriting your HJSON: `None` (defer to the config) → `book_defined` → `paragraph_detected` → `None`. The AI pane title's `lang=` chip reflects the outcome so you always know which way a prompt was resolved — `ru (book)` when the book language decided it, `ru (paragraph)` when detection did — and the status bar echoes the new mode as you toggle.

● ● ● *The lang chip records how the language was chosen*

```
AI · gemini · done · infer=Full · lang=ru (paragraph)
you Перепиши этот абзац живее.
ai Дождь хлестал наискось с гавани, и Мара ...
```

## WHAT YOU LEARNED

- The AI feature is a *pair*: the AI pane (answers stream in, history scrolls) and the AI prompt bar (you type). `Ctrl+I` focuses the prompt, `Ctrl+3` the pane, and `Esc` *bounces* between them; `Ctrl+B K` fullscreens the pane.
- *Scope* (`F9`) decides what the model sees. Six *manuscript* scopes — `None` / `Selection` / `Paragraph` / `Subchapter` / `Chapter` / `Book` — attach prose and auto-reset after one send (`Book` is retrieval-grounded). Four *structured* scopes — `Facts` / `Socrates` / `Editor` / `Graph` — open sticky conversations with the reading intelligences (Chapter 10).
- *Mode* (`F10`) decides how much the model leans on its own training: *Local* uses only the supplied context, *Full* (the default) may augment with general knowledge. Help and grammar are pinned to Local. Local is *not* privacy.
- *Destination* chips land a done answer: `r` replace, `i` insert, `t` top, `b` bottom, `c` copy (raw markdown), `g` grammar-apply. Editor applies convert markdown → Typst and set the buffer dirty.
- Six providers are bundled — Gemini, Claude, OpenAI, DeepSeek, Grok, Ollama — plus any model genai routes; the `llm` block sets `default`, per-provider models and key env vars, and `auto_fallback`. External providers *see your prompts* Ollama keeps them on-device.
- Prompt-language resolution (`Ctrl+B Shift+N`) picks the language a prompt answers in: *book\_defined* (the project language) or *paragraph\_detected* (`whatlang` over the live paragraph, above a 50-char floor); the title's `lang=` chip records which decided.

## CHAPTER 10

## 10

Chat With Your Book

---

An assistant that has never read your book can only ever guess at it. Ask a general model whether Mara doubts the captain and it will write you a fluent, confident paragraph about a character it has never met — plausible, ungrounded, and wrong as often as not. The panes you met in Chapter 3 already give you a conversation; this chapter is about what that conversation is **grounded in**. The whole point of writing inside your own material is that the assistant can be made to answer **from** it — to retrieve the passages that actually bear on your question, quote them back by name, and refuse to invent what the text does not say. That is the difference between chatting **with** a model and chatting **with your book**.

The control that governs all of this is a single key, **F9**, which cycles the AI pane's **scope**. You have already met the near end of that dial — the scopes that send the model a slice of prose around your cursor. This chapter is about the far end: the **structured scopes**, where the pane stops shipping raw text and starts **retrieving**, grounding each answer in evidence it can cite. Book, Facts, Graph, and the two reader conversations all live there, and each grounds in a different part of what Inkhaven knows about your work.

## The scope dial — F9 and what an answer stands on

Every question you send the AI pane carries a scope, and the scope decides what context travels with it. Press **F9** from anywhere and the scope advances one notch; the pane's title and the status bar both name the current one ( `scope=Book` ). Ten stops sit on the ring, and they fall into two families.

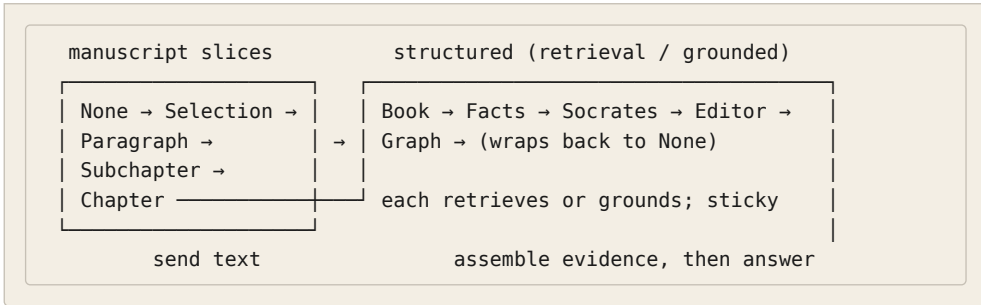
### TERM AI scope

The **scope** is what the AI pane sends the model alongside your prompt. **F9** cycles it. The first five stops are **manuscript slices** — a literal span of prose around your cursor. The last five are **structured scopes** — Book, Facts, Socrates, Editor, and Graph — which retrieve or assemble grounding rather than sending text verbatim, and are **sticky**: the conversation stays in that scope, reasoning over the same evidence, until you cycle away.

The first family is the plain one. **None** sends nothing but your words; **Selection** sends the highlighted span; **Paragraph**, **Subchapter**, and **Chapter** send exactly the prose those names enclose. What you see is what the model gets. These are covered where they belong, in the editor chapters; they are unchanged and unsurprising.

The second family is the subject of this chapter, and it behaves differently on purpose. A Book-scope question does not send your book — it **searches** it. A Facts-scope question loads the world's invariants as ground truth. A Graph-scope question folds in how your book's parts connect. Socrates and Editor seat a particular reader across the table from you. None of these send a fixed span of text; each **builds** its grounding from a different store, and each is worth understanding on its own.

● ● ● F9 walks the scope ring; the last five are structured



### STICKY SCOPES RETRIEVE ONCE

A structured scope holds its ground across a whole conversation. It gathers its evidence when you enter it (or send the first question), and your follow-ups reason over that same evidence rather than re-gathering on every turn. To make it look again — a new line of inquiry, or freshly edited prose — clear the chat history, and the next question re-grounds from scratch.

## Book scope — retrieval-augmented conversation

Book scope is the flagship of this chapter, and the feature it implements has a name worth learning, because it is the same idea that quietly powers half the tool.

### TERM Retrieval-augmented generation (RAG)

Instead of handing a model everything and hoping it finds the relevant part, **retrieval-augmented generation** first **retrieves** the handful of passages that actually bear on the question — by semantic similarity, not keyword match — and hands the model only those, as labelled evidence to reason from. The answer is grounded in real, quotable text rather than in the model's diffuse memory of everything it has ever read.

Book scope used to be the tool's weakest surface. It sent your whole manuscript along with every question — fine on a ten-page draft, ruinous on a real book. The model skimmed a hundred pages, grounded in none of them, and answered from a blur. Modern Inkhaven replaces that with RAG, and the change is total: Book scope

is retrieval now. There is no switch to flip and no “send the whole book” mode to fall back to, because the whole-book mode never worked at scale.

Here is what happens behind a Book-scope question. Inkhaven runs a semantic search for your query over the same on-save vector index every other semantic feature uses, keeps the most similar paragraphs, widens each one with a neighbour or two so it reads in context rather than as a fragment, drops duplicates, and packs the result into a focused, token-budgeted block of evidence — best hits first, ordered by their position in the book. That block, and only that block, is what the model sees. Then it answers.

● ● ● *A question in Book scope, and a cited answer*

AI · Scope: Book —  
 | > does Mara ever doubt the captain?\_ |

She does, twice. Her first real doubt surfaces on the crossing, when the captain orders the night run despite the glass falling — she calls it "a wager with other men's lives" [act-two/the-night-run]. By the storm chapter that doubt has hardened into open defiance [act-three/storm-at-sea].

You asked a question about your book and got back specific passages, by name, instead of a plausible paragraph the model invented. Every claim it makes about the book is anchored to a citation — and those citations are the heart of why the answer is trustworthy.

### Citations you can trust — and the ones you can't

Each bracketed label is a passage's **location path** — [chapter-slug/scene-slug] — not an opaque database id. You can read **where** a claim comes from and jump straight to it. The system prompt makes the contract explicit and ships localised for the five baseline languages (English, Russian, Spanish, French, German, with English as the fallback): ground every claim about the book in a retrieved passage, cite it by repeating its location label exactly, and never invent a label you were not given.

That last clause is not left to the model's good behaviour. Inkhaven scans the answer for citation-shaped tokens and checks each against the passages it actually

retrieved. Any location the model cites that **was not** in what it was handed is flagged inline, right where it appears:

● ● ● *A hallucinated citation, caught and flagged*

She returns in [act-two/the-storm] and again later in [act-three/the-reckoning] [citation could not be validated: act-three/the-reckoning].

The first citation was in the evidence, so it stands untouched. The second was not — the model reached past what it was given — and the flag says so plainly. A Book-scope answer whose citations are all clean is one you can trust down to the paragraph; one with a flag is telling you exactly which sentence to distrust. This is a structural guarantee, not a stylistic nicety: the check runs on every answer, and it leaves ordinary bracketed prose (a [note], an [aside]) alone, because a real citation contains a slash and no spaces and an aside does not.

When nothing in your book addresses the question, the model is instructed to say so — “the retrieved passages don’t address that directly” — and then either ask you to sharpen the question or offer general knowledge **clearly marked as not from the book**. It confabulates far less readily when the honest answer is built into its instructions.

### Seeing what it retrieved — the transparency toggle

You never have to take the grounding on faith. Above the conversation, a collapsed line reports how many passages grounded the answer; press **p** to expand it (and **p** again to collapse). Expanded, it lists each passage’s similarity score, a ★ for a **direct hit** versus a context-expansion neighbour pulled in around one, the passage’s location path, and its opening line.



● ● ● *p expands the retrieved-passages transparency section*

```
▼ Retrieved passages (6) · p to collapse
★ 0.851 manuscript/act-ii/storm-at-sea
    On the ninth night the storm took them.
    Waves the height of...
0.806 manuscript/act-i/the-harbour
    The harbour at Vell was a forest of masts...
...
(retrieved once for this chat – clear history to
retrieve again)
```

That closing line is the rule that keeps a conversation coherent. Retrieval runs **once per chat session**, so your follow-ups reason over a stable set of passages instead of yanking the ground out from under the thread with every question. The transparency section is the honest window onto that set: if an answer surprises you, expand it and see the evidence the model was actually standing on.

### Re-grounding after you edit

Because a chat holds its retrieval, editing the book underneath a live conversation creates a small hazard — the passages the model has are now the **pre-edit** prose. Inkhaven watches for exactly this. Save an edited paragraph while a Book conversation is open and it nudges you, once:

● ● ● *A gentle nudge when the book moves under a chat*

```
book changed since retrieval – clear chat to re-ground
on the new text
```

It is a reminder, not an interruption. The open conversation stays valid and usable; when you want the model to see the new prose, clear the history and the next question retrieves afresh. Re-grounding is always your call, never a surprise mid-thread.

## What's in the pool — the retrieval scope

Retrieval is anchored to the **user book your cursor is in** — its chapters, subchapters, and paragraphs. But a question about your story often wants your **notes**

about the story, so a curated set of author-content system books joins the pool. By default that means **Notes, Research, Places, Characters, Artefacts, World, and Language**. Ask “what is the significance of the brass key?” and a Characters or Artefacts entry can ground the answer right alongside the prose that mentions it.

The internal and meta system books stay out. Scripts, Prompts, Typst, Help, Intent, Sources, Glossary, and Snippets never enter retrieval — they hold the machinery of the project, not its content, and grounding a story question in a Bund script or a Typst template would only add noise. The split between “author content, searchable” and “project machinery, excluded” is what keeps a Book-scope answer about the **book**.

#### ONLY PARAGRAPHS GROUND AN ANSWER

Retrieval considers paragraph nodes only. A chapter or book node is a container, not prose, so the pool is always the actual writing — never a heading standing in for the scene beneath it. Each hit is expanded with its sibling paragraphs, so a retrieved passage arrives with the lines that surround it in the book.

## Tuning retrieval — the `book_rag` config block

Every dial on the retrieval sits in one config stanza, `book_rag`. There is no on/off knob — Book scope is RAG — so the block only shapes **how** it retrieves, never **whether**. Five keys, each doing one thing:

● ● ● *The `book_rag` block, shown with its defaults*

```
book_rag: {
  top_k: 5
  context_expansion: 1
  max_context_tokens: 8000
  include_system_books: [notes research places
    characters artefacts world language]
  exclude_system_books: [scripts prompts typst help
    intent sources glossary snippets]
}
```

`top_k` is how many paragraphs the semantic search keeps as direct hits — raise it to ground a broad question more widely, lower it to keep the evidence tight. `context_expansion` is how many sibling paragraphs are pulled in **around** each hit (a value of `1` means one on either side), so a passage reads in context instead of as an orphaned line. `max_context_tokens` is the hard ceiling on the assembled evidence block, estimated at roughly four characters to the token; when hits would overflow it, the best-scoring ones win and the rest are dropped. The two book lists are named by each system book's lowercase tag: `include_system_books` is the author-content set that joins the manuscript in the pool, and `exclude_system_books` is the meta set that never does — and exclusion wins, so a tag in both lists stays out. Trim the include list to a leaner pool, or extend it if you keep story content in a book not listed by default.

#### THE PERMISSIVE PRINCIPLE, HERE TOO

None of these are guardrails. They shape cost and focus — a smaller `top_k` and a tighter token budget make a cheaper, narrower call — but nothing in `book_rag` will ever refuse a question or cap a conversation. As everywhere in Inkhaven, the numbers inform; they never block.

## Inspecting retrieval from the terminal

Before you spend a single model call, you can see precisely what a question would retrieve — and confirm the grounding is sound — with `inkhaven book-rag retrieve`. It runs the **exact same** retrieval core the pane uses (they share one function, so they can never drift), but stops before the LLM: no answer is generated, no config is written, nothing is charged. It is the inspect-what-the-model-sees tool.

● ● ● *book-rag retrieve — the passages, no model call*

```
$ inkhaven book-rag retrieve \
  "what does Mara fear about the voyage?"
Book-RAG retrieval — `The Long Road`
(6 passages, 3 direct hits, ~288 tokens)

★ 0.806 manuscript/act-i/dawn-departure
    id: 019efc4b-ff42-7603-9a5d-4f83eedf50cb
    The ship slipped its moorings before sunrise...
    0.806 manuscript/act-i/the-harbour
        id: 019efc4c-05b9-7033-b7a2-7623b3112421
        The harbour at Vell was a forest of masts...
★ 0.851 manuscript/act-ii/storm-at-sea
    id: 019efc4c-0c27-7d61-9c1e-8df95471c961
    On the ninth night the storm took them...
```

Three flags shape the run. `--book-name <name>` picks the book in a multi-book project by title or slug (optional when there is only one). `--top-k <n>` overrides `book_rag.top_k` for this run only, leaving your config untouched — useful for feeling out how wide a question’s grounding gets before you commit. And `--context` prints the literal composed evidence block the model would receive — passage headers and full bodies, the exact `— Retrieved passages —` prefix — rather than the summary listing. When you want to know why an answer came out the way it did, that block is the ground truth it was built on.

## Facts scope — chat against the world's invariants

Cycle **F9** one past Book and you reach **Facts**, the odd sibling among the scopes. Where Book scope retrieves a **slice** of the manuscript, Facts scope loads the **whole** Facts system book — every established invariant of your world: its climate and geography, its distances and seasons, its chronology — as ground truth, and seeds the conversation with a fact-analysis system prompt. It is not a part of the book; it is the reference the whole book answers to, which is why it sits **after** Book on the ring rather than among the manuscript slices.

Reach for Facts scope when your question is about the world’s rules rather than its prose. “How long is the ride from Vell to Rillmark?” “Which season is it when the fleet sails?” “Is it plausible for snow to fall in the southern reach?” The answer

is grounded in what you have **established**, so the model reasons from your canon instead of a genre cliché.

#### TWO SIBLINGS SHARE THE FACTS GROUNDING

The same fact grounding powers two nearby chords. Ctrl+B Shift+X runs a one-shot **fact-check** of the open paragraph against every world fact, flagging any claim that contradicts the canon. Ctrl+B Shift+S opens a semantic **Facts search**: query the Facts book, mark the handful that matter, and send just those into a targeted chat — the scalable path when the Facts book grows too large to load whole. Both reuse the fact-analysis grounding the F9 Facts scope is built on. The world, and everything you can do with it, has its own chapter.

## Graph scope — chatting with how your book connects

The last stop on the ring is **Graph**, and it answers a different kind of question than any other scope: not what your book **says** but how its parts **connect**. It is the conversational face of the knowledge graph — the typed-edge layer over your nodes that records which fact contradicts which source, which paragraph links to which, how a scene is sourced. A flat manuscript cannot tell you what contradicts what; the graph can, and Graph scope lets you ask it in plain language.

A Graph-scope question retrieves the relevant passages exactly as Book scope does — same semantic search, same location-path citations, same hallucination flag — and then, beneath each passage, folds in the graph edges touching it: **contradicts**, **sourced\_from**, **links\_to**, **cites**, and the rest. The answer stands on both the prose **and** those relations. Like Facts, it is a sticky scope that retrieves once per chat and re-grounds when you clear history, and pressing **p** expands a transparency section showing the passages **and** the subgraph the answer was built on.

### Walking the graph — one hop, or many

Graph scope reads **one hop** — the edges immediately touching each retrieved passage. When a question needs the model to **follow** the graph — chaining from node to node, several relations deep — there is a heavier instrument: **graph ask**, the traversal loop.

● ● ● *graph ask — the model walks the graph turn by turn*

```
$ inkhaven graph ask \
  "which of my claims about the harbour
  contradict each other?"

(walk prints to stderr; the answer to stdout)
```

Given a question, **graph ask** searches for seed nodes, then issues read-only graph queries turn by turn — neighbours, contradictions, loci, paths — until it can answer, grounding the reply in what it actually observed. The exploration transcript prints to stderr so you can watch the path it took; the answer prints to stdout so you can pipe it. It is honest by construction: when the relations do not record what you asked, it says so rather than inventing a connection — the graph is only as complete as you have made it.

The same walk runs **inside the editor**, streamed live. Type a question in the AI prompt, then press **Ctrl+B z → w**, and the AI pane shows each step unfold — a search, a neighbours query, a contradiction check — before the grounded prose answer lands as a normal chat turn; **ESC** stops the walk at any point. Because a walk is several model calls where the one-hop Graph scope is one, it is an explicit action you opt into per question. Both the CLI and the in-editor walk are bounded by two knobs — **ask\_max\_steps** (the most LLM turns before a forced answer) and **ask\_search\_width** (seed nodes per search) — which, true to the permissive principle, cap the cost of a question without ever refusing it.

## The reader conversations — Socrates and Editor

Two scopes on the ring are not about grounding in data at all but in a **perspective**. Cycle past Facts and you reach **Socrates**, then **Editor** — the conversational modes of two of Inkhaven's inner readers. Where Book, Facts, and Graph ground the model in your **material**, these ground it in a particular way of **reading** that material, seated across the table to talk with you about the paragraph in front of you.

**Socrates** scope seats the active Reader Persona. The model reads the open paragraph in that persona's voice, carries in whatever questions the Inner Socrates has already raised about it, and discusses them with you — asking, never prescrib-

ing, and never rewriting your prose. It is the scope for interrogating a passage's assumptions rather than polishing its surface.

**Editor** scope seats the Inner Editor instead. It carries in that reader's literary and stylistic observations of the open paragraph — its notes on richness, tautology, vocabulary, and craft — and talks them through with you as a thoughtful editor would, observing rather than commanding. When you cycle into either scope, the status bar names it and tells you how many of that reader's findings are queued for the conversation; both are sticky, and re-cycling **F9** refreshes the seed from the paragraph you are now on.

#### THE READERS HAVE THEIR OWN CHAPTERS

Socrates and Editor scopes are the **conversational** face of the Inner Socrates and Inner Editor. What those readers notice, how they engage a paragraph, and how you tune them each fill their own chapters. Here it is enough to know that **F9** seats them for a chat: the same reader you meet as a pass over your prose, now willing to talk about what it found.

## Living with the chat — cost and habits

Every structured-scope answer is one retrieval (cheap and local — embeddings and a vector query, no network) plus one grounded model call, far less than shipping a whole manuscript ever cost. Those calls are tagged in the cost dashboard — **inkhaven cost**, or **Ctrl+B \$** — under the **book\_rag** category, so you can see across a day what chatting with your book actually costs. As with every figure in Inkhaven, it **informs**; no cap will ever refuse a question.

A few habits make the whole surface fluent. Cycle **F9** deliberately and read the scope off the status bar before you send — the same question means very different things in Book, Facts, and Graph. Press **p** when an answer surprises you and read the evidence it stood on. Clear the history when you change subject or edit the prose, so the next question re-grounds. And cycle **F9** back toward **None** when you want the model's plain help with no grounding at all — the far end of the dial is always one key away.

**F9** Cycle the AI scope one notch; the status bar names it.

**F9 → Book** RAG chat over the manuscript + author-content system books.

|                       |                                                                      |
|-----------------------|----------------------------------------------------------------------|
| <b>F9 → Facts</b>     | Chat grounded in the whole Facts book (the world's invariants).      |
| <b>F9 → Socrates</b>  | Converse with the active Reader Persona about this paragraph.        |
| <b>F9 → Editor</b>    | Converse with the Inner Editor about this paragraph's craft.         |
| <b>F9 → Graph</b>     | Chat grounded in prose + the graph edges that connect it.            |
| <b>p</b>              | In the AI pane: expand / collapse the retrieved-passages section.    |
| <b>Ctrl+B z → w</b>   | Walk the graph to answer the AI prompt, streamed live.               |
| <b>Ctrl+B Shift+X</b> | One-shot fact-check of the open paragraph against the Facts book.    |
| <b>Ctrl+B Shift+S</b> | Semantic Facts search → ground a chat in a chosen handful.           |
| <b>Ctrl+B \$</b>      | The cost dashboard — see the <code>book_rag</code> category's spend. |

## Scripting retrieval — the `ink.book_rag.*` words

The same retrieval that grounds the pane is reachable from a Bund script, so an automation can retrieve passages, compose the exact grounding block, inspect the config, and even validate an answer's citations without touching the TUI. Every word is **read-only** — nothing here mutates the store or calls a model; the retrieval is local, and generating the answer is left to the script author. All eight gate under the default-allowed `store_read` category, so a cautious project can disable the whole surface at once.

● ● ● *Retrieve and compose the grounding block from Bund*

```
"manuscript" "does Mara doubt the captain?"
  ink.book_rag.context .          # the composed evidence block

"manuscript" "the harbour market"
  ink.book_rag.retrieve          # list of {id, breadcrumb,
                                #          body, score, is_hit}
  dup ink.book_rag.cited_ids     # the valid citation tokens
```

`ink.book_rag.retrieve` takes an anchor (any book, or a node inside one) and a query and returns the passages as dicts; `ink.book_rag.context` returns the composed grounding block a model would receive; `ink.book_rag.scope`



returns the node ids in the retrieval pool; and `ink.book_rag.config` returns the live `book_rag` block as a dict. Four pure helpers round it out: `ink.book_rag.system_prompt` gives the localised grounding contract for a language, `ink.book_rag.estimate_tokens` the rough token count of a string, `ink.book_rag.cited_ids` the valid citation tokens of a passage set, and `ink.book_rag.validate_citations` runs the same hallucination check the pane runs — flagging, in a candidate answer, any bracketed location the passages do not support. The knowledge-graph side is scriptable too, through the `ink.graph.*` words, though the LLM-driven `graph ask` is not a synchronous word.

## WHAT YOU LEARNED

- The AI pane's **scope**, cycled with **F9**, decides what an answer is grounded in. The first five stops send a **slice of prose**; the last five — **Book, Facts, Graph, Socrates, Editor** — are **structured scopes** that retrieve or assemble evidence and are **sticky** until you cycle away.
- **Book scope** is retrieval-augmented generation: it semantically searches your manuscript, keeps the top passages, expands and token-budgets them, and grounds the answer with **location-path citations** — flagging inline any citation the passages do not support.
- Retrieval runs **once per chat**; press **p** to see the passages it stood on, and clear the history to **re-ground** after you edit — Inkhaven nudges you once when the book changes under a live conversation.
- The pool is your **user book** plus the **author-content system books** (Notes, Research, Places, Characters, Artefacts, World, Language); the meta books are excluded. Tune it all in the `book_rag` block — `top_k`, `context_expansion`, `max_context_tokens`, and the two book lists.
- `inkhaven book-rag retrieve` shows exactly what a question would retrieve with **no model call**; `--top-k` and `--context` inspect the grounding, and the CLI shares the pane's retrieval core so they never disagree.
- **Facts** scope grounds in the world's invariants; **Graph** scope grounds in how the book's parts connect (with `graph ask / Ctrl+B z → w` to **walk deeper**); **Socrates** and **Editor** seat an inner reader for a conversation about the open paragraph.
- The whole surface is scriptable through the read-only `ink.book_rag.*`. Bund words — retrieve, compose the grounding block, inspect the config, and validate citations — and its cost is tagged under `book_rag`, informing but never blocking.

## CHAPTER 11

## 11

Reusable Prompts

---

You will find yourself typing the same instruction to the assistant again and again. *Tighten this without changing the meaning. Make the tone darker but keep every fact. Read this paragraph for places I am telling instead of showing.* A good instruction is a small piece of craft — it took you three tries to word it so the model stops adding new plot — and retyping it from memory each time is both a chore and a slow erosion of the phrasing that worked. Inkhaven's answer is the **reusable prompt**: write the instruction once, give it a short name, and from then on summon it in two keystrokes, with the passage you are working on already stitched in. This chapter is the whole of that system — where prompts live, how you reach them, what gets

substituted into them, and how the same machinery quietly powers the assistant's built-in passes like grammar-check and fact-check.

## Why keep a prompt around

A reusable prompt is nothing more than a saved block of text with two conveniences bolted on. First, it has a **name** you can type — `/tighten`, `/darker` — so you never scroll a history buffer looking for the wording you liked. Second, it carries **placeholders** that Inkhaven fills in at the moment you send it, so the same template works on whatever paragraph is open without you pasting anything. Between those two, a prompt you tuned once becomes a verb: you *tighten* a passage the way you *save* a file.

### TERM Prompt

A named, reusable instruction to the AI assistant. It is plain text plus the placeholders `{{selection}}` and `{{context}}`. When you dispatch it, the placeholders are replaced with the passage you are working on and its location in the book, and the result lands in the AI prompt bar ready to send — nothing is sent until you press Enter.

Prompts are advisory, like everything the assistant does. Expanding a prompt fills the input bar; it never edits your prose on its own. You see the fully substituted text, you can edit it, and only your Enter sends it. That is the same contract the whole AI layer honours (Chapter 9): the model proposes, you dispose.

## Two libraries: the file and the book

Inkhaven draws prompts from two places at once, and the picker shows them merged into a single list.

The first is a plain file, `prompts.hjson`, in the root of your project. These are your **system prompts** — portable, versionable templates you carry between books. `inkhaven init` seeds the file from a shipped default, and you edit it with any text editor (or the built-in editors described later). In the picker they wear a cyan `[ system ]` chip.

The second is a system book named **Prompts**, sitting in the Tree alongside Notes, Facts, and the rest. Every paragraph you file under it becomes a prompt. These are your **book prompts** — project-local, authored in the TUI like any other paragraph, and carried inside the manuscript itself. They wear a green [ book ] chip.

#### TERM The Prompts system book

One of Inkhaven’s built-in system books. Paragraphs nested under it are not prose — they are prompt templates. A paragraph’s **slug** supplies its `/name`; its **title** supplies the description shown in the picker; its body is the template. System books are excluded from the manuscript corpus (word counts, concordance, exports), so your prompts never leak into the book.

The two libraries serve different instincts. Keep in `prompts.hjson` the **signature** templates you want in every project — your house style for tightening, your favourite critique framing. Keep in the Prompts book the templates that only make sense *here*, because they lean on this manuscript’s voice, its glossary, its world. When the two disagree — a `/tighten` in both — the book copy wins, on the principle that the project-local, hand-authored version is the more deliberate one. The exact ladder is spelled out under *Localized prompts* below, because it interleaves with language; the short form is **Prompts book, then `prompts.hjson`, then Inkhaven’s embedded default**.

## Opening the picker

There is no chord for the prompt picker, and you do not need one: it is bound to the character that already means “I am about to name something,” the forward slash. Focus the AI prompt bar with `Ctrl+I` and type `/`. A floating magenta panel opens just above the bar, listing every prompt from both libraries. Keep typing and the list filters live — the text after the slash is matched, case insensitively, against both the name and the description of each prompt.

● ● ● The prompt picker, filtered to `/ti`

```
+-- Prompts -----+
| [ system ] /tighten |
|           Tighten the prose without changing meaning |
| [ book ]   /tighten-quay-voice |
|           Tighten, in the harbour-chapter register |
| [ system ] /timeline-health |
|           Review the story timeline for consistency |
+-----+
AI prompt > /ti|
```

The panel is routed to the prompt bar — it has no separate focus, so the same keys that edit the bar also drive the list.

**Ctrl+I** Focus the AI prompt bar (from any pane).

**/** As the first character, opens the picker; refiltered as you type.

**Up / Down** Move the selection through the list.

**Enter / Tab** Expand the selected template into the bar (substituted).

**Esc** Close the picker without expanding anything.

Selecting a prompt with Enter or Tab does **not** send it. It expands the template — running the substitutions, stripping any editor chrome — and drops the finished text into the prompt bar, leaving the picker. Now you can read exactly what the model will receive, edit it (append a one-off note, delete a clause), and press Enter when you are satisfied. The status line confirms the pick: `loaded prompt tighten [system] – Enter to send`.

When the filter is empty the list keeps its natural order — system prompts first, then book prompts. As you type, matches are ranked: a prompt whose **name** begins with what you typed sorts above one where a later **description** word begins with it, which sorts above a mere substring hit anywhere. So `/ti` surfaces `tighten` before it surfaces a prompt merely *described* as “review the timeline.”

## Direct invocation: slash a name

Once a prompt’s name is in your fingers, you can skip the picker entirely. Type the whole name yourself and send:

● ● ● Dispatching a prompt by name

AI prompt > /tighten

Enter looks the name up — system library first, then the Prompts book by slug or title — expands the template against the current selection, and sends in one motion. If nothing matches, the input is left alone and the status line says **no prompt tighten — type / to see the list**, so a typo never fires a blind inference.

Two subtleties are worth knowing. First, text you type *after* the name is ignored by the resolver — the template is what gets sent, not your trailing words. To add a one-off note, expand the prompt into the bar (Enter inside the picker) and edit it there before submitting. Second, two prefixes are special and are **not** prompts: a leading **Help!** (case-sensitive) routes the rest of the line through the grounded Help search (Chapter 9), and a bare line with no leading slash is simply sent as an ordinary question.

## The substitutions

A template is plain text until you dispatch it, at which point Inkhaven replaces two placeholders. There are exactly two — no others — and both are filled from the editor state at the instant you expand the prompt.

### TERM {{selection}}

The text you are working on. If you have an active selection in the editor, that selected span is substituted verbatim. If nothing is selected, the **entire open paragraph** is substituted instead. This is the working text most prompts operate on, so a prompt “just works” whether you highlighted a sentence or want the whole paragraph.

**TERM {{context}}**

A breadcrumb of titles locating the open paragraph in the book, joined with `>` — for example `Rain > The City > the-quay`. It lets you tell the model *where* in the manuscript you are without flooding it with the surrounding prose. Cheap orientation, a few tokens, no retrieval.

You can use either, both, or neither. A template frames them however you like; a common shape wraps the working text in visible fences so the model can see exactly where it starts and stops:

● ● ● *A template using both placeholders*

```
Context: {{context}}
```

```
Tighten the passage below without changing meaning or
voice. Aim for 10-20% fewer words. Keep Typst markup
verbatim.
```

```
--- begin
{{selection}}
--- end
```

The substitution happens once, in memory, the moment you expand or dispatch the prompt. What the model receives is finished plain text with the placeholders already gone; the leading and trailing whitespace of the whole template is trimmed. Inkhaven never rewrites your `prompts.hjson` or your book paragraphs to do this — the source templates are untouched.

**SELECTION BEATS PARAGRAPH**

Because an empty selection falls back to the whole paragraph, you rarely need to select anything for a paragraph-level task. Reach for a selection only when you want the model to work on *part* of the paragraph — a single line of dialogue, one clause — and leave the rest out of its view.



## System prompts in prompts.hjson

The system library lives at `<project-root>/prompts.hjson`. It is HJSON — JSON with comments, unquoted keys, and triple-quoted multi-line strings — and its shape is a single `prompts` array of entries:

● ● ● *prompts.hjson — one entry*

```
{
  prompts: [
    {
      name: "darker"
      description: "Make the tone darker, keep facts"
      template: '''
        Rewrite the passage in a darker tone -- somber
        palette, unease, understated dread. Keep every
        fact, name, and timeline intact. Preserve Typst
        markup.

        Context: {{context}}

        --- begin
        {{selection}}
        --- end
      '''
    }
  ]
}
```

Each entry has three required fields and one optional one.

|                    |                                                                          |
|--------------------|--------------------------------------------------------------------------|
| <b>name</b>        | The /name identifier. Lowercase, no spaces; hyphens are fine.            |
| <b>description</b> | The one-line blurb shown in the picker. Keep it scannable.               |
| <b>template</b>    | The prompt body. Use <code>'''</code> triple-quotes for multi-line text. |
| <b>language</b>    | Optional ISO 639-1 code (en, ru, fr, de, es) — see below.                |

To add a prompt, append an entry to the array and save. Inkhaven reads `prompts.hjson` when the TUI launches; it does **not** watch the file for live edits, so close and reopen the project (or restart the binary) to pick up a change. A malformed

file is reported at load and the library falls back empty rather than crashing — fix the HJSON and relaunch.

Because the file is plain text in your project root, it versions with the rest of your work: commit it, diff it, branch it. That is exactly why it is the right home for the templates you want to *travel* between books.

## Book prompts in the Prompts book

A book prompt is authored the way you author anything in Inkhaven — as a paragraph. Navigate to the **Prompts** system book in the Tree, add a paragraph (+, or **P** to insert after the cursor), give it a title in the Add modal, and write the body in the editor. Save with **Ctrl+S** and it appears in the picker immediately — no restart, because the book prompts are read live from the store on every keystroke in the picker.

Two fields fall out of the paragraph automatically. Its **slug** — the filesystem-safe form of the title — becomes the `/name`. Its **title** becomes the picker description. A title of `Worldbuilding consistency pass` slugs to `worldbuilding-consistency-pass`, so typing `/world` filters straight to it.

### THE HEADING IS STRIPPED FOR YOU

Inkhaven seeds a new paragraph with a `= Title Typst` heading line. When a book prompt is expanded, that leading heading — and the blank lines after it — are removed before the template is used, so the model never sees editor chrome, only your instruction. Write the prompt body under the heading and forget the heading is there.

A book prompt earns its place over a file entry in four ways. It **travels with the manuscript**: anyone who clones the project gets your prompts for free, no side file to remember. It is **editable in the TUI**, with no round-trip through an external editor. It is **indexed like prose**, so semantic search can surface it. And it is **snapshot-able** — **F5** captures a version before you tune a prompt, **F6** picks an older one back out — so your best wording is never one careless edit from lost.

## Localized prompts

Every AI feature in Inkhaven keys off the project’s working language, and prompts are no exception. A prompt can be tagged with the language it was written for, and the resolver prefers the tag that matches the language you are writing in.

On a `prompts.hjson` entry the tag is the optional `language` field, an ISO 639-1 code. On a book prompt it is a `lang:<code>` tag on the paragraph — for instance `lang:ru`. Untagged prompts (every prompt from before this feature existed, and every one you never bother to tag) are treated as language-neutral and always remain eligible.

When a name is resolved — whether from `/name`, the picker, or a built-in pass — Inkhaven walks three passes, and within each pass it checks the Prompts book before `prompts.hjson`, and tries both the hyphenated and the spaced form of the name (`grammar-check` and `grammar check`):

### ● ● ● The three-pass resolution ladder

```

Pass 1  strict  prompt tagged for the ACTIVE language
                Prompts book -> prompts.hjson
Pass 2  untagged prompt with NO language tag (back-compat)
                Prompts book -> prompts.hjson
Pass 3  any     prompt tagged for ANY other language
                Prompts book -> prompts.hjson
                ... then the embedded default (English floor)

```

So a Russian `grammar-check` beats an untagged one, which beats an English-tagged one, which beats Inkhaven’s built-in English fallback — and at every rung a book prompt outranks a file prompt. This is the full form of the *book, then file, then embedded* precedence mentioned earlier: language sorts first, then source.

Which language counts as “active” is itself controllable. `Ctrl+B Shift+N` cycles the prompt-language mode through three settings: deferring to the `editor.prompt_language_mode` value in your project HJSON, using the project’s declared `language` for every resolution (`book_defined`), or detecting the language of the live paragraph and resolving to *that* (`paragraph_detected`). The AI pane’s title bar shows a `lang=` chip — `ru` (book) versus

`ru (paragraph)` — so you always know which language the next prompt will resolve against.

#### BOOTSTRAPPING A WHOLE LANGUAGE AT ONCE

You do not have to translate the built-in prompts by hand. From the command line, `inkhaven prompts bootstrap russian` asks your configured model to produce Russian variants of the eight embedded prompts and prints a ready-to-paste HJSON snippet; add `--update` to merge them into `prompts.hjson` in place (with a timestamped backup, never clobbering your own entries). Covered under *From the command line* below.

## Named flows and the resolution ladder

The same resolver that answers `/tighten` also supplies the prompts behind Inkhaven's built-in AI passes — and that is deliberate, because it means every one of them is **customisable** by exactly the mechanism you have just learned. A pass looks up a well-known name, and if you have defined a prompt of that name (in the Prompts book, or in `prompts.hjson`, in the language you are working in), yours is used instead of the shipped default.

The clearest example is `fact-check` (`Ctrl+B Shift+X`, Chapter 14). It resolves the name `fact-check` through the ladder: a `fact-check` paragraph in the Prompts book, then a `fact-check` entry in `prompts.hjson`, then the embedded multilingual default. Override the first two to bend the house style of the check; leave them absent and the built-in prompt runs. Grammar-check (`F7`) works the same way against the name `grammar-check`, as do the show-don't-tell scan, the sentence-rhythm rewrite, the diagnostic explainer, the critique passes, and the timeline-health review.

● ● ● *The eight embedded prompt names you can override*

|                         |                                        |
|-------------------------|----------------------------------------|
| grammar-check           | F7 copy-edit pass                      |
| show-dont-tell          | Ctrl+B Shift+T telling-vs-showing scan |
| sentence-rhythm-rewrite | Ctrl+B Shift+M rhythm rewrite          |
| explain-diagnostic      | Ctrl+F12 explain a Typst error         |
| critique-edit           | F12 weakest-elements critique          |
| critique-changes        | F12 evaluate a revision                |
| critique-compare        | F12 compare two paragraphs             |
| timeline-health         | Timeline modal consistency review      |

Two flows deliberately stand outside this system. The Help question (F1, or a **Help!** prefix) uses a fixed strict-grounding prompt pinned to Local mode, so the model can never invent a feature — it is not yours to override. And the grammar-check apply key expects the corrected text between `<<<CORRECTED>>>` / `<<<END>>>` markers; if you *do* override the `grammar-check` prompt, keep those markers in your version or the one-key apply will have nothing to extract.

## From the command line

Two subcommands touch prompts from outside the editor.

`inkhaven prompts bootstrap <language>` is the localizer described above. It sends the eight embedded prompt names to your configured model, asks for faithful variants in the target language (preserving Typst tokens and the `{{selection}}` / `{{context}}` placeholders verbatim), and prints an HJSON snippet to standard output. Nothing is written to your project unless you pass `--update`, which merges the results into `prompts.hjson` in place — overwriting only same-name, same-language entries, appending the rest, and leaving a time-stamped backup under `.config-backups/`. An optional `--genre` biases word choice toward the register you write in.

`inkhaven prompts-editor` launches a standalone four-pane workbench for `prompts.hjson` alone: the prompt list on the left, an editor with the same chord set as the main paragraph editor in the centre, an AI response pane on the right, and a prompt bar across the bottom — a focused place to draft and review your file prompts without the whole manuscript around them.

## EDITING THE FILE INSIDE THE TUI

You can also open `prompts.hjson` — and the rest of the project config — from inside a running session with `Ctrl+B 0`, the in-app HJSON editor. It syntax-highlights, saves atomically, and warns you that prompt changes take effect on the next launch. For a book prompt there is no such delay: it is live the moment you save the paragraph.

## Scripting: setting the system prompt

There is no `ink.` Bund word for the prompt *picker* — the picker is a keyboard affordance, not a scripted surface. The one prompt-adjacent scripting hook is `ink.ai.set_system_prompt`, which sets the **system** message the assistant runs under (distinct from the reusable templates this chapter is about). It takes one string off the stack:

● ● ● *Overriding the system prompt from Bund*

```
"You are a terse copy-editor. Never explain."
ink.ai.set_system_prompt

"" ink.ai.set_system_prompt \ empty string clears it
```

An empty string clears the override and the assistant falls back to the default system prompt for the current inference mode (Local versus Full, Chapter 9). This is a runtime, session-local change — it is not persisted, and it does not touch your prompt libraries. Use it in a Bund macro that puts the assistant in a specific frame of mind before a batch of work; reach for reusable prompts when you want a named, substitutable *instruction* instead.

## How prompts sit beside scope and mode

Reusable prompts are one of three orthogonal dials on what the assistant sees, and it helps to hold them apart. A **prompt** supplies the instruction and, via its placeholders, the working text and its location. **Scope** (`F9`) prepends a separate context

block — the selection, the paragraph, the chapter, the whole retrieval-grounded book — *above* your prompt; the model sees both. **Inference mode** (F10) swaps the underlying system prompt, forbidding or permitting general knowledge.

For a one-shot task — tighten this, run a grammar check — the prompt’s own `{{selection}}` is usually all you need, and you can leave scope at None. For an ongoing conversation about the work — brainstorm a subplot, plan a chapter — lean on scope and let chat history accumulate instead, clearing it with `Ctrl+B C` when you want a fresh thread. Prompts and scope compose cleanly; you will reach for one, the other, or both as the task asks.

### WHAT YOU LEARNED

- A **reusable prompt** is a named instruction with two placeholders; expanding it fills the AI prompt bar but never sends until you press Enter.
- Prompts come from two libraries: portable **system prompts** in `prompts.hjson` (cyan `[ system ]`) and project-local **book prompts** under the Prompts system book (green `[ book ]`).
- Open the picker by focusing the bar with `Ctrl+I` and typing `/` — there is no chord; `Enter / Tab` expand, `Esc` closes — or dispatch a template directly by typing `/name`.
- There are exactly two substitutions: `{{selection}}` (the selection, or the whole open paragraph) and `{{context}}` (the title breadcrumb).
- Resolution runs **Prompts book, then prompts.hjson, then the embedded default**, and within that a language ladder of `strict` → `untagged` → any keys the active-language variant first.
- Built-in passes like `fact-check` and `grammar-check` resolve named prompts through the same ladder, so you can override any of them; `inkhaven prompts bootstrap` and `inkhaven prompts-editor` manage the file from the command line.

CHAPTER 12

# 12

## Watching the Cost

---

Every tool that talks to a large language model spends money doing it, and most of them are quiet about it — the meter runs somewhere you cannot see, and the bill arrives at the end of the month with no way to trace which feature ate what. Inkhaven takes the opposite stance. It keeps almost everything free, makes the handful of paid touchpoints explicit, records every model call the moment it happens, and gives you one screen that shows the day's spend at a glance. This chapter is about that visibility: the philosophy behind it, the dashboard that surfaces it, the daily caps that warn you before a big pass, and exactly which features cost anything at all. Read it once and you will never again wonder where your model budget went.



## Cost informs, it never blocks

The single principle that governs money in Inkhaven is the one that governs everything else in the tool: it is a **permissive** program built for one writer who owns their own decisions. It informs; it does not gate. There is no feature Inkhaven will refuse to run because you have “used too much” today, and no hidden ceiling that silently drops a request. When you cross a budget, the tool tells you plainly and then does the thing you asked for anyway.

This is a deliberate reversal of how quotas usually work. A daily cap in Inkhaven is not a wall — it is a **speed bump**. It exists so that a runaway loop, or a habit you did not notice forming, becomes visible before it becomes expensive. The number in the config is a threshold for a warning, not a limit on your agency.

### THE PERMISSIVE PRINCIPLE

Cost caps **inform**, they never **block**. Past a daily budget the model-using passes print a one-line notice and continue. The only things Inkhaven ever truly refuses are matters of safety — a sandbox boundary, a destructive operation — never a matter of spend. Your money is your business.

Two consequences follow, and they run through the rest of the chapter. First, because nothing is blocked, the caps can afford to be honest rather than conservative — they are set where a normal day of writing never reaches them, so that touching one actually means something. Second, because the tool never surprises you, it has to be scrupulous about **showing** you: every inference is counted, every counted call is visible, and the dashboard is one keystroke away.

## Almost everything is free

The reason the bill stays small is architectural, not accidental. The overwhelming majority of what Inkhaven does for you is **deterministic** — it runs locally on your machine, from your text and your data, with no network call and no model behind it. Deterministic work is free, it is instant, it works with no network, and — the quiet virtue — it is **reproducible**: run it twice on the same input and you get the same answer, because there is no model in the loop to drift.

**TERM Deterministic**

A computation whose output is fixed by its input — the same paragraph in gives the same result out, every time, computed on your machine with no model and no network. The opposite of a **model call**, which asks a language model for a judgement and costs money to make. Inkhaven's default posture is deterministic; the model is reached only where a determinate answer is genuinely impossible.

Look across the reading intelligences and you find the same pattern again and again: a free deterministic **core** that runs all the time, and an optional model **pass** you invoke by hand when you want the deeper reading the patterns cannot give. The continuity ledger checks co-location, timeline, numbers, and character-fact drift with no model at all; its cross-paragraph coherence pass is a separate key you press only when you want it. The read-through measures your prose's intensity curve and walks the book for confusion and info-dumps deterministically; its synthetic first-read is opt-in. The voice profiler, the poetry scanner, the theologian's fast signals — all free, all local, all reproducible.

So the honest one-sentence summary of Inkhaven's cost is this: **the tool is free until you deliberately ask a model a question**. The rest of this chapter is about those deliberate asks — where they are, what they cost, and how to keep them in view.

## Where the money goes

There is no long list to memorise, because the paid surfaces are few and each one announces itself. Every model touchpoint in Inkhaven is one of two shapes: a **slow track** you engage on a paragraph or a book, or a **conversation** you type into. Here is the whole territory.

### The slow tracks

A slow track is the model pass that sits behind a deterministic reader and supplies the judgement the patterns cannot. Each is opt-in — you press its engage key or pass its flag — and each is capped and counted.

- **World fact-check (slow)** — reads a paragraph against your world's declared facts for contradictions a rule cannot catch. Default cap: 200 calls a day.

- **Inner Socrates (slow)** — the `J → E` engage pass, which surfaces the assumptions, tensions, and framings a paragraph rests on as questions. Default cap: 150 calls a day.
- **Inner Editor engagement** — the editor’s own coaching pass. Default cap: 200 engagement calls a day (and a session and monthly ceiling besides).
- **The coherence and synthetic passes** — the continuity ledger’s `k` coherence pass, the read-through’s `k` synthetic first-read, the Inner Stylist’s `E` coaching, the Inner Theologian’s session, and a `--deep` flag on the CLI where one exists. Each is a small number of model calls you asked for by name.

Every one of these is a single short model call per paragraph — one request, one response. With the default caps, an ordinary writing day never comes near the ceiling; you would have to engage the slow track on a hundred and fifty separate paragraphs to reach the Inner Socrates cap, and the tool would still let you.

### The conversations

The other paid surface is anything you type into and get an answer back from: the AI pane’s chat, chat-with-your-book, the research assistant’s session, grammar and explain and continuation on a selection, the graph’s `ask`. These are not capped by a daily ceiling — they are metered instead, counted by **category** so you can see how many of each you have run today. The research assistant goes one step further and shows a running dollar estimate in its status bar as you work, priced from the model you are actually using.

#### WHAT A MODEL CALL COSTS

Inkhaven ships a per-model price table (USD per million tokens) so the research assistant can turn tokens into dollars — a Sonnet-class model is priced around \$3 in / \$15 out per million, a Haiku-class one far less, an Opus-class one more. A single slow-track paragraph pass is a few hundred tokens; the arithmetic is pennies. The caps exist to catch a **runaway**, not a normal session. Prices are informative and live in config — adjust them as the market moves.

### The cost dashboard

Everything counted lands in one place. There are two doors to it — a command and a keystroke — and both render the exact same view, because the CLI and the modal share a single aggregator under the hood.

From a shell, `inkhaven cost` prints today's tally and exits. Inside the editor, the chord `Ctrl+B $` opens the same report as a scrollable modal titled **AI cost · Ctrl+B \$**. The report is strictly read-only: it changes no caps, enforces nothing, and computes its numbers fresh each time you open it.

**Ctrl+B \ \$**            Open the AI cost dashboard modal (read-only).

The report is in three bands. First the **daily budgets** — the capped slow tracks, each shown as calls-used over its cap with a twenty-cell usage bar and a percentage. Then **other AI calls** — every uncapped conversation, listed by category with a count and no bar, because there is nothing to measure against. Last a **total** line and the standing reminder that the budgets inform rather than limit.

● ● ● *inkhaven cost — a day's tally*

AI cost — LLM calls today (2026-08-05)

daily budgets (informative):

|                         |          |                        |    |
|-------------------------|----------|------------------------|----|
| world fact-check (slow) | 12 / 200 | <div><div></div></div> | 6% |
| inner socrates (slow)   | 3 / 150  | <div><div></div></div> | 2% |

other AI calls today:

|         |   |
|---------|---|
| chat    | 7 |
| explain | 2 |
| grammar | 1 |

total AI calls today            25

Budgets are informative, not limits — past a budget the slow tracks warn and continue. The tally resets per goals.day\_boundary.

A few things are worth reading off that screen. The bar fills proportionally and clamps at full, but the percentage always tells the truth — cross a cap and you will see the bar solid with, say, `150%` beside it, which is exactly the signal you want. The capped rows and the counted rows never double-count: the slow tracks record only against their own budget, so they do not also appear in the per-category list below. And the dashboard is **extensible by design** — any new analytical thread that records its calls under its own sub-budget key shows up here automatically, with no change to the dashboard itself. The Inner Socrates and Inner Editor entries you see are exactly that mechanism: sub-budgets keyed by feature, surfaced dynamically.

**ONE AGGREGATOR, TWO FACES**

`inkhaven cost` and `Ctrl+B $` are the same code. The CLI is for a quick check from a shell or a script; the modal is for a glance without leaving your paragraph. Neither can change anything — to move a cap you edit `config` (below), and the next report reflects it.

## The daily caps

The caps are the numbers the dashboard measures against, and understanding what they do — and do not — is the heart of the chapter. A cap is a per-day ceiling on one slow track’s model calls. It is **informative**: the tool checks it before a call, and if you are over it, it warns and proceeds.

**TERM Daily call cap**

A per-day threshold on a slow track’s model calls — 200 for world fact-check, 150 for Inner Socrates, 200 for Inner Editor engagement by default. Crossing it triggers a one-line warning, not a refusal. “Today” is defined by `goals.day_boundary`, so the caps, the streak, and the usage tallies all agree on when the day rolls over.

Before a slow-track call, a **preflight** estimates the request’s size and checks your standing against the cap. When you are under budget, it prints a quiet line naming the model, the rough token count, and where you stand — then reads:

● ● ● *A preflight, under budget*

```
inner socrates (slow) · model: claude-sonnet · ~420 tokens
· 3/150 calls today · reading...
```

When you are **over** budget, the preflight does not stop. It prints the notice and continues into the very same call — the permissive principle made literal:

● ● ● *A preflight, past the cap — it continues*

```
inner socrates (slow): past today's slow-track budget
(150/150 calls) — continuing (the cap is informative,
see `inkhaven cost`).
```

That is the whole behaviour: a warning, a pointer to the dashboard, and the work done regardless. There is a second, separate guard worth knowing about — a per-call **soft cap** on estimated tokens, which catches a single pass that is unusually large (a very long paragraph, say) and asks you to confirm with `--force`. That one can decline a call, because its job is to stop a single accidental monster request, not to ration your day; but it too yields the moment you insist.

#### THE ONE EXCEPTION, AND IT IS NOT A WALL

The per-call soft cap is the only place a model pass will pause rather than proceed, and it pauses for size, not for your daily total. Re-run with `--force` and it goes through. The daily caps never pause at all.

### The ``cost`` config block

The caps and the retention window live in one small config block. Edit your project's config to move any of them; the change takes effect on the next report and the next preflight.

● ● ● *The cost block — defaults shown*

```
cost: {
  world_daily_call_cap: 200           // world slow-track ceiling
  inner_socrates_daily_call_cap: 150 // Inner Socrates ceiling
  usage_retention_days: 30           // days of tallies kept in
                                     // .inkhaven/ai_usage.json
}
```

The Inner Editor keeps its own richer set of ceilings under its own block — a per-session cap, a “confirm above” threshold, a per-day cap of 200, and a monthly cap — because a coaching conversation has more shapes to bound than a one-shot paragraph pass. All of them are informative in the same way.

Two of these numbers deserve a note. `usage_retention_days` governs how much history the per-category tally file keeps before the oldest days are pruned — it is a housekeeping bound on a small JSON file, not a spend control. And the day boundary that all the caps share is `goals.day_boundary`: `utc` by default, or `local` so an evening session far from UTC is attributed to the right day and the caps reset at your midnight, not Greenwich's.

## How usage is tracked

The dashboard can only show what the tool bothered to record, so the recording is deliberate and complete. Every inference Inkhaven makes — chat, grammar, explain, continuation, critique, the retrieval-augmented answers, all of it — passes through a single chokepoint that records one tally under a `(day, category)` key the instant the call is made. Nothing that reaches a model goes uncounted.

### TERM The usage ledger

A small JSON file at `<project>/.inkhaven/ai_usage.json`, keyed `day → category → calls`. Every model call increments its category's count for the current day. The dashboard reads it; the slow tracks keep their capped counts in their own stores so they are not double-counted here.

The file is plain and human-readable — you can open it and see exactly what ran:

● ● ● `.inkhaven/ai_usage.json`

```
{
  "2026-08-05": {
    "chat": 7,
    "explain": 2,
    "grammar": 1
  }
}
```

Three properties keep this trustworthy. It is counted by **calendar day** on the same clock the caps use, so a day's tally resets exactly when the caps do. It is written **atomically** and under a lock, so two inferences racing to record at once never lose a tally — the count is exact, not approximate. And it is **pruned on write** to the

retention window, so the file never grows without bound; the oldest day falls off once you pass `usage_retention_days` of history.

One boundary is worth stating plainly: the ledger counts **calls**, not tokens or dollars, because a call count is exact and free to keep while a token count would need the model to tell it the truth. For dollars you have the research assistant’s live session estimate and the price table behind it; for volume you have this. Together they answer the two questions a writer actually asks — “how much am I leaning on the model today?” and “what is this session costing me?” — without either one pretending to a precision it does not have.

#### BEFORE A PROJECT IS OPEN

The tracker is pointed at your project at startup, so a call made headless or before any project is open is simply not recorded — there is nowhere to write it. In practice every call you make from inside a project lands in that project’s ledger, which is exactly the scope you want: cost is accounted per-book, not globally.

## A field guide to free versus paid

To close, the practical map — what you can lean on all day at no cost, and what to reach for deliberately. When in doubt, remember the rule: if you did not press an engage key or pass a `--deep`-style flag and type into a conversation, you did not spend anything.

**Free, always — the deterministic core:** the editor and the tree; snapshots, search, and the semantic index; the continuity ledger’s structural checks; the read-through’s measured intensity curve and forward walk; the prose voice profiler; the poetry scanner; the theologian’s fast signals; the graph’s structural queries; assembly to PDF, EPUB, and the web. None of it touches a model.

**Paid, on purpose — the deliberate asks:** the world and Socratic and editor slow tracks; the continuity coherence pass; the synthetic first-read; the stylist and theologian engagements; the research assistant’s session; chat and chat-with-your-book; grammar, explain, and continuation on a selection. Each is capped or counted, each shows in the dashboard, and each is one keystroke or one flag you chose to press.



That is the whole of watching the cost: a tool that stays free until you ask it not to, tells you the moment you do, and never once stands between you and the work.

#### WHAT YOU LEARNED

- Cost **informs, never blocks** — the daily caps are speed bumps, not walls; past a budget the slow tracks warn and continue.
- Almost everything is **deterministic and free** — the model is reached only where a determinate answer is impossible, and only when you engage it by hand.
- `inkhaven cost` and `Ctrl+B $` render the **same read-only dashboard**: capped slow tracks with usage bars, uncapped calls by category, and a daily total.
- The default caps are **200** (world), **150** (Inner Socrates), and **200** (Inner Editor engagement) calls a day; they and the `usage_retention_days` window live in the `cost` config block.
- Every inference is recorded by category in `.inkhaven/ai_usage.json`, keyed by day on the `goals.day_boundary` clock, written atomically and pruned to the retention window.

PART IV

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# **The World & Its Facts**

CHAPTER 13

# 13

## Places and Characters

---

A long book is held together by two things it can never afford to get wrong: **where** it happens and **who** it happens to. Names of towns and rivers, of protagonists and the people they wrong, recur across hundreds of pages, and the reader keeps a quiet ledger of every detail. A city founded in one chapter must not be founded again in another; a character with grey eyes on page 12 must not have brown ones on page 280. Inkhaven gives you two dedicated places to keep that ledger — the

**Places** book and the **Characters** book — and then it does something no notebook can: it reads your prose against them, lights up every mention, and lets you ask an assistant a question grounded in your own canon.

This chapter is the tour of that machinery. It covers what the two system books are and how you fill them, the coloured overlays that make your world visible inside the editor, the multilingual stemming that matches an inflected name to its entry, the two RAG chords that turn a selected name into a grounded question, and — at an operator’s level — the character-arc tracker that watches a declared arc across a whole manuscript. It closes on the quiet part: the Characters roster is not a private notebook feature. Half of Inkhaven’s reading intelligences read it too.

## Two books, seeded for you

Every project Inkhaven creates carries a small block of **system books** — Notes, Research, Prompts, Places, Characters, Help, and a handful more depending on your version. They are ordinary books in every visible respect: they hold chapters, subchapters, and paragraphs, and you edit their prose exactly as you edit your manuscript. What sets them apart is that Inkhaven seeded them at `init`, treats them as metadata rather than manuscript, and wires specific behaviour to their contents.

### TERM System book

A book Inkhaven creates and reserves — marked internally by a *system tag* (`places`, `characters`, `facts`, and so on). You can add, edit, group, and delete the paragraphs *inside* it freely, but you cannot delete or rename the book itself. Its contents drive a feature rather than appearing in your finished document, and the concordance and other prose tools deliberately exclude it from the corpus.

Places and Characters are the two worldbuilding system books. They sit near the bottom of the Tree pane, below your own books and above Help:

● ● ● *The Tree — your books above, the system block below*

|            |                             |
|------------|-----------------------------|
| My Novel   | (your book)                 |
| Interludes | (your book)                 |
| <hr/>      |                             |
| Notes      |                             |
| Research   |                             |
| Prompts    |                             |
| Places     | ← cyan overlay · Ctrl+B P   |
| Characters | ← yellow overlay · Ctrl+B C |
| Help       |                             |

The two books are structurally identical twins. Everything this chapter says about a Place is true of a Character with three substitutions: the book, the overlay colour (cyan versus yellow), and the RAG chord (`Ctrl+B P` versus `Ctrl+B C`). We will lean on the Places book to teach the mechanics, then note where Characters differs.

## Recording a place, recording a person

An entry is a paragraph, and **the paragraph title is the entity's name**. That single rule is the whole data model. To add one:

1. Move the tree cursor to the **Places** row and press `→` to expand it.
2. Optionally press `C` first to add a chapter for grouping — **Cities**, **Regions**, **Buildings** — if you expect many entries.
3. Press `+` to add a paragraph. The Add modal asks for a title. **Type the place name:** **King's Landing**, **Apartment 12**, **Москва**, **San Francisco — 1906**.
4. Press Enter; the new paragraph opens in the editor.
5. Type whatever you want to remember — physical description, history, who lives there, why it matters to the plot. This is ordinary Typst prose: headings (= History), bold (*\*important\**), lists all work.
6. `Ctrl+S` to save.

You can put paragraphs directly under the book or nest them as deep as your project's depth allows. Every paragraph anywhere in the subtree is picked up — there is no separate “register this entry” step; saving is registration.

● ● ● *A Places book that mixes flat and grouped entries*

|                 |                                    |
|-----------------|------------------------------------|
| Places          |                                    |
| Cities          | (chapter)                          |
| Major           | (subchapter)                       |
| Москва          | (paragraph)                        |
| Санкт-Петербург | (paragraph)                        |
| Minor           |                                    |
| Воронеж         |                                    |
| Regions         | (chapter)                          |
| Сибирь          |                                    |
| Дача на Волге   | (paragraph, directly under Places) |

The Characters book works the same way, and the same grouping instinct applies — most authors organise it by role (Protagonists, Antagonists, Supporting), by house or family, or by the part of the story where a character enters. The title is the character's **canonical name**: Aragorn, Anna Karenina, Robb Stark, Дмитрий.

#### FICTION

Your Characters book is the cast list, and your Places book is the map's index. Put voice notes, relationships, secrets, and the chapters a character enters and exits into the body — the assistant reads it back when you ask, and the arc tracker checks against it.

#### NON-FICTION

For non-fiction the same two books hold recurring **entities**: the people your argument keeps naming and the sites, institutions, or regions it returns to. The overlay then doubles as a consistency check — a name you spelled two ways stops lighting up on the variant.

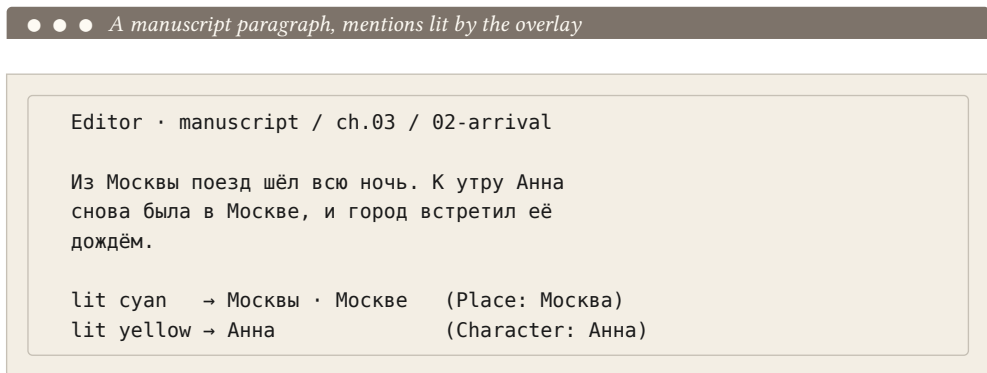
#### ONE ENTRY PER PARAGRAPH

Keep entries atomic. If you find yourself describing two locations in one paragraph, split them — the overlay, the RAG lookup, and the arc tracker all key off the paragraph title, so a paragraph that covers two things is only ever found under one name.

## The highlight overlays

Open any paragraph in your manuscript and look at it. Every word that matches a recorded Place name renders in **cyan and bold**; every word that matches a Character name renders in **yellow and bold**. Your world becomes visible in the prose without any tagging on your part — the lexicon is compiled from the two books' titles and applied on render.

Because a monospace page cannot show colour, picture the overlay as a highlight laid over the matching words:



Note what happened there. The Place entry is titled **Москва**, but the prose says **Москвы** and **Москве** — two inflected forms — and both light up anyway. That is the **stemming**, covered next. Note too the collision rule: when a Place name and a Character name would land on the same word, **Place wins by design**. This matters for surnames that are also place names; if it bites you, give one of the two a more distinctive title.

The overlay refreshes at three moments, so it is always current:

- **Live as you type** — each render re-checks the visible buffer against the lexicon, so a sentence you just wrote lights up immediately.
- **On every save** — adding a new entry and pressing **Ctrl+S** starts highlighting that name everywhere else at once.
- **On project open** — the lexicon is compiled from scratch at startup.

The colours are yours to change. In `inkhaven.hjson`:

● ● ● *Overriding the overlay colours*

```
theme: {
  places_fg:      "#a6e3a1"  # green instead of cyan
  characters_fg:  "#fab387"  # peach instead of yellow
}
```

The defaults are `#89dceb` (sky blue) for Places and `#f9e2af` (yellow) for Characters. Restart the TUI to pick up a change.

### Stemming: one entry, every form

Names inflect. English is gentle — `city` and `cities`, `Aragorn` and `Aragorn's` — but a language like Russian gives a single name six or more surface forms: `Анна`, `Анне`, `Анной`, `Анну`, `Анны`. You cannot be expected to file a separate entry for every grammatical case. So the matcher does not compare words literally; it compares **stems**.

Inkhaven runs Snowball stemmers (from the `rust-stemmers` crate), which reduce an inflected word to its root. The entry `Москва` and the prose word `Москве` stem to the same root, so the overlay lights the prose word even though it never matches the title character-for-character.

The stemmer language is driven, in order, by:

1. The top-level `language` field in `inkhaven.hjson` — it wins whenever it is non-empty. The default is `"english"`.
2. `editor.stemming.languages` — a fallback list, used only when `language` is empty, that runs several stemmers at once.

Set `language` to the dominant language of your manuscript:

● ● ● *A Russian project — one line does it*

```
language: russian
```

The supported set is `arabic`, `danish`, `dutch`, `english`, `finnish`, `french`, `german`, `greek`, `hungarian`, `italian`, `norwegian`, `portuguese`, `romanian`, `russian`, `spanish`, `swedish`, `tamil`, and `turkish`. If you write in a language not on the list, set `language: ""` to disable stemming — the overlay then falls



back to **exact** word matches, which is correct for an uninflected language and safe for any.

Multi-word titles are handled as a **sequence of stems**. `King's Landing`, `North Tower`, `Анна Каренина` — the matcher splits the title into tokens, stems each independently, and looks for the same run of stems in the prose. So `Анна Каренина` matches the full name in any case. It does **not**, however, match the parts in isolation: an entry for `Anna Karenina` lights the full name but not a standalone `Anna`. If both forms appear often, add a second short entry (`Anna`) or title the entry with the form the prose actually uses most.

#### DOES IT WORK IN RUSSIAN?

Yes — that is the point of stemming. The overlay, the multi-word matcher, and the RAG lookups all key off the project `language`, so a Russian, German, French, or Spanish project gets inflection-aware matching with no extra configuration. For a bilingual project (say, translation work) set `language: ""` and list both stemmers under `editor.stemming.languages` so a `Moscow` entry lights English inflections and a `Москва` entry lights Russian ones.

## Asking the AI: Ctrl+B P and Ctrl+B C

The overlay is passive — it shows you what you have recorded, but it never talks back. To **ask** the assistant about an entry, using your own canon as the ground truth, use the two RAG chords. They are the active half of this feature.

#### TERM RAG context

Retrieval-Augmented Generation: before the model answers, Inkhaven retrieves the relevant entries from your Places or Characters book and prepends them to the prompt. The model answers from *your* text, not from whatever it half-remembers about a real or famous place. It is how you keep the assistant inside your canon.

The flow is the same for both chords:

1. In the editor, select the name in your prose, or just place the cursor inside the word.

2. Press **Ctrl+B P** for a Place, or **Ctrl+B C** for a Character.

Inkhaven sweeps the matching book for every paragraph whose title contains your term (case-insensitive), builds a context block from their bodies, and then behaves one of two ways depending on the AI prompt bar:

- **If the prompt bar is empty**, the context is **stashed** as the next RAG prefix and focus jumps to the prompt bar. The status line reads **Place RAG armed for 'Москва' – type your question and Enter.** Type your question and press Enter; the model answers using the stash.
- **If the prompt bar already has text**, the inference fires **immediately** with the context prepended, and focus moves to the AI pane so you can watch the answer stream in.

The block the model receives looks like this:

● ● ● *The context block prepended to your question*

```
— Place context for `Москва` (1 match(es)) —

— Place: Москва —
= Москва

= History
Столица России. Основана в 1147 году.

= Geography
На реке Москве, в Центральной России.
— end place —
```

Two dials sharpen the answer. **F10** toggles the inference mode: with **Local**, the model is constrained to use **only** the context you supplied — ideal when you want a faithful summary of your canon and nothing invented from training data. Switch to **Full** if you want the model to reach beyond the entry. And **F9** sets the **scope** — combine it with the RAG chord to give the model both the surrounding chapter and the entry: put the cursor in a city's name, press **F9** to widen scope to Paragraph or Chapter, then **Ctrl+B P**, and ask “how does she feel about being back?” The model sees the scene **and** the city's recorded history.

If your selection matches no entry, the status reports **Place RAG: no entry titled like 'XYZ' in the Places book** and nothing fires — add the

entry first. The chords route through the same meta-prefix dispatcher, so they work from any focus, but selecting in the editor first is the natural path.

|                 |                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------|
| <b>Ctrl+B P</b> | Place RAG — sweep the Places book for the selected name, arm or fire a grounded question. |
| <b>Ctrl+B C</b> | Character RAG — the same against the Characters book.                                     |
| <b>F10</b>      | Toggle inference mode Local ↔ Full (Local = answer only from the stashed context).        |
| <b>F9</b>       | Cycle the AI scope; stack it under a RAG chord for scene + entry grounding.               |

#### PRONOUNS ARE NOT NAMES

The lexicon is a noun-phrase index, not a coreference resolver. It highlights literal titles (stemmed), so pronouns — he , she , они — are never lit and aliases need their own entries. For a “who is she here?” question, widen F9 scope so the model has the passage to reason over, then ask in the chat.

## Character arcs — a first tour (CHAR-1)

The Characters book earns a second life as the ground truth for Inkhaven’s **character-arc tracker**. This is a large feature with its own deep treatment in the *Know Your Book* companion; here we give it an operator’s tour — enough to run it and read its output, with pointers to where the full story lives.

The premise: you **declare** the arc you intend, and Inkhaven watches the manuscript to see whether the prose delivers it. You declare an arc by adding a `character_arc` block to a Characters-book entry, written as HJSON in the paragraph body:

● ● ● *A declared arc in a Characters entry body*

```
{
  character_arc: {
    arc_type: "positive_change"
    desired_state_start:  "Mara defers to her family."
    desired_midpoint_state: "Mara's first open defiance."
    desired_state_end:     "Mara acts without permission."
  }
}
```

The `arc_type` names one of five structural shapes the tracker understands — `positive_change` (a false belief overcome), `flat` (the character holds firm while the world changes), `corruption`, `fall`, and `disillusionment` — plus any other string, which falls back to a generic probe. Only `desired_state_start` and `desired_state_end` are required; the midpoint is optional. Omit the whole block and the entry simply has no arc to check.

Everything else is computed from the prose. Three commands drive it from the terminal:

- `inkhaven character refresh` recomputes the **agency score** (fast, deterministic) and re-extracts the chapter-by-chapter **observable-state chain** (an LLM pass, run lazily — only chapters whose text changed are re-extracted). `--name` limits it to one character.
- `inkhaven character check` runs the **completeness checks** against the declaration and is built to be a pre-submission gate: it exits `1` on any gap or stall and `2` if the ending or the earned-arc check fails, so you can wire it into a script. `--json` emits machine-readable findings.
- `inkhaven character plan` reports **Planning-Board coverage gaps** deterministically — a declared arc that no scene card names, or one confined to the book's first half. It exits `1` on any gap.

**TERM Agency score**

A deterministic, zero-cost number between 0 and 1 for a character in a chapter: the share of their appearances where they *act* (their name precedes a transitive action verb) versus where they are *acted upon* (a passive construction, or their name follows the verb). Computed from a per-language action-verb list — English, Russian, German, French, and Spanish ship built in, and you can fold genre verbs in via `char.extra_action_verbs`. A protagonist whose agency sags for a stretch of chapters is a protagonist going passive.

The stall detector is the other deterministic half: it finds the longest run of chapters where the character shows **no observable change** after their baseline appearance, and flags it when the run reaches `char.stall_threshold` (default 4). The four alignment and earned checks — start, midpoint, end, and “was this arc earned?” — are the LLM half, judged only on observable evidence in the prose and run only for characters with a declaration and at least `char.min_chapters_for_check` (default 3) chapters of extracted state.

`inkhaven character arc <name>` prints the whole picture read-only from the cached `char.duckdb` — declaration, state chain, agency, stalls, checks, and planning gaps together:

● ● ● *inkhaven character arc Mara*

```
Character arc – Mara · `My Novel`


---


Declared arc: positive_change
  start      : Mara defers to her family.
  midpoint   : Mara's first open defiance.
  end        : Mara acts without permission.

State chain (6 chapters):
  · ch. 1  agency 0.42  (3a/4p)  Silent at dinner.
  · ch. 2  agency 0.38  (2a/3p)  Runs an errand.
  + ch. 4  agency 0.61  (5a/3p)  Refuses to lie.
  · ch. 5  agency 0.55  (4a/3p)  Holds the line.

Arc checks:
  [earned] Arc earned – the refusal in ch.4 is
           seeded by the small evasions before it.


---


```

A **♦** marks a chapter where the character’s state changed; a **•** marks one where it held. Inside the editor the same view lives one chord away: **Ctrl+V Shift+N** opens the arc modal for the nearest character (one named in the open paragraph, else the first tracked one) — the declared arc, the state chain with its agency scores, the completeness checks, and any planning gaps, scrollable with **↑↓**, **Esc** to close. It is read-only over the cache, so populate the cache first with a **refresh / check**, or run the zero-AI **Ctrl+B Shift+C** review pass, whose deterministic arc findings also land in the Output pane’s **character** category.

#### WHERE THE FULL TREATMENT LIVES

This is the operator’s tour. The arc taxonomy in depth, the agency windows and cross-system enrichment signals, the exact wording of the LLM checks, and the Planning-Board integration are covered in the *Know Your Book* companion and in Chapter 17 (Continuity and Knowledge). The `char:` config block tunes every threshold; see the configuration reference.

## How the roster feeds the rest of Inkhaven

Here is the quiet payoff. The Characters book is not a private notebook that only the overlay and the RAG chord read. The list of entry titles — the **roster** — is the canonical cast of the whole project, and a surprising number of Inkhaven’s reading intelligences read it:

- **Continuity** (SENTINEL, Chapter 17) builds its “referenced-before- introduced” invariant on the roster, and the status-bar POV chip names the most-mentioned rostered character in the open paragraph.
- **Knowledge** (KEN, Chapter 17) grants each rostered character an epistemic timeline — who could know what, and when — keyed off their presence in timeline events, so a name absent from the roster gets no grant.
- **Voice at scale** (CHORUS, Chapter 18) resolves a scene’s POV to the most-mentioned rostered character and checks each one’s distinct voice against the others.
- **Dialogue attribution** resolves “who said this” to a roster-canonical name, and the arc tracker’s agency and stall scoring rides on the same roster.

The lesson is that filling in the Characters book is not busywork for one feature — it is the single act that turns on continuity, knowledge, voice, dialogue, and arc

tracking together. A name you never record is a name none of them can watch. This is why the roster is drawn from the entry **titles** directly: record a character once, as a paragraph, and the whole reading apparatus knows who they are.

#### WHAT YOU LEARNED

- The **Places** and **Characters** system books are seeded at `init`; you record an entity as a paragraph whose **title is its name**, and saving is registering.
- Mentions light up in the editor — **cyan** for Places, **yellow** for Characters — with **Snowball stemming** keyed to the project `language`, so one entry matches every inflected form; Place wins a collision.
- `Ctrl+B P` and `Ctrl+B C` sweep the matching book and **arm or fire** a question grounded in your own canon; `F10 Local` keeps the answer inside that context, and `F9 scope` stacks scene grounding on top.
- The arc tracker (CHAR-1) checks a declared `character_arc` block against the prose: `inkhaven character refresh / check / plan / arc` and the `Ctrl+V Shift+N` modal, with a deterministic **agency score** and **stall** detector plus LLM completeness checks.
- The Characters **roster** is shared ground truth — **continuity, knowledge, voice, dialogue, and arcs** all read it, so one entry switches on many readers at once.

## CHAPTER 14

## 14

The World

---

A long story happens somewhere. The somewhere has a sky with a certain number of moons, a climate that makes some weeks bitter and others sweet, distances that take real days to cross, and cities of a certain size in certain places. Get any of it wrong and a reader who is paying attention feels the floor tilt: snow in the desert, a moon that was full last night and full again tonight, a courier who rides four hundred miles between breakfast and supper. Inkhaven's answer is to let you **declare** the physics of your world once, in a single file, and then do two things with that declaration — build the rest of the world out from it by deterministic simulation, and watch your prose for the moment it forgets what you declared.



This chapter is the operator’s tour of that machinery: how you write the definition, what the compiler derives from it, how the map is drawn, the interactive worldbuilder you can shape a world in, the coherence checker for imagined societies, and the live fact-checker that reads over your shoulder in five languages. It is deliberately breadth-first. The full treatment — every field, the physics behind each layer, worked worlds from scratch — lives in the companion book **Building the World**; this chapter tells you what each piece is, how to reach it, and how to run it.

## The two halves

The whole subsystem turns on one file at the root of your project: `world.hjson`. It is opt-in. A project without it behaves exactly as before; drop one in and the world layer wakes up. Everything then divides into two halves that share that one declaration.

### TERM `world.hjson`

The single file, at the project root, where you declare a world’s **physics** — its star and planet and moons, and optionally its geology, geography, hydrology, economy, and any magic that breaks the rules on purpose. Only the world’s **name** and its **astronomy** are required; every other block is optional with sensible defaults. Scaffold one with `inkhaven realworld new <name>`.

The first half is **the compiler**. Give it the definition and a seed, and it derives a coherent world — astronomy, geology, climate, hydrology, demographics — in five layers, then materializes the result into a **World** system book, one chapter per layer, plus a rendered map. The second half is **the fact-checker**. As you write, Inkhaven reads each paragraph against the world you built — against its climate zones, its distances, its populations, its sky — and flags the sentences that contradict it, in whatever of five languages you wrote them in.

**DETERMINISM**

Everything the compiler produces is a pure function of (definition, seed). The same inputs give the same continents, the same rivers, the same cities, every run — driven by an in-tree SplitMix64 keyed to your seed, never the rand crate. The astronomy layer has no randomness at all; it is closed-form physics, re-derived each run. Keep the seed and your world — and its map — are reproducible forever; change it and the whole world regenerates.

## Declaring the physics

You write the definition in HJSON — JSON with the ceremony removed: no quotes required on keys, comments allowed, trailing commas forgiven. There is one catch worth stating first, because it bites everyone once.

**THE HJSON QUOTING RULE**

In HJSON an unquoted string runs to the **end of the line**. So an inline enum value must be quoted — write `class: "G2V"` and `kind: "city"`, not `class: G2V`. Multi-line blocks are fine; it is only inline scalar strings that need the quotes. When a value comes out looking wrong, this is almost always why.

Only two blocks are load-bearing: the world's `name`, and its `astronomy`, which is required because everything about seasons, tides, and the calendar is derived from it. The rest are optional, and each one you add makes the world richer and gives the fact-checker more to check against.

● ● ● *world.hjson — the shape of a definition*

```
{
  name: "Aldoria"
  seed: 0x1A2B3C           // int or "0x..." hex; drives every layer
  primary_language: "en"   // sets the fact-checker's language

  astronomy: {             // the only required block
    star: { class: "G2V", luminosity_solar: 1.0, mass_solar: 1.0 }
    planet: { axial_tilt_deg: 23.4, day_length_hours: 24.0 }
    orbit: { semi_major_axis_au: 1.0, year_length_days: 365 }
    moons: [ { name: "Luna", period_days: 27.32 } ]
    calendar: { months: 12, month_length_days: 30, weekdays: 7 }
  }

  geology: { generated: { plates: 7, continents: 4,
    sea_level: 0.40 } }
  geography: { landmarks: [ { name: "Caer Dath", kind: "city",
    climate_zone: "tundra", population: 27000 } ] }
  economy: { tech_level: "medieval", currency: "silver mark" }
  magic: { enabled: false }
}
```

Read the blocks as a table of what each one drives:

|                    |                                                    |
|--------------------|----------------------------------------------------|
| <b>name / seed</b> | Identity; the seed drives every procedural layer.  |
| <b>astronomy</b>   | Required. Seasons, tides, calendar — closed-form.  |
| <b>geology</b>     | generated (procedural) or dem (a real heightmap).  |
| <b>geography</b>   | Named regions + landmarks → Setting + gazetteer.   |
| <b>hydrology</b>   | Named waters + rainfall → Setting chapter.         |
| <b>economy</b>     | Tech / currency / trade / resources → known goods. |
| <b>magic</b>       | Declared exceptions the fact-checker respects.     |

A few blocks earn a word here because they change what the checker will accept. A `geography.landmark` that carries a `climate_zone` becomes a **gazetteer** entry the checker resolves by name — so it knows Caer Dath's weather even before you **geology**.

compile and accept the procedural cities. The `sea_level` under `generated` is a **threshold on the generated heightmap, not a literal ocean fraction**; the

terrain is not uniform, so a value near `0.40` gives an Earth-like, ocean-dominant world — raise it for a wetter one, lower it for more land. And `economy.resources` (like `geology.notable_minerals`) extend the checker's list of known goods, so trading your world's own metals is never flagged as an anachronism.

The `magic` block is the pressure valve. When your world breaks physics on purpose, you declare the exception so the checker respects it instead of flagging it in every chapter forever.

● ● ● *A magic rule — a declared exception to the physics*

```
magic: {
  enabled: true
  rules: [
    {
      kind: "messenger_birds"
      covers: ["travel_time"] // categories this rule excuses
      description: "Royal pelicans fly day and night with relays"
      applicable_to: { roles: ["royal_messenger"] }
    }
  ]
}
```

Each rule names a `kind`, the categories it `covers`, and an optional `applicable_to` scope (`roles` / `regions` / `seasons`). The checker consults the ledger **lazily** — only after it already has a candidate warning — and a covered, applicable rule suppresses the finding with a note rather than hiding it silently. The rule materializes into the World book's **Magic Ledger** chapter, and `inkhaven realworld magic` lists what is on the books.

#### START FROM EARTH

The repository ships a complete, heavily-commented real-Earth definition at `examples/realworld/Earth.hjson` — the exact Sun, Earth, and Moon, the Gregorian calendar, Earth-tuned geology, and a populated geography. Copy it in as `world.hjson` and compile to begin in a world you already recognise, then edit outward toward your own.

## The compiler and what it derives

`inkhaven realworld compile` runs five layers in dependency order, each a pure function of the ones before it. You can run one layer or all of them, print a human summary or the full structured JSON, and — with `--materialize` — write the result into the World system book.

|                     |                                                            |
|---------------------|------------------------------------------------------------|
| <b>Astronomy</b>    | Kepler year, insolation by latitude, lunar periods, tides. |
| <b>Geology</b>      | seed → plates → heightmap → continents, ranges, minerals.  |
| <b>Climate</b>      | Zonal model → Köppen biomes, rain shadows, winds.          |
| <b>Hydrology</b>    | D8 flow over the terrain → rivers, lakes, watersheds.      |
| <b>Demographics</b> | Carrying capacity → a rank-size hierarchy of towns.        |

### ● ● ● *Compiling the layers from the CLI*

```
$ inkhaven realworld compile --layer demographics
$ inkhaven realworld compile --layer climate --json
$ inkhaven realworld compile --layer geology --materialize
```

```
Geology (seed 0x1A2B3C)
  plates 7 · continents 4 · sea level 0.40
  highest cell 4210 m · 3 mountain ranges clustered
  materialized → World / Geology
  wrote assets/world/heightmap.png
```

`--layer` is `all` (the default — the whole world) or one of `astronomy` / `geology` / `climate` / `hydrology` / `demographics`. The astronomy layer also does one check worth knowing about: if you **declare** an `orbit.year_length_days`, the compiler computes Kepler's value and warns when the two disagree by more than a day — your calendar and your orbit are allowed to drift, but you will be told they have.

When you materialize, the writes are **idempotent** — re-running never duplicates, it updates in place — and they land in the World system book as one chapter per layer, plus the author-declared `Setting` chapter, the `Magic Ledger`, and the normalized heightmap under `assets/world/`. Nothing touches your manuscript.

## Proposals and Places

The compiler models settlements, but it never writes places into your book behind your back. Instead its cities become **proposals** you accept or reject one at a time. Only an accepted proposal becomes a real **Place** record, and each one records a Place-to-World cross-reference.

● ● ● *The proposal queue closes the loop*

```
$ inkhaven realworld propose          # seed the queue
$ inkhaven realworld proposals list
$ inkhaven realworld proposals accept-all
$ inkhaven realworld places          # the accepted places
```

Re-running **propose** never re-offers a site you already resolved. These accepted Places are exactly what the fact-checker resolves place names against, which is what closes the loop: **compile** → **accept cities** → **write** → **check**. In the TUI the whole cycle is under one chord, which the next section names.

## The World hub — Ctrl+B W

Every CLI operation has a home in the TUI under **Ctrl+B W**, the World hub. It opens a read-only, scrollable overview of the world — the definition, the compiled astronomy layer, and whether it has been materialized yet — and from there a handful of second keys drive the whole subsystem without leaving the editor.

|                           |                                                                 |
|---------------------------|-----------------------------------------------------------------|
| <b>Ctrl+B W</b>           | Open the World overview (definition + astronomy + status).      |
| → <b>C</b>                | Compile + materialize all five layers, seed the proposal queue. |
| → <b>P</b>                | Open the Place proposal queue (Enter accept · r reject).        |
| → <b>F</b> → <b>P/B/R</b> | Fact-check the paragraph / book / recent edits.                 |
| → <b>M</b>                | Render the world map with plakats.                              |
| → <b>S</b>                | Toggle the idle auto slow-check (off by default).               |

**Ctrl+B W** → **C** is the one-key path through the whole compiler: it compiles and materializes all five layers **and** seeds the proposal queue in a single step, so a fresh world is one chord from existing. The other keys open the queue, arm the fact-

checker’s scope picker, draw the map, and toggle the background slow check — each covered in its own place below.

## Maps — the plakat cartographer

`inkhaven realworld map` (or `Ctrl+B W → M`) turns the compiled layers into a **MapSpec** — a structured description of the map — and hands it to **plakat**, a separate cartographer you install once with `cargo install plakat`. `plakat` loads the spec and skips its own map-generation AI entirely, so the drawn map stays a pure function of the world and its seed, exactly like everything else.

● ● ● *Rendering the world map*

```
$ inkhaven realworld map
  features: assets/maps/world.features.png # the rendered map
  geojson:  assets/maps/world.geojson     # coast/rivers/roads
  spec:     assets/maps/world.mapspec.json # the emitted MapSpec
```

Mountains come from clustering the heightfield’s high cells; rivers run their real D8 watercourse; landmarks are your largest settlements, with coastal cities promoted to ports. `plakat`’s resolved landmark positions are read back to **refine each accepted Place’s coordinates**, so the map and the gazetteer agree. Two flags tune the run — `--spec-only` writes the MapSpec without invoking `plakat`, and `--no-ingest` renders without touching Place coordinates.

### PLAKAT IS OPTIONAL

These are called “`plakat`” maps after the cartographer that draws them. The binary is a genuinely external dependency, and it is the **only** one Inkhaven reaches for. A missing `plakat` never fails the run — it degrades to a notice, and everything else about the world still works. The rendered map is a convenience, not a load-bearing part of the pipeline.

## The interactive worldbuilder — at a glance

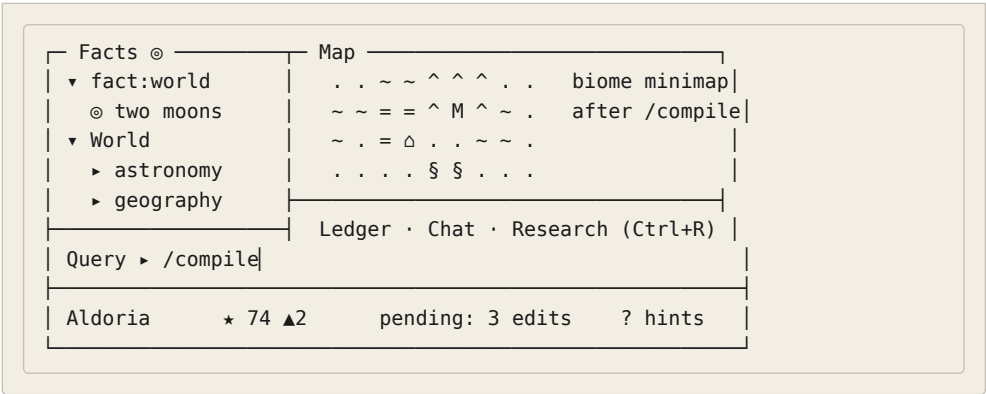
Everything above is reachable from the CLI and the `Ctrl+B W` hub, editing `world.hjson` by hand. When you would rather **build** a world interactively — shaping the sky, adding nations, drawing the map, watching a plausibility score move as you go — there is a whole separate application for it: `worldbuilder`, a full-screen TUI that is a third sibling beside `research` and the linguistic workbench.

### TERM The worldbuilder

A companion TUI (introduced in 1.9.0) that is a **front-end** to everything in this chapter. Every change it makes lands in the same `world.hjson`, compiled by the same chain, checked by the same checker. It never generates your prose — the author decides, and the worldbuilder measures, validates, and records.

Its window is four regions: a **Facts** tree over a **World** tree down the left; a wide right pane that cycles **Chat** / **Research** / **Map** / **Ledger**; a full-width **Query** prompt; and a status bar carrying the world’s name and its live **plausibility score** (★ NN, with ▲ / ▼ deltas). Plain text in the Query prompt is a question to the AI; a line that opens with a slash is a command.

● ● ● *inkhaven worldbuilder — the four regions*





The commands typed at that prompt are the shape of the whole session. A few of the load-bearing ones: `/interview` walks a guided five-stage world interview; `/set <dot.path> <value>` sets any `world.hjson` key, with shortcuts like `/star`, `/tilt`, `/moon`, and `/nation` for the common ones; `/wfact` records an author fact into the Facts book; `/compile`, `/validate`, and `/map` run the same chain, score, and cartographer you already met; and `/roll` compares candidate worlds on derived seeds so you can `/adopt` the one you like.

Crucially, shaping commands do not touch the file as you type. They accumulate into a **pending delta** — accepted-but-uncommitted edits you preview with `/diff`, undo with `/undo`, and only fold into `world.hjson` atomically with `/write`. The plausibility score, though, moves the moment an edit is **accepted**, before it is committed, so you can feel a change's effect before you keep it.

### The map editor

Cycle the right pane to **Map**, press `e`, and you can draw the world directly — and every mark is another pending `world.hjson` edit, reviewed with `/diff` and written with `/write` like any other. A brief tool list:

|                        |                                                                       |
|------------------------|-----------------------------------------------------------------------|
| <code>t / n</code>     | Place a town / named landmark → <code>geography.landmarks</code> (⬢). |
| <code>r</code>         | Draw a river source → mouth → <code>hydrology.rivers</code> (≈).      |
| <code>g / o</code>     | Region from the cell's biome (§) / road between towns (=).            |
| <code>+ / - , .</code> | Raise / lower terrain under a brush · brush size.                     |
| <code>d / f</code>     | Delete the feature under the cursor · jump to each flaw.              |

Drawing the terrain and running `/terrain` writes a sculpted grayscale heightmap under `assets/maps/` and points `geology.dem` at it, so the next compile rebuilds the whole world from the shape you drew. This is only the glance; the worldbuilder and its map editor get a chapter of their own in **Building the World**.

## FICTION

For a **secondary world** — the invented planet of a fantasy or SF novel — the worldbuilder is where the whole setting is born: shape a plausible sky and geology, draw the map, and let the checker hold your chapters to it.

## NON-FICTION

For **history, travel, or reportage** set on Earth, copy examples/  
in `realworld/Earth.hjson` and lean on the fact-checker's real climate, distance, and demographics knowledge to catch a wrong season or an impossible journey.

## Coherence for imagined societies — utopia

There is a second, narrower checker for a particular kind of world: the imagined **society** — a utopia or dystopia whose premises are supposed to hang together as an argument. You declare its premises as tagged paragraphs in the World book — `para:utopia-premise`, `-mechanism`, `-consequence`, `-elimination` (glyphs  $\vdash \odot \Rightarrow \emptyset$ ) — and run the checker over them.

### ● ● ● The utopia coherence checker

```
$ inkhaven world utopia-check --stage 1      # chain logic (free)
$ inkhaven world utopia-check --stage 2      # pairing (explicit)
$ inkhaven world utopia-check --stage all
$ inkhaven world utopia-model                # the extracted claims
```

Stage 1 (chain logic) is deterministic and free; Stage 2 (pairing premises for tension) is explicit-only because it spends LLM calls per pair; Stage 3 scans the prose for entailment. It reasons about **logical and systemic** structure only — does the mechanism actually produce the claimed consequence, does one premise quietly contradict another — and leaves moral and theological coherence to the Inner Theologian. Findings are advisory, cached in `.inkhaven/utopia.duckdb`, surface in the `Ctrl+B Shift+C` review pass, and ground the `utopian-architect` Socratic persona. `utopia-check` exits `1` on a chain-logic finding and `2` on an entailment violation, so it can gate a pre-submission script.

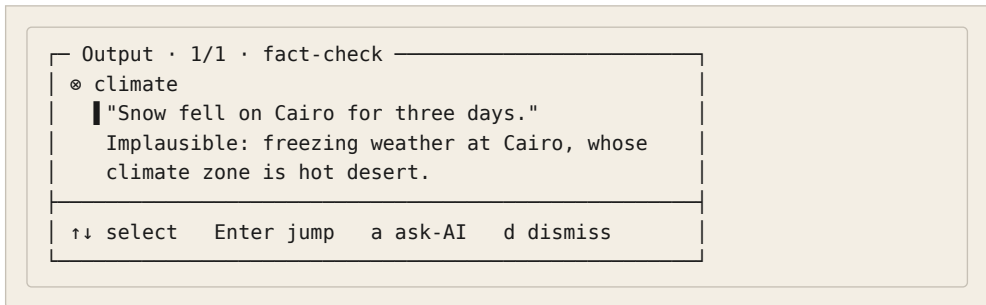
## The live fact-checker

Now the second half of the whole subsystem: the reader that watches your prose. When a project has a `world.hjson`, a **fast track** runs automatically — pause a few seconds on a paragraph and any findings appear in the Output pane, with no chord and no focus stolen. Re-checking a paragraph replaces its own prior findings, so the board never accretes stale warnings.

|                     |                                                    |
|---------------------|----------------------------------------------------|
| <b>travel_time</b>  | A distance + duration implying an impossible pace. |
| <b>climate</b>      | Weather at a known place against its climate zone. |
| <b>demographics</b> | A population diverging sharply from the model.     |
| <b>astronomy</b>    | A moon count disagreeing with the world's sky.     |
| <b>economy</b>      | A metal worked that the geology does not yield.    |

Climate, demographics, and economy resolve place names through the **gazetteer** — your accepted Places plus any `geography.landmarks` you declared — so the checker knows which Cairo you mean and what its weather should be.

● ● ● *A fact-check finding in the Output pane*



From the CLI the same check runs on demand: `inkhaven fact-check --text "..."` checks a literal string, and `--paragraph <id>` checks a stored paragraph. In the TUI, `Ctrl+B W → F` arms a scope picker — **P** for the open paragraph, **B** for the enclosing book, **R** for the twelve most recently edited paragraphs.

## Five languages, and graceful degradation

The checker works in **English, Russian, Spanish, French, and German**. It detects the paragraph’s language, renders its warnings in that language, and resolves place names in their grammatical cases — Russian `В МОСКВЕ` matches `МОСКВА`, a German genitive matches its nominative. The detector is a built-in heuristic that needs no model, and when it is not confident it **degrades** — rendering in English rather than guessing wrong — and never panics. An optional enhanced parser can be pointed at with `INKHAVEN_LANG_MODEL`, but nothing ever requires one.

## The slow track and coherence

The fast track is pattern-based and free. For the subtle contradictions patterns miss, `fact-check --slow` adds an LLM pass, and it is scrupulously cost-controlled: a preflight prints the estimated tokens and the day’s call tally; a per-call soft cap (`--max-cost`, default 6000; `--force` overrides) refuses an oversized call; a daily ceiling holds; and a missing provider or a reached cap degrades to a notice. An opt-in **idle auto** variant runs it in the background after about 45 seconds of quiet — toggled with `Ctrl+B W → S`, off by default because it spends tokens.

Where a single-paragraph check asks “is this sentence possible,” `inkhaven realworld coherence <node>` asks whether a **run** of paragraphs holds together — a character in two places without the travel between, a fact asserted then reversed, a timeline that cannot follow. Give it a book or chapter node; it gathers the paragraphs in document order and runs one cost-capped call, citing the paragraph numbers.

## Timeline-aware checking

When your project also has a **timeline**, the fast checker learns **when** a paragraph happens, and its guesses become grounded facts. A paragraph tied to a dated event gains a **calendar-grounded season** — snow in a paragraph the timeline places in summer becomes a flat contradiction, not a guess — plus event-derived travel time (a prose “three days” against a 35-day gap between the traveller’s events), a `date_coherence` check on seasonal hints like a midsummer feast, and `co_location` for a character whose events put them in two places at once. These run automatically, in all five languages, and respect the magic ledger exactly as the world checks do; timeline-derived findings carry a small calendar marker in the `inkhaven realworld co-location` .

## Facts, and grounding the AI

There is one more sense of “the world” in Inkhaven, and it is worth keeping distinct from the simulated one. Alongside your manuscript lives a **Facts** system book — a place to write down the settled truths of your world in plain prose, whether or not you ever declare a `world.hjson`. These facts do two jobs: they give the AI ground to stand on, and they give a second fact-checker something to check against.

### TERM The Facts book

A system book of established truths about your world, written as ordinary paragraphs. Distinct from the `world.hjson` simulation: the simulation **derives** a world from physics; the Facts book **records** the ones you have decided, including the ones no simulation could produce (“the queen has no heir,” “iron is taboo in the north”).

`Ctrl+B Shift+X` fact-checks the open paragraph against that Facts book. It locks the AI scope to the local paragraph, grounds the check against every established fact — climate, geography, seasons, distances, chronology — and streams a verdict into the AI pane, flagging any claim that contradicts the world you have written down. (With an empty Facts book it degrades to a generic local fact-check.) When it flags contradictions, `Ctrl+B Shift+J` cycles the editor cursor through them one at a time, showing the violated fact on the status bar. Its mnemonic is X for fact eXamination.

The same facts ground the assistant proactively through the **Facts** AI scope. As Chapter 3 described, `F9` cycles the AI scope; **Facts** is one of the **sticky** conversation scopes, so once selected it stays selected across follow-ups. With it active, your prompts are answered against the Facts book — the assistant reasons from your world’s established truths instead of inventing fresh ones. For a large Facts book, `Ctrl+B Shift+S` opens a semantic search over it: type a query, mark the handful of relevant facts, and send just those into a targeted Facts chat, so you ground in the passage that matters rather than the whole book.

**FICTION**

In **fiction**, the Facts book is the series bible in miniature — the settled truths the AI must honour and the checker must guard, so the assistant never quietly retcons your world mid-conversation.

**NON-FICTION**

In **non-fiction**, it is the ledger of claims you have verified — dates, figures, definitions — that the assistant answers from and the checker holds your prose to, keeping the manuscript honest to your own research.

## The ink.world Bund surface

Scripts written in Inkhaven's embedded Bund language (Chapter 34) can read the timeline-aware checker's view of the world, read-only, through the `ink.world.fact_check.timeline.*` family. It exposes exactly what the checker sees — the events near a point in world-time, the events for a character or a place, the season of a point, and a paragraph's effective date — so a script can reason about **when** things happen without re-deriving the calendar.

**events\_near**            Timeline events near a point in world-time.

**events\_for\_character**    The events that place a character in time.

**events\_for\_place**        The events tied to a named place.

**season\_for**              The calendar season at a given point.

**effective\_date**          A paragraph's effective world-date.

These are all `store_read` words — safe, side-effect-free, allowed by default in the Bund sandbox. The utopia checker has its own small read-only surface, `ink.utopia.*` (`model`, `findings`, `violations`, and a `suppress` action), for scripting the coherence pass. Between them a script can fold the world's own knowledge into any automation you build.

## WHAT YOU LEARNED

- The world layer is opt-in on a single root file, `world.hjson`, and has **two halves**: a deterministic **compiler** that derives a world from declared physics, and a **fact-checker** that reads your prose against it.
- Only `name` and `astronomy` are required; `geology`, `geography`, `hydrology`, `economy`, and `magic` are optional and each gives the checker more to check. Everything the compiler makes is a pure function of `(definition, seed)`.
- `inkhaven realworld compile` derives five layers (astronomy, geology, climate, hydrology, demographics); `--materialize` writes them into the **World** book, and settlements become **proposals** you accept into **Place** records.
- `Ctrl+B W` is the hub — `C` compiles all layers, `P` triages proposals, `F` arms the fact-check scope, `M` draws the **plakat** map, `S` toggles the idle slow check.
- `inkhaven worldbuilder` is a whole companion TUI for building a world interactively — pending deltas, a live plausibility score, a map editor — all writing the same `world.hjson`. `inkhaven world utopia-check` grades an imagined society's **logical** coherence.
- The live fact-checker flags travel, climate, demographics, astronomy, and economy contradictions automatically in **five languages**, degrades gracefully, and grows timeline-aware when a timeline exists; the **Facts** book and the `F9 Facts` scope ground the AI in your world's settled truths.
- Bund scripts read the world through `ink.world.fact_check.timeline.*` and `ink.utopia.*` — read-only, sandbox-safe. The deep dive lives in the companion book **Building the World**.

## CHAPTER 15

## 15

## The Knowledge Graph

---

Every chapter so far has added **nodes**. A paragraph is a node; so is a chapter, a book, an image, a script, a fact, a character, a place, a source, a glossary entry. They all live as one uniform kind of thing — a UUID in one database, embedded for semantic search — and Inkhaven has quietly stored them that way since your first project. What it did not store, until now, was the **lines between them**: a first-class, persisted, typed way to say “this fact contradicts that source”, “this paragraph links to that one”, “this word means this concept”. That layer of lines is the knowledge graph, and this chapter is its complete tour — the shape of an edge, the three ways the graph gets filled, the `graph` command that reads it, and the two ways you can **talk** to it.



The graph does not replace the manuscript tree or the vector index you met in the search chapter. It **overlays** them, threading the nodes you already have into one interrogable whole — so that a question the flat text cannot answer, like “which of my claims about the harbour contradict each other?”, becomes a walk across the lines instead of a guess.

#### YOU NEVER LOSE DATA TO THE GRAPH

An edge is an annotation **over** your nodes, never a container of them. Deleting an edge never touches a paragraph; deleting a node cascades its edges away so none is left dangling. And most edges are **derived** — you can throw the entire graph away and rebuild it from your manuscript, your sidecars, and your installed dictionaries. The graph is powerful precisely because it is disposable.

## The edge — a typed line between two nodes

Everything in the graph is built from one small object. An **edge** is a directed or symmetric relation from a source endpoint to a destination endpoint, carrying a kind and a little metadata about why it exists and how much to trust it.

#### TERM Edge

A single line in the graph: `(src) --kind--> (dst)`, plus a handful of fields — a **kind** (the typed relation), a **weight** (confidence), a **reason** (the rationale, when there is one), an **origin** (how it came to exist), and kind-specific **attrs**. Symmetric kinds are undirected and are found from either end.

### ● ● ● *The anatomy of an edge*

```
(src) --kind--> (dst)    + weight, reason, origin, attrs

src, dst    the two endpoints (a node, or an extern ref)
kind        the typed relation: links_to, contradicts, ...
directed    false for symmetric kinds (found either way)
weight      confidence in [0,1]    (1.0 = asserted)
reason      the human / LLM rationale, when there is one
origin      how it arose – & whether rebuild recomputes it
attrs       kind-specific extras (a locus key, a role, ...)
```

## Endpoints – nodes, and the things that aren't nodes

Most endpoints are ordinary **nodes**: a paragraph, a fact, a character, addressed by its UUID. But an edge sometimes needs to point at something that is **not** a node and should never become one – you do not want ten thousand citations to force ten thousand phantom nodes into being. So an endpoint can also be an **extern reference**: an addressable non-node entity the graph can name without materialising it.

### ● ● ● *The endpoint kinds an edge can reach*

|          |                                                  |
|----------|--------------------------------------------------|
| Node     | any manuscript node, by UUID                     |
| Source   | a Sources-book entry / @cite key                 |
| Work     | an external work id (OpenAlex/arXiv/Wikidata...) |
| Locus    | a canonical primary-source locus (bible:Jn 3:16) |
| Sense    | a WordNet synset in a language (en: s-dog)       |
| Ili      | an interlingual index id – cross-lingual pivot   |
| Grade    | a fact-check verdict bucket (inaccurate)         |
| Evidence | a labelled confront / relate evidence item       |
| Declared | a declared world entity (character/symbol/...)   |

Under the hood the two endpoint columns are real, indexed columns – not a JSON blob – because the **reverse** index is the whole point. Every edge is stored so that “what points at this node?” is a single indexed query rather than a scan of the table, which is what makes a one-hop neighbourhood instantaneous no matter how large the graph grows.

## Edge kinds – the vocabulary of relations

An edge’s **kind** is the typed relation it encodes, and the vocabulary is fixed and language-neutral. Each kind has a natural pair of endpoint types and a place it

comes from in the rest of the tool — the kinds are, in effect, the five implicit ways Inkhaven already encoded relationships, unified into one table.

● ● ● *The edge kinds, and where each one comes from*

|                          |                  |                                  |
|--------------------------|------------------|----------------------------------|
| links_to                 | ¶ → ¶            | the editor's paragraph links     |
| event_involves           | event→char/place | timeline event markers           |
| sourced_from             | fact→source/work | fact provenance                  |
| graded_as                | fact → grade     | /factcheck verdicts              |
| contradicts              | claim ↔ evidence | confront / /contradict           |
| in_tension               | fact ↔ fact      | /relate                          |
| qualifies, agrees        | " "              | confront / /relate               |
| cites                    | work → work      | snowball / OpenAlex              |
| cites_locus              | ¶ → locus        | @key[locus] citations            |
| hypernym/hyponym/antonym | sense↔sense      | WordNet taxonomy                 |
| translates               | sense → ili      | cross-lingual pivot              |
| mentions                 | ¶ → sense        | manuscript→lexicon               |
| declares                 | book → entity    | cast/symbols/motifs              |
| similar_to               | ¶ ↔ ¶            | embedding similarity (see below) |

The last one, `similar_to`, is special: it is deliberately **not** stored. Two paragraphs are “similar” only in the sense that a live nearest-neighbour query over the vector index says so this instant, and that answer changes every time you edit. Materialising it would be a cache that is stale the moment it is written, so embedding similarity stays a live HNSW query and never lands as an edge. Everything else in the list is a real, persisted line.

### Origin and durability — will a rebuild keep it?

The single most important field on an edge is its **origin**, because origin answers the question that governs the whole graph’s economics: **if I run graph rebuild, does this edge survive?** Your decisions must survive; a recomputable cache need not.

● ● ● *Origins, ordered from most durable to recomputed*

|            |                             |                    |
|------------|-----------------------------|--------------------|
| authorial  | you asserted it directly    | kept, never GC'd   |
| promoted   | a judgement you accepted    | kept               |
| judged     | an LLM judgement, advisory  | kept until dismiss |
| imported   | reference data (WordNet...) | its own command    |
| structural | derived from your fields    | RECOMPUTED         |
| derived    | recomputable (similarity)   | kept               |

`graph rebuild` clears and recomputes only the **structural** projection — the edges it can re-derive from your project’s own durable data. Your assertions (`authorial`), the judgements you have accepted (`promoted`), the LLM judgements still awaiting your triage (`judged`), and imported reference data (`imported`, rebuilt by its own command) are all preserved. This is why a rebuild is safe to run at any time and as often as you like: it can only touch the part of the graph that is, by definition, a function of your manuscript.

## Populating the graph

A fresh project’s graph is empty. Three actions fill it, and between them they cover the structural spine, the lexical bridge, and the judgements you accumulate while you work.

### Structural edges — graph rebuild

● ● ● *Deriving the structural projection*

```
inkhaven graph rebuild
```

`graph rebuild` walks your project and derives every **structural** edge from it: the paragraph links you drew in the editor, the timeline event involvements, the provenance of each fact, the `/factcheck` verdicts, and the `@key[locus]` primary-source citations. It is idempotent — it clears the structural projection and rebuilds it whole — so the discipline is simply to run it whenever the manuscript has changed enough that you want the graph current. It never touches your authorial, promoted, judged, or imported edges.

### The lexical bridge — graph lexical

● ● ● *Importing the WordNet lexical net*

```
inkhaven wordnet fetch en      # once, per language
inkhaven graph lexical
```

`graph lexical` imports the **WordNet lexical bridge** for the project language, if you have installed one. It links the words your manuscript actually uses to

their senses (the `mentions` edges), lays in the local semantic net between those senses — `hypernym`, `hyponym`, `antonym` — and rides the interlingual index with `translates` edges so a concept in one language meets its counterpart in another. Run `inkhaven wordnet fetch <lang>` first to install the dictionary; `graph lexical` is idempotent and imported, so it survives a structural rebuild and is refreshed only by re-running it.

### Judged edges, live — confront and graph link

The third source fills the graph **as you work**. The editor’s `confront` (`Ctrl+V ?`) and `/relate` persist their judged stance as `judged` edges — a `contradicts`, an `in_tension`, a `qualifies`, an `agrees` — accumulating rather than repeating, so a second `confront` of the same material does not re-assert the first. And `graph link`, below, asks an LLM to propose stance edges from a fact to its related facts. All of these land as **advisory** `judged` edges: the graph records them, but they wait for your verdict. You **promote** the ones that stick and **dismiss** the rest, and the place they wait is the edge inbox.

#### TERM The edge inbox

The set of advisory `judged` edges awaiting triage — the stance edges proposed by `confront`, by `graph link`, and by deep research, none of them yet accepted. You read it with `graph pending` or the `Ctrl+B z` hub’s `i` view, **promote** what you agree with (it becomes `promoted` and survives rebuilds), and **dismiss** the rest (the edge is deleted).

## The graph command

The read surface is one command with a verb, mirroring the shape of the rest of the CLI. Every verb but `link` and `ask` is deterministic and free; those two need a language model.

● ● ● *The graph verbs*

inkhaven graph <verb>

|                      |                                         |
|----------------------|-----------------------------------------|
| stats                | node + edge counts, per-kind breakdown  |
| rebuild              | (re)derive the structural edges         |
| lexical              | (re)build the WordNet lexical bridge    |
| neighbors <node>     | one-hop neighbourhood as a tree         |
| contradicting <node> | stance clashes touching a node          |
| loci <node>          | the primary-source loci it cites        |
| paths <from> <to>    | a bounded path between two nodes (≤8)   |
| pending              | the judged edge inbox (advisory)        |
| promote <edge>       | judged → promoted (kept across rebuild) |
| dismiss <edge>       | delete a stance edge                    |
| link <node>          | propose stance edges (needs an LLM)     |
| ask <question>       | answer by WALKING the graph (LLM)       |

### stats — the shape of the graph at a glance

**graph stats** is the first thing to run after a rebuild: it reports the node and edge counts and a per-kind breakdown, so you can see at once how much of the graph is structure, how much is lexical, and how many judged edges are waiting. It is the fastest way to answer “did the rebuild actually find anything?”

### neighbors — the one-hop neighbourhood

**graph neighbors <node>** is the workhorse read. It renders a node’s immediate neighbourhood as a tree, grouped by kind, with an arrow on each line telling you its direction — → outgoing, ← incoming, ⇆ symmetric. Large groups truncate, and the neighbourhood is hard-capped so a hub node with five hundred **mentions** edges cannot flood the view.

● ● ● *graph neighbors — a node's one hop, grouped by kind*

```

◆ 003. Quiet hour
├─ contradicts (1)
│   ┆ evidence fact: The lantern was lit at dusk – §3
├─ links_to (2)
│   ┆ ← 002. The tide returns
│   ┆ → 004. A tally of names
├─ mentions (500)
│   ┆ → sense en:enowen-01929162-a
│   ┆ ...
│   ┆ ... +492 more

```

### contradicting, loci, paths — the targeted reads

Three verbs answer sharper questions. `graph contradicting <node>` pulls only the recorded stance clashes touching a node — its `contradicts` and `in_tension` edges, in either direction — which is the query you want when you are hunting for a fact that fights another. `graph loci <node>` lists the canonical primary-source loci a node cites, the scholarly companion to `neighbors`. And `graph paths <from> <to>` finds a bounded citation-or-link path between two nodes — up to eight hops — and prints it, or tells you plainly when no path exists within the bound. A path is how you answer “is this claim connected to that source at all, and by what chain?”

### promote, dismiss, pending — triaging judgements

`graph pending` prints the edge inbox — the advisory `judged` stance edges awaiting your verdict. For each, `graph promote <edge>` accepts it: the edge’s origin becomes `promoted`, and it is thereafter kept across every rebuild. `graph dismiss <edge>` rejects it, deleting the stance edge outright. These three are the CLI face of the same triage you can do from the editor’s hub, and they are the only way a machine judgement ever becomes a durable part of your graph — nothing is promoted without your say-so.

### link — proposing stance edges from a fact

`graph link <node>` asks a language model to look at a fact and propose stance edges from it to its related facts — the contradictions, tensions, and agreements a human might notice reading the two side by side. It writes them as `judged` edges into the inbox; you then triage them with `pending` / `promote` / `dismiss` exactly as with any other judgement. It is the one write verb that reaches for the LLM, and it never asserts anything directly — its whole output is advisory by construction.

### ask — answering by walking the graph

`graph ask` is the verb that made the graph worth building, and it gets its own section below, because it is not a read of the graph so much as a **conversation** with it.

## Chatting with your graph

The graph is not only queried by verb. You can **converse** with it — ground a language model in the graph’s **relations**, not just the prose — so that you can ask how your book connects: what contradicts what, what grounds a claim, how a scene

is sourced. There are two surfaces for this, and they differ in how far they let the model travel: one reads a single hop, the other walks.

### The Graph AI scope — one hop, in the editor

You met the AI pane's **scopes** in the assistant chapters — the setting, cycled with **F9**, that decides what the model is grounded in before it answers. The last scope on the cycle, after Editor, is **Graph**.

#### TERM The Graph scope

An AI-pane scope (**F9**) that grounds the model in the graph. It retrieves the passages relevant to your question — the same semantic retrieval Book scope uses — and, beneath each passage, folds in the graph edges touching it (**contradicts**, **sourced\_from**, **links\_to**, **cites**, ...). The answer stands on both the prose **and** those relations, under the same citation contract as Book scope. A **sticky** scope: it retrieves once per chat and re-grounds when you clear history.

A prompt in Graph scope answers from one hop out — the passages it retrieved and the edges directly on them. It carries the same discipline as Book scope: every claim cites a passage's **[location/path]**, and an invented label is flagged rather than trusted. Press **p** in the AI pane to expand the "Retrieved passages

1. graph relations" transparency section and read the exact subgraph the answer was built on — the graph never asks you to take its grounding on faith.

### graph ask — the traversal loop on the CLI

● ● ● Asking a question the flat text cannot answer

```
inkhaven graph ask \  
"which of my claims about the harbour contradict?"
```

Where the Graph scope reads one hop, **graph ask** lets the model **walk**. It searches for seed nodes matching your question, then issues read-only graph queries — neighbours, contradictions, loci, paths — turn by turn, following the lines wherever they lead, until it has seen enough to answer. The exploration transcript prints to **stderr** so you can watch the path it took; the grounded answer goes to **stdout** so you can pipe it. It is honest by construction: when the relations simply do not



record what you asked, it says so rather than inventing a connection. The graph is only ever as complete as `graph rebuild`, `confront`, and `graph link` have made it, and `ask` will not paper over a gap.

The cost of a question is bounded and tunable — this is the permissive principle, where the caps inform and never block. Two settings govern the walk.

● ● ● *Bounding the walk (hjson config)*

```
graph: {
  ask_max_steps: 8      // max LLM turns before answering
  ask_search_width: 6   // seed nodes per search
}
```

### Walking the graph in the editor — `Ctrl+B z`

The same traversal runs **inside** the editor, streamed, through the graph hub. `Ctrl+B z` opens a small menu onto the graph with three entries, and it is the one place all of the graph's in-editor surfaces gather.

#### TERM The graph hub

The `Ctrl+B z` overlay — a three-way menu onto the knowledge graph. Press `n` for the **neighbourhood** of the paragraph you are editing (the same tree as `graph neighbors`, scrollable, `Esc` to close); `i` for the **edge inbox** (the advisory `judged` edges, where `P` promotes and `d` rejects the selected one); or `w` to **walk the graph** and answer the question typed in the AI prompt. Populate the graph first with `graph rebuild` / `graph lexical`.

The `w` walk is the streamed twin of `graph ask`. Type a question into the AI prompt, then `Ctrl+B z` and `w`. The AI pane shows the walk unfold live — each step as the model takes it, 🔍 `search`, 🔗 `neighbours`, ⚖️ `contradicting` — and then streams the grounded prose answer, which lands as a normal chat turn. The status bar reads `graph walk · turn k/N` as it goes, and `Esc` stops the whole walk at any moment. The bounds are the same `ask_max_steps` and `ask_search_width` as the CLI. It is deliberately an **explicit** action — a walk is several model calls, not the single hop of the Graph scope — so the depth, and its cost, is something you opt into per question.

## Triaging findings without opening the hub

One more pair of keys lives outside the hub. When a **confront finding** is sitting in the Output pane — a stance the fact-checker or continuity watch has proposed — you can act on its edge in place: **P** promotes the finding's stance edge to a kept decision that survives a rebuild, and **d** dismisses the finding and deletes its edge. It is the same promote/dismiss verdict as the inbox, on the finding where you first meet it.

|                 |                                                                 |
|-----------------|-----------------------------------------------------------------|
| <b>F9</b>       | Cycle the AI scope; Graph sits last, after Editor.              |
| <b>Ctrl+B z</b> | Open the graph hub (then n / i / w).                            |
| <b>z then n</b> | Neighbourhood of the open paragraph (tree, ↑↓, Esc).            |
| <b>z then i</b> | Edge inbox — P promotes, d rejects the selection.               |
| <b>z then w</b> | Walk the graph to answer the AI-prompt question.                |
| <b>P / d</b>    | On an Output-pane confront finding: promote / dismiss its edge. |
| <b>p</b>        | In the AI pane (Graph scope): expand the grounding subgraph.    |

## From a script — the `ink.graph` words

The graph is scriptable from the embedded Bund language, which you will meet in full in the scripting part. The read and triage verbs are exposed as deterministic words; the LLM walk (`graph ask`) is deliberately **not** a sync word, because a word that fans out to a model is not something a script should call as if it were pure.

### ● ● ● *The `ink.graph` Bund surface*

```
ink.graph.stats      ( -- dict )  counts + per-kind
ink.graph.neighbors  ( node -- list )  one-hop edges
ink.graph.contradicting ( node -- list )  stance clashes
ink.graph.loci       ( node -- list )  cited loci
ink.graph.paths      ( from to -- list|nil )  a path
ink.graph.pending    ( -- list )  the judged inbox
ink.graph.rebuild    ( -- dict )  re-derive structure
ink.graph.promote    ( edge -- bool )  judged→promoted
ink.graph.dismiss    ( edge -- )  delete a stance edge
```

The read words are classified `STORE_READ` and the three that change the graph — `rebuild`, `promote`, `dismiss` — are `STORE_WRITE`, so the sandbox policy knows exactly what a script may touch. This is the surface you reach for when you want to bake a graph check into a headless pipeline: `rebuild`, read `pending`, and act on what it returns without opening the editor at all.

## Multilingual by construction

The graph is multilingual because its **kinds** are language-neutral. A `contradicts` edge between a Russian fact and a French source is the very same kind of edge as any other — the relation does not know or care what language its endpoints are written in. Locus canonicalisation collapses `John 3:16`, `Иоанна 3:16`, and `Joh 3.16` to a single endpoint, so a citation in any language lands on the same node. And the lexical bridge is cross-lingual by design: because `translates` edges ride the interlingual index, a Russian sense and a German sense of one concept meet at the same ILI, and “what is the German word for this Russian concept?” becomes a reverse-index lookup rather than a translation call.

### FICTION

For a novel with an invented world, the graph’s value is the stance edges — the `contradicts` and `in_tension` lines confront lays down — and the neighbourhood view that shows a scene’s links and mentions at a glance.

### NON-FICTION

For non-fiction, the spine is `sourced_from`, `graded_as`, `cites`, and `cites_locus`: graph paths and graph ask let you trace a claim back to the source that grounds it, across the whole corpus.

The graph also feeds the **Inner-family readers** you met in the intelligences part. Their shared grounding now reports what the graph has established that no single declared-data store computes on its own — recurring character pairings (who keeps appearing together across your scenes) and the running count of unresolved factual contradictions. Relational context, in other words, surfaced where the readers can use it.

## Storage and crash-safety

The graph lives in its own DuckDB store, `edges.db`, sitting beside `metadata.db`, `blobs.db`, and the `vectors/` directory under your project. Keeping it separate is deliberate: the graph is a distinct data layer with its own durability rules, and it can be rebuilt without touching anything else.

### IT SURVIVES A KILL -9

Writes to `edges.db` are atomic — a batch of edges either lands whole or rolls back — so the store survives a `kill -9` mid-write and reopens consistent, to the 1.2.15 stability bar. Deleting or moving a node cascades its edges away, so no edge is ever left pointing at a node that is gone. And the `structural`, `derived`, and `imported` edges are provably rebuildable — so even a corrupted graph is never **lost data**. Run `graph rebuild` (and `graph lexical`) and it comes back.

That disposability is the quiet theme of the whole chapter. The graph is powerful — it holds relations the flat manuscript cannot express, and it lets a model walk them to answer questions the prose alone cannot — and yet the only part of it you can truly **lose** is the part you authored: your assertions and the judgements you chose to promote. Everything else is a function of your book, and a function can always be recomputed. Build the graph freely; it will never hold your book hostage.

## WHAT YOU LEARNED

- The knowledge graph is a **typed-edge** layer over the UUID nodes you already have — `(src) --kind--> (dst)` with a weight, a reason, an **origin**, and `attrs` — stored in its own `edges.db` and indexed on **both** endpoints so a neighbourhood is a query, not a scan.
- An edge's **origin** decides its durability: `authorial`, `promoted`, `judged`, and `imported` survive a rebuild; only the `structural` projection is recomputed — so `graph rebuild` is always safe.
- `graph`  
Three actions fill the graph: `graph rebuild` (structural edges), `lexical` (the WordNet bridge), and **judged** edges accumulated live from `confront / /relate / graph link` into the **edge inbox**.
- The `graph` command reads it — `stats`, `neighbors`, `contradicting`, `loci`, `paths`, `pending` — and triages it — `promote`, `dismiss` — while `link` and `ask` reach for an LLM.
- You can **chat** with the graph: the **Graph** scope (F9) grounds an answer in one hop of relations, and `graph ask` — on the CLI, or streamed in-editor via `Ctrl+B z → w` — lets the model **walk** the graph, honestly saying so when the relations do not record what you asked.
- `ink.graph.*` scripts the read and triage verbs; the graph is multilingual by construction; and `edges.db` is atomic, cascade-safe, and — for everything but your own assertions — rebuildable, so it is never lost data.

CHAPTER 16

# 16

## The Timeline

---

A long book keeps two clocks. One is the order in which the reader meets the scenes — the order of the pages. The other is the order in which the events actually happened in the world of the story, which a book with flashbacks, parallel storylines, and a prologue set three ages earlier will scramble on purpose. Most of the mistakes a novel makes about time are mistakes about the gap between those two clocks: a messenger who arrives before he could have ridden the distance, a character standing in two cities on the same afternoon, a midsummer feast three chapters after the first snow. Inkhaven's **timeline** is the machinery for keeping the second clock — a record of what happened **when**, in a calendar you invent, that the reading intelligences can hold up against the prose.

This is an opt-in layer. A project has no timeline until you ask for one, and nothing you have read so far changes when you do — the timeline sits over the paragraph hierarchy you already have, adding a thin skin of **event** metadata to selected paragraphs rather than a new place to store anything. This chapter covers the whole of it: what an event is and how you record one, how to define a calendar of your own devising and how Inkhaven reads dates written in it, the critique that catches the timeline’s internal slips, and — the reason the feature earns its keep — how the events you record feed the continuity watch and the knowledge tracker so that **when** becomes a fact those systems can check.

## Turning the timeline on

The feature is off by default so that existing projects upgrade without surprise, and so that a book that has no need of a calendar is never nagged about one. You turn it on with a `timeline` block in the project’s `inkhaven.hjson`, and the smallest possible form is three lines and a preset.

● ● ● *The minimal timeline block in inkhaven.hjson*

```
timeline: {
  enabled: true
  default_track: "main"
  calendar: { preset: "gregorian" }
}
```

Two settings do the everyday work. `enabled` is the gate: with it `false` (or absent) every timeline command refuses politely rather than seeding events into a project that never opted in — `inkhaven event add` errors with a one-line reminder to set the flag. `default_track` names the storyline an event belongs to when you do not say otherwise; think of a track as a swim lane — a POV character, a parallel plot, a “flashback” line — that lets two events at the same moment sit on different rows instead of colliding. The `calendar` block is the substance, and the rest of this chapter’s first half is about filling it in.

**WHY OPT-IN**

A timeline only pays for itself once you have events in it, and events are work to record. Inkhaven would rather ask you to turn the feature on deliberately than have every project carry an empty calendar it never uses. A non-fiction manuscript, a book of poems, a reference manual — none of these wants a story clock, and none of them is asked to.

**What an event is**

An event is not a new kind of object with its own file and its own database table. It is a **paragraph** — an ordinary node in your book’s tree — that carries an extra parcel of metadata marking it as something that happened at a moment in story-time. That parcel is small and entirely legible.

**TERM Event**

An **event** is a paragraph tagged with a start on the calendar and, optionally, an end, a precision, a track, and links to the characters and places it involves. Under the hood it is a block of **EventData** hung on a normal node: a start tick, an optional end tick, a **precision** (how exact the date is), a track label, a list of character ids, and a list of place ids. Everything else about the paragraph — its title, its prose, its position in the tree — is unchanged.

The heart of the parcel is a single number. Every moment on the timeline is stored as one signed 64-bit integer — a count of **ticks** since the calendar’s epoch, negative for anything before it, so a prologue set in a prior age is just a negative tick. All the arithmetic the timeline ever does is integer arithmetic on that number; every scrap of calendar complexity — months of uneven length, named moons, seasons that wrap the year — lives in the **display** layer that converts a tick to a human string and back. An event that is an instant carries a start tick and no end; an event with duration carries both, and the span it occupies is the closed interval between them.

**Where events live — the Timeline chapter**

The first time you add an event under a given book, Inkhaven creates a chapter called **Timeline** inside that book to hold it, and every later event under the same book goes into the same chapter. It is created lazily — a book you never add an



event to never grows one — and it is marked internally with a system tag rather than by its title, so you can rename it with **F2** and the timeline machinery still finds it. Different books each get their own.

● ● ● *The Timeline chapter, created on the first event add*

```
Aerin Saga/
├─ Chapter 1/
├─ Chapter 2/
└─ Timeline/           ← lazily created on first event add
    ├─ Birth of Aerin  ← an event paragraph (carries EventData)
    ├─ Storm of Year 1
    └─ Marketplace scene
```

This is worth a moment's thought, because it explains a distinction you will lean on. The event paragraph in the Timeline chapter is the **record** that the event happened; it is not usually the same paragraph as the **scene** that depicts it, which lives out in Chapter 4 where the reader meets it. The two are joined by a **link** — the event's list of linked paragraphs — and that link is what lets a scene inherit its date from the event, and what lets the swim-lane view show a Storm event under Chapter 4 even though the event record sits in the Timeline chapter. An event with no such link, and no characters or places either, is an **orphan**: a moment recorded but attached to nothing, which the tooling marks so you can decide whether it wants a scene or was a stray.

## Recording an event

There are three ways to record an event, and they differ only in where you are standing when you do it. The most explicit is the command line, which is also the clearest way to learn the shape of the thing.

● ● ● *Adding events from the command line*

```
# An instant event – precision inferred from the date shape.
inkhaven event add "Marketplace scene" \
  --start "1A.2.8" --book-name "Aerin Saga"

# A duration event – a start and an end.
inkhaven event add "The Storm" \
  --start "1A.2.3" --end "1A.2.5" \
  --track main --book-name "Aerin Saga"

# --precision overrides what the parser would infer.
inkhaven event add "Spring of Rain" \
  --start "1A.spring" --precision season \
  --book-name "Aerin Saga"
```

The `--start` string is a date written in your calendar, and its **shape** decides the event's precision unless `--precision` overrides it — a point we return to below. `--end` makes the event a duration; leave it off and the event is an instant. `--track` places the event on a storyline, defaulting to your `default_track`. `--book-name` names the book (by slug or title, case-insensitively) and is required only when the project holds more than one. Inkhaven refuses an end that falls before its start — events do not run backwards — and prints back what it recorded, the date rendered in your calendar's own display form.

Inside the editor the everyday way is a chord. `Ctrl+V Shift+E` records an event from wherever you are: with text selected in the Editor, the selection becomes the event's title; inside the timeline view it drops the event at the cursor's tick. This is the flow that fits actual writing — you are drafting a scene, you select the sentence that anchors it in time, you strike the chord, you type the date, and the event is on the timeline without your hands leaving the keyboard. Inside the swim-lane view the bare `n` key does the same at the cursor's position.

## FICTION

A novelist records events as the plot demands them: the coronation, the siege, the two-week ride. The track label is where parallel POV lines and flashbacks earn their keep — give each its own track and the swim lanes stop colliding.

## NON-FICTION

A non-fiction writer rarely needs a story clock at all. Where a work is genuinely chronological — a history, a biography, a project retrospective — the same events + calendar apparatus will hold real dates on the gregorian preset, and the critique still catches the orphan and the impossible overlap.

### Linking, characters, and places

A freshly added event is bare — it has a date and nothing else — which is why it starts life as an orphan. You give it substance by linking it to the scene that depicts it: with the event's paragraph open, press `Ctrl+V A` and pick the manuscript paragraph. The link is recorded and the orphan mark clears. From the swim-lane view, `Enter` on an event does **not** create a link — it **navigates** to the scene an event is already linked to (opening the one paragraph, or a picker when several point at it). The link is also scriptable, through the `ink.event.link_paragraph` word covered at the end of this chapter.

An event can also name its **participants** — the characters and places it involves. These explicit lists are exactly what KEN reads for its **presence** grants (a character present at an event knows what happens there), so you attach them by hand: `inkhaven event link-character <event-path> "<name>"` and `inkhaven event link-place <event-path> "<name>"` add an entry from your Characters or Places book to the event. The continuity **co-location** check is more forgiving — it **also** derives participants from the characters and places **named in the event's linked scenes**, so it can flag a character in two places at once even before you list anyone by hand; only KEN's knowledge grants insist on the explicit list, so a merely-mentioned name never grants knowledge it shouldn't.

**ORPHANS ARE A NUDGE, NOT AN ERROR**

An orphan event — one with no linked scene — is drawn with a hollow ◯ glyph everywhere it appears, and opening one lands a hint on the status bar naming the one chord (Ctrl+V A) that will link it. It is a soft signal that a recorded moment might want a scene, not a mistake to be corrected. Deliberate backstory that will never get a scene is free to stay orphaned forever.

## A calendar of your own devising

The calendar is the part of the timeline you actually design. Its job is to convert between the single tick number the timeline stores and the dates a person writes and reads, and Inkhaven gives you three ways to specify one: two presets for the common shapes and a **custom** form for a world of your own.

### ● ● ● *The three calendar flavours*

|                     |                     |   |                       |
|---------------------|---------------------|---|-----------------------|
| preset: "gregorian" | real-world dates    | → | 1917.11.7 · 2026.5.20 |
| preset: "sols"      | days since day zero | → | Sol 1 · Sol 142       |
| preset: "custom"    | anything you invent | → | 1A.Highsun.15         |

The **gregorian** preset is the real world: years of twelve named months, months of thirty days (a deliberate simplification — the timeline measures spans, not leap-years), and the four seasons already wired up, so a scene dated to month seven reads as summer without any further declaration. The **sols** preset is the opposite extreme — a single unit, “days since day zero”, displayed as **Sol N** — which suits a survival log, a voyage, or any story that counts days from a fixed beginning and needs nothing finer.

### Building a custom calendar

A custom calendar is a **stack of units**, declared base-first. The first unit is the base — one tick is one of these — and each unit above it declares how many of the level below make one of itself. A day, then a month of thirty days, then a year of twelve months, is the familiar shape; but the counts are yours, and so are the names.

```

calendar: {
  preset: "custom"
  base_unit: "day"
  units: [
    { name: "day", names: [] }
    { name: "month", per_parent: 30,
      names: ["Frostmoon", "Snowfall", "Greenstart",
              "Bloomtide", "Highsun", "Goldfall",
              "Mistwane", "Stormrise", "Coldgate",
              "Longnight", "Hearthlit", "Yearfall"] }
    { name: "year", per_parent: 12, names: [] }
  ]
  seasons: [
    { name: "winter", start_month: 1, span_months: 3 }
    { name: "spring", start_month: 4, span_months: 3 }
    { name: "summer", start_month: 7, span_months: 3 }
    { name: "autumn", start_month: 10, span_months: 3 }
  ]
  epoch_label:      "A"      # 1A.3.15 = First Age, year 1
  epoch_before_label: "BA"    # negative years, e.g. -1BA
  display_format:    "{year}{epoch_label}.{month}.{day}"
}

```

Four pieces here repay attention. A unit's `names` list gives its values display names — the twelve moons above turn month 5 into **Highsun** — and an empty list falls back to a bare number, so a numeric month renders `1.5.15` while a named one renders `1A.Highsun.15`. The `seasons` list names spans of months and is what gives a date its season; each season declares the month it starts on and how many months it covers, and a season may wrap the year boundary (winter over the turn) without confusing the arithmetic. The two `epoch_label` fields are the suffixes for years at or after the epoch and before it — leaving `epoch_before_label` empty makes the calendar reject negative years outright, which is the right choice for a world with no prehistory. And `display_format` is the template every date is rendered through, with `{year}`, `{month}`, `{month-name}`, `{day}`, and the epoch tokens as its placeholders. A `parse_aliases` list, not shown, lets you name landmark dates — **Founding**, **Day Zero** — that resolve to a fixed tick.

### How a date is read — precision from shape

Here is the idea that makes the calendar more than decoration. When you write a date, Inkhaven infers not only **which** moment you meant but **how exact** you were being, and it reads that exactness from the **shape** of what you typed. A date

that stops at the year is a year-precise date; one that names a month is month-precise; one that runs down to the day is day-precise. That inferred precision — `Tick`, `Hour`, `Day`, `Week`, `Month`, `Season`, or `Year` — is stored with the event and decides how wide a **window** the event occupies when the critique asks whether two events could overlap.

**TERM Precision**

A date’s **precision** is how exact you were when you wrote it, inferred from its shape. A day-precise date is a point — a one-day window. A season-precise date is a whole season — a window ninety days wide, in a thirty-day-month calendar. Precision is why “sometime that spring” and “the fifteenth of Highsun” behave differently: the first can collide with other vague dates; the second is a pin. `--precision` overrides the inference when you want to be explicit.

The parser walks the dotted segments of a date under the year — month, then day, then hour — and the last segment you supply sets the precision. A season **name** in the month slot is read as season-precision; a landmark alias resolves to its fixed tick at day-precision. The table below reads the Aerin calendar above.

● ● ● *Parsing the Aerin calendar — shape sets precision*

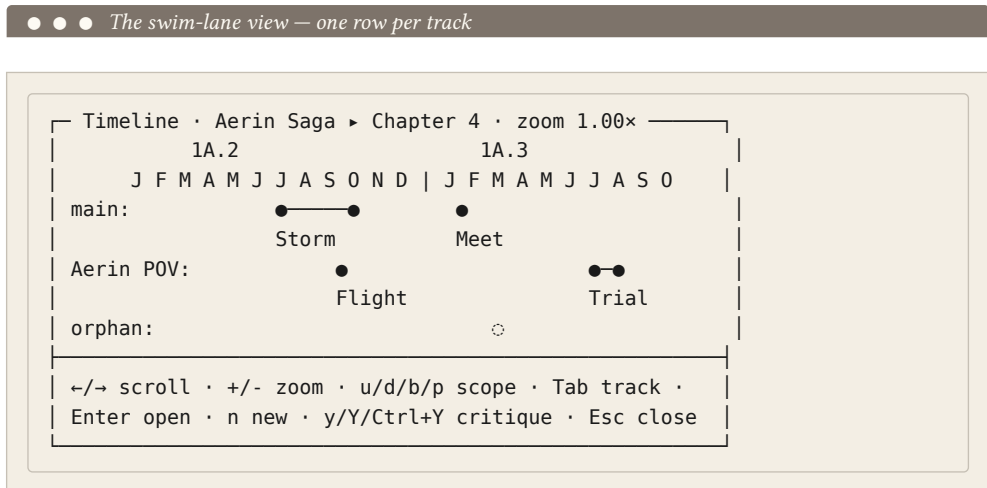
| you type  | parsed as   | precision | ticks |                |
|-----------|-------------|-----------|-------|----------------|
| -----     | -----       | -----     | ----- |                |
| 1A        | year 1      | Year      | 0     |                |
| 1A.3      | year 1 mo 3 | Month     | 60    |                |
| 1A.Frost  | Frostmoon   | Month     | 0     | (prefix match) |
| 1A.spring | spring      | Season    | 90    |                |
| 1A.3.15   | day 15      | Day       | 74    |                |
| -1BA      | year -1     | Year      | -360  |                |
| Founding  | alias       | Day       | 0     |                |

Two conveniences fall out of this. Month names match on a unique prefix, so `1A.Frost` finds **Frostmoon** without your typing it out; and a date is symmetric through parse and format, so a tick rendered to a string and read back lands on the same tick. The one value the calendar forbids is a year of zero — there is no year 0 between `1A` and `-1BA`, the epoch being year 1 — and Inkhaven says so plainly rather than guessing.

## Seeing the timeline

Two views show you the events you have recorded, and both are reached with the view prefix. This manual keeps them brief — the companion book **Building the World** walks every chord — but you should know they exist and what each is for.

**Ctrl+V e** opens a **chronological picker**: a vertical, time-ordered list of every event, filterable by track, from which **Enter** loads an event’s paragraph into the Editor. It is the fast way to jump to a known event. **Ctrl+V Shift+T** opens the **swim-lane view**, the headline UI: a horizontal timeline with one row per track, opened at your current paragraph’s nearest scope, which you scroll, zoom, and drill into by chapter. In both, the same three glyphs read the same way — a filled dot for an instant, a bar for a duration, a hollow ring for an orphan.



The one rule worth carrying out of these views is how **scope** filters events. At book scope every event shows. At a narrower scope — a chapter, a subchapter — an event appears if it **or any paragraph it links to** lives inside that scope. So a Storm event whose record sits in the Timeline chapter still shows when you scope into Chapter 4, because the scene it links to is there. That is the link doing its second job: not only dating the scene, but placing the event where the scene is read.

## The timeline critique

The critique is the timeline watching **itself** — the checks that depend on the timeline’s own semantics and belong to no other system. It once did more; over several releases, four of its original concerns moved to systems that do them better (we come to those next), leaving two genuinely timeline-internal checks. Both are pattern-based, deterministic, and free — no model is called — and both emit their findings to the Output pane alongside the fact-checker’s and the continuity watch’s, with the same severity glyphs and the same **Enter**-to-jump.

### Orphan events

The first check finds **orphans** — events linked to nothing, no paragraph, no character, no place — and grades how much the orphan matters. A trivial stub recorded five minutes ago is barely a note; a four-word, day-precise event that has sat unconnected for three months is a likely authorial slip. The grade combines **significance** — read from how concrete the date is, how rich the title, and how active the event’s track — with **staleness**, how long the orphan has gone unlinked past a grace window you can set. High significance is a contradiction-level finding at any age; a middling orphan warns once it goes stale; a low one stays a passing note.

### Fuzzy-precision overlap

The second check is the one the precision system was built for. Two events with coarse dates — a season here, a month there — each occupy a **window** rather than a point, and when two such windows collide **suspiciously** the prose probably cannot place both events consistently. Suspicion is not mere overlap; it is overlap plus a reason to worry — the two events share a track, or share a character or place. A same-track, shared-character collision is a warning; a loose cross-track brush is filtered out below the threshold. When three or more fuzzy events mutually overlap on a common window, Inkhaven reports the **cluster** once instead of drowning you in pairwise noise.



## ● ● ● Critique findings in the Output pane

```

┌─ Output · 3/3 · timeline-critique ───┐
│ ◎ orphan_event                       │
│   "The Lost Coronation Rite" – day-precise, 4 words, │
│   orphaned 92 days. Link a scene or retire it.      │
│ ▲ fuzzy_overlap                       │
│   "Training" and "Journey" (season-precise, main,   │
│   both with Mara) place in the same window.        │
│ ● overlap_cluster                       │
│   3 season events share a window at Velmaril.      │
└────────────────────────────────────────┘
↑↓ select · Enter jump · a ask-AI · d dismiss

```

You run the critique over a **scope**, and a few chords set how wide. Inside the swim-lane view, **y** runs it over the current view for the highlighted track only, **Y** over the current view for all tracks, and **Ctrl+Y** widens to the whole book; **F12** is a global shortcut for that same whole-book pass, reachable from any pane. There is also a unified pass: **Ctrl+B Shift+C** runs every fast, deterministic checker at once — the fact-checker and the Socratic reader over the open paragraph, and the timeline critique over the project — and drops all their findings into the Output pane together. On the command line, `inkhaven event critique` prints the same two checks, scoped with `--track` or `--book-name`, and its findings are localized to the project's working language.

## WHAT MOVED OUT OF THE CRITIQUE

Four checks that once lived here now live where they belong, and knowing where saves you looking in the wrong place. **Travel-time** (a journey the calendar says was too fast) and **co-location** (a character in two places at once) `inkhaven realworld fact-check` are the world fact-checker's, run by `--timeline-aware` and `inkhaven realworld co-location`. **Date coherence** (a midsummer feast in a winter scene) is the fact-checker's too. **Pacing** — how densely events pack the page — is the Inner Socratic reader's. The critique keeps only what is irreducibly about the timeline's own structure.

## How the timeline feeds the other intelligences

The timeline earns its place not by what it shows you but by what it lets the rest of the tool **know**. Once events carry dates, participants, and places, three of Inkhaven's reading intelligences can ask questions of the prose that no generic reader could — because the answers depend on a story clock that only you have declared.

### The fact-checker reads **when**, not just **what**

The world fact-checker knows your world's geography and climate; with a timeline it also knows **when** each scene happens. A paragraph linked to an event, or near one in world-time, gains an **effective date** — the date of the event it links to — and from that date, a **season**.

#### TERM **Effective date**

A paragraph's **effective date** is the world-time the checker reads it at: the start of the earliest event linked to it. From the effective date the checker derives the **season** per your calendar, and against that it can judge whether the weather, the light, and the festivals a scene describes belong to the time the timeline places it. A paragraph with no linked event has no effective date, and the timeline simply stays quiet about it.

So snow in a scene the timeline dates to summer stops being ambiguous and becomes a contradiction the checker can name; a three-day ride the calendar says took thirty-five is a warning it can raise. These timeline-aware findings carry a small calendar mark in the Output pane to tell you the clock was consulted, and they run in every project language.

### SENTINEL continuity uses event placement

The continuity watch — the system that watches a book for the things a long manuscript forgets — folds the timeline's placement into its deterministic findings. **Co-location** is the clearest case: from nothing but the events' participant and place lists, the watch finds a character whose overlapping events put them in two different places at once, and reports it without reading a word of prose. The same event placement backs the date-coherence and travel-time findings the continuity ledger surfaces. When your world genuinely allows the impossible — a teleporting

mage, an astral projection — a rule in the **magic ledger** excuses the conflict with a note instead of nagging.

### KEN uses event participation for who was present

The knowledge tracker — Inkhaven’s watch over **who knows what, and when** — draws one of its two grant sources straight from the timeline. Its principle is that a character **present at an event knows what happened there**, from the moment of the scene that depicts it onward. So every character in an event’s participant list is granted knowledge of that event’s subject, dated to the event’s first linked paragraph. That grant is what lets the tracker catch a character who refers to the coronation before any scene could have told them of it — **premature knowledge** — the referenced-before-introduced invariant extended from things to **knowledge**. The event’s title is the topic by default; a **reveals:** tag on the linked scene can name the topic more precisely when the title is terse. This is the timeline’s deepest contribution: presence at an event, which only your timeline records, becomes the ground truth for what a character could plausibly know.

## The CLI and the Bund word

Everything the timeline does interactively is also scriptable and headless, which is what lets a timeline live under version control and take part in a batch run.

The **inkhaven event** subcommand is the command-line surface: **add** records an event (with **--start**, **--end**, **--precision**, **--track**, **--book-name**), **list** prints every event in chronological order (filterable by **--book-name** and **--track**), **show** prints one event’s full detail by its slug-path, and **critique** runs the two retained checks. Every one of them refuses unless **timeline.enabled** is **true**, so you cannot seed events into a project that has not opted in.

● ● ● *inkhaven event list — the chronological view*

```
$ inkhaven event list --book-name "Aerin Saga"
  1A.1.1 ○ Birth of Aerin          track=main .../birth
  1A.2.3–2.5 – The Storm          track=main .../storm
  1A.2.8 ● Marketplace scene      track=main .../market
```

From Bund, the `ink.event` family gives scripts the same operations. Seven words cover the lifecycle — listing, adding, and refining an event — and each mutation fires a hook so other scripts can react.

● ● ● *The `ink.event.*` Bund words*

```

ink.event.list          ( -- list )
ink.event.list_orphans ( -- list )
"Saga" "Storm" "1A.2.3"
ink.event.add           ( book title spec -- uuid )
id "1A.2.5" ink.event.set_end      ( uuid spec -- )
id "season" ink.event.set_precision ( uuid prec -- )
id "main"   ink.event.set_track    ( uuid track -- )
id "saga/ch4/storm"
ink.event.link_paragraph ( uuid path -- )

```

`ink.event.add` takes a book name, a title, and a start spec, creates the event under that book's Timeline chapter (materialising the chapter if need be), infers precision from the spec's shape exactly as the CLI does, and returns the new event's id — which the `set_*` and `link_paragraph` words then refine. The `list` words return dictionaries carrying every field of an event: its id, title, slug, path, start and end ticks, precision, track, orphan flag, and its linked paragraphs, characters, and places. Reads are default-allowed; the mutating words sit behind Bund's `store_write` category, opt-in like every other write. A companion set of `ink.event.critique.*` words exposes the two checks to scripts, and two hooks — `hook.on_event_added` and `hook.on_event_orphaned` — fire on the lifecycle so a script can, say, tag every new event or flag one that has just lost its last link.

## WHAT YOU LEARNED

- The **timeline** is an opt-in layer (`timeline.enabled: true`) that records what happened **when**, as **events** — paragraphs carrying a start on a calendar, an optional end, a precision, a track, and links to characters and places.
- Every moment is one signed integer **tick**; a **calendar** — a **gregorian** or **sols** preset, or a **custom** stack of named units with seasons and epoch labels — converts ticks to dates and back, and a date's **shape** sets its **precision**.
- Events live in a lazily-created **Timeline chapter** per book; an event linked to nothing is an **orphan** (○), a nudge rather than an error. Record events with `inkhaven event add`, `Ctrl+V Shift+E`, or `n` in the swim-lane view.
- The **critique** keeps two timeline-internal checks — **orphan events** graded by significance and staleness, and **fuzzy-precision overlaps** that collide suspiciously — run by `y/Y/Ctrl+Y/F12` or `inkhaven event critique`; travel-time, co-location, date coherence, and pacing moved to other systems.
- The timeline **feeds the intelligences**: the fact-checker reads each scene's **effective date** and season, SENTINEL uses event placement for co-location and date coherence, and KEN turns **presence at an event** into who could know what.
- Everything is scriptable — the `inkhaven event` subcommands and the `ink.event.*` Bund words (with `hook.on_event_added` / `hook.on_event_orphaned`) — so a timeline works headless and under version control.

PART V

---

# The Intelligences

CHAPTER 17

# 17

## Continuity and Knowledge

---

A short book can be held in one head. A long one cannot, and it is the things you cannot hold that break it: a character who is standing in two cities on the same afternoon, a river that runs six days wide in chapter three and a morning's walk in chapter nine, a ferryman named twice before he is ever introduced, a detective who names the murder a chapter before he could possibly have learned of it. None of these are mistakes of prose. Every sentence is clean; the fracture is between the sentences, across a span no single reading holds in view at once. They are the errors a manuscript accumulates precisely because it is long — and they are the ones a careful reader will catch and you will not.

Inkhaven answers with two intelligences that read the whole book at once and watch for exactly these cross-page fractures. **SENTINEL** watches **continuity** — where and when, how many, what was established and then quietly changed. **KEN** watches **knowledge** — who could know what, and by when. They are siblings by design: KEN is SENTINEL’s cardinal invariant, **referenced before introduced**, carried one step further from **does this thing exist yet** to **could this person know this yet**. Both flag and never rewrite; both are deterministic and free at the core, spending nothing and running in a blink whatever the book’s length; both feed the same review pass and the same revision worklist you will meet in the next chapters.

This chapter is the operator’s tour: what each detects, how to run it from the command line and from the editor, and how to read what comes back. It is deliberately breadth-first. The full treatment — the theory of each detector, the design of the grant model, the worked examples — lives in the companion book **Know Your Book**, which this chapter points you toward wherever the depth runs past what an operator needs to get moving.

#### ADVISORY, ALWAYS

Neither intelligence ever touches your prose. They read the manuscript, the timeline, the world books, and the knowledge graph, and they write **findings** — anchored notices you can jump to and act on. Every actual edit stays yours, routed through the Editorial Pass of Chapter 19 with its snapshot-first, diff-reviewed contract. A finding is a pointed finger, not a red pen.

## SENTINEL — the book watches itself

Inkhaven could always **check** a manuscript’s continuity — but for years that meant knowing which of six separate commands to run, each with its own sidecar file and its own mental model. SENTINEL is the layer that folds them into **one always-watching concern**. One engine runs every deterministic continuity detector already in the tree, normalises each into a single finding shape, dedupes them, and ranks the result — contradictions first, then warnings, then information, and within a severity the earlier chapters first. What you get back is a single ordered ledger, not six piles to reconcile by hand.



## What it detects

Five detectors feed the ledger. Four were scattered across the tool before SENTINEL unified them; the fifth is the invariant nobody had.

### ● ● ● The five deterministic detectors

| detector    | break it catches                                    | reads                              |
|-------------|-----------------------------------------------------|------------------------------------|
| -----       | -----                                               | -----                              |
| co_location | a character in two places<br>at overlapping times   | the timeline<br>(magic suppressed) |
| timeline    | orphaned events · fuzzy-<br>precision overlaps      | the timeline<br>critique           |
| numeric     | a direction reversed · a<br>duration that conflicts | prose quantities<br>(EN / FR / ES) |
| char_facts  | an established fact changed<br>across chapters      | continuity.json<br>(the bible)     |
| introduce   | an entity referenced<br>before it is introduced     | the graph +<br>prose mentions      |

The first four are recognitions of a kind you would make yourself if you could hold the whole book in mind: two scenes that put the same person in two places at once, a timeline event with nothing anchoring it, a distance that shrinks between chapters with no reason given, an eye colour or a hometown that drifts. The `char_facts` detector leans on the **continuity bible** — the per-character, `inkhaven continuity` per-attribute record that `extract` builds with an AI pass and `inkhaven continuity list` dumps; once extracted, the drift check over it is pure comparison, deterministic and free.

## The invariant nobody had — referenced before introduced

The `introduce` detector is the one SENTINEL adds, and it is worth a moment. An entity's **introduction** is its first scene — the earliest paragraph any of its timeline events touch. A **reference** is any mention of it in the prose. When the first reference lands in a chapter earlier than the introduction, by more than a configured tolerance, SENTINEL flags it:

### ● ● ● The referenced-before-introduced finding

- ⊗ [introduce] "Aldous the ferryman" is referenced in ch. 2 but not introduced until ch. 5.

This is the fracture a first-time reader feels as a small stumble — **who?** — and that you, who know Aldous intimately, read straight past every time. It is language-safe everywhere, because the names come from your own Characters and Places books and the mention match is Unicode-aware; a Cyrillic or an accented name is matched exactly as a Latin one is.

#### TERM **Deterministic**

A check is **deterministic** when it computes its answer from structure — the timeline, the graph, the tagged prose — with no model in the loop, so it is free, instant, and gives the same answer every run. SENTINEL’s five detectors are all deterministic. The one fuzzy check it can invoke, the coherence pass, is kept explicitly out of the sweep for exactly that reason.

### Running it — **inkhaven continuity check**

The whole ledger is one command, and it is built to drop into a script or a CI gate as readily as into your own terminal.

● ● ● *The command and its flags*

```
inkhaven continuity check
  [--only DETECTOR]...    # run only these (repeatable)
  [--skip DETECTOR]...    # skip these (after --only)
  [--json]                # machine-readable array
  [--coherence]           # + the slow LLM pass
  [--max-cost 8000]       # soft token budget
  [--force]]              # ignore the cache
```

The detector names are exactly the five from the table — **co\_location**, **timeline**, **numeric**, **char\_facts**, **introduce**. Name one wrong and the command tells you so and lists the known ones, rather than silently checking nothing. A plain run prints the ranked findings with a severity glyph on each — ☹ a contradiction, ⚠ a warning, ● information — the detector that raised it in brackets, and the chapter it sits in.

● ● ● *A run with two findings*

```
$ inkhaven continuity check
⊗ [co_location] Mara is at the quay and at Rillmark
  on the third evening. (ch. 4)
● [introduce] "Aldous the ferryman" is referenced in
  ch. 2 but not introduced until ch. 5.

2 finding(s): 1 contradiction(s), 1 other.
```

The one behaviour to build a habit around: **the command exits non-zero when any contradiction survives**. A clean book, or one whose only findings are warnings and information, exits zero. That single rule is what lets `inkhaven continuity check` sit in a pre-commit hook or a CI pipeline as a gate — the build fails on a hard continuity break the way it fails on a compile error. Add `--json` and the same findings come back as an array of objects (`kind`, `severity`, `chapter`, `anchor`, `entities`, `message`, `source`) for a script to read.

#### `--ONLY` AND `--SKIP`

The two selectors compose in one order: `--only` narrows to the named detectors, then `--skip` removes from whatever remains. `--only introduce` runs the invariant alone; `--skip numeric` runs the other four. Reach for them when you are chasing one class of break, or when a detector's language coverage does not fit the book — `numeric` reads English, French, and Spanish quantities and skips cleanly for other languages, so on a German or Russian manuscript you lose nothing by skipping it explicitly.

### The ledger dashboard — `Ctrl+B Shift+I`

`Ctrl+B`

Inside the editor, the same ranked ledger is a keystroke away. `Shift+I` opens the **continuity ledger** — a scrollable modal of every deterministic finding, grouped by kind and ranked as on the command line. `↑↓` scroll it; `Enter` jumps straight to the offending paragraph so you can see the break in context; `Esc` closes it. It is the fastest way to walk a book's continuity without leaving your place in it.

● ● ● *Ctrl+B Shift+I — the continuity ledger*

```

┌ Continuity ledger · 3 findings ────────────┐
│ CONTRADICTIONS                             │
│ ④ co_location Mara: the quay & Rillmark, 3rd eve │
│   ch. 4 · the-crossing                      │
│                                              │
│ WARNINGS                                   │
│  △ numeric "six days' ride" → "a morning" ch. 9 │
│                                              │
│ INFO                                       │
│  ● introduce "Aldous the ferryman" ref. ch. 2  │
│───────────────────────────────────────────┘
└─ scroll · Enter jump · k coherence pass · Esc

```

The **k** key in this dashboard is the one that costs — it runs the slow coherence pass, described just below, over the open book. Everything else in the ledger is deterministic and free, which is why the dashboard opens instantly no matter how long the manuscript is.

### The slow coherence pass

There is one fuzzy check SENTINEL will **invoke** but will never run on its own. An LLM reads a run of paragraphs and flags the cross-paragraph contradictions the deterministic detectors cannot see — a fact asserted and then quietly reversed, a time of day that cannot follow the one before it, the kind of soft inconsistency that lives in the meaning rather than in a number or a name. Because it costs, it is **explicit, cost-capped, and opt-in** on both surfaces:

● ● ● *The two ways to run the coherence pass*

```

CLI    inkhaven continuity check --coherence
       [--max-cost 8000] [--force]

editor k  in the Ctrl+B Shift+I ledger dashboard
         (results land in Output · source coherence)

```

The pass respects the **magic:** ledger's declared exceptions — a world where someone genuinely can be in two places is not a bug — and it needs a configured LLM provider. Its cost is previewed against your daily cap before it runs. As everywhere in Inkhaven, that cost **informs**; it never blocks. **--max-cost** sets a soft

token budget for the preview, and `--force` ignores the cache to re-run a pass you have run before.

### The ambient watch

The last surface is the quietest. Turn on `continuity.ambient` and SENTINEL re-checks continuity on **every save** — but only over what the edit actually touched. It reads the edited paragraph’s entities and chapter from the graph, re-runs the deterministic detectors against just that scope, and surfaces the delta immediately in the Output pane. Because the core is deterministic and free, this happens inline, without a background job and without a pause you would notice. It is off by default; when you turn it on, the book watches itself as you write, and a break you introduce is flagged in the same breath you introduce it.

#### COOLDOWN

The ambient watch throttles itself with `ambient_cooldown_secs` (default 30) so a burst of rapid saves does not re-check on every keystroke-flush. The floor is a throttle, not a queue — it collapses a flurry of saves into one re-check rather than running the check that many times.

### Configuration

Everything above is governed by one config block, all fields optional; the values shown are the defaults.

● ● ● *The continuity: block*

```
continuity: {
  enabled: true           // review-pass ledger switch
  ambient: false          // re-check on every save
  ambient_cooldown_secs: 30 // throttle floor (seconds)
  co_location: true       // per-detector toggles
  timeline: true
  numeric: true
  char_facts: true
  introduce: true
  introduce_tolerance: 0   // chapters tolerated (0=strict)
}
```

Two knobs deserve a word. `enabled` is the master switch for the **review-pass line** only — turn it off and the deterministic ledger stops riding the `Ctrl+B Shift+C`

review pass, but the standalone `inkhaven continuity check` command still runs, because you invoked it explicitly. And `introduce_tolerance` is how much **referenced-early** the invariant will forgive: `0` is strict, flagging any reference before the introduction; raise it to allow a name to appear a chapter or two ahead of its scene without complaint, for a book that deliberately foreshadows.

### From a script — the Bund words

The ledger is readable from the embedded Bund language of Part VIII, so a hook or a script can gate on it. Two words, both read-only and deterministic — the coherence pass, because it costs, is deliberately **not** exposed to Bund.

● ● ● *ink.continuity.\* — read-only*

```
ink.continuity.findings ( -- list )
  the ranked, deduped findings as dicts:
  {kind, severity, chapter, source,
   message, entities}

ink.continuity.check      ( -- dict )
  summary counts:
  {total, contradictions, warnings,
   info, by_kind}
```

### What SENTINEL is, and is not

It is the **unification** of the continuity detectors you already had, plus the one invariant nobody had — not a new pile of them. It is **deterministic-first**: the fuzzy passes stay explicit and cost-capped, and the core is free. It is **advisory**: it flags, it never rewrites, and it adds no new runtime dependency to the tool. Every finding carries the detector that raised it, so the honest question — **does this work in Russian?** — answers itself per detector: `introduce`, `co_location`, `timeline`, and `char_facts` are multilingual as built, and `numeric` names its EN / FR / ES coverage and skips cleanly elsewhere.

### KEN — who knows what, when

Inkhaven watches several kinds of continuity — where and when (SENTINEL), whether an entity exists yet (the `introduce` invariant), whether the world stays consistent (Facts), whose head we are in (the voice reader’s POV). None of them watch the one axis that breaks **plots**: what a character **knows**. KEN is that watch.

It takes SENTINEL's cardinal move — flag a thing **named before it exists** — and carries it into knowledge: flag a character **acting on a fact before they could have learned it**. That is a mystery's cardinal sin and the most common invisible plot-hole in any book with a secret in it.

What makes KEN a **native** intelligence — a finding a generic AI could not produce — is that reconstructing who-knows-what across a whole book needs the timeline, the event-participant lists, the character bible, and scene POV all at once, and those are exactly what Inkhaven already maintains and a chat model does not have. KEN never guesses what a character knows. It **derives** it.

### The grant — when could they know it?

The heart of KEN is the **grant**: the earliest point at which a character could know a given topic. It is derived two deterministic ways, never guessed.

#### ● ● ● The two sources of a grant

```

PRESENCE    a character in a timeline event's
              participant list knows that event from
              the moment it happens — free, from the
              structure you already keep.

DECLARED    you mark it with a tag:
secret:<topic>    <topic> is a secret; a
                  reference by anyone ungranted
                  is a leak.
know:<topic>      grants the scene's POV char.
know:<topic>@<name> grants a named character.
reveals:<topic>   on an event's paragraph, binds
                  a terse title to the topic it
                  reveals (a matchable handle).
```

A **use** — a character referencing a topic — is caught the same deterministic way SENTINEL matches names: by Unicode-aware matching of the topic in that character's **attributed dialogue** or in the **narration of their POV scene**. Presence and the tags grant knowledge; a use before any grant is a break. You fill an event's participant list explicitly with **inkhaven event link-character** (Chapter 16) — presence then grants every listed character knowledge of that event, which is often less work than tagging each with **know:**.

**TERM Ken**

A character's **ken** is the range of what they know — the set of topics their grants cover, up to a given point in the book. KEN the feature is named for it. The whole check is a forward walk that builds each character's ken chapter by chapter and flags any use that outruns it.

**What it catches**

Four findings, three of them deterministic and free, the fourth an opt-in LLM pass for the subtle case the structure cannot see.

● ● ● *The KEN findings*

|                                                |                                                                                                                           |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <code>premature_knowledge</code><br>(☉ break)  | a character references a topic before their earliest grant.<br>"Bob names the murder in ch. 4; he learns of it in ch. 6." |
| <code>leaked_secret</code><br>(☉ break)        | a secret: topic referenced by a character never granted it.                                                               |
| <code>dropped_reveal</code><br>(● notice)      | a declared know: reveal whose topic never surfaces again — dangling knowledge, the epistemic unpaid setup.                |
| <code>implied_irony</code><br>(--deep, opt-in) | a character acts informed or ignorant without naming the topic — the subtle case only an LLM can see.                     |

The first two are **breaks** — the hard errors. `dropped_reveal` is a softer notice: you told the tool a reveal happens, and then the topic never comes back, which is either a thread you forgot to pay off or a tag you can retire. `implied_irony` is the only one that costs, and the only one that is not on by default.



## Running it — inkhaven knowledge

• • • *The command and its flags*

```
inkhaven knowledge      # the deterministic check
inkhaven knowledge --json # findings as JSON
inkhaven knowledge --deep # + the implied_irony pass
    [--max-cost 8000]    # soft budget for --deep
    [--book-name SLUG]   # restrict to one book
```

A plain run prints each finding with its severity glyph — ☒ a break, ● a notice, • information — its kind in brackets, and its message.

• • • *A run with two breaks*

```
$ inkhaven knowledge
☒ [premature_knowledge] Bob speaks of "the murder"
  in ch. 4 — before learning it in ch. 6.
☒ [leaked_secret] Sella references "the heir's true
  name" in ch. 3 — never established to know it.

2 finding(s): 2 break(s), 0 other.
```

Like `continuity check`, `knowledge` **exits non-zero on any hard break** (`premature_knowledge` or `leaked_secret`), so it is a CI gate too. And it is **self-gating**: with no `secret:` or `know:` tags and no timeline events, there is simply nothing to check, so it does nothing and costs nothing. You pay for KEN only in proportion to the epistemic structure you have actually declared. `--book-name` narrows a multi-book project to one book; without it, the whole project is checked.

### The knowledge dashboard — Ctrl+B Shift+Z

In the editor, `Ctrl+B Shift+Z` opens the **knowledge dashboard** — the findings grouped by kind, the same shape as the continuity ledger. `↑↓` scroll; `Enter` jumps to the offending paragraph; `Esc` closes.

● ● ● Ctrl+B Shift+Z — the knowledge dashboard

```

— Knowledge · who knows what, when · 3 —
BREAKS
| ④ premature_knowledge Bob: "the murder" ch. 4
|   grant: ch. 6 · the-confession
| ④ leaked_secret Sella: "the heir's true name"
|   ch. 3 · never granted
|
| NOTICES
| ● dropped_reveal "the ledger" declared, never
|   surfaces again
|
| ↑↓ scroll · Enter jump · Esc close

```

### The deep pass — implied\_irony

The whole KEN core is a forward walk, set membership, and Unicode mention matching: **no model, about zero cost, independent of book length**. Cost scales with declared topics and scenes, not pages. The **only** LLM touchpoint is the opt-in `--deep` pass, which hunts the case the structure cannot reach — a character who **acts** on knowledge, or **acts** ignorant, without ever naming the topic, so no mention match can catch it. It runs cost-capped under the daily cap, on the same rail as the world fact-checker, never automatically and never over the whole book at once. There is, by deliberate design, no **have the AI judge what everyone knows** pass — that would be guessing, and KEN does not guess.

### From a script — the Bund words

Three read-only words expose the grants and the findings. The `--deep` pass, because it costs, is not among them.

● ● ● *ink.knowledge.\* — read-only*

```
ink.knowledge.grants ( -- list )
  the who-could-know-what ledger:
    {character, topic, chapter, source}

ink.knowledge.findings ( -- list )
  the deterministic breaks:
    {kind, severity, chapter, character,
     topic, message}

ink.knowledge.check ( -- dict )
  {premature, leaked, dropped, clean}
```

The `clean` field of `ink.knowledge.check` is `true` when no hard break stands — a one-word pre-submit gate for a hook.

### What KEN is, and is not

It is not an all-knowing oracle: it reasons only over what it can ground — events, tags, named mentions — and stays **silent** where it cannot, rather than inventing a break. It is not a fact-checker: Facts asks **is the world consistent**, KEN asks **could this character know this yet** — a different axis entirely. It is not LLM-first, and it is not a rewriter. Its multilingual reach it inherits from SENTINEL and the dialogue-attribution layer: the mention matcher is Unicode-aware, and attribution ships English, Russian, German, French, and Spanish conventions, so topics and names are matched in the project’s language.

### One core, two feeds

The two intelligences share a shape worth naming plainly, because it is what makes them cheap enough to leave on. Both are **deterministic and free at the core** — a forward walk over structure you already keep, no model, the same answer every run, a cost that does not grow with the book. Both keep their one fuzzy pass **explicit, opt-in, and cost-capped** — SENTINEL’s coherence pass behind `--coherence` and the `k` key, KEN’s implied-irony pass behind `--deep` — so the price is only ever paid on purpose.

And both pour their findings into the same two downstream surfaces you will meet next. The first is the **review pass**, `Ctrl+B Shift+C`, which runs every fast checker at once: SENTINEL’s ledger rides it under the `continuity` source, and

KEN’s breaks join it too, each finding anchored so you can jump straight to it in the Output pane. The second is the **Editorial Pass**, `Ctrl+V Shift+R` — the revision worklist of Chapter 19 — where continuity and knowledge findings take their place in one ranked list beside every other reader. There, a knowledge break is routed as a **decision (fix the leak, move the reveal, or add a grant?)** rather than a mechanical rewrite, because resolving one is a choice about the story, not a phrasing. `inkhaven revise` synthesises that same worklist into an editorial letter. The intelligences watch; the Editorial Pass is where watching turns into an edit — under the snapshot-first, diff-reviewed contract that means no word of your prose ever changes without your say-so.

**FICTION**

For **fiction** this pair is close to the whole reason the world and timeline exist. Keep the timeline’s event-participant lists honest and tag your secrets with `secret:` / `know:` / `reveals:` , and KEN will catch the leaked confidence and the too-early deduction that a mystery lives or dies by — the errors beta readers find and you cannot.

**NON-FICTION**

For **non-fiction** SENTINEL still earns its place: the `introduce` invariant flags a term, a person, or an acronym used before it is defined — the reference-before-introduction that trips a reader of a manual or a monograph exactly as it trips a reader of a novel. KEN’s grant model is fiction-shaped and simply finds nothing to check.

|                       |                                                                                   |
|-----------------------|-----------------------------------------------------------------------------------|
| <b>Ctrl+B Shift+I</b> | Open the SENTINEL continuity ledger — deterministic findings, ranked and grouped. |
| <b>Enter</b>          | In the ledger, jump to the finding’s paragraph.                                   |
| <b>k</b>              | In the ledger, run the opt-in, cost-capped LLM coherence pass over the open book. |
| <b>Ctrl+B Shift+Z</b> | Open the KEN knowledge dashboard — who knows what, when, grouped by kind.         |
| <b>Ctrl+B Shift+C</b> | The review pass — runs every fast checker; SENTINEL and KEN ride it.              |
| <b>Ctrl+V Shift+R</b> | The Editorial Pass — the revision worklist both intelligences feed.               |

## WHAT YOU LEARNED

- **SENTINEL** watches **continuity** — five deterministic detectors (`co_location`, `timeline`, `numeric`, `char_facts`, and the **referenced-before-introduced** introduce invariant), deduped and ranked into one ledger.
- `inkhaven continuity` check runs it — `--only` / `--skip` select detectors, `--json` for scripts, `--coherence` for the opt-in LLM pass — and **exits non-zero on any contradiction**, so it gates CI.
- `Ctrl+B Shift+I` opens the ledger dashboard (`↑↓` scroll, `Enter` jump, `k` runs the coherence pass); `continuity.ambient` re-checks the edit's scope on every save, deterministic and inline.
- **KEN** watches **knowledge** — a character's **ken** is derived from timeline **presence** plus the `secret:` / `know:` / `reveals:` tags, never guessed; it flags `premature_knowledge`, `leaked_secret`, and `dropped_reveal`.
- `inkhaven knowledge` runs it (also non-zero on a hard break, and self-gating); `--deep` adds the opt-in `implied_irony` LLM pass; `Ctrl+B Shift+Z` opens the dashboard.
- Both are **deterministic and free** at the core, keep their one fuzzy pass **explicit and cost-capped**, expose read-only `ink.continuity.*` / `ink.knowledge.*` Bund words, and feed the `Ctrl+B Shift+C` review pass and the `Ctrl+V Shift+R` Editorial Pass. The full treatment is in **Know Your Book**.

CHAPTER 18

# 18

## The Read-Through and the Voices

---

The last chapter's readers watch a manuscript for what it must not do — a fact contradicted, a secret spilled early, a name used before it is met. This chapter is about a softer, harder question: not “is the book **wrong**” but “how does the book **read**.” Four intelligences answer it, and they divide the labour by zoom. LECTOR reads the whole book forward, once, as the one reader who matters most — the **first** reader, who does not know the ending. CHORUS holds the whole cast's voices in mind at book scale and asks whether any two of them blur. DIALOGUE reads speech as its own craft, tracking who is talking and whether the page ever loses them. And the narrator's own voice — the person telling the story — is profiled across the draft so you can see chapter forty drift away from chapter one.

None of the four edits a word. They are **reading** instruments: they measure, they report, and they hand any fix they imply to the Editorial Pass (`Ctrl+V Shift+R`, Chapter 19). Almost all of what they do is deterministic and free — no model call, no cost — and where a model **does** help, the call is explicit, cost-capped, and never automatic. Read this chapter as the tour of how a long book **sounds** and **feels** from the outside, and how Inkhaven lets you hear it without leaving the terminal.

#### TERM The outer readers

Where the **inner readers** (Chapter 20) read a paragraph at a time, these four read **out**: LECTOR the whole arc, CHORUS the whole cast, DIALOGUE every exchange, NARR-1 the whole narrator over time. They are advisory to the last one — every finding is a signal to consider, never a change to your prose.

## LECTOR — the read-through

Every other intelligence in Inkhaven reads **small**: a paragraph, a chapter break, a character's lines. Nothing read the manuscript the way a person actually reads it — cover to cover, in order, once, carrying forward everything met so far and knowing nothing of what is coming. That reader is the one whose experience the book lives or dies by, and LECTOR is Inkhaven's model of them. It reports two different things about the read, and it is worth keeping them apart in your mind: the **shape** of the read (its structure and pacing) and the **experience** of the read (its clarity, attention, stakes, and payoff).

### The Shape half — the intensity curve against a framework

LECTOR measures each chapter's dramatic **intensity** from the prose itself — no tagging, no plan required. It reads the signals a reader feels as rising tension: dialogue density, a per-language stakes-and-conflict lexicon, sentence-rhythm acceleration, a penalty for passages that summarise rather than dramatise, and a bonus for a chapter that ends on a turn. That gives a realised curve — how the book **actually** rises and falls — which it then lays against the **intended** curve of a story framework, and it flags the places where the shape wants a rise but the prose reads flat.

● ● ● *inkhaven readthrough — the measured curve vs the framework*

Read-through — 12 chapter(s) · Hero's Journey

measured 

expected 

ch 1 ■ ► Arrival  
 ch 2 ■ ► The City  
 ch 5 — ● Old Debts  
 ch 11 ■ ► The Breakwater  
 ch 12 ■ ● After

△ [shape\_sag] the Hero's Journey shape wants rising tension around ch. 5 (~55%) but the prose reads flat (~12%).

3 reader finding(s): 1 concern(s).

The framework is not something you must configure. If you set `lector.framework` it is used; otherwise LECTOR **suggests** one from your project `genre` — fantasy and epic and quest lean to the Hero's Journey, thriller and romance to Save the Cat, mystery and crime to the Seven-Point, character and coming-of-age to the Story Circle, slice-of-life and literary and vignette to Kishōtenketsu — and if it can infer nothing it falls back to Three-Act, so the overlay works on any manuscript with zero setup.

### TERM The six frameworks

LECTOR ships six intended-shape curves: **Three-Act**, **Save the Cat**, **Story Circle**, **Hero's Journey**, **Seven-Point**, and **Kishōtenketsu**. The last is the odd and important one: the four-movement East-Asian structure (ki-shō-ten-ketsu) whose energy peaks at the **ten** — the recontextualising twist — rather than at a conflict climax. It is **conflict-optional**, so a quiet, tension-light book is not scored as if it failed to be a thriller.

Alongside the curve, LECTOR classifies every chapter on the **scene** ⇌ **sequel** axis. A **scene** is the forward, external unit — goal, then conflict, then disaster; a **sequel** is the reflective, internal one — reaction, then dilemma, then decision. In the report a scene chapter carries ► (forward), a sequel ● (reflective), and a chapter that is neither clearly ■. The value is in the **rhythm**: a long run of pure scene reads



breathless with no room to feel the cost, and a long run of pure sequel sags, so LECTOR flags the **arrhythmia** rather than any single chapter's kind.

#### MEASURED, NOT DECLARED

The Shape half **measures** the curve the Planning Board (Chapter 16) **declares**. If you never touched the Board, LECTOR still works — the intended curve comes from the framework, the realised curve from your prose. The two are independent tools that happen to speak the same language of beats.

### The Audience half — the forward walk

The second half is the reader's **experience**, and its defining discipline is that it is **forward-only**. LECTOR walks the book from the first chapter, carrying the state a first reader would carry — which characters and places they have met, which threads are hanging open, how the energy has been running — and every finding it raises uses only the chapters read **so far**. A payoff in chapter thirty can never reach back and cancel a dip it noticed in chapter four, because the reader in chapter four had not read chapter thirty. That single rule is what makes this a **reader** rather than an **analyst**.

Five deterministic findings come out of the walk, and all of them are zero-AI:

#### ● ● ● The five forward-walk findings

|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| confusion     | an entity used before it is introduced — "who is this again?"       |
| info_dump     | too many new names to meet in one chapter                           |
| attention_dip | a flat, eventless chapter where attention drifts                    |
| put_down_risk | a RUN of flat chapters — a likely place a reader sets the book down |
| unpaid_setup  | a setup raised early and never paid off                             |

The **confusion** and **info\_dump** findings lean on the same Unicode-aware mention matcher SENTINEL uses (Chapter 17), so they work in every script Inkhaven supports, not only English. The intensity signals behind the Shape half key off the project language exactly as the other detectors do — the stakes and reflection

lexicons ship for English, Russian, German, French, and Spanish and skip cleanly elsewhere, while sentence rhythm is language-agnostic and always contributes.

### The synthetic first-read — the one, explicit LLM pass

Some things the deterministic walk simply cannot judge: whether a passage is **genuinely** confusing to a human, whether the stakes are **legible** on the page, whether engagement is **really** flagging. For those, a model helps — and LECTOR's one LLM feature, the **synthetic first-read**, reacts to each chapter as a first reader who does not know the ending. It is forward-only by construction: each call sees only a recap of what has been read plus the current chapter, never the whole book at once.

It is never automatic. You ask for it — `inkhaven readthrough --deep`, or the **k** key in the dashboard — and before each chapter the estimated cost is previewed against your daily cap. Per Inkhaven's permissive principle the cost **informs**; it never blocks. Its findings arrive tagged `source: reader` and land alongside the deterministic ones, marked so you always know which came from a model.

● ● ● *The synthetic first-read, cost-capped and explicit*

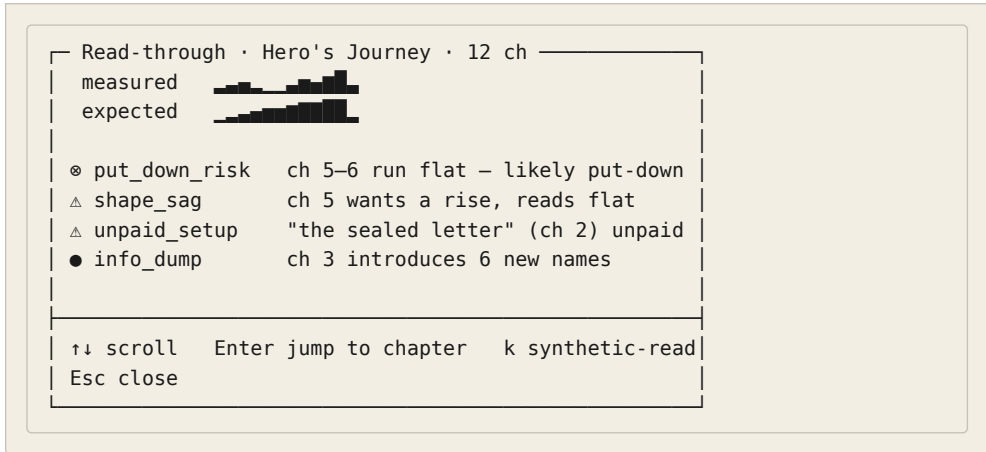
```
inkhaven readthrough --deep [--max-cost 8000] [--force]

ch 4 synthetic first-read · est. ~1,240 tokens (cap ok)
→ the reader can follow the heist, but the guard's
   betrayal reads as a surprise the setup never earned.

ch 5 synthetic first-read · est. ~1,180 tokens (cap ok)
→ the funeral lands, but two new cousins arrive unnamed
   and the reader loses the thread of who inherits.
```

### Running it — the command, the dashboard, the Bund

The command line prints the whole read in one shot: the measured-vs-expected curve, the per-chapter scene/sequel beat, and the ranked reader findings. `--deep` folds in the synthetic first-read; `--json` gives you the structured form for tooling. In the editor the read-through has its own dashboard.

● ● ● *Ctrl+B Shift+A — the read-through dashboard*

**Ctrl+B Shift+A** opens that scrollable modal — the curve, the beats, and the ranked findings together. Arrow keys scroll it, **Enter** jumps straight to the flagged chapter in the editor, **k** runs the cost-capped synthetic first-read (its results post to the Output pane's **readthrough** category), and **Esc** closes. The deterministic findings also ride the unified review pass: **Ctrl+B Shift+C** adds a **read-through** line, each finding anchored to its chapter, in the Output pane.

● ● ● *The LECTOR surface at a glance*

|                             |                                      |
|-----------------------------|--------------------------------------|
| inkhaven readthrough        | the full report                      |
| inkhaven readthrough --deep | + synthetic read                     |
| inkhaven readthrough --json | structured output                    |
| Ctrl+B Shift+A              | the dashboard (k = synthetic read)   |
| Ctrl+B Shift+C              | read-through line in the review pass |
| ink.readthrough.report      | the ranked findings                  |
| ink.readthrough.curve       | per-chapter measured/expected        |
| ink.readthrough.check       | counts by kind + severity            |

The three **ink.readthrough.\*** Bund words expose only the deterministic read — the synthetic first-read is deliberately not scriptable, because a cost-bearing model call has no place firing silently from a hook.

**WHAT LECTOR IS NOT**

It is not a rewriter — it reports the read and hands fixes to the Editorial Pass. It is not a per-paragraph reader — it is whole-book, forward, once. And it is not an oracle of taste: it reports legible, grounded signals — a flat stretch, an unmet name, an unpaid setup — never a verdict on whether your book is **good**.

**CHORUS — the voices at book scale**

A novel is not narrated in one voice but in many: the narrator's, and every character who speaks. Inkhaven already profiles the narrator (NARR-1, below). What makes a **cast** hold together — do the characters sound like distinct people, does each one stay themselves, does the book keep the point-of-view rules it set, does its register hold — is CHORUS. Three measurement pillars feed a seventh inner reader, the Inner Stylist, and like everything here CHORUS **measures and reports**; it never touches your prose.

**Pillar A — character voice and the distinctiveness matrix**

CHORUS profiles each character's dialogue through the **same** metric engine that profiles the narrator — sentence rhythm, lexical diversity, hedging, interiority — so a character's voice is measured on the same axes as the book's. From those fingerprints it builds the **distinctiveness matrix**: it z-scores every voice across the cast (the baseline is your own book's spread, so the comparison is genre-relative) and looks for pairs that read alike. The headline finding is the one every novelist fears — **"Mara and Joren read identically."**

● ● ● *inkhaven chorus voices — a signature card + the matrix*

Character voices — `The Sunken Throne` [en]

|             |                 |                  |
|-------------|-----------------|------------------|
| Mara        | confidence ●●●○ | Δ from cast mean |
| rhythm      | ■■■■            | varied +0.4      |
| diversity   | ■■■■            | high +0.6        |
| hedging     | ■               | asserts -0.3     |
| interiority | ■■■             | moderate 0.0     |

Distinctiveness

- △ Mara ↔ Joren read alike (z-dist 0.31 < 0.50)
- ✓ all other pairs distinct

Drift

- Aldric ch 9 drifts from his ch 1 voice

Two safeguards keep it honest. Sparse speakers — a character with only a handful of lines — are profiled but never **flagged**: CHORUS reports a confidence badge and refuses to judge a voice it cannot measure. And deliberate look-alikes — twins, a uniform chorus of guards — are silenced with `chorus.distinct_ignore_pairs`, so you tell it once that Mara and Joren are **meant** to echo and it stops raising them. Per-character drift is measured against a character's **first** chapter, answering “does Aldric still sound like Aldric in Act III?”

## Pillar B — POV and tense discipline

Two classic errors survive line editing because they are **structural**, not local. The first is **head-hopping**: a named character other than the scene's viewpoint shown accessing their own inner life — `Joren wondered...` in a scene that belongs to Mara. A scene's POV is either **declared** with a paragraph tag or **inferred** from who is mentioned most.

● ● ● *Declaring a scene's POV with a paragraph tag*

|                             |                                                            |
|-----------------------------|------------------------------------------------------------|
| <code>pov:Mara</code>       | single POV — anyone else's interiority is a leak           |
| <code>pov:first</code>      | first person — ANY named character's interiority is a leak |
| <code>pov:omniscient</code> | deliberately multi-POV — head-hop off                      |
| <code>pov:multi</code>      | (an alias for <code>pov:omniscient</code> )                |

CHORUS is honest that this is a heuristic: there is no parser in the paragraph tree, so it catches interiority attributed to a **named** subject, not pronoun antecedents — `she thought` where “she” is not the POV would need antecedent resolution CHORUS deliberately does not attempt. The second error is a **tense slip**: a manuscript that lapses out of the tense it established. Each narration sentence is classified past or present from its copula and auxiliary anchors, the scene’s dominant tense is the majority, and the sentences that break it are flagged.

#### THE TENSE GATE IS ENGLISH-GATED — RUSSIAN IS EXCLUDED BY DESIGN

Tense-slip detection covers **English, German, French, and Spanish** — languages that share the “hold one narrative tense” convention. **Russian is excluded on purpose**. Russian narrative tense is governed by **aspect**: the historical present and the perfective/imperfective interleaving are legitimate devices, not slips, and nothing in the tree models aspect — so a past-to-present heuristic would be **wrong** for Russian. `chorus scan` says so plainly rather than false-flagging. Character voice and head-hop **do** work in Russian.

#### Pillar C — register and diction

The third pillar tracks the narrator’s **register** — formal or plain, contracted or measured, plain or archaic — per chapter, so **drift** becomes visible: “the prose gets casual in Act III.” It bundles a contraction rate, an archaism density, a formality balance, and (for English) a latinate-diction proxy; chapters that drift from the opening past `chorus.register_drift_threshold` are flagged. It is solid for English and Russian and degrades gracefully — to whatever its word lists cover — for the other languages rather than guessing.

● ● ● *inkhaven chorus scan — POV, head-hop, tense, register*

Voice-discipline scan — `The Sunken Throne` [en]

POV / head-hop

- △ ch 4 · Mara-POV — "Joren wondered" (interiority leak)
- ch 7 · POV inferred (Seren) — no pov: tag

Tense

- △ ch 6 · 3 present-tense slips in a past-tense scene

Register

- ch 10 drifts casual vs ch 1 (Δ 0.11 > 0.08)

### The Inner Stylist — the coach (Ctrl+B J → Y)

The three pillars produce numbers; the **Inner Stylist** turns them into judgement. It is the seventh member of the inner-reader family — alongside the Editor, Socrates, the Theologian, the Poet, Rigor, and Grounding — and it does not measure, it **synthesises**: it reads all three pillars and offers a few grounded **Praise / Note / Concern** observations, in the inner-family's own voice ("I notice...", never a rewrite).

You reach it from the family hub. **Ctrl+B J** opens the Inner Socrates overview, and **Y** from there opens the Inner Stylist. Inside, **F** synthesises the pillars to the Output pane's **stylist** category, **E** engages the AI coach into the Thoughts pane (grounded coaching, never a rewrite), and **R** opens the scrollable voice report. It also rides the **Ctrl+B Shift+C** review pass.

● ● ● *Ctrl+B J → Y — the Inner Stylist overview*

```
┌ Inner Stylist · voice at scale ────────────┐
│ Praise                                     │
│   the cast reads distinct — 9 of 10 pairs separate │
│ Note                                     │
│   Aldric's voice drifts formal after ch 9 │
│ Concern                                  │
│   Mara ↔ Joren read alike — consider sharpening one │
└───────────────────────────────────────────┘

F synthesise → Output   E engage AI coach → Thoughts
R report dashboard     Esc close
```

Each finding carries a stable **key**, and `--suppress <key>` silences it for good — persisted in `inner_stylist.db`, this reader’s own store — so a judgement you have considered and rejected does not nag you again.

## The CHORUS command line and Bund

Four subcommands cover the surface. `chorus voices` prints the signature cards and the distinctiveness summary; `chorus scan` runs the POV, head-hop, tense, and register checks; `chorus stylist` synthesises the Praise/Note/Concern (with `--coach` for grounded LLM coaching and `--suppress / --unsuppress` to manage findings); and `chorus report` is the one-screen dashboard that folds the narrator profile, the cast voices, and the Stylist’s synthesis together.

### ● ● ● The CHORUS surface

```

inkhaven chorus voices  [--character NAME] [--json]
inkhaven chorus scan    [--json]
inkhaven chorus stylist [--coach] [--suppress KEY]
inkhaven chorus report  [--json]

ink.chorus.voices      per-character fingerprints
ink.chorus.distinct    the distinctiveness matrix
ink.chorus.drift       per-character voice drift
ink.chorus.headhops    POV / head-hop findings
ink.chorus.tense       tense summary (or the honest reason)
ink.chorus.register    per-chapter register + drifts
ink.stylist.findings   the synthesised Praise/Note/Concern

```

Every `ink.chorus.*` word is deterministic and returns a list or dict; the LLM coaching is not a Bund word (a cost-bearing call again stays off the script path) — use `chorus stylist --coach` for it.

### WHAT CHORUS IS NOT

It is not a style **corrector** — it flags, it never rewrites. It is not a grammar checker — there is no parser; the tense check is an honest, four-language heuristic. And it is not a “good writing” score — it measures **consistency** and **distinctiveness**, not quality. Statistical voice is not literary voice, and every surface states its own limits.



## DIALOGUE — speech as its own craft

Dialogue is the most technically demanding prose mode: a reader must always know who is speaking, said-bookisms (he ejaculated) degrade the page, and an unbroken run of talking heads loses the physical scene. The Inner Socrates flags **unattributed** speech in passing, but DIALOG-1 measures dialogue as a **mode** with its own properties — deterministically, with no AI and no parser, in the five languages.

`inkhaven dialogue scan` detects every dialogue span and raises three findings: **zero-attribution** (a speech span with no tag and no nearby character name — though an established two-speaker exchange is allowed to alternate untagged for a run, default eight turns, before it fires, because readers track that fine), **said-bookism density** (a chapter whose non-neutral tag verbs — `whispered`, `growled` — run above the book's own baseline), and **talking heads** (a run of dialogue-only paragraphs with no action beat to ground the scene).

● ● ● *inkhaven dialogue scan — a pre-submission gate*

Dialogue scan — `The Sunken Throne` [en]

|                         |    |
|-------------------------|----|
| dialogue spans detected | 34 |
| zero attribution        | 3  |
| talking-head sequences  | 1  |

[ch.12 · 7fla...] unattributed speech — no tag or  
character name within range

```
scan      exits    non-zero    if      any      zero-attribution    span
is      found,    so      it      doubles    as      a      CI      gate      —
inkhaven dialogue scan --findings zero-attribution || echo "fix
attribution first"
```

Detection is deliberately **not** uniform across languages, because quotation convention is not: English and German use paired marks ("...", „...", »...«), French and Russian use guillemets and em-dash openers (with the French inline **incise**, `, dit-il,`, stripped before measuring), and Spanish detects all three additively.

## Fingerprints and the fingerprint view (Ctrl+V Shift+Q)

`inkhaven dialogue profile` builds a six-metric voice signature for each character from the lines confidently attributed to them — are they terse or expansive, do they ask or declare, do they hedge or assert?

● ● ● *inkhaven dialogue profile — per-character signatures*

| Character | Utts | Avg words | MATTR | Quest. | Excl. | Hedge |
|-----------|------|-----------|-------|--------|-------|-------|
| Mara      | 47   | 11.3      | 0.74  | 0.31   | 0.08  | 0.019 |
| Aldric    | 83   | 11.8      | 0.69  | 0.12   | 0.22  | 0.041 |
| Seren     | 31   | 7.1       | 0.61  | 0.44   | 0.04  | 0.009 |

In the editor, `Ctrl+V Shift+Q` opens the fingerprint view for the **nearest** character — one named in the paragraph you are in, else the most-speaking — as ASCII bars with a compare line for the next speakers. It is built from confidently-attributed dialogue, so run the `Ctrl+B Shift+C` review pass (or `inkhaven dialogue scan`) once to populate it. The mnemonic is **Q** for **Quote**; `Ctrl+V D` was already taken.

### THE THEATERGOER PERSONA

DIALOG-1 also added an Inner Socrates persona, **theatergoer** — cycle to it with `Ctrl+B J → S`. It reads a dialogue scene as if the narrator's private glosses were invisible and asks whether the subtext is legible from the **words and visible action alone** — the test a stage or screen adaptation would impose.

The dialogue **findings** ride the `Ctrl+B Shift+C` review pass into the Output pane, navigable to the flagged paragraph; the check is zero-AI and hash-lazy, so only the chapter you just edited is re-detected. From Bund, `ink.dialogue.stats`, `ink.dialogue.fingerprint`, `ink.dialogue.violations`, `ink.dialogue.spans`, and `ink.dialogue.refresh` expose it all read-only. The `dialogue:` config block tunes the windows and — importantly for SF and fantasy — marks genre speech verbs as neutral with `extra_neutral_verbs` so telepathy's `transmitted` is not counted as a bookism.

## NARR-1 — the narrator's voice over time

The fourth reader is the oldest and the quietest: `inkhaven prose` profiles the **narrator's** voice as a statistical property of the whole book over time — sentence rhythm, lexical diversity, epistemic hedging, interiority, sensory balance, passive ratio — per chapter, deterministically, with no model, no parser, and no external dependency. It answers a question nothing else in Inkhaven does: **does chapter forty read like the person who wrote chapter one?** It measures where the voice moved; it never says the move was wrong.

The always-on tier is language-agnostic rhythm — the sentence-length distribution, the coefficient of variation (a falling CV across chapters is the primary signal of a voice **narrowing**), a bounded burstiness index, and MATTR, a length-corrected lexical diversity whose late drop can mean vocabulary fatigue. Two language-sensitive metrics ride on top — modal (hedging) density and an interiority ratio — with complete curated word lists for all five first-class languages. A deep pass (`--deep`) adds sensory-channel balance and a per-language active/passive ratio.

● ● ● *inkhaven prose — the narrator profiler*

```
inkhaven prose profile [--deep] [--json] [--language de]
inkhaven prose refresh                recompute, summary
inkhaven prose drift    [--mode baseline|rolling]
inkhaven prose suggest                how to read them

Voice profile — `The Sunken Throne` [en] · 12 ch
sentence CV      0.42   varied
MATTR            0.71   healthy
modal density    0.018
interiority      0.22

Drift vs ch 1
  ▲ ch 9  CV 0.42 → 0.28   the voice is narrowing
```

In the editor, `Ctrl+V V` (“Voice”) runs the profiler in the **background** — deterministic, content-hash lazy, so only edited chapters recompute and there is no cost. Any chapter metric that drifted past its threshold versus the baseline chapter is emitted to the Output pane as an informational `prose` finding that navigates to the chapter. `Ctrl+V Shift+V` toggles **ambient** mode (off by default): the check re-runs after an editing pause, gated by a cooldown floor. From

Bund, `ink.prose.{profile, drift, violations}` read the stored profiles and `ink.prose.refresh` recomputes; language-sensitive metrics return `null`, not zero, on an unsupported-language book.

#### NARR-1 AND CHORUS ARE ONE ENGINE

NARR-1 profiles the **narrator**; CHORUS reuses the very same metric engine per **character** to build the dialogue fingerprints and the distinctiveness matrix. When you read a character's rhythm/diversity/hedging/interiority card, you are reading the narrator's own instrument turned on that character's lines.

## Where the four readers meet

These four instruments overlap on purpose, and they converge in two places. The first is the review pass, `Ctrl+B Shift+C`: one keystroke runs the fast checkers together, and the LECTOR read-through line, the DIALOGUE findings, the Inner Stylist's synthesis, and the prose-drift notices all land in the Output pane alongside the fact and continuity checks — a single board of everything the book noticed about itself. The second is the **Editorial Pass**, `Ctrl+V Shift+R` (Chapter 19): every reader here is advisory, and the Pass is where their findings become a ranked worklist you can **act** on — a diff-reviewed rewrite, a guided decision, or a written brief — under the confirmed-diff, snapshot-first contract that means no reader ever changes your prose without your explicit say-so.

## WHAT YOU LEARNED

- **LECTOR** reads the whole book forward, once, as a first reader: the **Shape** half measures a prose-derived intensity curve against one of six frameworks (including conflict-optional **Kishōtenketsu**) and flags `shape_sag` plus scene/sequel arrhythmia; the **Audience** half walks forward-only for `confusion`, `info_dump`, `attention_dip`, `put_down_risk`, and `unpaid_setup`. Free at the core; the **synthetic first-read** is the one explicit, cost-capped LLM pass (`inkhaven readthrough --deep`, or `k` in `Ctrl+B Shift+A`).
- **CHORUS** profiles the cast at book scale: character voice fingerprints + the **distinctiveness matrix** (“Mara and Joren read identically”), POV/head-hop and a tense check (English/German/French/Spanish — **Russian excluded by design**), and register drift. The **Inner Stylist** (`Ctrl+B J → Y`) synthesises it into Praise / Note / Concern. Declare a scene’s POV with `pov:<name>`.
- **DIALOGUE** (DIALOG-1) reads speech as a mode — attribution, per-character fingerprints, said-bookism density, talking-head runs — deterministically in five languages. `inkhaven dialogue scan` doubles as a CI gate; `Ctrl+V Shift+Q` opens the fingerprint view.
- **NARR-1** (`inkhaven prose`) profiles the **narrator’s** voice over time — rhythm, diversity, hedging, interiority — to catch a voice narrowing or drifting. `Ctrl+V V` runs it in the background; `Ctrl+V Shift+V` makes it ambient. It shares its metric engine with CHORUS.
- All four are **advisory** — they measure and report, never edit. They converge in the `Ctrl+B Shift+C` review pass, and their findings become actionable only in the Editorial Pass (`Ctrl+V Shift+R`), under a snapshot-first, confirmed-diff contract.

## CHAPTER 19

## 19

Revision and History

---

Every reader you have met so far in this part **diagnoses**. SENTINEL finds the continuity break; LECTOR finds the saggy act and the point a reader would put the book down; the Inner Editor finds the paragraph that tells where it should show; CHORUS finds the two characters whose voices read alike. That is where most writing tools stop — with a list of what is wrong. But a diagnosis is not a revision. Between the finding and the fixed book lie two questions no detector answers on its own: **what do I do about this one?** and, a draft later, **did any of it actually help?**

This chapter is about the two intelligences that answer those questions. **RED-LINE** is the revision partner — it gathers every reader's findings into one worklist and gives each the right kind of help, turning a diagnosis into an author-confirmed

change without ever touching your prose on its own. **CHRONICLE** is the draft historian — it remembers what your book measured at each milestone and trends it, so the question “did the revision work?” finally has a number behind it. The two are a matched pair: **REDLINE** closes the loop, **CHRONICLE** measures whether it stayed closed. Read this chapter as the operator’s tour of both; the companion book **Know Your Book** holds the fuller treatment of the readers they draw on.

## REDLINE — the revision partner

You have, by the time you reach a revision, a great deal of diagnosis scattered across the tool. The **doctor** has its editorial classes. The Facts scan has flagged a contradiction. Semantic drift has noticed a fact that changed shape. The Planning Board’s **plan check** has a structural note. The prose-style detectors have underlined a bit of telling, a filter word, an anachronism. **SENTINEL** has a co-location error, **LECTOR** a put-down risk, the Inner Editor a craft observation, **CHORUS** a voice that has drifted. Acting on all of that by hand means visiting a dozen surfaces and holding a dozen mental lists. **REDLINE**’s first job is to **end the scavenger hunt**.

### One worklist, every reader

**REDLINE** gathers every reader’s findings into a single ranked list — the same unified **collect** the Editorial Pass and **inkhaven revise** both share. The **doctor**’s editorial classes, the Facts contradictions, semantic drift, the **plan** structure notes, the prose-style detectors, **SENTINEL** continuity, the **LECTOR** read-through, the Inner Editor’s craft notes, and **CHORUS** voice findings all land in one queue, errors first, each carrying a note of **how it can be acted on**. Nothing new is computed here — **REDLINE** reads what the readers have already found — which is why the pass opens instantly and needs no live model to show you the list.

#### TERM Worklist

The single ranked list of **every** reader’s findings, gathered by the shared **collect**. Each row is one finding — its severity, its category, where it lands in the book, and its **response kind**: the honest form of help for that particular problem. The Editorial Pass, **inkhaven revise**, and **CHRONICLE**’s milestone capture all read this same list, so what you act on is exactly what gets measured.

### Three kinds of help

The insight that makes REDLINE more than a to-do list is that different problems want **different kinds** of help, and pretending otherwise is how automated editors do damage. A repeated word wants a small, local rewrite. A fact collision the machine cannot adjudicate wants **you** to decide which version is true. A whole act that sags cannot honestly be fixed by rewriting one paragraph at all — it wants advice, not a patch. So every finding carries a **response kind**, shown as a glyph, and only one of the three ever touches prose.

- ✍ **Rewrite**      A diff-reviewed local prose fix — an honest single-paragraph change: de-echo, tighten pacing, show-don't-tell, cut a filter word, period-fit an anachronism, or carry out an Inner-Editor craft note.
- ⇒ **Decision**      A guided authorial choice. The AI cannot know which fact is right — which scene Mara is in, which value is canon. You state what is true; REDLINE reconciles the paragraph to your decision as a confirmed rewrite.
- ✉ **Brief**      A concrete revision brief, for a structural or book-level problem one paragraph cannot solve — a saggy act, a likely put-down point. The AI advises; it never rewrites. The brief lands in the Thoughts pane.

The marquee case is the Inner Editor's own craft note. When it becomes a ✍ Rewrite, its observation — the specific thing it noticed about **this** paragraph — is handed to the model as the instruction, so the fix addresses that note and not some generic recipe. That is the difference between “make this better” and “the tension slackens because the verb goes passive in the second clause; fix **that**.”

### The Editorial Pass — Ctrl+V Shift+R

In the editor, **Ctrl+V Shift+R** opens the **Editorial Pass**: the worklist as a cockpit you can walk. It sits under the view prefix — **Ctrl+V** reaches a tool rather than acting on the book's structure — and the mnemonic is simply **R** for **Revision**. Each row shows its response glyph, so you can see at a glance what acting on it will do before you do it.



● ● ● *The Editorial Pass — one worklist, each row tagged how to act*

```

Editorial Pass · 14 findings (2 err · 5 warn · 7 info)
x ≠ co_location ch. 3 Mara is in the tower and the
                        courtyard in the same breath
△ 🔍 echo ch. 3 "about" repeats five times in two
                        sentences — de-echo
△ 🔍 editor ch. 7 the verb tense wobbles mid-paragraph
· ☒ shape_sag ch. 5 the Three-Act shape wants a rise
                        here; the prose reads flat
· 🔍 filter ch. 2 filter word: "very" — consider
                        cutting

↑↓ move · [ ] filter · ⌕ jump · f act · F fix-all ·
s skip · d defer · D clear · Esc close

```

Moving through it is quick. **↑ ↓** move the cursor, and the selected finding's full message and any hint expand in place below it. **[ ]** cycle a category filter, so you can walk all the echoes, then all the continuity errors, one class at a time. **⌕** jumps the editor straight to the paragraph so you can read the finding in its context. The three keys that follow are where the work happens.

- f** Act on the selection. **⌕** streams an AI rewrite into a diff review; **↔** asks what is true, then reconciles the paragraph to your answer; **✉** writes a developmental brief to the Thoughts pane.
- F** Fix-all — walk every **⌕** Rewrite in turn, each one diff-reviewed. Decisions, Briefs, and finding-aware editor notes are never batched; they are handled one at a time.
- s / d** Skip for this session (s), or defer (d) — persisted, hidden until the prose changes so a finding you have judged and set aside stays gone.
- D / Esc** Clear every deferral (D), or close the pass (Esc).

The distinction between **f** and **F** is worth internalising. A single **f** acts on one finding of any kind. The **F** batch is deliberately narrower: it walks **only** the **⌕** Rewrites, one diff at a time, and **Esc** stops it wherever you are. Anything that requires your judgement — a Decision you must answer, a Brief that only advises, an editor note whose rewrite is finding-specific — is left for you to take one at a time. The batch is a convenience for the mechanical fixes, never an autopilot for the ones that need a human.

## The safety contract

REDLINE will not edit your prose on its own. This is not a policy bolted on top; it is the shape of the code. Every prose change REDLINE can make — a single  fix, a  $\rightleftarrows$  decision-reconcile, an editor note, or one step of the **F** batch — flows through exactly one path, and there is no other.

● ● ● *Every REDLINE prose change takes the same three steps*

```

the model's rewrite streams into the AI pane
      |
      ▼
an AI diff-review modal shows the exact change
  accept (a / e / ) or reject (r)
      | accept
      ▼
your pre-rewrite prose is SNAPSHOTTED first,
with a labelled entry — then it is replaced
      |
      ▼
recover it any time with F6
  
```

Read the three steps as promises. First, you always see the exact diff — the change is shown, never applied blind. Second, nothing is written until you accept it; reject and the paragraph is untouched. Third, on accept your old prose is snapshotted with a labelled entry **before** the new text lands, so the version you had is one **F6** away for as long as the project lives. There is no code path in REDLINE that writes prose without all three.

### THE BATCH IS REWRITE-ONLY BY CONSTRUCTION

The **F** queue can hold nothing but  Rewrites. A Decision or a Brief cannot acquire a fix recipe in the first place, so it can never slip into the prose-writing path — the type of the finding forbids it, and a guard test locks the invariant in place. When you press **F** you are never one wrong keystroke away from an unconfirmed edit; the queue **cannot** contain one.

## The editorial letter — inkhaven revise

The Editorial Pass is the row-by-row cockpit. Sometimes, though, you want the opposite view first — the **overview** a good editor opens a revision with, before touching a single line. That is **inkhaven revise**. It runs the same worklist, then

synthesises the whole of it into one developmental letter: the big picture first — the one or two things a reader will feel first — then grouped by theme (continuity, structure and pacing, voice and character, line and prose), most important first, each with a brief **why it matters** and **what to do**. It advises; it never rewrites.

● ● ● *inkhaven revise — the letter, or the findings for tooling*

```
inkhaven revise          # the editorial letter over the
                        # whole project
inkhaven revise --book-name X  # restrict to one book (slug or
                        # title)
inkhaven revise --json        # the findings as JSON: category,
                        # severity (high/med/low),
                        # response, location, message
```

The letter is written in the manuscript’s language — REDLINE inherits every source reader’s language coverage and claims no more than they do — and each finding keeps its `source`, so the perennial question “**does it work in Russian?**” answers itself per detector rather than as one blanket promise. The `--json` form is the machine face of the same worklist: the AI letter and every prose rewrite stay out of it, but the ranked findings come through in full, ready for a script or a hook.

### REDLINE from a script

Two Bund words expose the worklist read-only, for a revision-readiness check you can wire into a hook. The editorial letter and every rewrite are deliberately **not** scriptable — advice and prose changes stay author-initiated — but the deterministic findings are fair game.

● ● ● *The REDLINE Bund surface — read-only*

```
ink.revise.findings ( -- list )
  the ranked findings as dicts: { category, severity,
  response, location, message, source }

ink.revise.check    ( -- dict )
  summary counts: { findings, high, med, low, clean,
  by_response, by_category }
```

`check.clean` is a plain pass/fail gate — `true` when there are no high-severity findings — the thing a “is this draft ready?” script asks. The `high / med / low` vocabulary matches `revise --json`, so a report and a gate speak the same words.

## CHRONICLE — the draft history

REDLINE helps you act. But once you have acted — cleared a dozen findings, moved a scene, rewritten a flat chapter — a harder question arrives, and until CHRONICLE nothing in the tool could answer it: **did the book actually get better?** Every reader diagnoses the draft **in front of it**. None of them remembered what the last draft measured. So the single question a reviser most wants answered had no answer at all. CHRONICLE is the memory that closes that gap — and it is **pure measurement**: there is no prose-write path anywhere in it.

### A milestone — chronicle mark

A **milestone** is an explicit capture of the current draft — the whole diagnostic state frozen in one shot. You stamp one whenever a draft is worth remembering: after a big pass, before you send it to a reader, at the turn of a version.

#### ● ● ● *Stamping and listing milestones*

```
inkhaven chronicle mark "draft-2"
inkhaven chronicle mark "beta-1" --ref v0.9    # record a git ref
inkhaven chronicle list                        # newest first
inkhaven chronicle list --json                 # for tooling
```

Marking runs the unified worklist once — the same `collect` REDLINE uses — and captures the total finding count, the tallies by severity, category, response-kind and source, and, crucially, the **fingerprint of every finding**, so that a later diff can name exactly which ones cleared and which appeared. The `--ref` string is stored verbatim for your own bookkeeping; CHRONICLE never resolves, creates, or enumerates git refs — a milestone is a decision you make, not a tag it invents.

**TERM Milestone**

One named, timestamped capture of the draft's whole diagnostic state: the metric vector (counts by severity, category, response, and source) plus the fingerprint of every finding. Milestones are deliberate — you stamp them with `chronicle mark` — and they are the fixed points every trend and diff measures between. They live in the project's own `chronicle.db`, separate from your prose and your snapshots.

**The trend — every count fewer is better**

Run `inkhaven chronicle` bare and CHRONICLE captures the **live** state of the book right now and diffs it against your most recent milestone — the “did it get better since I last looked?” view. Every number it trends is a count of findings the readers raised, which means every one is **fewer-is-better**: a fall is an improvement, a rise a regression, and the regressions sort to the top of the category list so the worst news never hides.

● ● ● *inkhaven chronicle — the trend since your last mark*

```
Chronicle — since "draft-1" (2026-08-03) → now

findings          1 → 2  ▲
warnings          0 → 1  ▲ NEW
infos             1 → 1  ·

by category:
  put_down_risk    0 → 1  ▲ NEW
  shape_sag        0 → 1  ▲ NEW
  attention_dip    1 → 0  ▼ cleared

✓ 1 cleared      ▲ 2 introduced      · 0 unchanged

introduced (new since the last mark):
  ▲ put_down_risk  ch. 3  ch. 1-3 run flat and eventless...
  · shape_sag      ch. 2  the shape wants rising tension...
```

The arrows carry the whole polarity at a glance: ▼ a count that fell (good), ▲ one that rose (a regression), · one that held. A category that went from some findings to none is tagged **cleared**; one that went from none to some is tagged **NEW**. You are never left to work out whether a number moving is welcome — the direction is scored for you, and the layout puts the regressions where you will see them first.

## Cleared versus introduced — the signature

The move that makes CHRONICLE worth having is the **cleared** / **introduced** split. Because every finding carries a stable fingerprint, CHRONICLE can do a simple set difference between two milestones' finding sets and sort the result into three piles.

- ✓ **cleared** Findings that were there last milestone and are gone now — the ones your revision actually resolved. The proof that the work landed.
- ▲ **introduced** Findings new since the last milestone — the ones your edits, or the ripple around them, created. The early warning on collateral damage, before it ships.
- **unchanged** Findings present in both — still standing, still waiting for you.

This is what closes REDLINE's loop. When you spend an afternoon in the Editorial Pass clearing echoes and reconciling a continuity error, the **cleared** list is the receipt: those exact findings are gone. And the **introduced** list is the thing no amount of careful rewriting can see from inside a single paragraph — the new confusion your cut created three chapters downstream, the sag that opened up when you tightened the scene before it. The introduced findings are itemised, not just counted, so you can act on them rather than merely worry about them.

## Diffing two named milestones — chronicle diff

The bare trend measures **now** against the last mark. To compare two fixed points in your history — beta-1 against the release draft, say — name them both.

● ● ● *chronicle diff — two milestones head to head*

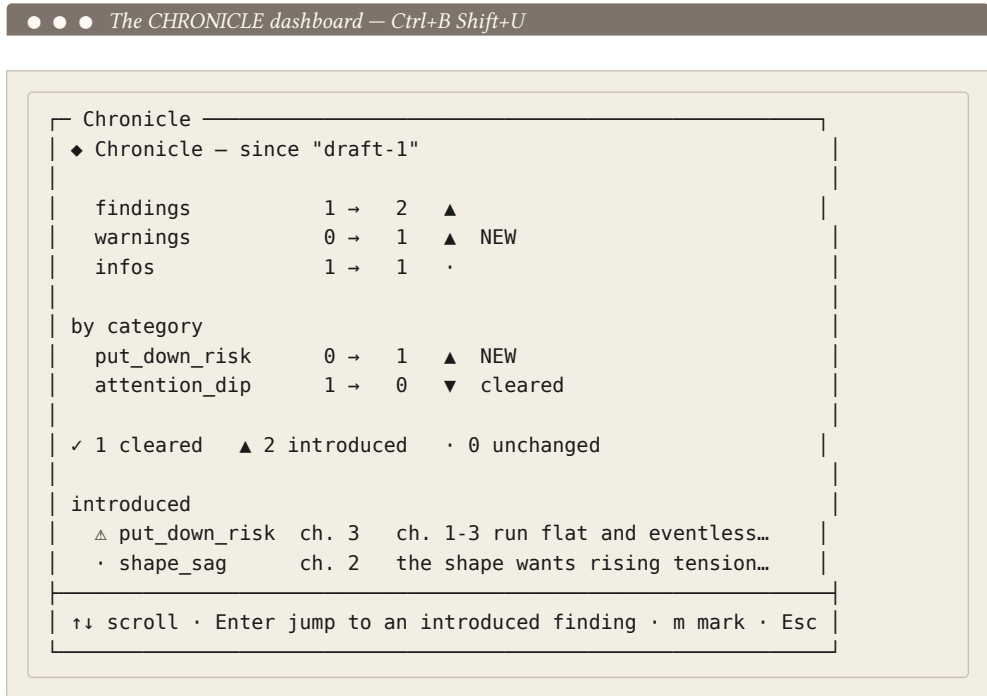
```
inkhaven chronicle diff draft-1 draft-3
inkhaven chronicle diff beta-1 rc-1 --json    # deltas + the three
                                              # finding lists
```

`chronicle diff <from> <to>` runs the same trend machinery between two stored milestones rather than against the live book, and `--json` on either the trend or a diff emits the deltas plus the cleared, introduced, and persisted lists for a script to read.

## The dashboard — Ctrl+B Shift+U

Inside the editor, `Ctrl+B Shift+U` opens the CHRONICLE dashboard — the trend and the cleared/introduced split, live, without leaving the window. It sits under the

meta prefix because it is a whole-book intelligence, and it reads the same numbers the CLI prints.



**↑ ↓** scroll the panel. **↵** on an introduced finding jumps the editor straight to its paragraph — the same jump the read-through and continuity ledgers give you, so a regression is never more than one keystroke from the prose that carries it. **m** marks the current draft on the spot, labelled by today's date; for a chosen name, **inkhaven chronicle mark** mark from the CLI instead ( "**<name>**" ). Each mark is a fresh milestone, so pick the date-stamped **m** or the named CLI form — not both. **Esc** closes. Marking from the dashboard is the one place the tool writes a milestone interactively; everything else the dashboard does is read-only.

- ↑ ↓** Scroll the dashboard.
- ↵** Jump to an introduced finding's paragraph.
- m** Mark the current draft now — labelled by today's date; for a chosen name, mark from the CLI (inkhaven chronicle mark "**<name>**") instead.
- Esc** Close the dashboard.

## CHRONICLE from a script

Three Bund words expose the history, all read-only. Marking is **not** scriptable — stamping a milestone is a deliberate act, the same shape as taking a snapshot — so scripts read the history, they do not write it.

● ● ● *The CHRONICLE Bund surface — read-only*

```
ink.chronicle.marks ( -- list )
  { label, ts, book, findings, errors, warnings, infos }

ink.chronicle.trend ( -- dict )
  { marked, since, headline, categories, cleared,
    introduced, persisted }

ink.chronicle.check ( -- dict )
  { baseline, cleared, introduced, introduced_errors,
    clean }
```

`check.clean` is `true` when your latest edits introduced **no** error-severity finding since the last mark — a pre-commit or pre-submit gate that says “you did not break anything new.” With no milestone yet, `check` is vacuously clean and `trend` reports `{marked: false}`, so a script that runs before you have ever marked a draft degrades gracefully rather than erroring.

### TWO DATABASES, TWO JOBS

CHRONICLE keeps its milestones in the project’s own `chronicle.db`, beside your prose but never mixed into it. It is deliberately not a git tool and not an auto-capture daemon: it will not create tags, and it will not stamp a milestone behind your back. A draft is a decision, so a milestone is a keystroke — `chronicle mark` on the CLI, or `m` in the dashboard.

## Closing the loop

The two intelligences in this chapter are one workflow read from two ends. Set them side by side and the shape is plain.

**REDLINE acts.** It gathers every reader’s diagnosis into one worklist, hands each finding the honest kind of help — a diff-reviewed rewrite, a decision it asks you to



make, or a brief it can only advise — and moves every prose change through the one safe path: shown as a diff, applied only on your accept, snapshotted before it lands. When you finish a pass in the Editorial Pass, you have turned a pile of findings into a set of confirmed changes, and not one word changed without your say-so.

**CHRONICLE measures.** Mark a milestone before the pass and another after, and the cleared list is the receipt for the findings REDLINE helped you resolve, while the introduced list is the warning about the ones your edits created three chapters away. The counts trend fewer-is-better; the split names names. Neither touches your prose — CHRONICLE only ever reads.

That is the loop. REDLINE closes findings; CHRONICLE proves they stayed closed and catches what opened up in their place. Do the work in the pass, stamp a milestone, and the next time you open the dashboard the book tells you, in its own numbers, whether the revision was a revision — or merely a rearrangement.

## WHAT YOU LEARNED

- **REDLINE** gathers **every reader's findings** into one ranked worklist — the shared `collect` behind the Editorial Pass, `inkhaven revise`, and **CHRONICLE**'s capture — errors first, each tagged with how it can be acted on.
- Each finding carries a **response kind**:  **Rewrite** (a diff-reviewed local fix),  $\Leftrightarrow$  **Decision** (you state what is true, REDLINE reconciles), or  **Brief** (advice for a structural problem, never a rewrite). Only a Rewrite touches prose.
- The Editorial Pass is `Ctrl+V Shift+R`: `f` acts on one finding, `F` walks **only** the  Rewrites one diff at a time, `s / d` skip or defer. Every prose change is a **confirmed diff plus a pre-change snapshot**, `F6` -restorable — there is no unconfirmed write, and the `F` batch is Rewrite-only by construction.
- `inkhaven revise` synthesises the same worklist into one **editorial letter** (big picture, then grouped by theme); `--json` and `ink.revise.*` expose the findings read-only. It advises; it never rewrites.
- **CHRONICLE** snapshots every reader's verdict per milestone.  
`chronicle mark`  
`<label>` stamps one; bare `inkhaven chronicle` trends the live book against the last mark, every count **fewer-is-better** ( $\blacktriangledown$  good,  $\blacktriangle$  a regression); `diff` compares two named marks.
- The signature is the **cleared vs introduced** split — set difference over stable fingerprints: what your revision **resolved** and what it **created**. This is how **CHRONICLE** closes the loop **REDLINE** opened.
- `Ctrl+B Shift+U` opens the dashboard ( jumps to an introduced finding, `m` marks the draft); `ink.chronicle.*` and its own `chronicle.db` keep the history. **CHRONICLE** is **pure measurement** — no prose-write path anywhere.

CHAPTER 20

# 20

## The Inner Family

---

Every reader you have met so far in this part answers a question you can put plainly: is this fact consistent, does this reference come before its introduction, does the pacing sag on page one. The readers in this chapter are different in kind. They do not answer; they **ask**. They read your prose the way a careful editor, a sceptical friend, or a thoughtful stranger would — and hand back not corrections but the questions those readers would raise. The choice of what to do about each one stays entirely yours.

Inkhaven calls them the **Inner family**, and there are five: the **Inner Socrates** (the dialectician who surfaces the assumptions and framings your prose treats as given), the **Inner Editor** (who notices craft — what the words are doing to

themselves and to the reader), the **Inner Theologian** (who reads the moral and theological weight of what you depict, through eleven traditions and belonging to none), the **Inner Poet** (who measures verse against the form it declares), and the **reasoning-rigor reader** (who scans arguments, deterministically, for the classic weaknesses). This chapter is the operator’s tour of all five: what each one is for, how to reach it, and how to run it. When you want the full treatment — the theory behind each, the worked case studies — the companion book **Know Your Book** devotes its Part VI to exactly this family.

#### THE ONE PROMISE

Not one member of the Inner family ever edits your prose. Every finding is a question, an observation, or a flag — surfaced to the Output pane or printed to your terminal, and nothing more. There is no confirm-this-rewrite path anywhere in the family; that machinery belongs to REDLINE (Chapter 19), not here. The family reads; you write.

## What the whole family shares

Before the five, the shape they hold in common. Learn it once and each member is a variation on a form you already know.

They are **advisory**. A finding is a candidate for your attention, never a verdict on your book. The word “should” is, by structural commitment, absent from what they say — they report what a reader would notice or ask, and leave the judgement to you.

They surface to the **Output pane**. Findings land as messages on the notice board you met in Chapter 3, each carrying its own glyph so you can tell the families apart at a glance:  for a Socratic question,  for an Editor observation,  for a theological question,  for a reasoning-rigor flag. Press **f** in the Output pane to filter to one source; press **o** on a row to expand it to the evidence it grounds in; press **d** to dismiss one.

Most of them split into a **Fast track** and a **Slow track**. The Fast track is deterministic — pattern-matching and the language detector, no model, no network, instant and free. The Slow track calls your configured LLM for the questions patterns cannot reach, and is always cost-capped: each call prints a preflight, respects a per-

call soft cap and a daily ceiling, and degrades to a quiet notice when no provider is configured. The Fast track always works; the Slow track is where you spend.

They are **multilingual**. Every deterministic detector keys its cue tables off the paragraph's language across the five Inkhaven supports — English, Russian, German, French, Spanish — and every LLM question is posed in the book's language with an English fallback. The question a Russian paragraph raises is raised in Russian.

And most of them honour a **shared intent ledger**. When a finding names something you did on purpose — an ambiguity you are holding, a motif of repetition, an unreliable narrator whose prose should not believe him — you declare that choice once and the matching findings fall silent. Inner Socrates and the Inner Editor write into the same ledger, so a declaration is a first-class part of your examined-authorship record, not a per-tool mute.

## Inner Socrates — the dialectician

The oldest member of the family, and its spine. Inner Socrates reads alongside you and surfaces **questions** about your prose — the assumptions it treats as given, the framings it presupposes, the tensions inside it, what each scene does for the work. It makes exactly one claim, and never another: **you have written something; here are the questions a careful reader would ask about it**. If a surface would ever say “the prose should be X,” it does not ship.

● ● ● *A first question, from the terminal*

```
$ inkhaven inner-socrates check --text \
    "The regent had to declare war; the council left no choice."

◇ Inquiry [Asserted Necessity]
  This passage treats an outcome as inevitable ("had to").
  What alternatives did you decide to leave out?

1 question(s) · persona: Inner Socrates
```

That is the whole spirit: a question a careful reader would ask, in a persona's voice, about a choice the prose made. There is no fix offered — only the question, and the space it opens.

### The two tracks

The **Fast track** is deterministic and instant, needs no provider, and runs in all five languages. It has seven categories, each a pattern a careful reader would catch on a first pass:

● ● ● *The seven Fast-track categories*

|                    |                                              |
|--------------------|----------------------------------------------|
| Asserted Necessity | an outcome treated as inevitable             |
| Hedging            | authorial hedging                            |
| Pattern            | a run of same-opening / same-length lines    |
| Speaker            | dialogue running on with no speaker tag      |
| Length             | a very long sentence                         |
| Tense Shift        | a slip between past and present (EN)         |
| Reference          | a pronoun with two possible antecedents (EN) |

The **Slow track** reaches the questions patterns cannot. Over a paragraph it adds five LLM categories — Hidden Assumption, Internal Tension, Framing, Significance, Echo — and over a project **timeline** it adds three more (Dramatization Gap, Implication, Temporal Density), reading the prose against the events the timeline declares. You reach the Slow track with `--slow` on the CLI or with the **E** sub-key in the hub; it is cost-capped and, in the editor, runs in the background so it never blocks your typing.

### Notice, Inquiry, Probe

Every Socratic finding carries one of three severities, and they are worth knowing because they set what you see by default.

**TERM** Notice · Inquiry · Probe

A **Notice** is a surface observation — hidden by default. An **Inquiry** is a question that invites reflection — the bulk of the output, and visible. A **Probe** is a rare, structural question about the work as a whole — always visible. The default visible threshold is **Inquiry**, so Inner Socrates is quiet unless it has a real question to raise.

### The persona roster

The same paragraph reads differently to different readers, so Inner Socrates reads as a chosen persona — a distinct careful-reader perspective whose per-category

emphasis weights scale what it notices (a weight of **0.0** mutes a category outright). Fourteen ship bundled, in four groups.

Five read for **fiction**: Inner Socrates itself (the default), the Careful Editor, the Skeptical Reader, the First-Time Reader, and the Slow Reader. Four (added in the AUDIENCE work) read for **nonfiction** — the Skeptical Practitioner for technical writing, the Domain Newcomer for general nonfiction, the Expert Reviewer for academic prose, the End User for documentation; these mute the narrative-only categories and lean instead on what a step assumes the reader already knows. Three read **ideas-driven** work: the Dialectician for philosophy (the unstated premise, the equivocation, the unanswered objection), the Theological Reader for theology (deliberately non-empiricist — it probes coherence and scope, not proof), and the Utopian Architect for utopia and dystopia (a hybrid that reads the narrative as fiction while pressing on what the imagined society assumes and what it costs).

### The two adversaries

The last two break the family's spine on purpose. Inner Socrates is otherwise **always** a neutral questioner — it never praises, never charges, only asks. The two **verdict personas** are one-sided by design:

#### FICTION

The **Defender** ⚔ is counsel for the defence. It states **only praise** — what works in the passage and why it is worth protecting. The steelman.

#### NON-FICTION

The **Prosecutor** ⚔ is the prosecution. It states **only concern** — the charge, what fails, stated as an indictment. The devil's advocate.

This is governed by a persona **stance**: **question** (the neutral default that every other persona keeps), **praise** (the Defender), or **concern** (the Prosecutor). Two things follow. The neutral interrogator is untouched — the verdict prompt is reached only for a one-sided stance, so the twelve other personas ask exactly the question they always did. And a verdict is **LLM-only**: the deterministic Fast track can neither praise nor charge, so with a verdict persona active the Fast chord simply points you to the Slow track (E). You can give your **own** persona a stance by adding **stance: praise** (or **concern**) to its HJSON file — the way to author a one-sided reader of your own.

## AUTHORING A PERSONA

A persona is a small HJSON file — an id, a name, a voice summary and notes, an optional `stance`, and a block of per-category `emphasis` weights. Drop it in `~/.config/inkhaven/personas/` to share it across projects or in the project's `books/intent/01-personas/` to keep it local; project wins over user wins over bundled. `Ctrl+B J → N` runs an AI wizard that scaffolds one for you.

### The intent ledger

Some of what Inner Socrates flags is deliberate, and the **intent ledger** is how you say so. It is the prose counterpart of the world simulator's magic ledger, and it shares its whole vocabulary — an Entry with a Kind, a Coverage (which categories it may suppress), and a Scope (project, chapter, a paragraph range, a character, a scene, or a timeline range). A matching entry **suppresses** a would-be finding with a note instead of nagging you again.

Entries accumulate two ways. You declare them by hand, or the **promotion mechanism** offers one after you have dismissed the same kind of finding repeatedly — `inner-socrates suggestions list` shows the patterns, and `suggestions promote <category>` turns one into an entry. Carry a series' worth of declared intentions into the next book with the `.isl` bundle (`inner-socrates bundle export / bundle import`).

### The hub — Ctrl+B J

In the editor the whole Socratic family lives on one chord. `Ctrl+B J` opens the **Inner Socrates overview** — the active persona, the recent questions, and a way into the ledger — and from there a single sub-key does each thing.



● ● ● *Ctrl+B J — the Inner Socrates overview*

```

┌ Inner Socrates ◇ ───────────
│ Persona: Inner Socrates (fiction) · stance: question
│ Ambient auto-check: off   (A toggles)
│ Today: 4 / 150 slow calls (cap informs, never blocks)
│ Recent questions
│   ◇ Inquiry [Framing]   ch03 · the-quay
│   ◇ Inquiry [Hedging]  ch03 · the-inn
│
│ F fast ¶ · E slow · S persona · C converse · N new
│ L ledger · A ambient · T theolog · P poet · R rigor
└────────────────────────────────

```

|                 |                                                                                     |
|-----------------|-------------------------------------------------------------------------------------|
| <b>Ctrl+B J</b> | Open the Inner Socrates overview (persona, recent questions, ledger).               |
| <b>J → F</b>    | Fast-check the open paragraph — deterministic, instant, free → Output.              |
| <b>J → E</b>    | Engage the Slow pass — the LLM deep questions or verdict, run in the background.    |
| <b>J → S</b>    | Cycle the active persona through the fourteen.                                      |
| <b>J → C</b>    | Open a conversation with the persona in the AI pane (it discusses, never rewrites). |
| <b>J → R</b>    | Run the deterministic reasoning-rigor reader over the open paragraph.               |
| <b>J → N</b>    | AI wizard to author a new persona.                                                  |
| <b>J → L</b>    | View the intent ledger.                                                             |
| <b>J → A</b>    | Toggle the ambient auto-check (off by default; runs on a writing pause).            |

Two of those sub-keys are the branch points to other family members: **J → T** opens the Inner Theologian and **J → P** the Inner Poet, both covered below. (**J → Y** opens the Inner Stylist, the voice-at-scale coach — that one belongs to CHORUS and Chapter 18.) The hub letter is **J** because **Ctrl+B I** was already book-info; think of **J** as the door to the readers who question.

## The command line

Everything the hub does, and more, is scriptable under `inkhaven inner-socrates`.

● ● ● *The Inner Socrates CLI*

```
inner-socrates check [--text "..."] [--paragraph <uuid>
                    | --path <slug-path>] [--slow]
                    [--max-cost <n>] [--force]
inner-socrates timeline [--max-cost <n>] [--force]
inner-socrates ledger
inner-socrates persona list | show <id> | activate <id>
inner-socrates suggestions list | promote <cat> | dismiss <cat>
inner-socrates bundle export [--scope-level series|project|all]
inner-socrates bundle import <path> [--conflict skip|override]
```

Target a paragraph three ways: `--text "..."` for literal prose, `--paragraph <uuid>` for an explicit id, or `--path <slug-path>` for the convenient path printed in the bracket by `inkhaven list` (for example `manuscript/03-rain/01-opening`, order-prefixes tolerated). `check` `--slow` and `timeline` are the two that call the provider; both are cost-capped and both degrade to the Fast track — or to nothing — when no provider is set.

## Inner Editor — the reader of craft

Where Inner Socrates asks about your **choices**, the Inner Editor observes your **craft**. It reads the prose the way a thoughtful editor reading over your shoulder would — a row of repeated words, a sentence whose rhythm earns its weight, a register that shifts halfway through, a paragraph that insists on a feeling its texture does not quite support — and it tells you what it **sees**, not what to fix. It is LLM-only, works at paragraph scope, and it never rewrites your prose.

### Praise, Note, Concern

The Editor grades every observation on its own three-level scale — `Praise < Note < Concern` — distinct from the Socratic Notice/Inquiry/Probe. The visible threshold is **Note**, which means **Praise is hidden by default**. That is deliberate: praise is a first-class output, but only when it is **earned** — generic encouragement

is forbidden by design, so a paragraph that genuinely does something well earns a specific note you can reveal ( `--severity praise` ), and everything else stays quiet.

It observes across eight categories in three modes — literary (Richness, Tautology, Style, Style-instability), vocabulary (Dictionary richness), and a few that cross the grain: the Belief stance (the subtle “does the prose believe its own message?” reading), Craft praise, and Editorial suggestions. Every observation grounds in actual textual evidence, and the Editor speaks in observations and qualified suggestions — **I notice, you might consider, if intentional** — never **should** or **must**.

● ● ● *Editor observations in the Output pane*

```

Output · inner-editor
├─ Note  ch01 · p007  Tautology
│   "Restates one idea four ways — 'quiet hush', 'noise-
│   less and without sound'. If the chant is meant, it
│   works; if not, you might thin it."
├─ Note  ch01 · p007  Belief
│   "The prose protests the silence so strenuously the
│   texture argues against it. Ironical, or played
│   straight? If straight, is that intended?"
└─
↑↓ select · o expand · i intent · d dismiss

```

## Reaching it — Ctrl+V O

The Inner Editor lives under the **view** prefix, not the meta one — it is a way of **looking** at your prose rather than acting on your book. `Ctrl+V O` (the **O** is for **Observe**) opens its overview; from there four sub-keys do the work.

- |                 |                                                                                        |
|-----------------|----------------------------------------------------------------------------------------|
| <b>Ctrl+V O</b> | Open the Inner Editor overview — status, tuning, today's tally.                        |
| <b>O → E</b>    | Engage the open paragraph — one LLM pass → Output (Praise/Note/Concern).               |
| <b>O → C</b>    | Open an Editor conversation about the paragraph (also <code>F9</code> → Editor scope). |
| <b>O → A</b>    | Toggle the ambient auto-engage — reads on a writing pause (opt-in; LLM cost).          |

**O → F** Jump to the Editor's findings in the Output pane.

The ambient auto-engage (**A**) is the Editor reading as you write: a pause on a paragraph runs one engagement in the background. It is off by default because it spends LLM calls, it has a same-paragraph cooldown, and any edit re-arms the timer — so it never interrupts you mid-sentence. When an observation deserves more than a glance, **C** turns the AI pane into the Editor itself, opening with what it noticed and ready to discuss in the same non-prescriptive voice; it helps you think, and will not draft the scene for you.

### The command line, and living together

The CLI surface is `inkhaven inner-editor engage / findings / config / usage`, plus `intent` (declare a category deliberate — it writes into the **same** ledger Inner Socrates consults) and `suggestions` (the same dismissal-driven promotion mechanism). Each engagement is one LLM call, tallied under the `inner_editor` budget in `inkhaven cost` and `Ctrl+B $`; as everywhere in Inkhaven, the caps inform and never block.

#### SOCRATES AND THE EDITOR, SIDE BY SIDE

Both run on the same paragraph and render distinctly in Output — **purple** ◇ Socratic questions, **warm-earth** 🌱 Editor observations. Filter Output to one with `f`, disable either independently, or run both. Socrates examines your **choices**; the Editor examines your **craft**. Together they are the texture half of Inkhaven's examined-authorship system.

## Inner Theologian — the moral reader

Fiction carries moral weight whether or not the author chose it: a novel enacts a theodicy, a view of what suffering means and what transformation costs, even when no one sat down to pick one. The Inner Theologian is the family's third axis — moral and theological **seriousness**, asking whether the work engages honestly with the weight of what it depicts.

It is a **tradition-neutral comparativist**. It reads any manuscript through the lenses of eleven moral and theological traditions — Catholic, Protestant, Orthodox, Gnostic, LDS, Islam, Judaism, Hinduism, Buddhism, Confucianism, and secular

moral philosophy — **not to judge the work by any of them, but to ask what each of them sees**. It belongs to no tradition, advocates none, and never delivers a verdict. Everything it produces is a question, and it never edits your prose. Crucially, it is **silent without a moral frame**: a passage with no moral weight to read draws nothing, so it does not manufacture significance where there is none.

### The two tracks

The **Fast track** is deterministic and zero-AI: three `info` signals in the Output `theologian` category (the `⚡` glyph), folded into the `Ctrl+B Shift+C` review pass. **Moral invisibility** — a named character harms another with no acknowledgment in the paragraphs that follow. **Consequence gap** — lethal or severe violence with no depicted consequence. **Sacred levity** — sacred or ritual vocabulary sharing a paragraph with comic markers (the most cautious of the three, because comedy with sacred content is a legitimate style; it only flags for attention). All five languages, tradition-neutral word lists, and none ever rises above `info`.

The **Slow track** is the LLM session, and the reason to reach for the feature. `Ctrl+B J → T` engages the open paragraph: the reader poses two or three moral or theological questions through the lenses most illuminating for that passage, in the book's language, always naming which tradition raises which — and always inviting you to say a lens is irrelevant, because that too is useful information. The questions land in the Output pane (and its longer reflective text in the Thoughts pane).

● ● ● *A slow-track theological session*

```
⚡ theologian  ch07 · the-vigil
Buddhist lens — the scene frames the loss as pure
affliction from without. Does the narrative allow that
the clinging is itself part of the suffering?

⚡ theologian  ch07 · the-vigil
Secular moral philosophy — a duty is asserted ("he owed
them this"). To whom is it owed, and what grounds it?
```

### From the terminal

The CLI is `inkhaven theologian scan / session / suppress`. `scan` runs the fast-track signals (exit 1 if any, so it drops into a check pass); `session` runs a slow-track session, defaulting to Category 6 — the work's implicit theology — over the whole book. Narrow it with `--chapter`, pick a question family with

`--category 1-6` (moral weight, theodicy, redemption, the sacred, duty, implicit theology), or restrict to one tradition with `--lens <code>`.

Before a session the reader grounds itself in what is already known — world findings flagged with a theological dimension, character arcs that involve a change of belief, any open fast-track signals in scope — and opens there; each source degrades cleanly, none is required. Findings live in `.inkhaven/inner_theologian.db`, and the `theologian:` config block tunes the windows, the sub-budget, the disabled lenses, and the language (`enabled: false` turns the whole feature off).

## Inner Poet — the reader of verse

The Inner Poet is the family's specialist for verse. It **observes and measures** a poem against its declared form — metre, rhyme, syllable counts, structural completion — and, on the slow track, offers one LLM observation. It follows the cardinal rule harder than any other member: **it never generates or rewrites a line of verse**. You write the poem; it reads it. This section is the brief tour; Chapter 22 gives poetry its full treatment.

Verse lives in the `para:verse-*` structural family, and a poem is measured against a **declared form** — a `poem:` block beside the stanza. Open a verse paragraph and press `Ctrl+B J → P` for the Inner Poet, with its own sub-keys:

|                           |                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------|
| <code>Ctrl+B J → P</code> | Open the Inner Poet on the open verse paragraph.                                         |
| <code>P → F</code>        | Fast-scan metre + rhyme against the declared form → Output. Deterministic, free.         |
| <code>P → E</code>        | Engage the LLM slow track — an observation on sound, caesura, the turn. Never a rewrite. |
| <code>P → D</code>        | Declare a form — a picker that writes the language-localised poem: block.                |
| <code>P → T</code>        | The two-column translation view (source    translation) + the Form/Sound trilemma.       |
| <code>P → A</code>        | Ambient — auto fast-scan each verse paragraph as you open it. Free.                      |

The Fast scan grades against the form in the same Praise / Note / Concern vocabulary as the Editor, with the scansion drawn in glyphs — `/` stressed, `×` unstressed, `·` flexible. While a verse paragraph is open the status bar shows a live syllable readout

(J 8 syl · 12/4) and the outline shows completion chips (8/14, 14/14 ✓).

inkhaven poetry forms / syllabify /

The CLI surface is metre / rhyme / scan / status / trilemma; the whole of it, and the trilemma of verse translation, is Chapter 22's subject.

## The reasoning-rigor reader

The fifth member is the argument-side of the family. Where the Inner Socrates' Dialectician asks the hard logical question by hand, and one at a time, the reasoning-rigor reader (RIGOR) asks — deterministically, and at book scale — whether the **arguments themselves hold**. It scans manuscript prose for argument-rigor signals via language-keyed cue markers and surfaces each as an advisory finding with the glyph  $\vdash$  /.

It is the most austere member: **deterministic and free** (no LLM, no network, no persistence), **advisory** (a cue is a candidate weakness to weigh, never a verdict), **multilingual**, and **stateless** — findings are recomputed each pass from the manuscript files and emitted straight out. It reads every prose paragraph under a user book in reading order, strips the Typst to plain text, and matches against the language's cue tables, at most one finding per category per paragraph so a dense passage never floods the pane.

### The six signals

Each category is keyed to a conservative table of strong, rarely-innocent cues — a false positive costs a glance, but noise erodes trust, so the lists stay lean.

#### ● ● ● The six reasoning-rigor categories

|                    |                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------|
| false-dichotomy    | a forced binary — "either ... or", "the only alternative", "one or the other"                                |
| question-begging   | asserted as self-evident — "obviously", "of course", "needless to say"                                       |
| straw-man          | a view dismissed — "so-called", "would have us believe", "simplistic"                                        |
| overgeneralization | an unqualified universal — "always", "never", "without exception"                                            |
| non-sequitur       | a conclusion connective ("therefore", "thus") with NO warrant ("because", "since") anywhere in the paragraph |
| equivocation       | a Glossary term with $\geq 2$ senses used $\geq 2$ times in a paragraph, sense unpinned                      |

Two of those deserve a note. **overgeneralization** deliberately fires only on the strong absolutes — **always**, **never**, **without exception** — and **not** on the innocent **all** or **every**, which do honest work far too often to flag. **equivocation** is the one that needs more than prose: it projects from the scholarly Glossary, watching multi-sense terms that carry **watch\_equivocation**, and with no such Glossary entries it is simply inert. Each finding carries a localized advisory sentence that names the matched cue and poses the question you should answer.

### The command line — and the CI gate

RIGOR is reachable in the editor at **Ctrl+B J → R** — the deterministic reasoning-rigor reader over the open paragraph. It can also join the review-pass ambient surface (set **fast\_track: true** in the **rigor:** config block), but its fuller home is the command line.

● ● ● *Scanning for rigor, with the strict gate*

```
$ inkhaven rigor scan --strict

⊢/ [ch.2 · overgeneralization]
  A universal claim ("never") — would a single counter-
  example break it? If so, qualify it.

⊢/ [ch.5 · non-sequitur]
  "therefore" draws a conclusion, but no warrant ("because",
  "since") appears in the paragraph. What licenses the step?

2 signal(s) · exit 1  (--strict)
```

**rigor scan** runs across one user book (**--book NAME** to pick, default the resolved user book), **--signal CODE** filters to one category, and **--json** emits a machine-readable array. By default the command **exits 0** — it is advisory, and means only to inform. But **--strict** makes **any** signal a non-zero exit, which turns the reader into a pre-submission gate you can wire into CI: a build that fails when an unqualified universal or an unwarranted conclusion slips into the manuscript. It is the same conservative reader either way; the flag only decides whether it whispers or blocks the pipe.



## FIRST-CLASS IN FIVE LANGUAGES

RIGOR ships its own cue tables **and** its own advisory sentences in English, Russian, German, French, and Spanish — a Russian paragraph draws findings that name the matched Cyrillic cue. Matching is Unicode-aware and whole-word (so “art” never fires inside “start”). Spanish declares no clean “**either ... or**” correlative, so false-dichotomy there rests on the forced-binary phrases alone — the reader never claims coverage it does not have.

## The family, at a glance

Five readers, one promise. Here is where each one lives and what it costs to run.

### ● ● ● *The Inner family — reach and track*

| Reader           | Reach        | Track     | Glyph   |
|------------------|--------------|-----------|---------|
| Inner Socrates   | Ctrl+B J     | fast+slow | ◇       |
| Inner Editor     | Ctrl+V 0     | slow only | ⚡       |
| Inner Theologian | Ctrl+B J → T | fast+slow | ⚡       |
| Inner Poet       | Ctrl+B J → P | fast+slow | (P/N/C) |
| Rigor reader     | Ctrl+B J → R | fast only | ⚡/      |

Reach for the Fast tracks and RIGOR freely — they are deterministic, instant, and cost nothing, and they run in all five languages. Spend on the Slow tracks when you want the question a pattern cannot reach — a hidden assumption, a theological weight, an observation on the turn of a sonnet — knowing every call is capped, background, and degrades quietly when no provider is set. And whatever you run, remember the one promise that binds the whole family: they read your book and hand back questions. Not one of them writes a word of it.

## WHAT YOU LEARNED

- The **Inner family** is five readers that **question rather than answer** — Inner Socrates, Inner Editor, Inner Theologian, Inner Poet, and the reasoning-rigor reader — and every one is **advisory**: they surface findings, never edit prose.
- **Inner Socrates** asks about your choices — a deterministic **Fast track** (seven categories, five languages) and a cost-capped LLM **Slow track**; fourteen **personas** (fiction / nonfiction / ideas, plus the one-sided **Defender** and **Prosecutor**); the shared **intent ledger** suppresses what you chose on purpose. Hub: Ctrl+B J (F fast, E slow, S persona); CLI `inkhaven inner-socrates`.
- **Inner Editor** observes **craft** as **Praise / Note / Concern** — Praise hidden until earned — non-prescriptively, LLM-only. Reach: Ctrl+V 0 (E engage, A ambient, F findings); CLI `inkhaven inner-editor`.
- **Inner Theologian** reads moral weight through **eleven tradition lenses**, belonging to none and **silent without a moral frame**; Ctrl+B J → T, CLI `inkhaven theologian`. **Inner Poet** measures verse against a declared form and **never generates**; Ctrl+B J → P, CLI `inkhaven poetry` (Chapter 22).
- The **reasoning-rigor reader** deterministically flags six argument weaknesses (false dichotomy, question-begging, straw-man, overgeneralization, non-sequitur, equivocation) in five languages; `inkhaven rigor scan`, with `--strict` as a CI gate.

PART VI

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# Language, Verse & Scholarship

## CHAPTER 21

## 21

Constructed Languages

---

Two of Inkhaven's tools work on language itself rather than on your prose, and because they sit near each other in the interface it is easy to confuse them, so this chapter names them apart before it does anything else. The first is the **ConLang Suite** — a workbench for **inventing** a language from nothing: sounds, words, grammar, even a script and a font, all held inside the editor. The second is the **WordNet thesaurus** — a lookup over a **real**, existing language, for finding a sharper word when you are writing ordinary prose. One builds a language that has never been spoken; the other consults the accumulated senses of one that has. They share a chapter because they both live in the same window and both are reached by a chord, and they share almost nothing else.

**TWO TOOLS, ONE LINE TO KEEP STRAIGHT**

The ConLang Suite is for a language you are **making up** — the tongue your elves speak, the trade creole of your invented port. The WordNet thesaurus is for the **actual** language you are writing in — English, French, German, Spanish, Russian — and it never touches an invented language. If you are reaching for a synonym in your own prose, that is WordNet; if you are coining a word that does not exist, that is the ConLang Suite.

This chapter is the **operator’s tour** of both: what each is, where it lives, and how to run it. The ConLang Suite is deep enough to have its own companion book — **Developing a Constructed Language** — which is the full guide, from a first phoneme to a printed grammar; when a topic here is only sketched, that is where the long treatment lives.

## Part one — the ConLang Suite

A constructed language in Inkhaven is not a file format or a wizard. It is a **book** — an ordinary sub-book under the **Language** system book, with chapters you can open, read, and edit like any other. What makes it a **language** is that a set of engines reconstructs a working model of it from typed **HJSON blocks** you place in those chapters. The book stays the home of record: everything the suite knows about your language is text you wrote into it, and everything the suite produces you can trace back to a block.

**TERM HJSON block**

A small, human-friendly JSON dialect — commas optional, comments allowed, quotes often unneeded. Each block is one paragraph in the language’s book: a list of phonemes, a set of morphemes, a dictionary entry. You author them by hand or let a command write them; the engines read them to build the language.

### Scaffolding a language

You start a language with one command, which creates the sub-book and its chapters ready to fill:

● ● ● *inkhaven language init — the scaffold*

```
$ inkhaven language init Eldar

Created Language ▶ Eldar
  · Meta / overview      iso_code, alphabet, world context
  · Phonology            sounds, classes, templates, stress
  · Grammar              morphology, typology, varieties
  · Dictionary           one HJSON paragraph per word
  · Sample texts         register anchors for translation

Next: add a `phonemes` block under Phonology,
      then `inkhaven language add-word Eldar ...`.
```

Each chapter is a destination for a particular kind of block. The table below is the map you keep in your head: which block goes where, and which engine it feeds.

|                        |                                                                                                                                 |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Phonology</b>       | Phonemes, classes, templates, constraints, allophony, stress, tone, romanization, the font block — drives the phonology engine. |
| <b>Grammar</b>         | Morphemes, paradigms, typology answers, varieties, contact, idioms, metaphors — drives morphology and syntax.                   |
| <b>Dictionary</b>      | One HJSON paragraph per word — the lexicon, overlay, and generation target.                                                     |
| <b>Meta / overview</b> | The ISO-style code, alphabet, and world context — resolves `:lang:` and sorts the buckets.                                      |

#### QUOTE YOUR INLINE ENUMS

HJSON lets an unquoted string run to the end of the line, so an inline value like `kind: consonant` would swallow anything after it. Quote the short enum values — `kind: "consonant"`, `position: "suffix"` — and the parser reads them cleanly. It gives a clear error if you forget, but it is the one HJSON gotcha worth learning first.

### Phonology — the sounds and how they combine

The **Phonology** chapter is where a language gets its voice. A phoneme block lists the sounds (each with its IPA symbol, a romanization, and whether it is a consonant or a vowel), groups them into named **classes**, gives **templates** for building

syllables, and adds **constraints** that forbid the shapes the language does not allow. From this alone the suite can generate sayable words.

● ● ● *A phoneme block under the Phonology chapter*

```
{
  phonemes: [
    { ipa: "p", romanize: "p", kind: "consonant" }
    { ipa: "ʃ", romanize: "sh", kind: "consonant" }
    { ipa: "a", romanize: "a", kind: "vowel" }
  ]
  classes: { C: ["p", "ʃ"], V: ["a"] }
  templates: { root: [ { pattern: "C V (C)", weight: 1.0 } ] }
  constraints: [
    { kind: "max_cluster_size", value: 2 }
    { kind: "no_geminate" }
  ]
  stress: "penultimate"
}
```

That is enough to speak. `inkhaven language generate-word Eldar` draws a phonotactically valid form from the templates; `syllabify`, `ipa`, `stress`, and `romanize` each inspect one facet of a word you give it. Two further blocks deepen the phonology: an **allophony** list rewrites sounds in context (SPE rule notation, `k > tʃ / _ i` — “k becomes tʃ before i”), and a **tone** block gives the language pitch. The companion book teaches the full notation; here it is enough to know the block is where the sound lives.

### The lexicon — words, and where they come from

A word is an HJSON paragraph under **Dictionary**, and it carries more than a translation — a register, a domain, an era, whatever you want to filter on later:

● ● ● *One dictionary entry*

```
{ word: "makil", type: "noun", translation: "sword",
  register: "formal", domain: ["weapon"], era: "third_age" }
```

You add words by hand (`add-word`), import them from a CSV, or — the interesting path — have the AI generate a themed batch. The key idea is that **the forms are never the AI’s to invent**. The deterministic generator produces phonotactically

legal candidate forms from your own templates; the model only assigns **meanings** to them. And before any generated word is kept, it must pass the **dedup gate**.

### TERM The dedup gate

The consistency check every generated word must survive before it can join the Dictionary. It rejects a form that is **phonotactically illegal**, a **homophone** of a word you already have, a **duplicate meaning**, or a **near-synonym** that would crowd an existing sense. What survives is genuinely new — a legal sound the language did not already carry, for a meaning it did not already cover.

● ● ● *language generate-lexicon — meanings onto legal forms*

```
$ inkhaven language generate-lexicon Eldar \
  --topic seafaring --count 6

keep  toran    n.  tide
keep  miluva   n.  harbor
reject vesh          homophone of existing word
reject koran  n.  mast  near-synonym of "toran"
keep  dalmar   n.  hull

3 kept · 2 rejected by the dedup gate
Re-run with --yes to write the survivors.
```

Nothing is written without **--yes** — like every AI feature in Inkhaven, this one proposes and you commit. **language audit** hunts the existing lexicon for the same faults after the fact (illegal forms, homophones, duplicate meanings); **language query** filters it by the rich fields; **language gaps** tells you which core concepts you have not coined yet, ready to hand straight to **generate-lexicon**; and **language stats** prints a descriptive profile of the whole inventory.

### Morphology and grammar — briefly

Beyond sounds and words, a language has **structure**, and the suite models it — but this is the part the companion book carries, so here is only the shape. A **morphology** block in the Grammar chapter declares **morphemes** (prefixes, suffixes, infixes, circumfixes, and non-concatenative processes like ablaut and reduplication) and the **paradigms** that arrange them, so **language paradigm** can inflect a root through a full case-and-number table. A **typology** answer set (**language grammar Eldar --set word\_order=sov**) records the language's



big structural choices against a WALS-aligned catalog, and the **syntax** engine ( `language sentence` ) uses them to assemble a real clause — ordering the words, case-marking the nouns, running agreement — and prints it with an interlinear gloss. From there the suite reaches into diachronics (sound change and daughter languages), dialects, borrowing, translation, and script design. It is a large country; the CLI is how you travel it.

### The inkhaven language CLI

The suite's depth lives on the command line — the editor holds a viewer, the CLI holds the operations. There are far more verbs than fit here, grouped by what they touch; the companion book documents each in full. This is the operator's inventory: enough to know a verb exists and what family it belongs to.

**init · add-word ·** Create a language; add or import words.

**import**

**generate-word ·** Inspect a form — build one, break it into syllables, pronounce it, mark

**syllabify · ipa ·** its stress, romanize it, scan its verse.

**stress · romanize ·**

**scan**

**generate-lexicon ·** Grow and inspect the lexicon; find undefined conlang words in your  
**audit · query ·** prose.

**gaps · stats · scan-  
manuscript**

**paradigm · derive** Inflect and derive words; gloss a line.

**· agree · gloss**

**grammar · sen-** Set typology; assemble clauses of rising complexity.

**tence · relative ·**

**complement · co-**

**ordinate**

**compose** Generate names, prose, verse, or (AI) a blessing / curse / incantation from the lexicon.

**sound-change ·** Diachronics — evolve a proto into daughters.

**derive-lexicon ·**

**family-tree · cog-**

**nates**

**varieties · lect ·** Dialects, registers, and language contact.  
**dialects · borrow ·**  
**areal**

**translate · reverse** Rule-based translation and its memory.  
**· cross · remember**  
**· corpus · eval**

**dictionary · gram-** Render the language as a printable book.  
**mar-book · tutor-**  
**ial**

**export · import ·** Interchange with other tools; link speakers to your world.  
**link-place · link-**  
**character**

### Scansion — language scan

Because a phonology already knows syllables and stress, it can **scan verse** in the invented language. `inkhaven language scan Eldar --text "..."` takes one or more lines, syllabifies each word, marks every syllable stressed or unstressed, and names the metre — the same scansion engine the poetry tools use, pointed at your conlang's own sounds. Stress is resolved per word: an explicit accent mark in the text wins, then the lexicon's **stress** field, then the language's stress rule; an unmarked one-syllable word stays **flexible** and may bend to fit the beat.

● ● ● *language scan — verse in an invented tongue*

```
$ inkhaven language scan Eldar --text "makil toran dala"

scan · Eldar  (/ stressed × unstressed · flexible)

ma kil to ran da la
× / / × / .
→ trochaic · 3 feet
```

### In the editor — the hub, insertion, and translation

Three things bring the suite into the writing window itself, so you are not always at the CLI.

**The ConLang hub — Ctrl+B X.** From any pane, **Ctrl+B X** opens a read-only overview of every language under the Language book: for each, its phoneme inventory split into consonants and vowels, its template and constraint and allophony

counts, its prosody (the stress rule, whether it has tone), its romanization schemes, its lexicon size, and how many places and characters speak it. It is a dashboard, not a workbench — the deep operations stay on the CLI, and the hub's footer reminds you which ones. Scroll with the arrows; `Esc` closes.

- ● ● *Ctrl+B X* — the read-only ConLang hub

```

└─ ConLang suite
  Language: Eldar
  Phonemes      : 22 (16 C, 6 V)
  Templates     : 2 root · 4 constraint(s)
  Allophony     : 3 rule(s)
  Prosody       : stress penultimate · tone –
  Romanization  : 1 scheme(s)
  Lexicon       : 214 entr(y/ies)
  Speakers      : 3 place(s) · 5 character(s)

Ctrl+B Q translate · CLI: language audit ·
      generate-lexicon · query · scan-manuscript

```

**Inserting a word** — **:lang:** . While writing, type a colon, the language's name or ISO code, and a closing colon — **:Eldar:** — and a lexicon picker opens on that language. Choose a word and it is inserted in place of the trigger, so you can drop invented vocabulary into your prose without leaving the keyboard or memorising your own dictionary.

**Translating a paragraph** — **Ctrl+B Q** and **Ctrl+B Shift+Q**. In the Editor, **Ctrl+B Q** translates the open paragraph **into** an invented language: it gathers the target language's grammar, phonology, a filtered slice of its dictionary, and its sample texts into a prompt, and streams the result into the AI pane, wrapped in markers so the AI pane's apply chord lifts only the translation. **Ctrl+B Shift+Q** runs it in reverse — **from** the invented language back to your own. Translate a passage one way, paste the result into the next paragraph, and translate it back: when the round trip drifts, you have found an inconsistency in the grammar or dictionary the manuscript would eventually have tripped over.

## ONE CHORD, TWO MEANINGS BY PANE

Ctrl+B Q is **translate into an invented language** only in the **Editor**. With the **Tree** focused, the same chord is the imposition preview for hand-binding — a different tool entirely (Chapter 33). The two never collide because they belong to different panes; it is the pane-specific chord table from Chapter 3 at work.

## Part two — the WordNet thesaurus

Now the other tool, and the distinction from the first is the whole point. The WordNet thesaurus has nothing to do with invented languages. It is a **sense-based** thesaurus over a **real** one — the language your manuscript is actually written in — and its job is to hand you a better word when you are writing ordinary prose.

### TERM Sense-based

A plain thesaurus offers “big → large, huge, enormous” without asking which **big** you meant. WordNet organises words by **sense** — the distinct meanings a word carries — and offers synonyms, antonyms, and related words for the **specific sense you pick**. “Bank” the riverside and “bank” the institution are different senses with different neighbours, and you choose which you meant.

### Installing a dictionary

The thesaurus needs a language’s WordNet index installed before it can work. English fetches openly today; French, German, and Spanish arrive through the Open Multilingual WordNet sources. For a language whose data cannot be openly downloaded — Russian is the standing case — you build the index from a local WN-LMF file you supply.

● ● ● *inkhaven wordnet — install and inspect*

```
$ inkhaven wordnet list          # sources + what's installed
$ inkhaven wordnet fetch en      # download + index English
$ inkhaven wordnet import ru     russian-wordnet.xml.gz
```

The index is stored under your data directory, at `<data_dir>/inkhaven/wordnet/`, and is **shared across every project** — it is a property of your machine, not of one book, because the English language does not change from book to book. Nothing about a lookup ever leaves your computer; the fetch is the only step that touches the network, and after that the thesaurus is entirely local.

### Looking a word up

`inkhaven wordnet lookup <word>` prints a word’s senses, each with its own set of related words. This is the command-line face of the same index the editor uses.

● ● ● *wordnet lookup — two senses, two neighbourhoods*

```
$ inkhaven wordnet lookup bank

bank (noun)
· sense 1 — a financial institution
  syn: depository, banking company
  hyper: financial institution
· sense 2 — sloping land beside water
  syn: riverside, embankment
  hyper: slope
```

A `--lang` flag chooses which language’s index to search; without it, the lookup uses English.

### In the editor — Ctrl+V Shift+Y

The everyday way to use the thesaurus is not the CLI but the editor chord. Put the cursor on a word (or select it) and press `Ctrl+V Shift+Y`. The thesaurus opens on that word’s senses; pick the sense you meant, then a synonym, antonym, hypernym, or hyponym, and the chosen word **replaces the original in place** — no retyping, and the Typst markup around it is preserved. It is the fastest path from “this word is not quite right” to a better one, without leaving the line.

● ● ● *Ctrl+V Shift+Y — pick a sense, then a replacement*

```

— Thesaurus · "huge" —
| sense 1 — unusually great in size or extent
| | syn enormous
| |   syn immense
| |   syn vast
| |   ant tiny
| — sense 2 — enormous in scope or degree —
|   syn colossal
|
| ↕ select · Enter replace in place · Esc cancel

```

The chord is **Ctrl+V Shift+Y** — under the **view** prefix, because it opens a view onto your word rather than acting on the book. (It briefly lived on the meta layer in an early release; if an old note says **Ctrl+B** something, the documented home is **Ctrl+V Shift+Y**.)

### Across languages, and the AI fallback

WordNet carries an **interlingual index** that links senses across languages, so a lookup can cross from one to another: the German word for a concept your Russian text names is a lookup on the shared sense. This is why a non-English lookup still works — the relations expand through that shared index.

And when a language has **no** installed WordNet and nothing to download — Russian, if you have not imported one — the chord does not simply fail. It falls back to the AI, which offers sense-grouped alternatives keyed to your project language, so the workflow is identical everywhere. You always press **Ctrl+V Shift+Y**; what answers is WordNet where it exists and the model where it cannot.

### THE LINE, ONCE MORE

If you take one thing from this chapter, take the seam down its middle. The ConLang Suite (**Ctrl+B X**, `inkhaven language ...`, `:lang:`) **invents** a language and generates within it. The WordNet thesaurus (**Ctrl+V Shift+Y**, `inkhaven wordnet ...`) **consults** a real one for your ordinary prose. They are never the same operation, and neither reaches into the other's territory.

## WHAT YOU LEARNED

- The **ConLang Suite** builds an **invented** language as a **book** under the Language system book: **HJSON blocks** for phonology, lexicon, and morphology, scaffolded by `inkhaven language init` and read by the suite's engines.
- Words come from a **deterministic generator** (legal forms) plus **AI meanings**, and every generated word must pass the **dedup gate** — no illegal form, homophone, duplicate meaning, or near-synonym survives.
- The `inkhaven language` CLI is the deep surface — inspection, generation, morphology, syntax, diachronics, translation, and printable books; the companion **Developing a Constructed Language** is the full guide.
- In the editor: `Ctrl+B X` opens the read-only ConLang hub, `:lang:` inserts a lexicon word in place, and `Ctrl+B Q` / `Ctrl+B Shift+Q` translate a paragraph into and out of an invented language.
- The **WordNet thesaurus** is a separate tool for **real** prose: **sense-based**, multilingual (`fetch en/fr/de/es`, `import ru`), stored at `<data_dir>/inkhaven/wordnet/` and shared across projects.
- `inkhaven wordnet` handles `fetch` / `import` / `lookup` / `list`; in the editor `Ctrl+V Shift+Y` replaces the word under the cursor in place, crossing languages through the interlingual index and falling back to the AI when no index exists.

CHAPTER 22

# 22

## Poetry

---

Every other reader in this book watches your prose. This one listens to your verse — and it holds itself to a stricter discipline than any of them. Inkhaven will scan a line for its metre, weigh a rhyme, count a haiku's syllables, and tell you a sonnet is two lines short, but it will never write you a line, never finish your couplet, never suggest a rhyme to close the stanza. It observes and measures; it does not compose. That single refusal is the whole shape of the feature, and it is worth stating before anything else, because it is what makes the Inner Poet trustworthy: a tool that measures your verse is a mirror, and a mirror that started painting would be worse than useless.



This chapter is the operator’s tour — what verse **is** inside Inkhaven, how you declare the form you are aiming at, how the Inner Poet reads a stanza against it, and how the same machinery is reachable from a chord in the editor and from the `inkhaven poetry` command in your shell. It is deliberately breadth-first. The full treatment — the history of each form, the theory of scansion, the craft of translating verse — lives in the companion volume, **Poetry with Inkhaven**, and this chapter points you to it wherever the depth is more than an operator needs.

#### THE COMPANION

**Poetry with Inkhaven** is the deep dive: ten chapters on verse in the terminal, from the sound of a line to translating a stanza at the desk. This chapter tells you what the tools are and how to run them; the companion tells you what to do with what they show you.

## The iron rule — measure, never make

Inkhaven’s AI touches prose only under the strict advisory contract you met earlier: it proposes, you confirm, nothing is written without your say-so. Verse goes further still. The Inner Poet has **no write path at all** — not a confirmed one, not a guarded one. It cannot be made to emit a line of poetry, because the capability was never built. Its slow, LLM-driven track is prompted, in its own system instruction, to observe and never rewrite; its fast track is pure arithmetic over your syllables.

#### TERM The iron rule of verse

The Inner Poet **observes and measures** — it never generates or rewrites a line of verse. You write the poem; it reads it. Where a prose tool might offer a rewrite for you to accept, the poetry tools offer only a reading: a scansion, a rhyme verdict, a completion count, an observation. The composing is yours.

The reason is not timidity but respect for the form. A metre is a decision made across a whole poem; a rhyme is a bet on sound and sense at once; a substitution in a pentameter line is often the point, not the error. A machine that “fixed” these would flatten exactly the choices that make verse verse. So Inkhaven does the one thing a machine does better than a poet — it **counts, exactly, without tiring** — and leaves everything else to you.

## Verse is a paragraph family, not a project

Poetry is not a separate kind of project, a special editor, or a branch of the tree. It is a **family of structural paragraph subtypes**, each tagged `para:verse-*`, exactly as code blocks are `para:code` and mathematics is `para:math`. A poem is just a paragraph (or a run of them) that carries a verse tag. This has a quiet but important consequence: a stanza can live **anywhere** — an epigraph atop a chapter, a song a character sings, a recited fragment in the middle of a scene — without changing the manuscript's structure, and a poetry-only project is simply a book whose paragraphs happen to all be verse.

### TERM The verse family

Six `para:verse-*` subtypes, each with its own tree glyph: `verse-line` (`¶`), `verse-stanza` (`⌋`), `verse-couplet` (`⏟`), `verse-tercet` (`⋮`), `verse-quatrain` (`⋮`), and `verse-translation` (`↔`). Any of them marks a paragraph as verse; the Inner Poet scopes on the whole family, and the prose readers skip it.

Because every verse tag begins `para:`, the prose companions — the Inner Editor, the Inner Socrates, the narrative profiler — already know to leave it alone, and its words are kept out of the prose word count. You do not wire any of that up; tagging a paragraph as verse is enough to hand it to the Inner Poet and take it away from everyone else. You add the next stanza of a poem without leaving the writing flow: `Ctrl+B Shift+Y` creates a sibling verse paragraph of the same subtype, immediately below and open for editing — structure only, never a generated line.

## Declaring a form — the poem: block

The Inner Poet measures your lines against a **target**, and you set that target by declaring a form. A declared form is a small `poem:` block — an HJSON object naming the metre, the foot count, the rhyme scheme, and any structural rules — that sits beside the stanza as a sidecar. Everything the Inner Poet says is said **relative to** this declaration: without one it can only observe free-verse tendencies; with one it can tell you a line runs long, a rhyme fell to a near-rhyme, or a villanelle's refrain has drifted.

You rarely write a **poem:** block by hand. Inkhaven ships a catalogue of eighteen canonical forms, and you scaffold any of them — tuned for your project’s language — with one command.

● ● ● *A sonnet's declared form, printed by the forms library*

```
$ inkhaven poetry forms --form sonnet --language en

poem: {
  // 14 lines of iambic pentameter (generic English scheme)
  form: sonnet
  metre: iambic
  feet: 5
  metre_tradition: accentual_syllabic
  rhyme_scheme: "ABAB CDCD EFEF GG"
  language: en
}
```

Run **inkhaven poetry forms** with no arguments and it lists every form with a one-line description; add **--form <name>** to print that one’s block, and **--language en|ru|fr|de|es** to tune it. The tuning is not cosmetic. French verse is **syllabic** — you count syllables and elide the mute **e** — so a French form switches its tradition and sets **elide\_mute\_e**; Russian accentual-syllabic verse allows the pyrrhic substitution and expects a final stress, so a Russian form sets **allow\_pyrrhic** and **require\_final\_stress**. The same sonnet, asked for in Russian, comes back with the rules Russian prosody actually plays by.

#### TERM The declared form

A **poem:** block — the target the Inner Poet measures against. Its fields: **form** (the canonical name), **metre** (**iambic**, **trochaic**, **anapestic**, ...), **feet** (per line), **metre\_tradition** (**accentual\_syllabic**, **syllabic**, **free**), **rhyme\_scheme** (label string like **ABAB CDCD EFEF GG**), and structural counts (**stanzas**, **lines\_per\_stanza**). Localising it retunes the rules.

To attach a form inside the editor rather than the shell, open the Inner Poet on a verse paragraph and press **D** — a picker that writes the language-localised **poem:** **inkhaven poetry forms** sidecar for you. And if none of the eighteen fits, **--new --name my-form** prints a **custom** scaffold you edit and keep in **.inkhaven/custom-forms.hjson**.

## The Inner Poet — the reader for verse

The Inner Poet is the member of Inkhaven's inner-reader family that scopes on verse. Like its siblings it works on two tracks: a **fast** track that is deterministic, offline, and free, and a **slow** track that engages the LLM for an observation. You reach both from the editor at `Ctrl+B J → P`, and the fast track from the shell as `inkhaven poetry scan`.

### The fast track — metre and rhyme, counted

The fast track scans a stanza against its declared form and reports what it finds as a list of **findings**, each carrying one of three weights. Unusually for an Inkhaven reader, it does not only warn — it also **praises**: a line that scans cleanly earns a Praise finding, because in verse a thing done right is worth naming.

● ● ● *The fast-track scan of a sonnet from the shell*

```
$ inkhaven poetry scan --text "$(cat sonnet.txt)" \
--form shakespearean_sonnet
♪ Praise    [Metre] Line 1 scans cleanly as iambic pentameter.
♪ Note      [Rhyme] Lines 12↔14 (G-G): "gone" / "dawn" –
            near-rhyme. Intended?
♪ Concern   [Metre] Line 9 has 13 syllables; declared iambic
            allows 10 (±1 for a feminine ending). Long by 2.
```

#### TERM Praise · Note · Concern

The three weights of a fast-track finding. **Concern** is a real departure — a line long past tolerance, a rhyme pair that does not rhyme. **Note** is a question worth asking — a near-rhyme, a line that scans a touch short. **Praise** marks a line that meets its declared metre cleanly. A stanza with nothing to say draws silence, which is itself a verdict.

In the editor the same scan runs on `F`, and its findings land in the Output pane under the `inner-poet` category, each jumpable to its line. Turn on **ambient** mode with `A` and the fast scan re-runs automatically every time you open a verse paragraph — free, uncapped, always current, so the reading is simply **there** as

you move through a poem. Findings and your suppressions persist per project in `inner_poet.db`, so a finding you have chosen to live with stays quiet.

### The slow track — an observation, never a rewrite

Press **E** to engage the slow track. It sends the stanza to your LLM under a system prompt that pins it to the family’s register — “I notice...”, “you might consider...”, never “should”, never “must” — and asks for a few observations on what the verse is **doing**: whether the line breaks fall at phrase boundaries or cut across the syntax (enjambment), the sound texture (alliteration, assonance, consonance), and where the breath-pause falls (caesura). For a sonnet it also asks whether a real turn — a volta — is present at the expected place, after line eight in a Petrarchan sonnet, after line twelve in a Shakespearean one, noting whether the turn is substantive, partial, or absent, without prescribing what it ought to be. The observation streams into the Thoughts pane, the quiet reading surface, and never proposes a replacement word.

## Scansion — reading the beats

Underneath the fast track is a scanner that turns a line into a row of beats. It syllabifies each word, marks the stressed syllable, and renders the result in three single-width glyphs.

### TERM The scansion glyphs

**/** — a **stressed** syllable. **×** — an **unstressed** syllable. **•** — a **flexible** beat: a monosyllable that will promote or demote to fit the metre (English is full of them). An acute accent in the text (`compáre`) fixes a stress and overrides the rule — the way Russian verse is annotated.

You see the glyph row from `inkhaven poetry metre`, which scans one line, counts its syllables, and — when you name a form — checks it against that form’s declared foot pattern.

● ● ● *Scanning one line, then checking it against a form*

```
$ inkhaven poetry metre \
  --line "The curfew tolls the knell of parting day" \
  --form blank_verse
The curfew tolls the knell of parting day
· / x · · · · / x ·   (10 syllables)
→ detected: iambic pentameter (fit 0.90)
→ declared iambic (5 feet): 10 of 10 syllables, fit 1.00
```

The two “→” lines are two different questions. **Detected** asks the line what it is, with no target in mind, reusing the same scansion engine the conlang tools use — it may answer “irregular / free” for a line that fits no clean pattern. **Declared** checks the line against the foot you asked for and reports the fit as a fraction of the **fixed** (non-flexible) beats that land where the metre wants them, so a line of mostly monosyllables can still score a clean fit. The scanner knows about the two ways a line legitimately runs off its count: a **feminine ending** (one extra unstressed syllable past the last foot, normal in iambic verse) and **catalexis** (one syllable short) — both tolerated within  $\pm 1$ , so the Inner Poet does not cry wolf over a perfectly good line.

### Exact English scansion — the phoneme dictionary

English spelling lies about stress. The crude fallback rule stresses the penultimate syllable, which mis-scans every final-stressed word in the language — **compare**, **begin**, **above**, **return** all come out backwards. The fix is to install a pronouncing dictionary in CMUDict format, which gives the scanner each word’s **exact** stress and syllable count.

● ● ● *Installing the pronouncing dictionary makes English exact*

```
$ inkhaven poetry phonemes import cmudict.dict
✓ installed 135000 words → ../inkhaven/phonemes/en.dict
English metre + rhyme now use it (other languages are
near-phonemic already).

$ inkhaven poetry phonemes status
♪ pronouncing dictionary: 135000 words installed
```

Once installed, English metre **and** rhyme consult it: **compare** scans as the iamb it is, and — crucially for rhyme — **love** / **move** is correctly told apart as an **eye**

rhyme rather than the false “perfect” that spelling alone would report. The other project languages have regular enough orthographies that they are near-phonemic already; Russian scansion, in particular, is exact without any dictionary. An out-of-vocabulary English word falls back to the heuristic unchanged, so the dictionary only ever helps.

## The forms and their metres

The bundled catalogue holds eighteen forms, spanning the fixed forms, the open forms, and the syllabic traditions. Each declares the metre and rhyme the Inner Poet will hold a poem to.

### ● ● ● *The eighteen built-in forms*

|                                                   |                             |
|---------------------------------------------------|-----------------------------|
| sonnet / petrarchan_sonnet / shakespearian_sonnet |                             |
| 14 lines, iambic pentameter, the schemes differ   |                             |
| haiku / senryuu / tanka                           | syllabic – 5-7-5, 5-7-7-... |
| villanelle                                        | 19 lines, two refrains      |
| pantoum                                           | interlocking quatrains      |
| terza_rima / ottava_rima                          | chained / 8-line stanzas    |
| ghazal                                            | couplets + a signature      |
| blank_verse / heroic_couplet                      | unrhymed / rhymed iambic    |
| limerick                                          | 5 anapestic lines, AABBA    |
| ode / elegy                                       | elevated / lament, open     |
| free_verse / prose_poem                           | no metre – tendencies only  |

Foot counts read out in the familiar length-names — **monometer**, **dimeter**, **trimeter**, **tetrameter**, **pentameter**, **hexameter** — and the metre itself is one of the standard feet: iamb (weak-strong), trochee (strong-weak), dactyl, anapest, amphibrach, spondee. A syllabic form such as the haiku is counted, not scanned for stress; a free form is only **observed** — the Inner Poet reports a free-verse passage’s line-length tendencies and flags a rhythm gone monotone, but passes no metrical verdict where none was declared.

## Rhyme — quality and kind

The rhyme engine classifies a pair of words on two axes at once. Its **quality** is how true the rhyme is, and its **type** is where the stress falls in it.

**TERM Rhyme quality and type**

**Quality:** **perfect** (a true rhyme), **near** (a slant rhyme — close but not exact), **eye** (matched spelling, mismatched sound — only detectable with the phoneme dictionary), **none**. **Type:** **masculine** (stress on the final syllable), **feminine** (penultimate), **dactylic** (antepenultimate).

● ● ● *Classifying a rhyme from the shell*

```
$ inkhaven poetry rhyme flower power
  flower / power: perfect feminine rhyme on "-ower"
```

```
$ inkhaven poetry rhyme love move
  love / move: eye-rhyme only
  (matched spelling, different vowel — install the phoneme
   dictionary to tell these apart)
```

The engine is multilingual by construction. It works on the **rhyme tail** — from the stressed vowel onward — with per-language normalisation: German final consonant devoicing (so **Hund** rhymes with **bunt**), French mute-**e** elision, light Russian vowel reduction, and Spanish **assonance**, where a vowel-only match counts. This is exact for the regular orthographies — Russian above all — and for English it climbs to exact the moment the pronouncing dictionary is installed. In a fast-track scan, the engine walks your declared rhyme scheme, pairs up the lines that share a label, and reports any pair that falls to a near-rhyme (a Note) or fails to rhyme at all (a Concern), naming the two lines and the two words.

**Completion — the form still owed**

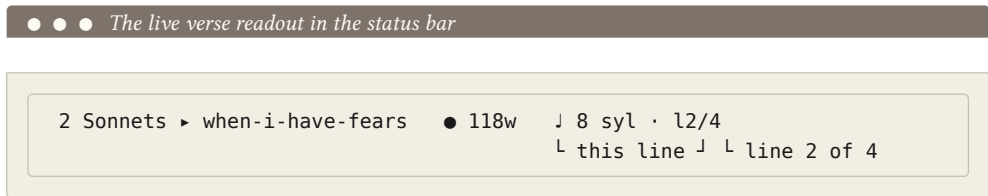
A fixed form is a promise about the **whole** poem, and the completion checker tracks how much of that promise the draft has kept: a line ratio, plus any structural component the form requires that the text is missing or breaking.

The checks are specific to each form's architecture. A **sonnet** owes fourteen lines. A **villanelle** owes its two refrains on a strict schedule — line one returns at lines six, twelve, and eighteen; line three at nine, fifteen, and nineteen — and the checker names any slot where the refrain has drifted. A **pantoum** interlocks: lines two and four of each quatrain must reappear as lines one and three of the next. A



**ghazal** traditionally signs its closing couplet — the **maqta** — with the poet's name, declared as the form's `signature_word`, and the checker looks for it. What the checker never does is fill the gap; it tells you what the form still owes and leaves the paying of it to you.

You see this live while you write. The Outline shows a completion **chip** on each poem — `8/14` in progress, `14/14 ✓` in green when a bounded form is exactly complete (an over-long `16/14` is **not** ticked, because too many lines is its own problem). And whenever a verse paragraph is open, the status bar carries a live readout of the current line's syllable count and its position in the stanza.



You reach the same completion report from the shell with `inkhaven poetry status --text "..." --form villanelle`, which prints the ratio, the drafting/complete state, and any structural issue as a list.

## The translator's trilemma

Translating verse forces a choice no translation can escape, among three things a poem is at once: its **Form** (the metre and rhyme), its **Meaning** (the sense), and its **Sound** (the texture — the alliteration and assonance). You cannot keep all three; every verse translation sacrifices something, and the interesting question is **what**. Inkhaven's answer is not to resolve the trilemma — it has no resolution — but to **make the trade visible** by scoring the two dimensions that are measurable and prompting the LLM for the one that is not.

TERM **The trilemma**

**Form** and **Sound** are scored deterministically: Form by scanning both texts for metre and rhyme preservation, Sound by comparing their alliterative density. **Meaning** is the AI's axis — engaged in the editor, it names which dimension a translation most preserved and which it most sacrificed. Nothing is judged; the trade is only shown.

A translation lives in its own subtype, `para:verse-translation`, whose body holds the source above a delimiter line and the translation below it — either the thematic `↔` glyph or a plain rule of three-or-more dashes (---) you can actually type. Press **T** on such a paragraph for the two-column view: source and translation side by side, with the Form and Sound bars beneath.

● ● ● *The trilemma scored — Form and Sound measured, Meaning to the AI*

```
$ inkhaven poetry trilemma --source "$(cat src.txt)" \  
  --translation "$(cat en.txt)" --form sonnet \  
  --language ru --to-language en
```

Translation trilemma (ru → en):

|         |                                                                                     |     |                                                        |
|---------|-------------------------------------------------------------------------------------|-----|--------------------------------------------------------|
| Form    |  | 70% | 3/4 lines keep the foot ·<br>2/3 source rhymes kept    |
| Meaning |  |     | (the AI axis — engage the<br>Inner Poet in the editor) |
| Sound   |  | 84% | alliteration density<br>0.31 → 0.24                    |

Read the bars as a portrait of the compromise, not a grade. A translation that scores high on Form and low on Sound kept the metre and let the music go; the opposite kept the music and bent the shape. The Meaning bar is deliberately empty in the shell — that axis is the LLM's, and it fills in the editor when you engage the Inner Poet on the paired paragraph.

## At the desk — the chords

Everything above is reachable without leaving the editor, through one branch of the inner-reader menu. Open a verse paragraph, press **Ctrl+B J** for the inner readers, then **P** for the Inner Poet, and the sub-keys fan out.

- Ctrl+B J → P → F** Fast-scan metre + rhyme against the declared form → Output (Praise / Note / Concern). Deterministic, free.
- Ctrl+B J → P → E** Engage the LLM slow track — an observation on enjambment, sound, caesura, and a sonnet's turn. Never a rewrite.
- Ctrl+B J → P → D** Declare a form — a picker that writes the language-localised poem: sidecar.
- Ctrl+B J → P → T** The two-column translation view: source || translation, with the Form / Sound trilemma.
- Ctrl+B J → P → A** Ambient — auto fast-scan each verse paragraph as you open it. Free, no cost cap.
- Ctrl+B Shift+Y** Next stanza — create a sibling verse paragraph of the same subtype, open for editing. Structure only.

The menu is pane-aware, so **P** opens the Inner Poet only when the open paragraph is verse; on prose it tells you there is no verse here. The **J** branch is shared with the Inner Poet's siblings — **T** for the Inner Theologian, **Y** for the Inner Stylist — which is why the poetry key is **P**, one level in.

## From the shell — inkhaven poetry

The same toolset is a command, for scripting, for a continuous-integration gate, or simply for a quick reading without opening the project. Every subcommand except **phonemes** (the English pronouncing dictionary) takes **--language en|ru|fr|de|es**, and several take **--json** for machine-readable output.

- poetry forms** List the forms; **--form** prints one's block, **--new** scaffolds a custom one.
- poetry syllabify** Show a word's or a --line's syllable boundaries and stress.

|                        |                                                                                |
|------------------------|--------------------------------------------------------------------------------|
| <b>poetry metre</b>    | Scan a --line's beats, detect its metre, and check it against a --form.        |
| <b>poetry rhyme</b>    | Classify the rhyme between two words — quality and type.                       |
| <b>poetry scan</b>     | The fast-track stanza scan against a --form. --fail-on-concern is the CI gate. |
| <b>poetry status</b>   | A poem's completion against its --form — ratio and missing components.         |
| <b>poetry trilemma</b> | Score a --source against its --translation on Form and Sound.                  |
| <b>poetry phonemes</b> | import / lookup / status — the English pronouncing dictionary.                 |

The `--fail-on-concern` flag on `poetry scan` is the one built for automation: it exits non-zero the moment any Concern-level finding is present, so a poetry manuscript can be gated in CI the same way a codebase is gated by its tests — the build fails if a sonnet's metre has slipped. Nothing here writes to your manuscript; the shell tools, like the editor's, only ever read.

#### WHERE TO GO DEEPER

This was the operator's tour. **Poetry with Inkhaven** takes each thread further: the sound of a line, the theory behind the scansion, the forms in their history, the craft of translating verse, and a closing chapter on writing a whole poem at the desk with the Inner Poet at your side. Reach for it when the question stops being “how do I run this” and becomes “what should I do with what it shows me”.

## WHAT YOU LEARNED

- The **iron rule**: the Inner Poet **observes and measures** verse — metre, rhyme, syllables, completion — and **never** generates or rewrites a line. The composing is always yours.
- Verse is a **paragraph family** (`para:verse-*`), so a poem lives anywhere in a book; the prose readers skip it and it stays out of the word count.
- You **declare a form** in a poem: block — from `inkhaven poetry forms` or the editor's D picker — and everything the Inner Poet says is measured against it, retuned for your language.
- The **fast track** (F, `poetry scan`) counts metre and rhyme deterministically and reports **Praise / Note / Concern**; the **slow track** (E) offers a non-prescriptive LLM observation in the Thoughts pane.
- Scansion renders beats as / stressed, × unstressed, · flexible; a CMUdict **phoneme dictionary** makes English metre and rhyme exact (love / move as an eye-rhyme), and Russian is exact already.
- The **translator's trilemma** — Form vs Sound vs Meaning — is **scored, not resolved**: Form and Sound measured deterministically, Meaning left to the AI, in a two-column view (T) or `poetry trilemma`.
- Reach it all at `Ctrl+B J → P` in the editor or `inkhaven poetry` in the shell; the full craft is the **Poetry with Inkhaven** companion.

CHAPTER 23

# 23

## Research and Scholarship

---

A book that makes claims about the world has to get them right, and a book with its own invented world has to stay true to itself. Both are the same problem seen from two sides: a manuscript is only as trustworthy as the facts under it, and facts left untended drift, contradict each other, and quietly rot. Inkhaven answers this with four distinct tools that share one job — giving a book a spine of fact you can stand behind. This chapter is the operator’s tour of all four: the **Research Assistant** for gathering and checking facts, the **Sources** book that turns your reading into a bibliography, the **Glossary** that keeps your terminology from wandering, and the **scholarly apparatus** a serious work of theology, philosophy, classics, or law is expected to carry.

The Facts themselves — the research books, the story bible, the graph that ties them together — belong to Part IV; this chapter is about the machinery that **surrounds** them: how a fact gets gathered and vetted, how a citation becomes a reference list, how a term is held to one spelling, and how a scholarly index is built. For the full, patient treatment of research as a craft — the trust ladder, cross-checking, computing facts, composing from a corpus — this manual defers to its companion, **Grounding Your Book in Fact**, which teaches the same tools as a workflow rather than a feature list. Here we tell you what each is, how to reach it, and how to run it.

## The Research Assistant — a room of its own

Most of Inkhaven lives in the four-pane editor. Research does not. The Research Assistant is a **separate full-screen program** you launch from the shell, a quiet room you step into to gather and check facts and step back out of to write:

```
inkhaven research
inkhaven research --thread "1920s Vienna"
```

It opens on its own two-pane screen — a **Facts tree** down the left (about 40% of the width), a **streaming chat** on the right, and a two-line query prompt beneath them — and it wants at least 80 columns. Nothing here is a pane of the editor; it is its own window with its own keys, and quitting it (**q**) drops you back at the shell exactly as you left it. Because it is separate, everything it does is insulated from your manuscript: it writes only to your research books, never to your prose.

### TERM Thread

A **thread** is one research conversation, saved. You keep one per line of inquiry — “the aqueduct”, “1920s Vienna”, “cell biology for chapter 9” — so unrelated research does not tangle.

`--thread <name>` opens or creates one; `--list-threads` shows them, `--export-thread <name> --format <fmt> --out <file>` writes one out.

Threads persist between sessions.

● ● ● *inkhaven research — the two-pane Assistant*

|                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> <div>Facts</div> <div> <div>▼ Fortifications</div> <div> <div>✓ The citadel wall...</div> <div>✓ Bastion spacing...</div> </div> </div> <div>▼ Logistics</div> <div> <div>? Grain reserves...</div> <div>* (novel: the north postern gate)</div> </div> </div> | <div> <div>thread: siege-of-1849</div> <div>&gt; how thick were the citadel's outer walls?</div> <div>Roughly 4 m at the base, tapering with height.</div> <div>— src: Wikidata Q... · ✓</div> <div>/fact → keep as a Fact?</div> </div> |
| <div> <div>[RAG: Facts+Full]</div> <div>\$0.03</div> <div>?:help</div> <div>q:quit</div> </div>                                                                                                                                                                      |                                                                                                                                                                                                                                          |

## Two targets: Facts and Notes

The left tree fills as you work, and every entry lands in one of two books. The distinction is the whole point of keeping research honest.

### TERM Facts and Notes

A **Fact** is something you trust enough to lean on; a **Note** is something you have written down but are not yet sure of. `/fact <claim>` keeps a claim as a Fact — it must cross a confirmation gate first; `/note <claim>` keeps it as a Note instead, no gate. When a Note earns its trust, `/promote` lifts it into the Facts book. A third book, **Sources**, holds the citations behind them, and the next section is entirely about it.

A fourth state sits on top of the Facts book, for fiction. Pressing **u** on a fact in the tree marks it **undisputed** — an authorial axiom, drawn with a ✱, that you have simply **decided** is true of your world. An undisputed fact is exempt from fact-checking against the real world (there is no outside source for “the north postern gate”), but it is still checked for internal coherence, as you will see below.

## Two ways to speak

You address the Assistant in one of two registers, and learning the difference is most of learning the tool. The first is **plain language** — you type a question the way you would ask a knowledgeable friend, and the answer is grounded on whatever you have already kept. This is for thinking: exploring, orienting, working out what you need to know. The second is a **slash command** — `/fact`, `/geonames`,



`/triangulate` — which does something precise and repeatable. Type a single `/` and a hint bar lists the commands matching what you have typed so far; `Tab` completes one, and `Ctrl+B h` opens the full on-screen reference. You never have to memorise the set.

### Where a fact comes from — the trust ladder

Not every fact is equally trustworthy, and the Assistant records **where each one came from** so you always know how firm the ground is.

#### TERM **Provenance**

**Provenance** is the record of a fact's origin — its rung on the **trust ladder**. From firmest to loosest: **computed** (you can re-run the sum), **structured** (a database id like a Wikidata Q-number), **scholarly** (a real paper with a DOI), **documents** (a source you imported), **web** (a cited page), and **model** (an educated guess from the language model). Higher rungs are more verifiable; the whole of research is a climb upward. Appendix B of **Grounding Your Book in Fact** is the full reference.

The source commands each start a fact higher than a bare guess and, where a citation exists, file it to your Sources book for free. `/wikidata <query>` returns structured facts by Q-id; `/geonames <query>` looks a real place up in a gazetteer (needs a free `research.geonames.username`); `/openalex` and `/arxiv` return the top scholarly work or preprint and auto-file its citation; `/gutenberg` (alias `/pg`) ingests a public-domain book; `/web <query>` searches and grounds a cited answer on real pages; and `/calc <expr>` **computes** a fact — unit conversions, great-circle distances, compound growth, domain formulas — to land it on the top rung, `computed`.

### Checking what you keep

Gathering is half the job; the other half is not trusting yourself.

|                           |                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>/triangulate</code> | Cross-check a claim against Wikidata and the two scholarly indexes at once; reports SUPPORTS / CONTRADICTS / SILENT. Alias <code>/tri</code> . |
| <code>/factcheck</code>   | Audit the whole Facts book — per-fact truth and cross-fact consistency — marking each with a verdict: ✓ accurate, ? dubious, ✗ inaccurate.     |
| <code>/whatswrong</code>  | Explain why the selected flagged fact failed, and what the correct information appears to be.                                                  |

|                     |                                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>/undisputed</b>  | Check your undisputed (authorial) facts for internal coherence: PLAUSIBLE / ODD / INCOHERENT, in the project language. |
| <b>/upgrade</b>     | Re-ground a model fact on a corroborating source and raise its provenance in place — the wording is never changed.     |
| <b>/stale</b>       | List model / web facts older than N days (default 90) for re-verification.                                             |
| <b>/deadsources</b> | Scan kept web sources for link-rot and flag the ones that no longer resolve.                                           |

The two glyph systems are worth keeping straight. **/triangulate** holds one claim up against three independent sources at once and reports whether they **support** it, **contradict** it, or stay **silent** — a claim all three support, with none against, is far firmer than any single lookup. **/factcheck** sweeps the whole Facts book after the fact and stamps each entry with ✓, ?, or ✗, the same glyphs you see in the tree; **/whatswrong** then explains a failure. Invented facts sidestep all of this — there is nothing outside your book to check them against — so **/undisputed** gives them their own audit: not “is this true?” but “does this hang together with everything else I decided?”, answered PLAUSIBLE, ODD, or INCOHERENT.

### Following the citations — snowball

When one paper is the right paper, its bibliography is a map of the field around it. **inkhaven research --snowball "<seed>"** follows a work’s citations backward and forward across OpenAlex and reports the neighbourhood, so you can pull in the few that matter rather than search blind. It is citation-chasing made a single command.

### Working headless — batch and the agentic mode

The Assistant does not need you sitting in front of it. Three headless modes let it run while you do something else.

|                                      |                                                                                                                                                                                                                 |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>--batch &lt;file&gt;</b>          | Research a list of questions unattended; proposes findings by default. Add <b>--auto-confirm --confidence &lt;0..1&gt;</b> to insert those above a threshold and report the rest. <b>--out</b> writes a report. |
| <b>--agentic<br/>"&lt;topic&gt;"</b> | Research a topic autonomously, emitting findings as untrusted Facts to triage later with <b>/review</b> . <b>--out</b> writes a run log.                                                                        |
| <b>--import / --sync</b>             | <b>--import &lt;path&gt;</b> ingests a document or folder; <b>--sync &lt;folder&gt;</b> re-imports only the files that changed since last launch.                                                               |

The `--batch` mode closes the loop opened by the Assistant's own `/gaps` command: ask for the open questions your corpus cannot answer, write them to a file, and hand the file back for it to research unattended. The `--agentic` mode goes further — it is the **deep** mode, an autonomous loop that chooses its own next question and files what it finds directly into the Facts book, marked **untrusted**. Nothing autonomous is ever silently trusted: you step through the results afterward with `/review`, where each candidate is **accepted** (a), **deleted** (d), or marked **undisputed** (u), and contradictions are flagged with `≠`.

#### NOTHING HAPPENS TO YOUR MANUSCRIPT BY ACCIDENT

The Research Assistant never touches your prose. It writes only to your research books — Facts, Notes, and Sources — and only the deliberate acts you take (keeping a fact, recording a source) change anything on disk. You can ask anything, chase any tangent, and run any command as a test drive, knowing the book on the other screen is exactly as you left it.

### The keys of the room

Inside the Research screen the navigation is its own small vocabulary, closer to the Tree's than to the editor's.

|                        |                                                                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tab / Shift+Tab</b> | Cycle focus: Facts tree → query prompt → chat.                                                                                          |
| <b>F10</b>             | Cycle the RAG mode a plain question is grounded on: Facts+Full → Facts only → Full only. (The <code>/rag</code> command does the same.) |
| <b>Ctrl+P (tree)</b>   | Pin / unpin the cursor fact as RAG context — up to three, marked <code>[?]</code> .                                                     |
| <b>n (tree)</b>        | Add a fact by hand: type a title, press Enter, type the body, then Ctrl+S to save.                                                      |
| <b>Ctrl+B h</b>        | The full quick reference — every key and every <code>/command</code> .                                                                  |
| <b>? / q</b>           | Toggle the hints bar · quit (Ctrl+C and Ctrl+Q also quit).                                                                              |

The chat also scrolls with `j / k / g / G` and searches with `Ctrl+F`; a confirm overlay (the gate a `/fact` crosses) switches title and body with `Tab` and inserts with `Ctrl+S`. That is enough to run the room; the companion book is the place to learn it as a craft.

## Sources and the bibliography

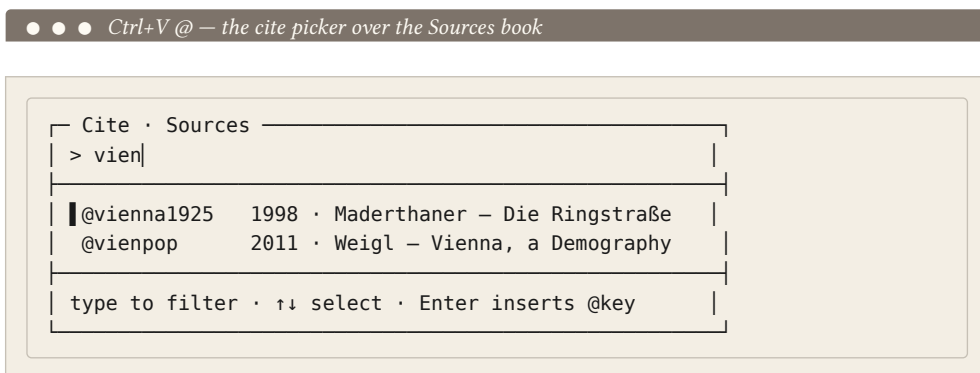
Every time the Assistant grounds an answer on a scholarly paper or a page, it files a proper citation into the **Sources** book without your lifting a finger. Sources is the third research book — Facts holds what you trust, Notes what you are unsure of, Sources the citable works behind them — and it is a **system book** like the others: a real book in your tree, one paragraph per citation, that Inkhaven manages. By the end of a project it is a bibliography you never sat down to write.

### TERM Citation key

A **citation key** is the short handle a source is cited by — `@vienna1925`, `@kant`, `@bible`. In your prose it appears with an `@` sigil, the same one Typst uses natively; at book assembly each key resolves to a full reference. A citation's key is the **title** of its paragraph in the Sources book, so naming a Sources paragraph `vienna1925` is what defines `@vienna1925`.

### Citing as you write — Ctrl+V @

You do not type citation keys from memory. In the editor, `Ctrl+V @` opens the **cite picker** — a fuzzy search over every entry in your Sources book. Type any part of a key, author, or title to narrow it; `↑↓` moves the selection, `Enter` inserts the bare `@key` at your cursor, `Esc` closes. Each row shows the key beside its year, author, and title, so you pick the right work by sight rather than by remembering its handle. If the list is empty, your Sources book has no entries yet — add a paragraph under Sources (its title becomes the key) or import a `.bib` file, and try again. When the chosen source carries a reference scheme — scripture, or a `scheme:` you declared — the picker inserts `@key[]` and leaves your cursor between the brackets for the passage, the `@key[locus]` form the index locorum is built from.



## The bibliography assembles itself

Because each Sources entry was filed from a real work with a real identifier — a DOI, a Q-number, a catalogue id — the reference list is not a guess at how a citation **should** look but the actual, resolvable reference. When you assemble the book (Chapter 24), Inkhaven collects the cited keys, generates a `sources.bib` BibTeX file, and emits a Typst `#bibliography` so the formatted reference list renders into the finished PDF. The citation you inserted with `Ctrl+V @` and the entry in the back matter are the same record, and they can never drift apart, because one is generated from the other.

## Managing Sources from the shell

For interchange with a reference manager, `inkhaven sources` works the book from outside the editor:

|                                                |                                                                                                                             |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>sources list</b>                            | List every citation defined in the Sources book.                                                                            |
| <b>sources check</b>                           | Validate the entries and find prose @keys with no matching source; exits non-zero on a problem, so it drops into a CI step. |
| <b>sources import &lt;f.bib&gt;</b>            | Bring citations in from a BibTeX file — a Zotero or Mendeley export — as new Sources paragraphs.                            |
| <b>sources export --format bibtex csl-json</b> | Write the citations out; --out <file> to a file. CSL-JSON closes the round-trip with Zotero, BibTeX suits LaTeX.            |

The `check` subcommand is the one to wire into a build. It reports every `@key` you cited in prose that has no matching entry in the Sources book — a dangling citation or a misspelled key that would otherwise surface only as a broken reference in the final PDF — and it fails the process when it finds one. A companion pass, `inkhaven sources coverage`, works the other way: it flags sentences that make

a checkable factual claim — a statistic, a date, an attributed finding — while carrying no citation at all (and with `--ai`, checks each against your Facts book), so a non-fiction manuscript can be gated on both undefined keys and unsupported claims.

### The Bund surface — `ink.sources.*`

Everything above is scriptable. The `ink.sources.*` namespace exposes the Sources book to the embedded Bund language (Chapter 27), all of it read-only: `ink.sources.list` pushes every entry as a dict, `ink.sources.get` takes a key and returns the one entry (or `NODATA`), `ink.sources.bibtex` returns the whole compiled BibTeX as a string, and `ink.sources.check` returns the list of undefined `@key` citations found in prose. They mirror the CLI exactly, for a script that gates a commit or splices a bibliography into a larger pipeline.

## Terminology governance — the Glossary

A long book slips its terms without noticing. The thing you called an **access token** in Chapter 2 becomes an **auth token** in Chapter 9 and an **authentication token** in the appendix, and a reader — or a reviewer — is left wondering whether you mean three things or one. The **Glossary** book, and the detector built on it, hold your terminology to a single canonical form.

### TERM Canonical term

A **canonical term** is the one spelling you have chosen for a concept; its **banned synonyms** are the variants you want caught and corrected. A Glossary entry pairs them — canonical `access token`, synonyms `auth token` and `authentication token` — with an optional definition, a note on why you chose that form, and a **scope** (project-wide by default, or one book). Entries are authored as small HJSON paragraphs in the Glossary system book.

### The banned-synonym overlay — `Ctrl+V z`

With a Glossary in place, Inkhaven watches your prose for the banned forms. The detector red-underlines every occurrence of a synonym — so “auth token” is flagged while the canonical “access token” is left clean — and when your cursor sits on a flagged word the footer names the fix. `Ctrl+V z` toggles this terminology overlay. It is **on** by default (nested within the master style toggle, `Ctrl+B Shift+F`) and it is **self-gating**: an empty Glossary flags nothing, so the feature costs you nothing until you decide to use it.

● ● ● *The terminology overlay — a banned synonym, underlined*

```

Editor · api-tokens [modified]
1 The service issues an auth token on login, and
2 the client presents that token on every call.

terms: "auth token" → use "access token"

```

### The escape hatch — Ctrl+V Shift+Z

Sometimes a “banned” form is the right one just here — a character says it in dialogue, a quoted document uses it. With the cursor on a red-underlined synonym, **Ctrl+V Shift+Z** declares the term **deliberate**: it records the canonical term as a deliberate variant in the intent ledger, and from then on the overlay and **inkhaven terms check** stop flagging its synonyms. It is the “I meant to write it this way” release valve, so the detector can stay strict without becoming a nag. (Move onto the red underline first; away from a hit, the command has nothing to declare and says so.)

### From the shell — inkhaven terms

Two subcommands work the Glossary from outside the editor:

- |                      |                                                                                                                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>terms check</b>   | Scan prose for banned synonyms and report each occurrence with its location; --book scopes to one book, --json for a machine report. Exits non-zero on any finding — a CI gate.                                   |
| <b>terms suggest</b> | Ask a model to propose Glossary entries for genuine terminology drift it finds in a book. --provider, --max-cost (default 8000), --force past the cap, --auto-create to write the drafts under the Glossary book. |

**terms check** is the deterministic, free gate: it walks every paragraph, matches one-, two-, and three-word banned forms Unicode-aware (so Cyrillic and accented terms behave exactly as Latin ones), prints **path line: "synonym" → use "canonical"** for each, and fails the run if any remain. **terms suggest** is the one command here that spends a model: it clusters the near-variants actually present in a book’s prose and asks for canonical entries, skipping mere inflection and proper names — a way to **bootstrap** a Glossary from a manuscript you already have.

## The Bund surface — `ink.terms.*`

The Glossary is scriptable through `ink.terms.*`: `ink.terms.list` and `ink.terms.get` read the entries, `ink.terms.check` takes a book slug and returns the banned-synonym findings as dicts, and `ink.terms.declare_intent` takes a canonical term and a scope and records the deliberate-variant suppression — the scripted twin of `Ctrl+V Shift+Z`. The three reads are default-allowed; `declare_intent` writes to the ledger, so it needs the store-write category enabled. The Glossary’s canonical terms also appear in the story-bible overview (`Ctrl+V Shift+L`), alongside your characters, places, and facts.

## The scholarly apparatus

A critical edition carries an apparatus at the back: an **index locorum** of every passage it cites, an **index verborum** of its key terms, a working lexicon of those terms’ senses, and — behind it all — an argument that holds together. Inkhaven builds each from the manuscript you already have, reading your prose and your Sources and Glossary books. They **measure and report**; they never rewrite. Three of the four are deterministic and free; only `argue` calls a model. This is the toolkit for theology, philosophy, classics, and law.

### inkhaven index-locorum — the index of places

An `@key[locus]` citation — `@bible[John 3:16]`, `@kant[A51/B75]`, `@quran[2:255]` — is a **primary-source** reference: a citation that carries a specific passage in its brackets. `index-locorum` harvests every one across the manuscript, groups them under their source, and sorts the passages **naturally** (so `3:2` precedes `3:16`), listing the chapters that cite each. A bare `@key` without a locus is an ordinary citation and belongs in the bibliography, not here.

```
inkhaven index-locorum [--book-name <NAME>]
                        [--format md|typst|json] [-o <FILE>] [--strict]
```

Scripture keys (`bible`, `quran`, `book-of-mormon`) validate and **canonicalize** with zero configuration — `Jn 3.16`, `иоанна 3:16`, and `John 3:16` collapse into one entry — and any other source can declare its own reference scheme. Malformed loci are reported on `stderr` with the format they should have taken; `--strict` turns that advisory into a failing exit, a gate you can put in CI.



● ● ● *An Index Locorum, grouped by source and sorted by passage*

#### Index Locorum

##### Augustine – Confessiones

1.1 ch. 2, ch. 7

8.12 ch. 7

10.27 ch. 4, ch. 9

##### The Bible

John 1:1 ch. 1

John 3:16 ch. 3, ch. 6

Romans 5:8 ch. 6

## inkhaven index-verborum — the index of terms

The term-level twin: every scholarly-lexicon term that **actually appears** in the manuscript, with its original-language form, its distinct senses, and the chapters that use each. It reads the lexicon terms from your Glossary — an entry counts only if it carries an `original_forms` field or `senses`, not a plain consistency entry — and a term you defined but never used is **dropped**, so the index cannot flatter you. Where you tag a use with the Typst superscript convention `term#super[N]`, the index records **which sense** was used where.

```
inkhaven index-verborum [--book-name <NAME>]
                        [--format md|typst|json] [-o <FILE>]
```

### USED VERSUS DECLARED

`index-verborum` reports the terms you **used**; `inkhaven lexicon list` reports the inventory you **declared**, whether or not each term has appeared yet. They answer different questions and will differ exactly when you have defined a term you have not yet written into the prose.

## inkhaven lexicon list — the sense inventory

The working sense-inventory behind the index: the terms your work tracks, each with its original-language form(s), numbered senses, and whether it is **watched for equivocation** — an argument sliding between a term’s senses. `list` is the only subcommand.

```
inkhaven lexicon list [--book <NAME>] [--watched] [--json]
```

A term is equivocation-watched only when it both sets `watch_equivocation` **and** declares at least two senses — declaring the senses is what lets the tool tell a legitimate polysemy from a genuine equivocation, so a single-sense term is never policed. Watched terms are marked `⊢/`, and `--watched` narrows the list to just them. This is the inventory the reasoning-rigor reader consults when it polices a paragraph.

### inkhaven argue — the argument outline

The one command here that calls a model. Per chapter, `argue` reports the load-bearing claims the chapter argues for and the support it gives each, and flags the two cheapest structural gaps: a **central claim with no support**, and an **orphan citation** — a source cited but supporting no identified claim.

```
inkhaven argue [--book-name <NAME>] [--provider <NAME>] [--json]
```

It hands the model each chapter’s plain prose plus the `@key` citations that chapter uses, and asks for the claims, their support, and any orphans. An anti-hallucination guard keeps it honest: every returned claim must be quoted from the chapter, and a claim whose words are not actually present is **dropped**, so the report can only ever contain arguments you really made. A **gap** is an unsupported claim or an orphan citation, and `argue` **exits non-zero when it finds one** — so, like `index-locorum --strict` and `sources check`, it drops straight into a CI step that will not let a structurally broken chapter pass.

#### FICTION

If you write **fiction**, the apparatus is rarely your deliverable, but the Research Assistant is: gather real texture with the source commands, keep the load-bearing facts fact-checked, and let **undisputed** guard the coherence of the world you invented.

#### NON-FICTION

If you write **non-fiction** or **scholarship**, this whole chapter is your deliverable’s spine — a fact-checked corpus, a bibliography that built itself, a glossary that held your terms, and, for a critical edition, indices and an argument audit you can gate a build on.

## WHAT YOU LEARNED

- The **Research Assistant** (`inkhaven research`) is a separate TUI: a Facts tree and a chat, keeping **Facts** (trusted, gated) and **Notes** (speculative), with `⌘ undisputed` authorial facts exempt from real-world checks.
- Every kept fact records its **provenance** — a rung on the trust ladder from **computed** down to **model** — and the source, checking, and headless commands (`/triangulate`, `/factcheck`, `--snowball`, `--batch`, the `--agentic` deep mode with `/review`) climb and defend it.
- The **Sources** book turns your reading into a bibliography: cite with `Ctrl+V @`, and assembly emits `sources.bib` and a `#bibliography`; `inkhaven sources check/list/import/export` and `ink.sources.*` manage it.
- The **Glossary** holds terms to one canonical form: `Ctrl+V z` red-underlines banned synonyms, `Ctrl+V Shift+Z` declares one deliberate, and `inkhaven terms check/suggest` and `ink.terms.*` gate and grow it.
- The **scholarly apparatus** builds a critical edition's back matter from your prose — `index-locorum`, `index-verborum`, `lexicon list` (free, deterministic) and `argue` (LLM) — measuring and reporting, never rewriting, and gating a build with `--strict` or a non-zero exit.
- Facts and the graph live in Part IV; the full research craft is the companion **Grounding Your Book in Fact**.

PART VII

---

# Producing the Book

## CHAPTER 24

## 24

Assembly and PDF

---

For twenty-three chapters your book has been a tree — books, chapters, subchapters, and paragraphs, each a plain `.typ` file that Inkhaven manages, indexes, and watches. This chapter is where the tree becomes a document. It is the shortest distance in the whole tool between **what you have written** and **something you can hold**: two keystrokes to a compiled PDF, three to drop that PDF in the folder you launched from. Nothing here is a black box. Every file the process writes is Typst you can open and read, laid out in a directory you can walk, and the whole of it is regenerated from your tree each time you ask — so you never have to trust that the output matches the source. It always does, because it was just built from it.

## The two chords, and the third

Producing a book is a small ladder of three chords, each doing everything the one before it did and one thing more. You reach them from anywhere — the Tree, the Editor, it does not matter — because they act on **the book your cursor is inside**, not on whatever pane has focus.

- |                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+B A</b> | Assemble — turn the tree into a Typst-compileable directory under the artefacts folder. No PDF yet, just the source a compiler can run. |
| <b>Ctrl+B B</b> | Build — assemble, then run `typst compile` on the result. The PDF lands next to the assembled source.                                   |
| <b>Ctrl+B O</b> | Take the book — build, then copy the finished PDF into the directory you launched Inkhaven from, timestamped.                           |

Read the ladder from the bottom. **Ctrl+B A** is assembly alone — useful when you want to inspect the generated Typst, hand it to `typst watch`, or drive the compile yourself. **Ctrl+B B** is the everyday chord: assemble and compile in one motion, and if the compile fails, hand the error straight to the AI. **Ctrl+B O** — **O** for the thing you **take** away — is the finishing move: it does everything **Ctrl+B B** does and then lifts the PDF out of the ephemeral artefacts cache and sets it down where your shell can see it.

### WHICH BOOK?

All three chords act on the **user book** your tree cursor currently sits inside. If the cursor is on a system book — Notes, Research, Places, and the rest — the chord refuses with a status message asking you to pick a user book. System books are author tools; they are never part of a manuscript.

Before any of the three does its real work it quietly saves. If the open paragraph — or the second paragraph in a split edit — has unsaved changes, they are flushed to disk first, so the thing you compile is the thing on your screen. A save failure is logged and stamped on the status bar but never aborts the build; you can always dismiss the splash with **Esc**.

## What assembly writes

Your manuscript lives as `.typ` files, but not in a shape Typst can compile directly. A paragraph file holds only its prose. A chapter is a **folder**, not a file, and it carries no text of its own. Nothing anywhere says “include this, then that, then wrap the whole in a book.” Assembly’s job is to synthesise all of that missing structure — the include tree, the wrappers, the page setup — into a fresh directory, leaving your source tree untouched.

## Where the output lands

Assembled books do not go in your project. They go in a per-user cache directory, outside the tree, so `git status`, your backups, and shell tab-completion never trip over build intermediates:

● ● ● *The artefacts directory for one book*

```
<cache>/inkhaven/artefacts/<project>/<book-slug>/
├─ <book-slug>.typ      ← the root; hand THIS to typst
├─ globals.typ         ← the wrap_* helper definitions
├─ settings.typ        ← page / fonts / layout config
├─ sources.bib         ← bibliography (if you cite)
├─ snippets/          ← reusable Typst includes
└─ book/
   ├─ index.typ        ← the book's include list
   ├─ 01-prologue/
   │   ├─ index.typ    ← this chapter's wrapper
   │   └─ 01-opening.typ
   └─ 02-arrival/
       ├─ index.typ
       └─ 01-the-quay.typ
```

The one file you ever name to a compiler is the root `<book-slug>.typ` at the top. Everything else it reaches by `#import` and `#include`. The `book/` subtree mirrors your manuscript’s shape one-to-one: every chapter and subchapter becomes a folder with its own `index.typ`, every prose paragraph becomes a `.typ` file, ordered by the same `NN-` prefix the Tree pane uses.

The `artefacts` location is configurable (`artefacts_directory` in `inkhaven.hjson`); left blank, it defaults to the OS cache path shown above. When a chord finishes, the status bar prints the absolute root path so you can copy it straight into a terminal.

## The include tree and the wrap\_ functions

Assembly does not concatenate your prose into one big file. It builds a **tree** of tiny files that include one another, and it threads three helper functions through them — `wrap_book`, `wrap_chapter` (and `wrap_subchapter`), and `wrap_paragraph`. Those helpers are defined once in `globals.typ`; the generated `index.typ` files just call them. This is the seam where your typography lives: change what `wrap_chapter` does, and every chapter in the book changes with it.

The root `<book-slug>.typ` sets the stage and hands the book to its wrapper:

● ● ● *The generated root file (abridged)*

```
// Auto-generated by inkhaven Book assembly.
// Book: The Salt Road

#import "globals.typ": *
#import "settings.typ": *

#wrap_book(include "book/index.typ")

#bibliography("sources.bib", style: "chicago-author-date")
```

That `book/index.typ` is a flat list of the book's top-level children. Because it is included at file scope — markup mode — every statement carries a leading `#`:

● ● ● *book/index.typ — a BookRoot index*

```
// Auto-generated by inkhaven Book assembly.
#import "../globals.typ": *

#metadata((node_id: "..."))
#include "01-prologue/index.typ"
#include "02-arrival/index.typ"
#wrap_paragraph(include "03-coda.typ")
```

A chapter's own `index.typ` is different in one important way. Its body sits **inside** the second argument of `wrap_chapter(...)`, which is code mode — so the inner calls drop their `#`:



● ● ● *A chapter index.typ — code mode inside the wrap*

```
// Auto-generated by inkhaven Book assembly.
#import "../globals.typ": *

#metadata((node_id: "..."))
#wrap_chapter("Arrival", {
  wrap_paragraph(include "01-the-quay.typ")
  wrap_paragraph(include "02-the-inn.typ")
})
```

### WHY THE MARKUP/CODE SPLIT MATTERS

An early bug rendered bare `{ include ... }` blocks as literal text in the PDF — braces and all — because file-scope statements need the `#` prefix and the code inside a `wrap_*` block must **not** have it. The assembler now tracks which mode each `index.typ` line is in. You never write these files, but if you ever read one and wonder about the inconsistent `#`, that is the reason.

Each generated `index.typ` also emits one invisible `#metadata` element carrying the tree node's id. It contributes nothing to the page — `#metadata` is query-only — but it lets the PDF-outline machinery correlate a chapter to the page it starts on. Empty branches get a `[]` placeholder so a childless chapter compiles instead of producing a parse-failing `wrap_chapter("X", {})`.

### The three seed files: globals, settings, root setup

Where do `globals.typ` and `settings.typ` come from? Not from your manuscript — from the **Typst system book**. Every project has one, and inside it a chapter named after each user book, seeded with three paragraphs: `globals.typ`, `settings.typ`, and `index.typ`. Assembly reads those three and maps them into the output:

- `globals.typ` is copied out verbatim — your `wrap_*` definitions, any custom helpers, image search paths.
- `settings.typ` is composed: Inkhaven synthesises a header of `#set page`, `#set text`, and `#set par` rules from your HJSON config (more on this below), then appends whatever free-form Typst you wrote in that paragraph.

- `index.typ` is **returned** and stitched into the root `<book-slug>.typ`, just before `wrap_book`, so any imports or setup you keep there run before the book renders.

To edit your book's typography, then, you do not touch the artefacts copies — they are overwritten every run. You edit the paragraphs under the Typst system book, and re-assemble. If assembly can't find a Typst chapter named after your book, it says so and tells you to open the book once to seed it.

### A clean slate, every run

Assembly wipes `<artefacts>/<book-slug>/` entirely before it writes a byte. This is deliberate: a chapter you deleted, a paragraph you renamed, a stale PDF from last week — none of them should linger to confuse the next `typst compile`. The cost is that anything you hand-edited in the artefacts tree is gone on the next `Ctrl+B A`. Treat that directory as disposable output, never as source.

The Timeline chapter and its event paragraphs are skipped at assembly — they are metadata about the manuscript, not prose, and nothing about them should leak into the PDF. The same is true across the Markdown, TeX, and EPUB export paths.

## Compiling — Ctrl+B B

`Ctrl+B B` runs assembly and then compiles its output. While `typst` works, a splash animates with a spinner and an elapsed-seconds counter; you can press `Esc` to interrupt a stuck compile. On success the status bar reports the PDF's path.

Inkhaven can drive two compile engines behind one interface. The default is **external**: it finds `typst` on your `PATH` and spawns it as a child process. The opt-in alternative is **in-process** (`typst_compile.engine = "inprocess"` in HJSON), which links the Typst compiler directly and needs no external binary at all — useful on a machine where you would rather not install one. The splash and the credits pane both print a one-line engine summary so you always know which is active and, for the external engine, exactly which binary would run.

### TYPST IS A SEPARATE TOOL

For the external engine you install `typst` yourself; Inkhaven only shells out to it. If it is missing, the compile fails with a clean error pointing at the install docs — or at the in-process knob. Assembly (`Ctrl+B A`) never needs Typst; only the compile step does.

## When the compile fails

A typesetting error should not send you hunting through a wall of stderr. When `typst compile` fails, Inkhaven captures its diagnostics and does two things without any further keystroke from you.

First it scans the error text for unresolved cross-references — a dangling `@fig:` or `@eq:` — and promotes each into a precise, actionable finding in the Output pane, because that is a real defect, not noise.

Then it opens a fresh AI chat. The chat history is cleared so the new context is clean, the inference mode is forced to Full, and a system prompt tuned for Typst errors — one that knows Inkhaven’s generated file layout — is loaded. The captured stderr is packaged with the book’s name, slug, and root path into a message that asks for the smallest concrete fix, and it is sent automatically. Focus jumps to the AI pane, and the answer streams in while you read the error.

● ● ● *A compile failure, handed to the assistant*

```
AI · llama · streaming · typst-error
| you typst compile failed with the following error. |
| Please diagnose it ... --- typst stderr --- |
| error: unknown variable: wrap_paragraph |
|   └─ book/02-arrival/index.typ:6:2 |
| ai The call on line 6 is misspelled — `wrap_ |
|    paragraph` should be `wrap_paragraph`. But you |
|    don't edit these generated files; the typo is |
|    in your globals.typ under the Typst book...
```

Because the assembled tree is kept — not deleted after a failed compile — you can also open the offending `index.typ` or paragraph file directly at the line Typst named, see it in context, fix the source in Inkhaven, and rebuild.

## Take the book — Ctrl+B O

`Ctrl+B O` is `Ctrl+B B` plus a delivery step. Once the PDF compiles, it is copied out of the artefacts cache and into the working directory you launched Inkhaven from, under a timestamped name:

● ● ● *What Take writes to your launch cwd*

```
salt-road-20260805-1432.pdf
```

The stem is the book slug; the stamp is `YYYYDDMM-HHMM`. The original stays in the `artefacts` directory too — Take **copies**, it does not move — so a later `Ctrl+B Q` imposition pass or a re-take still has the source PDF to work from. The status bar reports both the delivered path and the source.

Take can also emit **extra formats** alongside the PDF. If you have configured additional outputs (the `Ctrl+B 0` extras knob in HJSON — Markdown, TeX, EPUB), they are written next to the delivered PDF with the same stem. An extra that fails is reported on the status bar but never aborts the take: the PDF you actually asked for is already on disk before the extras run. See Chapter 25 for the full multi-format story.

## Images

Images are first-class tree nodes, and assembly gives each one the treatment its position calls for. When `write_branch` meets an Image child it emits a call to one of three helpers, chosen by the parent's level:

- an image directly under a **Book** becomes `wrap_image_book` — frontispiece and book-art treatment;
- under a **Chapter**, `wrap_image_chapter`;
- under a **Subchapter**, `wrap_image_subchapter`.

Each call carries the image's filename, its title, and — as Typst string literals or the bare keyword `none` — its caption and alt text. As with the paragraph wrappers, the call is `#`-prefixed at book scope and bare inside a chapter's code-mode block. You define what each helper **does** — figure frame, full-bleed, caption styling — in `globals.typ`.

**BDSLIB IS THE SOURCE OF TRUTH FOR IMAGE BYTES**

The image **bytes** are not copied from the working file under `books/...`. They are pulled from Inkhaven's embedded store, which holds the authoritative copy, and written into the assembled tree next to the `index.typ` that references them. A hand-edit of the on-disk working copy is therefore never accidentally re-ingested — the store is what ships into the PDF.

Inside the editor, `Ctrl+B P` with the cursor in an `#image("...")` path opens a sibling picker so you can point a figure at another image in the same folder without typing the path. Enter on an Image row in the Tree pops a terminal-native preview (kitty, sixel, iTerm2, or half-block).

## Reusable snippets — the REUSE-1 sidecar

Some Typst you write once and include in many places — a styled warning block, a recurring table header, a house-style callout. Inkhaven keeps those in the **Snippets** system book, one paragraph per snippet, and at assembly it copies each into a `snippets/` sidecar in the output:

● ● ● *Snippets copied at assembly*

```
<book-slug>/
├─ snippets/
│   └─ warning.typ
│   └─ house-callout.typ
└─ book/
    └─ 01-prologue/
        └─ 01-opening.typ ← #include ".../snippets/warning.typ"
```

Anywhere in your prose a `#include ".../snippets/<slug>.typ"` then resolves at compile time. The editor writes those includes for you: `Ctrl+V x` fuzzy-picks a snippet and inserts the correctly depth-relative path — `../../snippets/...` computed from the paragraph's place in the tree — or, with the cursor inside an existing snippet include, replaces the path in place. A save-time validator flags any include whose snippet slug isn't defined, so a renamed snippet can't silently break a later build.

The sidecar is written only when the Snippets book has content; a project that uses none pays nothing, and no `snippets/` directory appears. The output name is always `<slug>.typ` even when the source snippet was authored as a Jinja template — so the include resolves regardless of how the snippet was written.

## Jinja templates at assembly

A paragraph can be a **Jinja template** rather than plain Typst (the `?` flavour, added with the `e` key in the Tree). Assembly renders each such paragraph to Typst before the compiler ever sees it. Every Jinja snippet is registered under a name derived from its slug path — `snippets/macros/warning.jinja` — so templates can `{% include %}` one another, and each manuscript template renders against a context exposing its own title and slug, its enclosing book and chapter, the project language and genre, and a `linked` map of the HJSON data from any paragraphs it links to.

By default a template that fails to render aborts the build with a precise error naming the paragraph. Set `jinja.continue_on_error` and a failure instead writes a loud red error block into the PDF and moves on, so you can fix a batch of templates one at a time rather than one-per-rebuild. The full treatment of the template system — context, filters, the data-linking workflow — is in Chapter 5.

## The bibliography and the scholarly apparatus

If your book cites sources, assembly builds the bibliography for you. It walks the **Sources** system book, parses each entry, compiles a `sources.bib` file into the output directory, and — when there are entries and you have not disabled the auto-line — appends a `#bibliography("sources.bib", style: ...)` call to the root file, after `wrap_book`. Typst then resolves the `@key` citation tokens in your prose against it. The citation style is configurable; the scope (this book’s sources only, or every entry under Sources) is a one-line HJSON switch.

Three optional apparatus chapters can follow the bibliography, each gated by a config flag and each rendered only when it has content, in a fixed order:

- an **Index Locorum** — every primary-source locus you cited, resolved to its source and validated against that source’s reference scheme;

- an **Index Verborum** — the scholarly-lexicon terms the book uses, with their original-language forms, senses, and the chapters that use each;
- a **Glossary** — every defined term, alphabetical, with its senses or definition.

All three are localised to the project language (Glossary, Glosario, Glossar, Глоссарий, ...). This is a deliberately brief tour; Chapter 23 covers research, sources, and the apparatus in full.

## Templates, front matter, and page config

The `settings.typ` header is where your page shape, type, and spacing come from, and you set almost all of it without writing Typst. The `typst_page`, `typst_fonts`, and `typst_layout` blocks in `inkhaven.hjson` are synthesised at assembly into `#set page(...)`, `#set text(...)`, and `#set par(...)` rules — paper size and margins, body and monospace fonts with bundled-font fallbacks, language, justification, leading, first-line indent, heading numbering. The generated header is clearly marked **do not edit**; anything you add below the “user overrides” line in the `settings.typ` paragraph is preserved across rebuilds.

Separately, a **front-matter** block — a title-page treatment — can be prepended to the root file just before `wrap_book`, driven by the `frontmatter` config and the book’s title. Books that don’t opt in get an unchanged root. Both surfaces are covered exhaustively in the configuration appendix (Appendix C); this chapter only notes where in the pipeline they take effect.

## The same path from the command line

Everything the chords do is available headless, for CI, batch builds, and end-to-end verification that your synthesised `settings.typ` actually compiles. `inkhaven build` is the CLI mirror of `Ctrl+B A/B`:

 Building without the TUI

```
$ inkhaven build --book-name "The Salt Road"
Assembling `The Salt Road` (slug: salt-road)...
[12/44] book/01-prologue/01-opening.typ
Assembly OK · root: ../salt-road/salt-road.typ (44 files)

$ inkhaven build --book-name "The Salt Road" --compile
PDF: ../salt-road/salt-road.pdf
```

Without `--compile` it stops after writing the artefacts tree; with it, it runs `typst compile` and prints only the final PDF path to stdout (progress goes to stderr, so the command is pipe-friendly). The `--book-name` argument is optional when the project has exactly one user book, required otherwise. The older `inkhaven export typst` / `export pdf` subcommands remain for concatenated single-file output and simpler pipelines; the assembly path is the richer one, and the one the two chords use.



## WHAT YOU LEARNED

- Three chords climb one ladder: **Ctrl+B A** assembles the tree into a Typst-compileable directory, **Ctrl+B B** also compiles it to PDF, and **Ctrl+B 0** also copies that PDF, timestamped, into your launch directory.
- Assembly writes a **fresh, disposable** tree under the artefacts cache — a root `<slug>.typ`, `globals.typ`, `settings.typ`, and a `book/` subtree of `index.typ` files that call `wrap_book` / `wrap_chapter` / `wrap_paragraph`; it wipes the directory every run, so treat it as output, never source.
- Typography lives in the **Typst system book** (`globals.typ` / `settings.typ`) and in the `typst_page` / `typst_fonts` / `typst_layout` HJSON blocks — edit those and re-assemble, never the artefacts copies.
- A failed compile **routes its stderr into a fresh, Typst-aware AI chat** automatically, and promotes dangling cross-references to Output-pane findings; the assembled tree is kept so you can read the offending line in context.
- Images ship their bytes from the store and get `wrap_image_*` calls by level; **snippets** are copied to a `snippets/` sidecar; **Jinja** paragraphs render to Typst; and **Sources** becomes `sources.bib` with an optional index apparatus.

## CHAPTER 25

## 25

## EPUB, Web and More

---

The last chapter took your manuscript to the printed page — Typst to PDF, the destination a book has had since the first press. This chapter is about every **other** place the same words can go: an EPUB a reader opens on a phone, a website a reader opens in a browser, a Word document an agent opens on a submission desk, a LaTeX bundle a server compiles for a preprint archive, and an audiobook a reader listens to on a walk. Inkhaven treats all of these as **exports** — one manuscript, many destinations — and the point of this chapter is that reaching any of them is a single command with a small, shared set of flags. Learn the shape once and every format is a variation on it.

## One manuscript, many destinations

Almost every export in Inkhaven starts from the **same assembled source**. The tool walks your book in tree order — depth-first, top to bottom — and concatenates each paragraph’s `.typ` file into one long document, exactly the way it does for the PDF build. Branch nodes (books, chapters, subchapters) contribute nothing themselves; the headings live in the paragraphs, written by the `= Title` line each paragraph carries. The Timeline chapter and any event paragraphs are skipped, because a timeline is metadata about the manuscript, not manuscript prose. Whatever comes out of that walk is what every exporter converts.

Because the source is shared, so are the **filters** that decide which paragraphs reach it. The same four flags apply to every format the `inkhaven export` command produces, and to the reader-facing exports besides.

### TERM The export flags

`--book-name <name>` picks **which** user book to export (a project can hold several); it matches a book’s title or slug, case-insensitively, and is required only when the project has more than one user book.

`--status <rung>` keeps only paragraphs at or above a rung on the workflow ladder (`napkin` → `first` → `second` → `third` → `final` → `ready`).

`--tag <tag>` keeps only paragraphs carrying that tag.

`--profile <dim>=<value>` selects one arm of conditional (single-sourced) content. They **combine**: a paragraph must pass all of them to ship.

The general form is `inkhaven export <format>`, where the format is one of `typst` (the default — the raw concatenated source), `pdf`, `markdown`, `tex`, `epub`, or `html`. Everything else in this chapter is either a richer variant of one of these (the full `inkhaven epub`, the arXiv `--bundle`) or a destination with its own top-level command (`inkhaven docx`, `inkhaven audiobook`).

● ● ● *The export command and its shared flags*

```
$ inkhaven export markdown -o draft.md
wrote draft.md (markdown)

$ inkhaven export epub --book-name "Harbor" \
  --status final -o harbor.epub
wrote harbor.epub (epub)

$ inkhaven export html -o site/ --templates my-theme/
wrote HTML site to site/
```

**TWO BOOKS, ONE FLAG**

If a project holds only one user book, `--book-name` is optional and that book is used. The moment you add a second, every export refuses until you name one — and the error lists the books it found, so you never have to guess the spelling. System books (Help, Prompts, Places, Characters, Facts, Notes, and the rest) are never candidates; they are Inkhaven’s internals, not your manuscript.

## EPUB — the e-book readers actually open

An EPUB is the format e-readers consume: a zip of XHTML pages with a spine that orders them and a navigation document that indexes them. Inkhaven writes one directly — no external converter, no `pandoc` — because it already holds your prose as Typst and a `zip` library in the binary. There are two ways to reach it, and the difference matters.

The full path is the dedicated command:

● ● ● *Exporting a complete EPUB 3*

```
$ inkhaven epub --book-name "The Harbor Code" \
  -o harbor.epub
  embedding cover (image/jpeg, 184 KB)
Exporting `The Harbor Code` → EPUB (14 chapters)...
EPUB: harbor.epub (14 chapters · 612 KB)
```

`inkhaven epub` builds a standards-compliant EPUB 3 the way a reader expects it: **one spine document per Chapter node**, in display order, each holding that chapter’s descendant paragraphs converted to XHTML, with any subchapter titles surfaced as `<h2>` headings inside the flow. The archive it produces has the shape the EPUB spec demands — the `mimetype` entry first and uncompressed, a `META-INF/container.xml`, a package document listing every chapter in the manifest and spine, an EPUB 3 `nav.xhtml` plus an EPUB 2 `toc.ncx` for older readers, a stylesheet, and `chapter-001.xhtml`, `chapter-002.xhtml`, and so on.

● ● ● *Inside the .epub archive*

```
mimetype                (stored, first – the EPUB rule)
META-INF/container.xml
OEBPS/content.opf       (metadata + manifest + spine)
OEBPS/nav.xhtml         (EPUB 3 navigation)
OEBPS/toc.ncx           (EPUB 2 back-compat)
OEBPS/style.css
OEBPS/cover.jpg         (if a cover was found)
OEBPS/cover.xhtml
OEBPS/chapter-001.xhtml
OEBPS/chapter-002.xhtml
...
```

The metadata is filled in for you and overridable. The title defaults to the book’s title (`--title` overrides); the author to your `editor.comment_author` config, falling back to “Unknown Author” (`--author` overrides); the language to the ISO code mapped from the project’s `language` field; and the identifier to a fresh `urn:uuid`, stable within one export so a re-export replaces cleanly in a reader’s library. Without `--output`, the file lands at `<project>/<book-slug>.epub`.

### What the converter understands

The Typst-to-XHTML pass handles the markup Inkhaven prose actually uses, and is honest about the rest. Headings (`=`, `==`, `===`) become `<h1>/<h2>/<h3>`; `_emphasis_` becomes `<em>`; `*strong*` becomes `<strong>`; blank-line-separated blocks become `<p>` paragraphs. Each paragraph node’s leading `= title` line is **organisational scaffolding** — the scene label you see in the Tree — so it is stripped, and the reader sees flowing chapter prose rather than “001. Approach” labels. XML special characters are escaped, and an unpaired `_` (a filename, a stray underscore) passes through literally rather than opening an unterminated tag.

Footnotes get special care. A `#footnote[...]` in your prose becomes a proper EPUB 3 **popup footnote**: a numbered reference anchor in the flow, and a collected `<aside>` section at the chapter's end with a backlink. Apple Books and similar readers render these as tap-to-pop popups; readers that do not simply show the notes at the foot of the chapter.

### Covers and inline images

Drop a `cover.jpg` or `cover.png` in the project root and `inkhaven epub` finds it, embeds it as the library thumbnail, and makes it the opening page — the console line confirms the format and size when it does. Images referenced inline in a chapter's prose are carried into the archive too, written verbatim (already-compressed formats are stored, not re-deflated) and declared in the package manifest so they render where the prose places them.

#### TWO EPUB PATHS, ONE TO PREFER

`inkhaven export epub` is the **minimal** writer: it runs the prose through the Typst-to-Markdown converter and packs the result as a single-chapter EPUB — handy when you want the `.epub` and the `.md` to match byte for byte, and it is the variant wired into the batch-export flow. For a real e-book — chapters as separate spine documents, a cover, inline images, popup footnotes — use the dedicated `inkhaven epub` command. When in doubt, that is the one you want.

## Importing an EPUB

The inverse of the export is `inkhaven import-epub <file.epub>`, which turns an existing e-book into a user book you can edit. It parses the package, then materialises a **Book**, a **Chapter** per spine document, and one **Paragraph** per chapter holding that chapter's prose converted back from XHTML to Typst. Images referenced by a chapter become Image nodes under it; manifest images that no chapter references (cover art, orphans) are extracted to a `<book-slug>-images/` sidecar folder for you to place by hand. Image references in the prose are rewritten to Typst comments so an imported chapter still compiles cleanly while telling you where the picture belonged.

● ● ● *Round-tripping a book back in*

```
$ inkhaven import-epub harbor.epub
EPUB import complete — book `The Harbor Code`:
  chapters: 14
  paragraphs: 14
  images: 6 imported, 1 extracted
  author: A. Writer (set your book metadata ...)
  (1 unreferenced image(s) extracted to
   `the-harbor-code-images/` for hand-placement)
```

Two options shape the import. `--book-name` overrides the title of the created book (otherwise it takes the EPUB’s `dc:title`, falling back to “Imported EPUB”); `--dry-run` reports what **would** be created — chapter, paragraph, and image counts — without writing anything. The import never aborts on a single bad chapter: per-item failures are collected and printed, and a real (non-dry) run that hit any failure exits non-zero, so a scripted import can tell a clean run from a partial one.

## A website from your book

`inkhaven export html --output <dir>` renders your book as a **self-contained static website** — one HTML page per chapter, an index page from the book’s front matter, a navigation list linking them all, and a client-side search box, all written into the directory you name. It is the third consumer of the same node tree, localised to the book’s language, with images copied alongside and the profile and variable machinery applied exactly as the other exports apply them.

● ● ● *Building the site*

```
$ inkhaven export html -o public/
wrote HTML site to public/

$ ls public/
index.html          chapter-01.html    theme.css
appendix-index.html chapter-02.html    search.js
search-index.js     ...
```

The look is driven by **overridable templates**. Inkhaven ships a default set split into two concerns — the `functional/` machinery (the page skeleton, the nav, the search wiring) and the `theme/` look (the CSS, the visual choices) — and you replace either by pointing `--templates <dir>` at your own, or by setting `docs.html.template_dir` in config. A file you supply overrides the bundled default of the same name; anything you leave out falls back to the default, so a theme can be as small as one stylesheet. To start from the built-ins rather than a blank page, scaffold them:

● ● ● *Scaffolding the default templates to edit*

```
$ inkhaven export html --eject-templates my-theme/
wrote default HTML templates to my-theme/ - edit
them and export with --templates my-theme/
```

The `docs.html` config block governs the rest: `site_title` overrides the page title, `search` toggles the search box, and an `include` table decides which companion books become **appendix pages** of the same site — Sources, Glossary, Places, Characters, the Language, the World, Mythology, Notes, and a back-of-book index each fold in when enabled. One book, one contents list, one website.

#### THE COMPANION BOOK ON THIS

A whole companion — **Publish Your Book to the Web** — is devoted to the HTML export: your first site, the command line, styling, templates, variables, what to publish, and going live. This section is the map; that book is the territory. Reach for it when you want to make the site genuinely your own.

## The arXiv and scientific bundle

For an academic paper, LaTeX is the currency, and a preprint server compiles your source on its own machine — so it needs **everything the build touches** in one place. `inkhaven export tex --bundle <dir|zip>` writes exactly that: a self-contained submission holding the `.tex` (converted from your Typst via the `tylax` library), a `sources.bib` compiled from your Sources book, every figure your prose references (copied in flat, with its `\includegraphics` path rewritten to match),



and a `MANIFEST.txt` that explains the package and flags anything you must check by hand. A `.zip` extension writes one archive; otherwise it writes a directory.

• • • *Writing an arXiv-ready bundle*

```
$ inkhaven export tex --bundle submission.zip
wrote zip bundle to submission.zip – paper.tex,
12 bib entries, 3 figure(s)
```

The bundle quietly corrects the two things that would otherwise stop the submission compiling. It rewrites citation commands so a bibliography key `@smith2020` becomes `\cite{smith2020}` while a genuine cross-reference stays `\ref{...}`, and it maps your Typst citation style to a real LaTeX bibliography style (`ieee` to `IEEEtran`, `apa` to `apalike`, and so on, defaulting to `plain` for anything unknown, which the MANIFEST notes). Figures referenced but missing from disk are listed in the MANIFEST rather than silently dropped, so you know exactly what to add before you upload.

The bundle composes with the paper **front matter**. When a project defines a `frontmatter` block — authors, affiliations, abstract, keywords, funding — that title block is prepended to the Typst, `tex`, `pdf`, and `typst` exports. The `--blind` flag omits the identifying parts (authors, affiliations, ORCID, corresponding author, funding) while keeping title, abstract, keywords, and the availability statements — the shape a double-blind review wants — and it combines with `--bundle` for a blind submission archive.

## DOCX, Markdown, and the manuscript formats

Two more destinations serve the desk rather than the reader: the standard manuscript formats an agent or editor expects, and a plain-text dump for sharing or pasting.

### Word, in standard manuscript format

`inkhaven docx` writes a Word document in **Shunn standard manuscript format** — the layout submissions actually require. It is a title page (your contact block in one corner, a rounded word count in another, the title and byline centred), then double-spaced 12-point body with a one-inch margin and a half-inch first-line indent, each chapter beginning a fresh page, scene breaks rendered as a centred

`#`, and a `Surname / KEYWORD / page` running header from page two. The file is real OOXML, hand-built over the same `zip` library as the EPUB writer, so it opens cleanly in Word, LibreOffice, and Google Docs.

• • • *A submission-ready Word document*

```
$ inkhaven docx --title "The Harbor Code" \
  --author "Jane Writer" --font courier
wrote the-harbor-code-manuscript.docx
```

Its flags mirror the reader-facing exports: `--book-name`, `--output`, `--title`, `--author`, a `--contact` block (use `\n` for line breaks), and `--font` to choose the body typeface — `times` (the default) or `courier`, the two Shunn accepts. If

`inkhaven manuscript` produces the same Shunn layout as a Typst document you compile with `typst compile`.

### Markdown, and plain LaTeX

`inkhaven export markdown` runs the manuscript through a Typst-to-Markdown converter — headings, bold and italic, lists, images, and citations map across; anything it does not recognise is preserved verbatim in a fenced block so nothing silently disappears. It is **lossy by design**: Markdown cannot represent everything Typst can, and the goal is a readable dump good enough to share or paste, not a round trip. (It is also the intermediate the minimal EPUB writer converts from.) `inkhaven export tex` gives you the plain LaTeX without the bundle wrapper — the same `tylax` conversion, written to a file or standard output.

## The audiobook and reading aloud

Inkhaven can **speak** your prose — a paragraph at a time while you write, or a whole book synthesised to an audiobook. Both run through one text-to-speech engine, and the whole feature is off until you turn it on.

**TERM The TTS engine**

Inkhaven resolves one of two backends from your `editor.tts.engine` setting. **Piper** is a neural, cross-platform engine whose voice models are downloaded per project; **System** is the host's own voice (macOS `say`). `auto` (the default) prefers Piper and falls back to System; `piper` forces Piper and errors if it cannot resolve; `system` forces the host voice. Nothing speaks unless `editor.tts.enabled = true` — TTS is opt-in.

**Managing voices from the command line**

The `inkhaven tts` family manages the whole stack headlessly, mirroring the in-editor surfaces so you can script it. `tts engine` reports which backend is active and why; `tts binary status` and `tts binary download` manage the Piper binary; `tts voice list`, `tts voice download`, and `tts voice remove` handle per-voice CRUD against the Hugging Face catalog; `tts catalog refresh` clears the cached catalog; and `tts test "<phrase>"` synthesises and plays a phrase as a diagnostic (`--voice` to A/B a voice, `--output` to write a file instead of playing).

● ● ● *Browsing and fetching voices*

```
$ inkhaven tts voice list --filter en
inkhaven voices — .inkhaven/voices
catalog: fresh
count:      11

key                language quality status
en_GB-alan-medium  en_GB      medium [⚡ available]
en_US-lessac-medium en_US      medium [✓ downloaded]
en_US-ryan-high    en_US      high   [⚡ available]

$ inkhaven tts voice download en_US-ryan-high
downloading ... OK
```

**Reading aloud in the editor**

Three chords bring speech into the writing flow. `Ctrl+B S`, with a paragraph open in the Editor, **reads it aloud** through the resolved engine — a small modal shows the elapsed time, the voice, and the opening of the paragraph while it plays, and any key stops it. (In the Tree, `Ctrl+B S` keeps its structural meaning — add a subchapter — so the two never collide.) `Ctrl+B Shift+V` opens the **voice picker**:

the full catalog plus every voice already downloaded, sorted by language and quality, filterable as you type, with `Enter` fetching an available voice or selecting a

downloaded one and `d` removing one. And `Shift+R` **saves the open paragraph as an audio file** through the macOS `say` engine, opening a path picker pre-filled under `<project>/audio/` — the format follows the extension you give it.

**Ctrl+B S** Editor: read the open paragraph aloud. (Tree: add a subchapter.)

**Ctrl+B Shift+V** Open the Piper voice picker — browse, filter, download, select, remove.

**Ctrl+B Shift+R** Editor: save the open paragraph as an audio file (macOS `say`).

### The whole book as an audiobook

`inkhaven audiobook` synthesises a user book to a single `.m4b` — the resumable, jump-by-chapter format audiobook players expect — with **one chapter marker per Chapter node**. It synthesises each chapter's prose to a temporary audio file, measures the durations, and muxes the lot with chapter metadata. Two things must be in place: `editor.tts.enabled = true`, and `ffmpeg` plus `ffprobe` on your `PATH` (there is no pure-Rust muxer with chapter support, so this one external is required — the command says so clearly if it is missing).

#### ● ● ● *Producing a chapter-marked audiobook*

```
$ inkhaven audiobook --book-name "Harbor" -o harbor.m4b
audiobook: backend=piper voice=en_US-ryan-high
[1/14] synthesising `Arrivals`...
[2/14] synthesising `The Wharf`...
...
audiobook: muxing 14 chapters → m4b...
Audiobook: harbor.m4b (14 chapters · 8h12m04s · 214 MB)
```

It takes the same `--book-name`, `--output`, `--title`, and `--author` flags as the EPUB export. Synthesis is **roughly real time**: a long book is hours of audio and hours to produce, so this is a batch export — the command reports per-chapter progress on standard error and is not something you run interactively.

## THE TTS CONFIG BLOCK

Everything above keys off the `editor.tts` block: `enabled` (the master switch), `engine` (`auto/piper/system`), `voice`, `speed`, `voices_dir` (default `.inkhaven/voices`, sandboxed to the project), `auto_download` (fetch missing voices on first use), and `catalog_url`. A greeting and goodbye string, if set, are spoken at startup and shutdown. All of it is inert until `enabled` is `true`.

## Batch export — taking the book once

You rarely want just one format at a time. The `output.extra_formats` config list wires the multi-format build to a single chord: `Ctrl+B 0` first builds the PDF (the normal build), then feeds the **same assembled source** to the in-process converters and drops each result next to the PDF with a matching stem. List any of `markdown`, `tex`, `epub`, and `docx` there, plus the production entries `imposed_pdf` and `cover_pdf` that operate on the just-built PDF. Unknown entries log a warning and are skipped; a per-format error is reported but never aborts the build.

● ● ● *extra\_formats in inkhaven.hjson*

```
output: {
  extra_formats: ["epub", "markdown", "docx"]
  extras_step_pause_ms: 400
}
```

With that set, one `Ctrl+B 0` leaves you a PDF, an EPUB, a Markdown dump, and a Word document side by side — the whole distribution of a draft from one keystroke. It is the natural close to a writing session: press it, and every destination this chapter described is written at once.

## WHAT YOU LEARNED

- Almost every export starts from **one assembled Typst source** — a tree-order walk of your paragraphs — and shares four filter flags: `--book-name`, `--status`, `--tag`, and `--profile`.
- `inkhaven export <format>` reaches `typst`, `pdf`, `markdown`, `tex`, `epub`, and `html`; richer variants have their own commands.
- `inkhaven epub` writes a real EPUB 3 — a spine document per chapter, a cover, inline images, and popup footnotes; `inkhaven import-epub` reverses it into an editable book.
- `inkhaven export html -o <dir>` builds a self-contained website with overridable `functional/` and `theme/` templates and companion appendix pages; the **Publish Your Book to the Web** companion covers it in full.
- `inkhaven export tex --bundle` writes an arXiv-ready LaTeX package — `.tex`, `sources.bib`, figures, and a MANIFEST — with citation and style quirks fixed; `--blind` strips identifying front matter.
- `inkhaven docx` (and `manuscript`) produce Shunn standard manuscript format for submission; `export markdown` is a lossy plain-text dump.
- With `editor.tts.enabled`, `Ctrl+B S` reads a paragraph aloud, `Ctrl+B Shift+V` picks a voice, and `inkhaven audiobook` synthesises a chapter-marked `.m4b` (Piper or macOS say, `ffmpeg` required).
- `Ctrl+B 0` builds the PDF and every format in `output.extra_formats` at once — the whole distribution from one chord.

## CHAPTER 26

## 26

Technical Documentation

---

Every other reading intelligence in this book watches for the thing a **story** gets wrong — a contradiction, a broken promise, a character who knows a secret before it is told. Technical writing has a different enemy, and it is quieter. A reference manual, an API guide, a textbook, a standard-operating-procedure binder — none of these will contradict itself so much as **go stale**: a sentence that was true of last release and is now a small, confident lie about this one. The code example that no longer compiles. The cross-reference to a section you renamed. The URL that has rotted to a `404`. The chapter you swore you would finish before shipping and did not.

Inkhaven’s whole temperament is to check prose against a ground truth. The technical-documentation tools — the `docs` subsystem and the back-of-book index — turn that temperament on the documentation author’s **own** ground truths: the manual is made to answer to the **code** it shows, the **cross-references** it promises, the **release** it ships against, and the **vocabulary** it defines. None of it is AI. None of it edits your prose. Each check simply **reports**, and each one exits non-zero when it finds trouble, so it drops straight into the last gate before you publish.

## Who this chapter is for

If you write fiction, you can skim this chapter and lose nothing — your ground truths are the world and the cast, covered in Parts IV and V. This chapter is for the **other** author: the person writing non-fiction, reference, or technical documentation, whose claims must be right the way a compiler is right, and whose book has an appendix a reader turns to last.

It belongs to Part VII — **Producing the Book** — because these are the tools you run around the moment of shipping, alongside the PDF, EPUB, and web exports of the previous chapters. Where those chapters turn your manuscript into a finished artefact, this one makes sure the artefact is **true** before it leaves your hands: the examples run, the links resolve, nothing half-baked slips through, and the index that ships with it was built from the same source as the prose.

### TWO SUBSYSTEMS, ONE JOB

This chapter covers two separate commands that share a purpose. `docs` (with its three subcommands `verify`, `links`, and `review`) is the **currency** toolkit — it keeps a living document honest against its code, its cross-references, and its release. `inkhaven index` builds the **back-of-book index**. Both are deterministic, both are free of any model, and both read your **user** books only — system and companion books are skipped.



## The docs subsystem — three checks, three ground truths

The `docs` command is a small family of three checks, each answering to a different ground truth a manual can drift away from.

### TERM TDOC

TDOC is Inkhaven's shorthand for its technical-documentation checks:

`docs`

`verify` (the code examples still run), `docs links` (the cross-references still resolve), and `docs review` (the manuscript is current). All three are **deterministic** and **advisory** — they compute a report and change nothing. Only the opt-in external-link sweep touches the network; no model is ever consulted, anywhere in the subsystem.

Three properties hold across all three subcommands, and they are worth stating once so you can lean on them everywhere below. First, each scans your **user books only** — the system books (Facts, Notes, Timeline) and the companion reference books are never treated as manuscript prose. Second, each restricts to one book with `--book-name X`, taking the book's title or slug. Third — the property that makes them useful in a release script — each **exits non-zero** the moment it finds a problem. A clean run is silent and returns success; a run with findings prints them and fails, so `inkhaven docs verify && inkhaven docs links && inkhaven docs review` is a one-line pre-flight you can wire into `make release` or a CI job.

### docs verify — the examples still run

A code listing that no longer compiles is the most embarrassing kind of stale documentation, because the reader discovers it the hard way — by typing it in. `docs verify` catches it first, by actually running the examples through a compiler or interpreter you configure and reporting the ones that fail.

## Marking a listing for verification

Nothing is verified unless you ask for it, block by block. A fenced code listing inside a `para:code` paragraph is marked by adding the word `verify` to its fence **info string** — the text on the opening fence line after the language:

● ● ● *A verify-marked Rust listing in a para:code paragraph*

```
```rust verify
fn greeting(name: &str) -> String {
    format!("Hello, {name}!")
}
```
```

Only paragraphs tagged `para:code` are scanned, and within them only blocks carrying the `verify` flag are run — an ordinary example with no flag is left alone. The flag can be space- or comma-separated from the language — an info string of `rust verify` or `rust,verify` both work — and a single guard word, `no-verify`, always wins: a fence of `rust verify,no-verify` is skipped no matter what. That gives you an escape hatch for the one listing that is illustrative rather than runnable — pseudocode, a fragment, a deliberately broken example — without turning verification off everywhere.

## The two safety gates

Running code that lives in a manuscript is a real capability, so Inkhaven puts two deliberate gates between a fresh clone and an executed command.

● ● ● *docs verify with neither gate open — it refuses, safely*

```
$ inkhaven docs verify
Error: docs.verify is off. Enable it in inkhaven.hjson
(`docs: { verify: { enabled: true } }`) and configure
runners, then re-run. Use `--dry-run` to preview.
```

The first gate is the config switch `docs.verify.enabled`, which is `false` by default. On a project that has never turned it on, `docs verify` refuses and points you at the setting rather than running anything. The second gate is `--yes`: even with the feature enabled, the bare command **lists** the blocks and the exact runner commands that would execute with your privileges, then stops, requiring you to re-run with `--yes` to actually run them.

The safe way to look before you leap is `--dry-run`, which previews the resolved command for every block — with `{file}` shown in place — and never touches the config gate at all. It is the reviewer’s-eye view of what a project **would** do:

● ● ● *docs verify --dry-run — resolved commands, nothing executed*

```
$ inkhaven docs verify --dry-run
docs verify --dry-run: 2 block(s) would run:

api-basics/hello [rust] → rustc --edition 2021 \
  --crate-type lib <code>.tmp -o <code>.tmp.dir/out
guide/quickstart [python] → python3 <code>.tmp
```

## Running the checks

With the feature enabled and `--yes` supplied, `docs verify` pulls every verify-marked block from every user book (or the ones you narrow to), writes each to a temporary file, runs it through the runner configured for its language, and reports the outcome per block.

● ● ● *A real run — one pass, one failure, one skip*

```
$ inkhaven docs verify --yes
✓ api-basics/hello [rust]
✗ guide/quickstart [python]
  Traceback (most recent call last):
    File "...", line 3, in <module>
      NameError: name 'reqeusts' is not defined
- legacy/old-sample [go] — no runner configured for `go`

docs verify: 1 passed · 1 failed · 1 skipped
Error: 1 code block(s) failed verification
```

Two flags narrow the sweep when you do not want the whole book: `--book-name X` restricts to one book, and `--paragraph <slug>` restricts to a single paragraph named by its slug-path. Everything else about the run is unchanged — the same gates, the same runners, the same tally.

## The four outcomes

Each block resolves to exactly one of four states, and knowing them tells you how to read a report:

- **pass** — the runner exited `0`. Quiet; counted and moved past.
- **fail** — the runner exited non-zero **or** ran past its timeout. The report carries the last forty lines of the runner’s combined stdout and stderr — the informative tail, where the compiler’s actual complaint lives.
- **skip** — no runner is configured for that block’s language. This is **not** an error; it is how you leave a language unverified on purpose. Skips never fail the run.
- **errored** — the block could not be run at all (the temp file could not be written, or the runner could not be spawned). This **is** counted as a failure, because an example you could not check is not an example you can trust.

The command exits non-zero if any block failed or errored; skips and passes alone leave it green. A runner is capped by `docs.verify.timeout_seconds` (30 by default), so a listing with an infinite loop fails on the clock rather than hanging your release script — the timeout is reported as a failure carrying `timed out after Ns`.

#### HOW A BLOCK ACTUALLY RUNS

Each block is written to a temp file whose suffix comes from the language’s entry in `docs.verify.extensions` (rust  $\rightarrow$  `.rs`, python  $\rightarrow$  `.py`, and so on; an unknown language falls back to `.txt`). The runner command is executed through `sh -c`, with `{file}` replaced by that temp file’s path and `{dir}` by its parent directory. Output is captured to a file rather than a pipe, so a chatty compiler can never deadlock the run, and both temp files are cleaned up afterward. Nothing runs but the command **you** wrote in your own config.

## docs links — the cross-references still resolve

The second ground truth is the web of references a document makes — to its own other sections, and to pages out on the internet. `docs links` checks both.

● ● ● docs links --external — one dead internal link, one dead URL

```
$ inkhaven docs links --external
x guide/webhooks → 550e8400-... (linked paragraph no
  longer exists)
x reference/appendix → https://old.example/spec
  (404 Not Found)
(12 external URL(s) checked)

docs links: 1 internal · 1 external broken
Error: 2 broken link(s)
```

The **internal** check runs always, project-wide over every user book, and is purely deterministic: for each paragraph it walks the `linked_paragraphs` cross-references and flags any whose target paragraph no longer exists in the hierarchy — the classic broken link left behind when you rename or delete a node that something else pointed at. No network, no ambiguity.

The **external** check is opt-in, behind `--external`, because it reaches the network. It gathers every `http(s)` URL embedded in your prose — both bare URLs and those inside a `#link("...")` call — deduplicates them so each distinct URL is reported once (with the first place it was found), and checks each for link-rot. Here `--book-name X` narrows the external sweep to a single book, which is handy when one chapter cites the wider web and the rest do not.

#### CONSERVATIVE ON PURPOSE

The external sweep uses the same cautious classifier as the research assistant's dead-source check: only a hard 404 / 410 or an outright connection failure counts as **dead**. A host that is merely slow, rate-limited, or behind an authentication wall is **not** reported — the check would rather miss a genuinely dead link than cry wolf on a live one and train you to ignore it. It exits non-zero when any link, internal or external, is broken.

## docs review — the manuscript is current

The third check is not about correctness but about **readiness**. Every paragraph in Inkhaven carries a status on a seven-rung ladder, and a long document is never

uniformly finished — some sections are polished, some are still notes to self. `docs review` is the dashboard that shows you the distribution and, crucially, what is still below the bar you mean to ship at.

#### TERM The readiness ladder

Each paragraph’s status is one of seven rungs, lowest to highest: `none` → `napkin` → `first` → `second` → `third` → `final` → `ready`. `docs review` measures the whole manuscript against a **floor** on this ladder and lists every paragraph that sits below it — the “nothing half-baked ships” gate.

● ● ● *docs review — per-chapter breakdown and what is below the floor*

```
$ inkhaven docs review
docs review — API Reference

Getting Started
  ready 5 · final 1    [1 below `ready`]
Endpoints
  ready 6 · third 2 · napkin 1    [3 below `ready`]

Below `ready` (needs work):
- getting-started/authentication (final)
- endpoints/webhooks (third)
- endpoints/pagination (third)
- endpoints/rate-limits (napkin)

docs review: 15 paragraphs · 11 ready · 4 below `ready`
```

Per chapter, the report prints the status breakdown (how many paragraphs sit at each rung) and flags the count still below your floor; then it lists each of those paragraphs by slug-path with its current rung. Two flags shape the view:

- `--floor <rung>` sets the bar — `napkin`, `first`, `second`, `third`, `final`, or `ready` (the default). Everything below the floor is listed as needing work; run `--floor final` while drafting to ignore the last coat of polish and see only what is genuinely unfinished.
- `--since <ref>` marks every paragraph whose `.typ` file has changed since a git tag or commit — the “what do I need to re-read since the last release” view. A

changed paragraph in the below-floor list is tagged `← changed since ref`, and the summary line counts the changes. If the project is not a git repository or the ref is unknown, change detection is skipped with a note and the rest of the report still runs.

`docs review` exits non-zero whenever any paragraph is below the floor, so `docs review --floor final` in a release script is a hard stop on shipping a section you never finished.

## Verifying the open listing — Ctrl+B Shift+D

The CLI is the whole-book, CI-shaped tool; in the editor there is a single chord for the tighter loop of checking the one listing in front of you. Open a `para:code` paragraph and press `Ctrl+B Shift+D`.

**Ctrl+B Shift+D**      Verify the open code listing — run every verify-marked block in the current `para:code` paragraph and report to the Output pane.

Every verify-marked block in the open listing is run synchronously — one listing is quick — through the same configured runners the CLI uses, gated on the same `docs.verify.enabled` switch. Passing blocks are quiet. A failure lands in the **Output pane**, anchored on the paragraph, so it colours that paragraph's tree badge and answers the `t` (this-paragraph) filter, exactly like every other finding; the status line reports the tally.

● ● ● *A failed verification in the Output pane, on the paragraph*

```
Output · 1/1 · doc-verify —
┌ △ doc_verify
│   code example failed `python` verification
│   Traceback (most recent call last):
│     File "...", line 3, in <module>
│       NameError: name 'requests' is not defined
└─
┌ ↕ select  o expand  Enter jump  a ask  d dismiss
└─
docs verify · 0 passed · 1 failed · 0 skipped
```

Re-running the chord clears the paragraph's prior verify findings before checking again, so the Output pane always shows the current state rather than a pile of stale

failures. If the feature is off, or the open paragraph is not a code listing, or the listing has no verify-marked blocks, the status line says so and nothing runs. For a whole-

book or CI pass, drop back to `verify`.

## Configuring verification

Everything `docs verify` does is driven by one config block. It lives under `docs:` in `inkhaven.hjson`, and only the `verify` sub-block has any runtime behaviour — the surrounding `docs:` block also carries the HTML export settings (Chapter 25) and the index settings below.

● ● ● *The docs.verify block — off by default, runners you supply*

```
docs: {
  verify: {
    enabled: false           # master switch — nothing runs
    timeout_seconds: 30      # per-block wall-clock cap
    runners: {               # language → command, run via sh -c
      rust: "rustc --edition 2021 --crate-type lib
             {file} -o {dir}/out"
      python: "python3 {file}"
    }
    # extensions: seeded (rust→rs, python→py, sh→sh,
    #   go→go, ...); an unknown language falls back to .txt
  }
}
```

The pieces are few and each does exactly one thing. `enabled` is the master switch — `false` and nothing ever runs. `timeout_seconds` is the per-block cap. `runners` is the heart of it: a map from a fence language to a shell command, where `{file}` becomes the temp file holding the block and `{dir}` its parent directory. A language with **no** runner is skipped, never failed — which is the lever you use to decide, per project, which languages are verified at all. `extensions` maps a language to the temp-file suffix and comes seeded with the common ones (Rust, Python, shell, Go, JavaScript, C, and more); you rarely touch it, but it is there to override when a runner cares about the file name.



**RUNNERS RUN WITH YOUR PRIVILEGES**

A runner is an arbitrary shell command you wrote, executed as you. The two gates — `enabled` plus `--yes` — exist precisely because this is real execution, not a sandbox. Configure runners you trust, verify code you wrote, and treat a strange project's `docs.verify` block the way you would treat any script: read it (a plain `--dry-run` shows every resolved command) before you pass `--yes`.

**Single-sourcing — docs.variables**

One more staleness trap is the fact repeated in fifty places — most often the version number — where you update forty-nine and miss the fiftieth. `docs.variables` is Inkhaven's single-sourcing answer. It is not a `docs` subcommand; it is a substitution applied at **export assembly time**, across every export format.

● ● ● *A variable defined once, used anywhere in prose*

```
# in inkhaven.hjson
docs: {
  variables: {
    version: "3.0.0"
    api_base: "https://api.example.com/v2"
  }
}

# in any paragraph body
Install release {{version}} and point your client at
{{api_base}} to begin.
```

Anywhere `{{key}}` appears in a paragraph body, the export replaces it with the value of that key — so the version lives in exactly one place, and every PDF, EPUB, and HTML build carries the same current number. Because the substitution happens at build time and not in the stored `.typ` file, your source stays readable with the placeholders intact; only the **exported** artefact has the resolved value. Leave the map empty and nothing is substituted.

## The back-of-book index

A scholarly or technical book earns an index — the alphabetised appendix a reader turns to last, each term pointing at the chapters where it appears. `inkhaven index` builds one, and it builds it from vocabulary you already maintain rather than asking you to mark up the text.

### TERM The back-of-book index (INDEX-1)

`inkhaven index` builds a term → chapters index from your Glossary’s canonical terms (plus any extras you list), locating each in the manuscript by whole-word match, deduplicating its hits to the chapter, and rendering an alphabetised index in Markdown, Typst, or JSON. It is deterministic and free — pure text search, no model — and it is **not** the semantic-search index, nor the Index Locorum or Index Verborum, which are separate scholarly commands.

### Where the terms come from

The list of terms to index is the union of two sources. The first is your **Glossary**: when `docs.index.from_glossary` is on (the default), every canonical term becomes an index entry, and every Glossary synonym becomes a **see-reference** pointing at its canonical term. The second is `docs.index.terms`, a config list of extra terms — proper names, topics, anything you never put in the Glossary. If neither source yields a term, the command errors rather than emit an empty index.

### How a term is located

Each term is searched across every user book’s prose (or one book with `--book-name`). A paragraph’s Typst is first stripped to plain text, then matched **whole-word** and **case-insensitively**: the term “art” matches the standalone word and never “artist” or “start”. Multiple hits within the same chapter collapse to a **single** location, because the index points at chapters, not at every occurrence — this is an index, not a concordance. A term found nowhere in the manuscript is silently dropped, and the finished entries are sorted case-insensitively by term. A see-reference is emitted only when its canonical term actually made it into the index (a cross-reference to an absent term is useless), and a synonym identical to its canonical is skipped.

## Output formats

`--format` picks the renderer and `--out FILE` (short `-o`) writes to a file instead of standard output.

● ● ● *The three renderers of inkhaven index*

```
$ inkhaven index                # Markdown, all books
$ inkhaven index --book-name "Treatise"
$ inkhaven index --format typst -o appendix-index.typ
$ inkhaven index --format json

# md    → term – Chapter A, Chapter B
#       term – see canonical
# typst → term – Chapter A, Chapter B (under `= Index`)
#       term – see canonical
# json  → { "index": [ { term, see,
#                   locations: [ { chapter, anchor } ] },
#                   "count": N }
```

Markdown (the default, `md`) is the portable form. Typst (`typ` or `typst`) emits markup under an `= Index` heading, ready to `#include` as a printed appendix. JSON exposes the full structure — each location carrying both its `chapter` and an `anchor` — for downstream tooling.

## Anchor and the web index

Every located term carries an `anchor` alongside its `chapter`. In the standalone command the anchor resolves to the chapter (the CLI's unit is the chapter), and only the JSON format surfaces it as a field; the Markdown and Typst renderers print the chapter name alone. The anchor earns its keep in the **HTML static-site export** (Chapter 25): with `docs.html.include.index` on, the site build runs **the very same index builder** over the chapters and folds in an `appendix-index.html` page where each location is a live `<a href="...">chapter</a>` link into the chapter's HTML. So the index is clickable on the web and a clean alphabetised appendix in print — both produced by one pure function, so what the CLI emits and what the site folds in can never drift apart.

## Configuring the index

● ● ● *The docs.index block, and the HTML include toggle*

```
docs: {
  index: {
    from_glossary: true    # seed from Glossary terms + synonyms
    terms: []             # extra terms beyond the Glossary
  }
  html: {
    include: { index: true } # fold a hyperlinked index into
                           # the HTML site (off by default)
  }
}
```

`from_glossary` is on by default; `terms` starts empty; and the HTML site's `include.index` is **off** by default — turn it on when you want the appendix folded into the exported site.

## What these tools are not

It is worth naming the boundaries plainly, because they are the same boundaries every Inkhaven intelligence respects. None of this is AI — the cores are deterministic text and process work, and the single network touch is the opt-in external-link sweep, which still consults no model. None of it is an autopilot: `docs verify` runs nothing until you both enable it and pass `--yes`, and even then it runs only the commands you wrote. None of it is a rewriter — every check reports and none edit your prose. And `docs verify` is not a linter for arbitrary code: it runs only the blocks you marked `verify`, in the languages you gave a runner. The index, likewise, is not a concordance and not a model's guess at what matters — it is whole-word search over your own term list.

These are the tools of the last mile, the ones you run in the same breath as the export. Chapter 24 and Chapter 25 turned your manuscript into a PDF and a website; this chapter is how you know, before either goes out, that the examples in it still run, its links still resolve, no section shipped half-finished, and the reader has an index to find their way back in.

## WHAT YOU LEARNED

- The **docs subsystem** is three deterministic, advisory checks over your user books, each exiting non-zero on a finding so it fits a pre-release / CI gate: `verify`, `links`, and `review`.
- **docs verify** runs the fenced blocks whose info string you mark with `verify` (in `para:code` paragraphs) through per-language **runners** you configure; gated by `docs.verify.enabled` (off by default) **and** `--yes`, with `--dry-run` to preview. Outcomes: `pass` · `fail` (non-zero or timeout, last 40 lines) · `skip` (no runner) · `errored`.
- **docs links** flags internal `linked_paragraphs` cross-references that no longer resolve (always), and with `--external` checks `http(s)` URLs for link-rot conservatively (only `404 / 410` and hard failures).
- **docs review** is a currency dashboard over the readiness ladder (`none ... ready`); `--floor` sets the bar, `--since <ref>` flags paragraphs changed since a git tag.
- **Ctrl+B Shift+D** verifies the **open** code listing, posting failures to the Output pane on the paragraph; `docs.variables` single-sources `{{key}}` values at export time across every format.
- **inkhaven index** builds an alphabetised back-of-book index from Glossary terms + `docs.index.terms` (whole-word, chapter-deduplicated) in `md / typst / json`; the HTML export's hyperlinked index page uses the same builder.

PART VIII

---

# Scripting with Bund

## CHAPTER 27

## 27

## Scripting with Bund

---

Everything in this manual so far has been a command someone at Inkhaven already wrote for you — a chord, a subcommand, a scan. This chapter is the one place the tool hands you the keys and lets you write commands of your own. **Bund** is a small scripting language embedded inside Inkhaven: you can run a line of it against your project, bind it to a chord, or — most usefully — have it fire automatically the moment you save a paragraph, take a snapshot, or hit your daily goal. It is the escape hatch for the author whose workflow has outgrown the built-in chord set and wants the tool to do something nobody at Inkhaven thought to build.

You do not need Bund to write a book. The whole of Parts I through VII assumes you never touch it, and most authors never will. Reach for this chapter when you

catch yourself doing the same small chore by hand every day — “warn me whenever a scene mentions a character I haven’t introduced,” “tag every paragraph in this chapter as needing a second pass,” “keep a fresh story-graph PNG on my desktop” — and want it to happen on its own.

## What Bund is

Bund is a stack-based language Vladimir Ulogov wrote and open-sourced (it lives at [github.com/vulogov/bundcore](https://github.com/vulogov/bundcore)). Inkhaven embeds three of its crates — `bundcore`, `bund_language_parser`, and `rust_multistackvm` — and layers its own vocabulary on top. The language itself is Forth-shaped: postfix operators, two stacks, curly-brace lambdas. It is small enough to learn in an afternoon, and this section is enough to read every example in the chapter.

### Numbers go on, operators pull off

Bund is **postfix**. You push operands first; the operator pulls them back off. Where you would write `2 + 3` elsewhere, in Bund you write:

```
2 3 +           // push 2, push 3, + pops both, pushes 5
```

After that line the stack holds a single value, `5`. Longer expressions chain the same way, left to right:

```
2 3 + 4 *       // (2 + 3) then times 4 → 20
40 2 +         // → 42
```

There is one Forth gotcha worth naming up front: operands pop in stack order, so `a b /` computes `b ÷ a`, not `a ÷ b`. Trust the stack, not your infix instinct. Strings use double quotes, and `println` pops one value and prints it with a newline:

```
"Hello, Inkhaven!" println
```

Comments run from `//` to end of line, exactly as above.

### Shuffling the stack

A handful of words rearrange what is already on the stack. You will meet all of them in real hooks, because a hook is handed its arguments on the stack and has to move them into position.



● ● ● *The five stack-shuffling words*

|      |                    |                    |
|------|--------------------|--------------------|
| dup  | ( a -- a a )       | duplicate the top  |
| drop | ( a -- )           | discard the top    |
| swap | ( a b -- b a )     | swap the top two   |
| over | ( a b -- a b a )   | copy the second up |
| rot  | ( a b c -- b c a ) | rotate three       |

The parenthesised note is a **stack diagram**: what the word expects on the left of `--`, what it leaves on the right. The whole of Bund's vocabulary is documented this way, and so is every hook in this chapter. To square the top of the stack you duplicate it and multiply:

```
5 dup *           // → 25
```

### Lambdas and named words

Curly braces wrap a block of code into a **lambda** — a first-class value you can push around like a number. Give one a name with `register` and it becomes a word you can call:

```
"square" { dup * } register
5 square      // → 25
9 square      // → 81
```

This is the mechanism behind every hook in the chapter: a hook is simply a lambda you have registered under a well-known name like `hook.on_save`. When the matching thing happens, Inkhaven calls that name for you.

### TERM The Adam VM

Every line of Bund in your Inkhaven session runs against a single, process-wide virtual machine that bundcore calls **Adam** — the first VM constructed in the process. The CLI `inkhaven bund`, every `Ctrl+Z R` buffer run, every hook fire, and every keymap mutation all share it. Because Adam is one persistent instance, state carries across calls: a word you `register` once is available everywhere for the rest of the session. Adam is built lazily on the first evaluation, then frozen — `stdlib` loaded, policy applied, your scripts run — for the life of the process.

## What Inkhaven adds on top

Bundcore gives you the language: arithmetic, strings, conditionals, lambdas, math, time, the stack words above. Inkhaven adds two namespaces of its own on top of that base:

- The `ink.*` standard library — words that reach into **your project**: read a paragraph’s text, search the manuscript, add a tag, run a continuity sweep, write a file. This is the bulk of the chapter.
- The `hook.*` names — the well-known lambda names Inkhaven calls when something happens in the editor.

Everything else — how the language parses, how `if` and `while` work, how you build a list — is vanilla bundcore, documented upstream at `docs.rs/bundcore`.

## The four ways to run a script

There are four surfaces from which Bund reaches Adam, and they differ only in where the code comes from and when it runs.

|                      |                                                                       |
|----------------------|-----------------------------------------------------------------------|
| <b>inkhaven bund</b> | One-shot from the shell — evaluate, print the top of the stack, exit. |
| <b>Ctrl+Z E</b>      | Eval modal — type an expression, Enter, result on the status bar.     |
| <b>Ctrl+Z R</b>      | Run the whole open <code>.bund</code> buffer against Adam.            |
| <b>Ctrl+Z N</b>      | Create a new <code>.bund</code> Script node under the Scripts book.   |

Beyond those four, two surfaces run code **for** you without a keystroke: the `scripting.bootstrap` string in `inkhaven.hjson`, which runs once when the project opens, and every `.bund` Script node in the tree, which is evaluated at project open right after the bootstrap. Those two are where real hooks live, and they are the subject of the next two sections.

The quickest way to see Bund work at all is the shell:

● ● ● *One-shot evaluation from the terminal*

```
$ inkhaven bund "40 2 +"
42

$ inkhaven bund '"hello" println "world" println'
hello
world
```

Inside the TUI, `Ctrl+Z E` pops a one-line prompt, runs what you type against Adam, and puts the result on the status bar — the closest thing to a read-eval-print loop. Pair it with `println` when you want to watch a list go by:

```
"" ink.node.children dup println
```

The `dup` keeps the list on the stack so the status bar still shows it after `println` has already printed it once.

## The .bund Script node

A one-liner in the eval modal vanishes when you press Enter. Anything you want to **keep** — a hook, a custom word, a small automation — lives in a **Script node**: a first-class node in your project tree, stored on disk as a `.bund` file, edited in the normal editor, and backed up with the rest of the manuscript.

### TERM Script node

A Script node is a tree node of kind `Script`, holding Bund source instead of Typst prose. It sits alongside your paragraphs (`.typ`) and data nodes (`.hjson`) as a first-class citizen: it round-trips through backups, `inkhaven reindex` picks up edits made outside the TUI, and the editor knows to run it as Bund rather than typeset it. Its natural home is the `Scripts` system book, but it can live inside any user Book whose workflow it belongs to — nothing forces it under `Scripts`.

Press `Ctrl+Z N` from any pane and the Add modal opens pre-pointed at the `Scripts` system book. Give the node a title, and it opens in the editor as an empty Bund buffer. Write your script, save it like any paragraph, and run it on demand

with `Ctrl+Z R`. If you started a node as prose and want to convert it — or the reverse — the tree’s **morph** operation turns a `Paragraph` into a `Script` and back without losing the body.

The important behaviour is what happens at **project open**. When Inkhaven builds Adam, it walks the whole tree, finds every `Script` node in tree order, and evaluates each one. That is the mechanism that makes hooks persistent: you register `hook.on_save` inside a `Script` node once, and from then on every project open re-installs it before you have typed a word. A syntax error in one script logs a warning and is skipped; the others still load, and Adam still finishes building.

#### ORDER OF LOADING

At project open Adam is assembled in a fixed order: bundcore’s `stdlib`, then Inkhaven’s `ink.*` words, then the sandbox policy is applied, then the inline `scripting.bootstrap` runs, and **finally** every `Script` node is evaluated in tree order. So a `Script` node can rely on a word the bootstrap defined, but not the other way around.

## The trust gate

Auto-running code that ships inside a project is a real risk: open a project someone emailed you and, without a gate, its `Script` nodes would execute the instant you opened it. Inkhaven closes that door with a **trust decision**, set by `scripting.trust_decision` in `inkhaven.hjson`, with three values:

● ● ● *scripting.trust\_decision — three values*

|         |                                                                                                                                                                                              |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "ask"   | (default) run scripts ONLY when the file <code>&lt;project&gt;/.inkhaven/trust</code> exists and holds the marker line <code>`trust`</code> . Otherwise the scripts are pending your opt-in. |
| "trust" | run scripts unconditionally. Use only on projects you authored or audited.                                                                                                                   |
| "deny"  | never run scripts, trust file or not.<br>Good for opening a project to read only.                                                                                                            |

Under the default `"ask"`, a project you did not write opens with its scripts **inert**: you get a status-bar notice that scripts are pending, and nothing runs until you deliberately create `.inkhaven/trust` with the marker line inside it. The marker match is case-insensitive and ignores surrounding whitespace; a trust file that says anything else does not grant trust. Your own projects, whose scripts you wrote, are the ones where setting `"trust"` in the HJSON is appropriate.

#### THE HONEST LIMITATION

A malicious project's own HJSON could set `trust_decision: "trust"`. The gate is aimed at the author publishing their own work, not at a cryptographic guarantee against a hostile package. If you open a project you did not write, leave the default `"ask"` in place and read its Script nodes before you create the trust file. The `"deny"` value exists for exactly the read-only-review case.

## Hooks — code that runs when something happens

A hook is the natural home for a “warn me when...” rule. You register a lambda under a well-known name; Inkhaven calls that name **after** the matching editor action has already succeeded, pushing the action's details onto the stack for your lambda to inspect. The lambda's job is to observe and react — print a warning, add a tag, speak a line aloud — not to undo what happened.

Here is the complete set of hook points, each with the stack it is handed on entry:

#### ● ● ● Store-mutation hooks

```
hook.on_create      ( uuid kind -- )
hook.on_save        ( uuid -- )
hook.on_rename      ( uuid new_title -- )
hook.on_snapshot    ( parent_uuid snap_uuid -- )
hook.on_delete      ( uuid -- )
hook.on_status_promoted ( uuid from_status to_status -- )
```

● ● ● *Session and milestone hooks*

```
hook.on_goal_hit      ( today_words daily_goal -- )
hook.on_active_goal_hit ( active_secs goal_secs -- )
hook.on_streak_break  ( prev_streak_days -- )
hook.on_streak_milestone ( milestone_days -- )
hook.on_diagnostic    ( uuid count first_message -- )
```

● ● ● *Timeline and production hooks*

```
hook.on_event_added      ( uuid -- )
hook.on_event_orphaned   ( uuid -- )
hook.on_assemble ( uuid slug root_typ files_written -- )
hook.on_take             ( uuid slug pdf_dest -- )
```

A few of these repay a note. `hook.on_status_promoted` fires both when you cycle a paragraph's status by hand (Ctrl+B R) and when it auto-promotes on hitting its goal; the status strings arrive lowercased (`napkin`, `first`, ..., `ready`). `hook.on_goal_hit` fires the first time the day's word count crosses your `goals.daily_words` line and self-resets if you dip back below, so it never nags. `hook.on_streak_milestone` fires once per upward crossing of 7 / 30 / 100 / 365 days — a celebration, never a blocker. `hook.on_diagnostic` is debounced: it fires only when Typst's diagnostic state actually changes, not on every idle tick.

### How a hook behaves when it misbehaves

Three guarantees make hooks safe to write casually. They matter enough to state plainly:

- **A hook never breaks the editor.** Every failure path — a syntax error, a stack underflow, even a panic inside your lambda — is caught, logged at WARN, and swallowed. The save (or rename, or snapshot) that fired the hook still completes. You find the failure in `.inkhaven.log`, not in a corrupted file.
- **Hooks are bounded against runaway recursion.** A hook that triggers another mutation that fires another hook is capped at a depth of four; past that the dispatch is skipped and logged. And when a hook fires from inside a running `bund` evaluation — because your script itself called a write word — it short-circuits rather than re-enter the VM's write lock.

- **Hook output goes to the log, not the void.** Anything a hook `println`s is drained after the fire and emitted through tracing: it lands in `.inkhaven.log` in the TUI and on `stderr` from the CLI. Pre-1.2.6 that output vanished silently; now it is findable.

One consequence is worth internalising: hooks fire **synchronously** on the thread that did the save, holding Adam's write lock. A slow hook is a slow save, felt as a pause under your fingers. Keep hook bodies cheap; push anything expensive to a command you run on demand.

### Registering a hook

For a one-line rule, the `scripting.bootstrap` string in `inkhaven.hjson` is enough:

```
"hook.on_save" { drop "saved" println } register
```

The `drop` throws away the paragraph UUID this hook is handed — this trivial version does not need it. For anything longer, put the same `register` call in a `.bund` Script node so it lives with the manuscript and gets backed up. Both are evaluated at project open, bootstrap first.

## The `ink.*` standard library

The `ink.*` namespace is where Bund stops being a pocket calculator and becomes a way to **reach into your book**. It is large — dozens of families — because nearly every Inkhaven feature that has a CLI surface also has a Bund surface built on the same core, so a script, the CLI, and the TUI all agree. This section surveys the shape rather than enumerating every word; the per-feature chapters and `Documentation/Bund/BUND_TUTORIAL.md` give the exact stack diagrams.

## The families, by what they touch

### ● ● ● *The tree, prose, and metadata*

```
ink.node.*      list / get / children of nodes
ink.paragraph.* text / target / set_target / save /
                set_status
ink.tree.*      add / delete / rename / move / morph
ink.tag.*       list / search / add / remove
ink.event.*     story-timeline events + critique
ink.snapshot.list ink.path.to_uuid
```

### ● ● ● *Search, retrieval, and the AI*

```
ink.search.*    semantic search: text / load
ink.book_rag.*  Book-scope retrieval + grounding
ink.ai.*        history / send / send_blocking /
                poll / set_system_prompt
```

### ● ● ● *The reading intelligences (read-only)*

```
ink.continuity.* ink.knowledge.* ink.readthrough.*
ink.chorus.*      ink.stylist.*   ink.revise.*
ink.chronicle.*   ink.char.*      ink.dialogue.*
ink.prose.*       ink.graph.*     ink.myth.*
ink.utopia.*      ink.theologian.* ink.outline.*
ink.inner_editor.* ink.inner_socrates.*
ink.sources.*     ink.terms.*     ink.snippets.*
ink.review.*      ink.thread.list
```



● ● ● *Language, verse, output, and files*

```

ink.lang.*   the full conlang suite (dozens of words)
ink.poem.*   syllables / scansion / rhyme / findings
ink.io.*     print / log / notify / message.*
ink.pane.*   ink.input      the Bund output pane + modal
ink.editor.* cursor / text / find / goto / insert /
             replace / replace_all / scroll / delete
ink.fs.*     read / write      ink.tts.speak
ink.pdf.*    ink.export.* ink.typst.* ink.db.*
ink.story.render ink.key.* ink.theme.set

```

Most of the reading-intelligence families follow one pattern: a handful of read words that pull findings or a report and push them as dicts, plus at most one `suppress` word that records an author decision. So `ink.continuity.findings`, `ink.knowledge.check`, `ink.chorus.voices`, `ink.revise.findings` all behave alike — call it, get a list of `{id, ..., observation}` dictionaries back, walk them. The deliberately-costly parts (the LLM coherence pass, the AI editorial letter, any prose rewrite) are **not** exposed to Bund; the deterministic core is.

### Reading the result of a word

Every `ink.*` word that returns structured data returns a Bund list of dicts. From the CLI those pretty-print as JSON, which makes the terminal the fastest way to see what a word actually gives you:

● ● ● *Inspecting a word's output from the shell*

```

$ inkhaven --project ~/my-book \
  bund '"" ink.node.children'
[ ... JSON list of every root node ... ]

$ inkhaven --project ~/my-book \
  bund '"manuscript" "the storm" book_rag.context'
[ ... the exact grounding block Book chat would use ... ]

```

That second line is genuinely useful on its own: it prints the grounding block Bookscope chat would feed the model for that query, straight from the terminal, no TUI needed.

## The sandbox policy

The `ink.*` surface includes words that write files, mutate your tree, rebind your chords, and call the LLM. Left ungated, a save-hook you pasted from a tutorial could quietly do any of those. So every `ink.*` word is sorted into a **category**, and a conservative subset of categories is denied by default. You opt in explicitly, in HJSON, category by category.

### The categories

● ● ● *Default-ALLOWED — safe, non-destructive*

|                          |                                    |
|--------------------------|------------------------------------|
| <code>store_read</code>  | read nodes, tags, findings, search |
| <code>fs_read</code>     | read a file                        |
| <code>editor_read</code> | query the buffer / cursor / pane   |
| <code>ai_read</code>     | read chat history                  |
| <code>audio</code>       | TTS playback (also gated by HJSON) |

● ● ● *Default-DENIED — opt in per family*

|                                  |                                          |
|----------------------------------|------------------------------------------|
| <code>store_write</code>         | mutate tree, tags, events, status        |
| <code>editor_write</code>        | insert / replace / scroll the buffer     |
| <code>ai_write</code>            | send a prompt, set the system prompt     |
| <code>fs_write</code>            | write a file, export, render a PNG       |
| <code>theme_write</code>         | recolour the interface                   |
| <code>keymap</code>              | rebind chords via <code>ink.key.*</code> |
| <code>net shell code_eval</code> | reach outside the project                |

The rule of thumb is exactly what it looks like: **reads are open, writes are shut**. A word like `ink.continuity.findings` is `store_read` and works out of the box; `ink.tag.add` is `store_write` and errors until you enable it. You turn a family on with one line:

```
scripting: {
  enabled_categories: ["store_write", "fs_write"]
}
```

That grants every `store_write` word (tag and event mutations, tree edits, status changes) and every `fs_write` word (file writes, exports, `ink.story.render`) at once.

### Finer control, and the resolution order

Four knobs sit under `scripting` for cases where a whole category is too coarse. `enabled_categories` and `disabled_categories` flip a family on or off; `enabled_words` allows a single word even when its category is denied (grant `ink.fs.read` without opening all of `fs_read`); `disabled_words` denies a single word regardless of its category. When a word is looked up, the order is fixed: an explicit `enabled_words` entry wins, then `disabled_words`, then a denied category, otherwise allow. A power-user `no_default_deny: true` clears the built-in baseline entirely — off by default, and rarely what you want.

When a denied word runs, it does not silently do nothing — it returns a clean error, `script denied by inkhaven policy`, and the specific word name is written to `.inkhaven.log` at the moment the policy was applied. If a script mysteriously stops, the log names the offending word.

### Filesystem confinement

Enabling `fs_read` or `fs_write` answers “may this script touch the filesystem at all?” — a separate question from “**where?**” By default, paths handed to `ink.fs.read` and `ink.fs.write` are confined to the project root: a script cannot read your home directory or write outside the book, even with the category enabled. Setting `fs_unsandboxed: true` collapses that confinement to “anywhere this user can reach” — the pre-1.2.15 behaviour, recommended only for trusted projects that genuinely need a shared location outside the tree.

#### THE EVERY-WORD-CLASSIFIED GUARD

A word registered but **forgotten** from the category table would be silently allowed — escaping `disabled_categories` entirely. Inkhaven forbids that with a test, `every_registered_word_is_classified`, that walks every `ink.*` word Adam registers and fails the build unless each one is either given a category or listed in an explicit “pure — touches nothing protected” allowlist (the in-memory PDF ops, the `ink.lang.dict` constructor). A new word cannot ship un-gated by accident.

## Worked examples

Two complete scripts, each small enough to type, show the pieces working together.

### A save-hook that watches for unIntroduced names

This registers `hook.on_save` to pull the just-saved paragraph's text and check it — the sketch below keeps the parsing trivial and leaves the name-matching to `ink.search.text` against your Characters book:

```
"hook.on_save" {
  // ( uuid -- )   fired after a paragraph saves
  dup ink.paragraph.text    // ( uuid text )
  swap drop              // ( text )   keep the body
  // ... scan `text` for names, search Characters,
  //   ink.io.notify when one is missing ...
  drop
} register
```

Registered in a Script node, this reinstalls itself at every project open (once the project is trusted). It reads only — `ink.paragraph.text` and `ink.search.text` are both `store_read` — so it needs no category opt-in.

### Tagging scenes for a pass

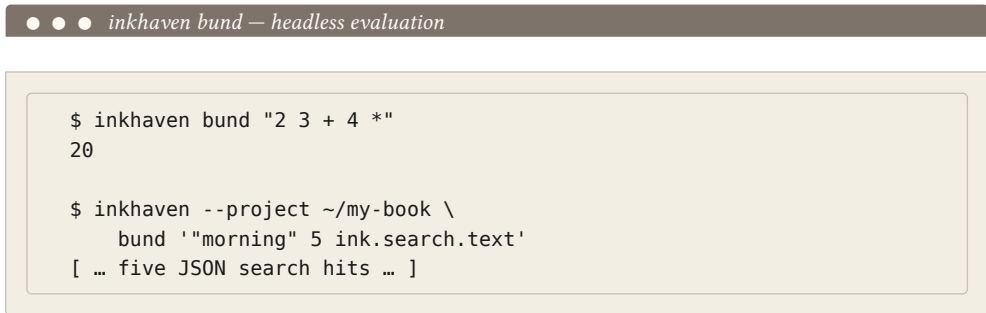
This one **does** write, so it needs `store_write` enabled. `ink.tag.add` takes a paragraph's **slug path** — the bracketed path `inkhaven list` prints — and a tag name, and adds the tag to that paragraph. Marking the scenes you mean to revisit is the kind of chore worth scripting:

```
// ( path tag -- ) — root-anchored slug path, then the tag.
"the-ninth-lantern/03-the-mole/opening" "editing-pass-2" ink.tag.add
"the-ninth-lantern/03-the-mole/the-turn" "editing-pass-2" ink.tag.add
```

Each line pushes a path and a tag and calls `ink.tag.add`, which resolves the path to a paragraph and records the tag. (To **discover** paths programmatically, `ink.node.children` walks the top-level books and each node dict carries its `id`, `kind`, `title`, and `slug`.) Run it once from `Ctrl+Z R`, or wrap it in a word you call when you start a pass.

## The inkhaven bund CLI

The `bund` subcommand runs a script headless, with no TUI, and prints the top of the stack when it exits:



```

● ● ● inkhaven bund — headless evaluation

$ inkhaven bund "2 3 + 4 *"
20

$ inkhaven --project ~/my-book \
  bund '"morning" 5 ink.search.text'
[ ... five JSON search hits ... ]

```

Two behaviours are worth knowing. First, whether `ink.*` words work depends on the working directory: if it (or `--project`) is an initialised Inkhaven project, the store is opened — which arms the scripting layer automatically — and the project words come alive. Against a plain directory the script runs on the bare Adam VM, so pure arithmetic and strings work but `ink.*` words error cleanly. (Opening a project loads the embedding model, a few-seconds cost the arithmetic-only path skips.) Second, output ordering is faithful: any `print/println` the script produced is emitted first, in script order, then the top-of-stack value on its own line — so a script's narration and its result never smear together. A script that leaves an empty stack and printed nothing reports `(no result)`.

The CLI is the way to fold Bund into a shell pipeline, a Makefile, or a cron job: because Adam and the store are the same as the TUI's, a headless `inkhaven bund` sees exactly the book your editor does.

## WHAT YOU LEARNED

- **Bund** is a small, Forth-shaped, stack-based language embedded in Inkhaven for hooks, custom rules, and automation — postfix operators, two stacks, curly-brace lambdas registered as named words.
- Every session shares one process-wide VM, **Adam**: state persists across calls, and it is built once at project open — `stdlib`, then `policy`, then `bootstrap`, then every Script node in tree order.
- A **.bund Script node** keeps a script with the manuscript; auto-loading it at open is gated by the **trust decision** (`ask` / `trust` / `deny` plus the `.inkhaven/trust` marker file), so a foreign project's code never runs unasked.
- **Hooks** are lambdas you register under well-known names — `hook.on_save`, `hook.on_snapshot`, `hook.on_streak_milestone`, and the rest — fired after the matching action; failures are always logged and swallowed, never breaking a save.
- The **ink.\* standard library** reaches into your project — tree, prose, tags, timeline, search, the reading intelligences, language, files, the AI — with reads open and writes shut by default.
- The **sandbox** sorts every word into a category, denies the destructive ones until you opt in per family in HJSON, confines `ink.fs.*` to the project root, and a build-time guard forbids any word from shipping un-classified.
- `inkhaven bund "<code>"` runs a script headless against the same Adam and store the TUI uses — the way to fold Bund into a shell pipeline.

PART IX

---

# Keeping It Healthy

## CHAPTER 28

## 28

Keeping It Healthy

---

A manuscript is the most valuable thing on your disk and the least replaceable. Everything else in this book is about writing it; this chapter is about not losing it. Inkhaven’s store is robust under ordinary use, but “robust” is not a plan, and the tool takes the view that a book you have spent a year on deserves several independent safety nets rather than one. This chapter walks the whole maintenance surface: the backup pipeline and how to restore from it, crash recovery, the `reindex` command that reconciles the database with the files on disk, the project **doctor** that scans for structural trouble, the advisory lock that keeps two sessions from corrupting each other, and the log you read when something drifts. None of it is glamorous. All of it is the difference between a bad afternoon and a lost book.



## THE SHAPE OF THE SAFETY NET

Four layers stand between you and a lost word, from most to least frequent: **focus-loss autosave** (every time attention leaves the Editor), **snapshots** (F5 / F6, per paragraph — chapter 6), the **crash mirror** (a side copy of every dirty buffer, chapter 6), and the **zip backup** (a whole-project archive, this chapter). Each catches what the one above it misses. This chapter is the outermost layer and the recovery tools that read the ones beneath it.

## Backups — a whole project in one archive

The backup pipeline answers one question: if this directory vanished, could you get the entire project back exactly as it was? The `backup` command produces a single dated `.zip` that says yes.

### ● ● ● Taking a manual backup

```
$ inkhaven --project ~/Books/my-novel backup --out ~/Backups
wrote backup: ~/Backups/blackinkhaven_20260805_143010.zip
```

What the command does, in order: it walks every file under the project root; it skips the runtime log (`.inkhaven.log`, uninteresting in an archive) and any directory you have configured as the backup output target (so an archive never tries to zip earlier archives of itself); it streams the rest into a deflate-compressed zip named `blackinkhaven_YYYYDDMM_HHMMSS.zip` inside `--out`; it updates `.inkhaven-backup.json` in the project root with the moment it finished, so the on-exit hook knows when the last good backup ran; and it prints `wrote backup:` with the path. That is the whole operation.

**TERM Filesystem-level backup**

The backup is taken at the level of **files**, not the database. It never opens the store — DuckDB and the vector index are copied as bytes, exactly as they sit on disk. That is deliberate: a backup must be safe to run while the TUI is closed, and a byte-for-byte copy reproduces an **exact** working tree — identical UUIDs, identical paragraph paths, identical embeddings — when it is restored.

**What an archive contains**

Relative to its root, every archive holds the marker config, the prompt library, the three database files, the vector index, your books, and the small session and backup-timestamp sidecars.

● ● ● *The layout inside a backup zip*

|                       |                                            |
|-----------------------|--------------------------------------------|
| inkhaven.hjson        | ← the "is this an Inkhaven backup?" marker |
| prompts.hjson         |                                            |
| metadata.db           | ← hierarchy: nodes, slugs, titles, tags    |
| blobs.db              | ← paragraph bodies + image bytes           |
| frequency.db          |                                            |
| vectors/...           | ← the HNSW embedding index                 |
| books/...             | ← your prose, as .typ files                |
| .session.json         | ← last-open paragraph + cursor (optional)  |
| .inkhaven-backup.json | ← the last-good-backup timestamp           |

Because the archive is a complete, portable tree, you can drag it to another machine, restore it, and the project comes back whole. The one thing that does not travel inside it is the embedding model itself — that lives in a per-user cache, and the first launch on a new machine re-downloads it (chapter 1).

**Where backups live by default**

You can always name a destination with `--out`. When you omit it, the location depends on `backup.out_dir` in your config: an **empty** value (the default) writes to a sibling of the project, `<parent>/inkhaven-backups/<project-name>/`, keeping archives out of the project tree so a backup never contains itself; a **relative** value is resolved against the project root; an **absolute** value is used verbatim. The manual `backup` command and the on-exit hook write to the same resolved place.

**A NOTE ON THE FILENAME**

The stamp is `YYYYDDMM_HHMMSS` — year, **day**, month — which does not sort chronologically by name across a month boundary. The on-disk modification time always does, so sort by `mtime (ls -t)` when you want the newest, and do not trust the filename’s lexical order alone.

**Keeping the backup directory from growing forever**

By default archives accumulate — every backup is kept, which is the safe choice but not always the tidy one. Set `backup.keep_last` to a positive number and each new backup prunes the oldest beyond that count, so a `keep_last: 10` holds a rolling window of the ten most recent. `0` (the default) keeps everything. The prune runs after a successful write, never before, so a failed backup can never delete a good one.

**FICTION**

Run a manual backup before anything irreversible — deleting a book, switching the embedding model, a big reorganisation — so the pre-change state is one restore away.

**NON-FICTION**

Point `--out` at a directory under version control (git-lfs handles the binary blobs) and you get an offsite, dated history of the whole corpus for free.

**Auto-backup on exit — the safety net for forgetting**

Manual backups depend on remembering to take them, so Inkhaven takes one **for** you when it notices the last one is stale. On every clean exit the TUI compares `.inkhaven-backup.json` against `backup.max_age`. If the last successful backup is older than that window, the exit sequence drops the App (which flushes the DuckDB checkpoints and the vector-index WAL), renders a small splash with a live progress bar, streams every project file into a fresh dated zip, updates the timestamp marker, and only then tears down the terminal and returns you to your shell.

● ● ● *The auto-backup splash on exit*

```
Inkhaven · backup
Performing database backup...
Project: /home/you/Books/my-novel
[████████....] 321/512 ( 63%)
```

The hook is a **net**, not a gate. If anything goes wrong mid-backup, the error is written to `.inkhaven.log` and you are still returned to the shell — a failed safety backup never traps you inside the editor. By default the splash waits for a keypress after it finishes (`backup.wait_for_key_after_backup`, default `true`) so you can read the result before it dismisses; set it `false` for the old auto-dismiss behaviour.

### Turning it off, and choosing the window

There are three ways to disable the exit hook. The explicit one is `backup.auto_backup_on_exit: false`, which turns off just the hook and leaves the manual command working. The two older switches still work as side effects: an empty `out_dir` or a `max_age` of `"0s"` also disable it. The window itself is a `humantime` string, so pick it by how fast you write:

|                            |                                                                |
|----------------------------|----------------------------------------------------------------|
| <code>"24h" / "12h"</code> | A daily writer — never more than a day at risk.                |
| <code>"7d"</code>          | The default — a comfortable weekly cadence.                    |
| <code>"30d"</code>         | Sporadic long-form, if you also keep a separate archive habit. |
| <code>"0s" / ""</code>     | Disable the exit hook (manual backup still works).             |

Remember that the exit backup is for project-level disaster recovery; the paragraph snapshots (F5) are your in-session versioning. They are different tools for different scales of loss, and you want both.

## Restoring from a backup

Restoring unpacks an archive into a **fresh** directory. The command takes the zip and a `--to` destination.

● ● ● *Restoring an archive into a new directory*

```
$ inkhaven restore ~/Backups/blackinkhaven_20260805_143010.zip \
  --to ~/Books/my-novel-restored
restored backup `...143010.zip` into ~/Books/my-novel-restored
```

Two guards protect you from mistakes. First, the archive must contain `inkhaven.hjson` at its root — the marker that says “this really is an Inkhaven backup” — or the restore aborts, so you cannot accidentally explode an unrelated zip over your work. Second, it refuses if `--to` already holds a project (its own `inkhaven.hjson`); pick a clean directory, or clear the old one first. The destination is created if it does not exist. The restored project is fully independent of the source: the UUIDs match, but the two share no state from that point on, which is exactly what makes restore double as a **branch** tool — restore the same archive twice into two directories and evolve them separately.

### Restoring in place, and rebuilding from prose alone

To overwrite the same directory there is no in-place flag by design; you do it by hand, taking a safety backup first, removing the old tree, and restoring onto it. And if the database is gone but the `books/` tree survived, you can rebuild from the `.typ` files alone — a fresh `init`, then a `reindex --adopt` that registers every orphaned file under the matching branch of the hierarchy (this is the next section’s `--adopt`, doing recovery duty).

● ● ● *Rebuilding a project from surviving .typ files*

```
$ mkdir ~/Books/rebuild
$ cp -r ~/Books/my-novel/books ~/Books/rebuild/books
$ inkhaven init --force ~/Books/rebuild
$ inkhaven --project ~/Books/rebuild reindex --adopt
```

This assigns fresh UUIDs — it is a new project, not the old one — but your prose and its chapter structure come back from the directory layout.

## Crash recovery — `inkhaven recover`

The backup answers “the directory is gone.” Crash recovery answers the smaller, commoner disaster: the process died — a panic, a lost power cable, a `kill -9` — between two keystrokes, with unsaved edits in a buffer. Chapter 6 covers the machinery in full; here is the maintenance-side summary.

While you edit, a background task mirrors every **dirty** buffer to disk on a short cadence (`editor.crash_mirror_seconds`, two seconds by default), so a sudden death can cost at most the last couple of seconds of typing. If Inkhaven panics, its handler writes a **crash report** (`inkhaven-crash-<UTC>.hjson`, an index of what was open and a manifest of rescued buffers) plus one **rescue file** per dirty paragraph (`<paragraph>.typ.inkhaven-rescue`, the actual flushed prose) — both written atomically, both silent on failure. See chapter 6 for exactly what the report does and does not record.

### Walking the rescued buffers

After a crash, reopen the project, then run `inkhaven recover` from a shell, pointing it at the report. It walks each rescued buffer, prints its size and the delta against what is currently on disk, and asks what to do — `y` to apply, `N` (the default) to skip, `d` to see a unified diff first and then re-ask.

● ● ● *inkhaven recover — the per-buffer prompt*

```
$ inkhaven recover ./inkhaven-crash-20260805T0321.hjson

Found 1 rescued buffer(s).

[1/1] books/ch2/the-quay.typ
  rescue : ...the-quay.typ.inkhaven-rescue (214 bytes)
  on-disk: books/ch2/the-quay.typ (188 bytes, delta +26)
  apply? [y/N/diff]: y
  applied.

Done: 1 applied, 0 skipped, 0 error(s).
```

Applying a rescue is careful about the copy it overwrites: before writing, it snapshots the current on-disk file as `<paragraph>.typ.pre-recover-<UTC>`, so the pre-rescue version is always one `mv` away. A buffer whose rescue is byte-for-byte identical to disk is reported as “no action needed” and left

untouched. `--yes` applies every rescue without prompting; `--keep` leaves the report and rescue files in place. Without `--keep`, a clean run moves them into `<project>/.inkhaven/recovered/` so your working tree is tidied. Two things worth knowing: `recover` **never opens the database** — it works purely on file paths, so it is safe to run while a fresh TUI session is open on the same project — and any buffer that fails to apply leaves the whole report in place and exits non-zero, so a `recover && rm report` habit can never silently discard unrecovered work.

#### PATH SAFETY

A crash report is just an HJSON file, so `recover` treats it as untrusted: every rescued path is resolved **within** the project root, and any `..` traversal is rejected rather than followed. A hand-crafted report cannot make `recover --yes` write outside the project.

## Reindex — reconciling the store with the disk

The database and the `.typ` files on disk are meant to agree. Most of the time they do, because the editor keeps them in sync on every save. They can drift when something changes files **outside** the TUI — you edited a paragraph in another editor, a `git checkout` brought back files the database had forgotten, you moved or deleted files in `books/` from a shell, or you switched the embedding model. `reindex` is the command that makes them agree again.

#### ● ● ● The three reindex modes

```
$ inkhaven --project ~/Books/my-novel reindex
reindex: 3 updated, 511 unchanged, 0 missing, 0 orphan(s)

$ inkhaven --project ~/Books/my-novel reindex --prune --adopt
reindex: 0 updated, 514 unchanged, 1 missing, 2 orphan(s)
  pruned 1 missing record(s) from the store
  adopted 2 orphan .typ file(s) into the hierarchy
```

Plain `reindex` re-reads every `.typ` file the database knows about; where the file's bytes differ from the stored content it updates the record and re-embeds the paragraph, leaving unchanged ones alone. That bare form is what you run after

switching `embeddings.model` (to re-embed the whole book with the new model), or after a restore, or simply as an “are my files and database aligned?” check. The two flags handle the drift the bare form only **reports**:

- prune** Remove store records whose `.typ` file is missing from disk — after deleting files or folders outside the TUI.
- adopt** Register `.typ` files on disk the store doesn't know about, under the deepest branch whose path matches their parent folder — after dropping new files into books/.

Run bare, `reindex` prints how many records point at missing files and lists any orphan `.typ` files, then tells you to re-run with `--prune` or `--adopt` to act on them; nothing is deleted or created until you ask. You can combine the flags — `--prune --adopt` does both passes — and the command is idempotent, so running it twice is harmless.

#### REINDEX VS. DOCTOR

`reindex` reconciles **content** — file bytes against stored bytes, plus add/remove of whole records. The **doctor** (next) inspects **structure** — broken parent links, dangling references, zero-byte files, corrupt sidecars. Reach for `reindex` after external file edits; reach for `doctor` when you suspect the store itself is bent.

## The project doctor

The doctor is Inkhaven’s health check. In its plain form, `inkhaven doctor` prints a `brew doctor`-style report — the binary version and toolchain, the Typst engine summary and whether an external `typst` is on `PATH`, the package cache path and size, the project’s shape and word count, and a **Notes** section flagging things worth acting on (engine set to external with no `typst` installed, both font sources disabled, and so on). That informational dump is useful but not the interesting part. The interesting part is the **scan**.

### Scanning for trouble — `doctor --scan`

`inkhaven doctor --scan` walks the hierarchy and the on-disk tree and emits structured **findings**, each carrying a **class**, a **severity**, an optional **path**, and a one-



line detail. `--json` emits the same report as JSON (it implies `--scan`), which is what makes the doctor a CI gate.

#### ● ● ● *A doctor scan with findings*

```
$ inkhaven doctor --scan
Project scan
  generated_at   : 2026-08-05T14:31:09Z
  project_root   : /home/you/Books/my-novel
  findings       : 2

[1] critical · broken-parent-ref          · ch3/lost-scene
    `Lost Scene` (id ...) has parent_id ..., which
    doesn't exist — the node is detached from the tree
[2] warning  · sibling-slug-collision      · the-quay
    2 siblings under `Chapter 2` share slug `the-quay`
    (The Quay, The Quay) — their files collide; rename one
```

The scan exits `0` when clean, and `2` when any finding lands at **Warning** or above — the conventional linter exit code — so a CI job can fail the build on a structural regression:

#### ● ● ● *Gating CI on a clean project*

```
$ inkhaven doctor --json | jq -e '.findings == []'
```

## The finding classes

The classes fall into three families. The first is **data integrity** — the ones that mean prose is at risk or already lost, and the ones the doctor can actually repair:

|                                |                                                                                                |
|--------------------------------|------------------------------------------------------------------------------------------------|
| <b>zero-byte-file</b>          | A .typ file is 0 bytes and the store has no content either — prose lost. CRITICAL.             |
| <b>orphan-paragraph-row</b>    | A store row points at a file that isn't on disk, and the store has no content for it. WARNING. |
| <b>missing-referenced-file</b> | As above, but the row's path is malformed (empty or contains ..). WARNING.                     |
| <b>bdslib-only</b>             | The disk file is missing or 0 bytes but the store still holds the prose — recoverable. INFO.   |

**corrupt-comments-sidecar** A paragraph's `.comments.json` doesn't parse — comments unreadable. WARNING.

The second family is **referential integrity** — the structural links inside the hierarchy that a partial delete or a botched restore can bend. These are detection-only today (no autofix; you resolve them by hand):

**broken-parent-ref** A node's `parent_id` points at a UUID not in the tree — the node is detached and invisible. CRITICAL.

**dangling-paragraph-link** A paragraph links to another that was deleted — the link dangles. WARNING.

**dangling-event-ref** A timeline event references a character or place that no longer exists. WARNING.

**sibling-slug-collision** Two children of one parent share a slug, so their files collide on save. WARNING.

**duplicate-system-book** Two Books carry the same system tag, making system-book lookups ambiguous. WARNING.

The third family is **author judgment** — plot- and prose-level observations (dropped character, pacing collapse, stalled thread, naming inconsistency, echo repetition, numeric contradiction, continuity drift, paragraph too long, stale submission). These are all **Info**, never carry an autofix, and belong properly to the Editorial Pass (`inkhaven edit`, chapter 27) rather than to project integrity — only you can decide whether a dropped character was deliberate. One class, `unresolved-tension`, is **opt-in**: it never runs on a plain `--scan` and only appears when you name it with `--class unresolved-tension`, because its tagging is approximate and an open thread may be a deliberate hook. You can scope any scan to a single class with `--class <name>`.

### Repairing — doctor `--autofix`

`--autofix` applies the repairs the doctor knows how to make. It prompts `y/N` per finding (add `--yes` to accept every one non-interactively), applies the fix, and logs each outcome.

● ● ● Autofix, one finding at a time

```
$ inkhaven doctor --scan --autofix
Autofix – applying repairs.

[1/1] warning · corrupt-comments-sidecar
      path: books/ch2/the-quay.typ.comments.json
      comments sidecar for `the-quay` doesn't parse as JSON
      apply repair? [y/N]: y
      applied: moved corrupt sidecar ... → ...corrupt-20260805T143210.bak

Autofix done: 1 applied, 0 skipped, 0 error(s).
```

The repairs are conservative and specific. A zero-byte, orphan, or missing-referenced finding is fixed by **deleting the dead row and file** (there is nothing left to save). A corrupt sidecar is **renamed** to a timestamped `.bak` rather than deleted, so you can inspect it. A `bdslib-only` finding is **rematerialised** — the store’s content is written back to the missing disk file, atomically, and only after re-checking that no real file has appeared in the meantime, so a concurrent save is never clobbered. Author-judgment and referential-integrity classes return “no autofix” and point you at the prose or the structure to fix by hand.

### The doctor panel — Ctrl+B Shift+0

Everything above has an in-editor twin. `Ctrl+B Shift+0` opens the **doctor panel** from any pane: it runs the same project scan synchronously on open and shows every finding in a modal table with its class, severity, path, and detail. Inside, `↑↓` move the selection, `r` repairs the highlighted finding, `R` repairs every finding it can, and `Esc` closes. A repaired finding drops out of the table so you watch the list shrink as you go; a finding with no autofix reports why on the status line and stays. Every repair — from the panel or the CLI — is appended to `<project>/inkhaven/doctor.log` as a timestamped `UTC | outcome | class | detail` line, so there is an audit trail of what the doctor touched.

● ● ● *The doctor panel — Ctrl+B Shift+0*

```

Project doctor · 2 finding(s)
● critical broken-parent-ref      ch3/lost-scene
  `Lost Scene` detached — parent_id doesn't exist
△ warning corrupt-comments...    ...the-quay.typ
  comments sidecar for `the-quay` won't parse

↑↓ select  r repair  R repair-all  Esc close

```

The chord lives on the digit-0 row deliberately: **Ctrl+B 0** opens the HJSON config editor and **Ctrl+B Shift+0** opens the doctor — the “inspect the system” cluster, paired in your muscle memory.

#### TWO DOCTORS, ONE WORD

Do not confuse the **project doctor** here (Ctrl+B Shift+0, inkhaven doctor) with the **thread doctor** of chapter 20 (Shift+D in the thread view) or the **Editorial Pass** of chapter 27 (inkhaven edit). This doctor is about **project integrity** — structure and files. Those are about the manuscript’s craft. They share findings at the edges (the author-judgment classes) but the jobs are different.

## The advisory project lock

Two TUI sessions open on the same project both write `metadata.db` and `.session.json`; interleaved writes can corrupt the store. Inkhaven guards against that with an **advisory** lock — and the word advisory is the whole design. In keeping with the tool’s permissive principle, the lock **informs**; it never hard-blocks a writer who has decided to proceed.

**TERM Advisory lock**

On launch, Inkhaven takes an OS-level advisory lock (`flock` / `LockFileEx`) on `<project>/inkhaven.lock`. If a live session already holds it, the launcher **warns** and — by default — asks whether to open anyway. The kernel releases the lock automatically when the holding process exits, **including on a crash or kill -9**, so there is no stale-lock cleanup to get wrong: a dead session never locks you out.

When a second session finds the project busy, the default behaviour prints who holds it (a friendly `PID ... since HH:MM`, read from the lockfile purely to make the message informative) and prompts:

● ● ● *The 'open anyway?' prompt*

```
△ Another inkhaven session may already have this project
  open (PID 4321 on mac since 14:02).
  Opening it twice at once can corrupt the project store.
  Open anyway? [y/N]
```

Answer `y` and you proceed **without** the lock, minding your own concurrent edits; answer anything else and the launch is cancelled, leaving the existing session untouched. The behaviour is set by `project_lock.on_conflict`: `"prompt"` (the default) asks as above and warns-then-proceeds when launched non-interactively; `"warn"` always warns and proceeds without asking; `"refuse"` never opens a second session. Setting `project_lock.enabled: false` disables the guard entirely. The same policy governs store-mutating **CLI** commands (a `reindex` or a `doctor --autofix` run while a TUI holds the project): they consult the lock, warn if it is busy, and proceed unless `on_conflict` is `refuse` — so concurrent writers serialise rather than race. A filesystem that does not support advisory locks (some network mounts) degrades to opening **unlocked** rather than locking you out. The lockfile itself is left in place afterward as a harmless, empty anchor reused on the next launch.

## Logs, drift, and first-run trouble

When something behaves oddly, the log is the first place to look. While the TUI runs, the `tracing` subscriber appends to `<project-root>/.inkhaven.log` (plain, no colour); CLI commands log to `stderr` instead, so they never disturb the TUI’s alternate-screen rendering. What lands there: provider warnings from the AI layer (some, like an empty streaming chunk from certain providers, are normal and ignorable), notable store operations, and the non-fatal background errors — a failed auto-backup, a focus-loss save that could not write — that would otherwise be invisible.

● ● ● *Reading and raising the log*

```
$ tail -n 200 ~/Books/my-novel/.inkhaven.log

# turn the log up to debug for one run:
$ RUST_LOG=inkhaven=debug,bdslib=info \
  inkhaven --project ~/Books/my-novel
```

The TUI honours a standard `RUST_LOG`-style filter through tracing’s `EnvFilter`, so you can scope the verbosity to the module you are chasing.

### Common drift and how to clear it

Most “the tool and the disk disagree” symptoms have the same small set of cures, and nearly all of them are a `reindex`:

**Tree and disk disagree**    `reindex --prune --adopt` — reconcile both directions at once.

**Editor shows stale content**    `reindex` — a previous session crashed without flushing; re-read the `.typ` files.

**Search looks wrong after a model switch**    `reindex` — re-embed every paragraph with the new model.

**Structure bent**    `doctor --scan` — inspect the hierarchy’s referential integrity.

**A database file is corrupt**    restore a backup, or rebuild from books/ with `init --force + reindex --adopt`.

### First-run model download

The one piece of setup that reaches the network is the embedding model: the first time you open a project, `fastembed` downloads the chosen model into a per-user cache (its location per OS is in chapter 1), and every later project reuses it. If that download hangs — a slow connection, a server hiccup — the startup splash shows the elapsed time, and `Ctrl+Q` aborts so you can retry later. For an air-gapped install, pre-seed the cache from another machine. Switching `embeddings.model` later downloads the new model on the next open and leaves the old one on disk until you clear the cache directory yourself. Chapter 1 has the full first-run walkthrough.

## The configuration that governs all this

Four small config blocks tune everything in this chapter. Here they are together, at their defaults, with the fields that matter for maintenance.

### ● ● ● The maintenance-related config blocks

```
backup: {
  out_dir: ""           // ""=sibling inkhaven-backups/; rel=under
                        // project; abs=verbatim
  max_age: "7d"         // exit-hook staleness window; "0s"/"" off
  auto_backup_on_exit: true
  wait_for_key_after_backup: true
  keep_last: 0          // 0=keep all; N=rolling window of N
  amber_threshold: 0.5  // status chip goes amber at this
                        // fraction of max_age
}

project_lock: {
  enabled: true
  on_conflict: "prompt" // "prompt" | "warn" | "refuse"
}

health: {
  enabled: false        // background health monitor (opt-in)
  auto_repair: {
    rescue_orphans: false // delete *.inkhaven-rescue files >30d old
  }
}
```

A few notes on the less obvious ones. `backup.amber_threshold` controls when the optional status-bar freshness chip turns amber (a soft “plan a backup soon” heads-up) before it turns to the hard warning past `max_age`; `0.0` disables the amber stage. The `health` block is an **opt-in** background monitor, off by default so no existing project inherits a new background task silently — and even with it on, every auto-repair is individually off until you enable it, so turning on the monitor never grants it permission to mutate your project. Its per-check cadences (a project scan every 90s, a backup-freshness check every 300s, a rescue-orphan sweep hourly) are tuned in code rather than exposed as HJSON. The only auto-repair that exists today, `rescue_orphans`, cleans up stray `.inkhaven-rescue` files older than thirty days — the debris of crashes you already recovered from.



## WHAT YOU LEARNED

- `inkhaven backup` writes a dated, filesystem-level `.zip` of the **whole** project — databases, vectors, and prose — that `restore` reproduces exactly; the **exit hook** takes one automatically when the last is older than `backup.max_age`, and `keep_last` bounds how many are retained.
- `restore` unpacks into a **fresh** directory (refusing to clobber a project or unpack a non-Inkhaven zip); the same command doubles as a **branch** tool, and a lost database can be rebuilt from surviving `books/` with `init --force + reindex --adopt`.
- `inkhaven recover` walks a crash report’s rescued buffers (`y/N/diff`, `--yes`, `--keep`), snapshots each file as `.pre-recover-<UTC>` before overwriting, never opens the database, and refuses path traversal — the crash mirror itself is chapter 6.
- `reindex` reconciles **content**: bare re-reads and re-embeds changed files; `--prune` drops records for missing files; `--adopt` registers orphan `.typ` files under their matching branch.
- The **doctor** inspects **structure**: `doctor --scan (--json, --class, exit 2 on Warning+)` reports data-integrity, referential-integrity, and author-judgment classes; `--autofix` repairs the safe ones; `Ctrl+B Shift+0` is the in-editor panel (`r/R` repair, logged to `.inkhaven/doctor.log`).
- The **advisory lock** (`.inkhaven.lock`) informs but never blocks — it warns and asks “open anyway?”, the kernel releases it even on a crash, and `project_lock.on_conflict` chooses `prompt / warn / refuse`.
- `.inkhaven.log` (raise it with `RUST_LOG`) is the first stop for drift; most disk/store disagreements clear with a `reindex`, and first-run model-download trouble is covered in chapter 1.

## CHAPTER 29

# 29

## Configuration

---

Every Inkhaven project keeps its settings in one plain-text file at its root: `inkhaven.json`. It is the single place you tell the tool which language you write in, which AI provider to reach for, what colours to paint, how often to back up, and a hundred smaller things. This chapter is the guided tour of that file — what it is, how Inkhaven reads it, how to change it without breaking it, and the handful of blocks you are actually likely to touch. It is deliberately **not** the exhaustive list: every field, its type, and its default lives in **Appendix C**, and this chapter points you there again and again. Read this for the shape of the thing; reach for the appendix when you need a specific knob.

The reassuring news, before any detail: you can write an entire book and never open this file. Inkhaven writes a complete, working `inkhaven.hjson` when you first create a project, and every value in it has a sensible default. The configuration is there for when you want to bend the tool to your hand — not a gate you must pass through first.

## What `inkhaven.hjson` is

When you run `inkhaven init`, the tool writes a full, annotated configuration file into the new project's root, copied verbatim from a template baked into the binary. That file is yours from then on — Inkhaven reads it but, with one narrow exception we will come to, never rewrites it behind your back. It sits beside your manuscript in the project tree, so it travels with the book: clone the folder, and the settings come along.

### TERM **HJSON**

Inkhaven's config is written in HJSON — a relaxed superset of JSON meant for humans to edit. It is JSON with the sharp edges filed off: comments are allowed, keys need no quotes, commas are optional, and strings can span many lines. Any strict JSON is already valid HJSON, so nothing you know about JSON stops working.

The relaxations matter because you **will** read this file more than you write it, and comments are how it explains itself. Here is a small but real slice:

● ● ● *inkhaven.hjson — the HJSON dialect at a glance*

```
// inkhaven.hjson — the whole project's configuration.
// Lines beginning with // are comments, ignored on read.
{
  language: english           // keys and values need no quotes
  genre:    literary_realism

  goals: {
    daily_words: 1500         // trailing commas are optional
    books: {
      story: { target_words: 80000 }
    }
  }

  // A multiline value uses triple-quote fences:
  note:
    ...
    Everything between the fences is kept verbatim,
    line breaks and all — handy for a long template.
    ...
}
```

Three habits carry you through almost every edit. Comment a line out with `//` rather than deleting it, so you can put it back. Leave keys and simple values unquoted; quote only when a value contains characters that would confuse the parser (a leading `#`, a stray colon) or when you want a string to stay exactly as typed — durations like `"7d"` and hex colours like `"#1e1e2e"` are quoted for that reason. And group related settings inside `{ ... }` blocks, exactly as the shipped file does.

## How Inkhaven reads it: the layered config

Configuration is read **once**, at launch, and then handed — as an immutable, cloned value — to every subsystem that needs it: the editor, the AI client, the theme renderer, the backup hook. Nothing re-reads the file mid-session, which is why a change only takes effect on the next launch. But “the file” is a small simplification. What Inkhaven actually assembles, through `Config::load_layered`, is a **stack** of sources, lowest priority to highest:

● ● ● *Precedence — each layer overrides the one above it*

```
built-in defaults      (compiled into the binary)
+-- project inkhaven.hjson      (the full config)
   +-- ~/.config/inkhaven/config.hjson      (partial)
       +-- ~/.config/inkhaven/conf/*.hjson  (partial,
                                           sorted by name; last wins)
```

Read the ladder from the top down. First come the **compiled-in defaults** — every field the tool knows about, already filled in. On top of those, your project's `inkhaven.hjson` is merged, key by key. Then, if they exist, your **user-global** override files are layered on last, so a global file wins over the project. That ordering is deliberate: because `inkhaven init` writes a complete project config, a project-wins rule would bury any personal preference you set globally. The global files are **partial** — you put in only the keys you want to change everywhere, and everything else falls through to the project. It is the way to carry one theme, or one keybinding, across every book you write without editing each project by hand.

Two properties of this merge are worth holding onto. First, **missing fields never break anything**. Every field is optional and falls back to its default, so a config written by an older release keeps working when a new release adds fields, and a partial global file is exactly that — partial. A typo'd key is simply ignored: harmless, but also silently ineffective, so if a setting seems not to take, check the spelling against Appendix C. Second, **a broken global file is skipped, not fatal**. A malformed override under `~/.config/inkhaven/` is logged as a warning and passed over — one stray brace there must never brick every project you own. A malformed **project** `inkhaven.hjson`, by contrast, is a hard error that stops the launch with a clear message, because that file is load-bearing for the one project in front of you.

## THE SECURITY FLOOR

The merge has one carve-out. Object blocks merge key by key, but a **list** replaces the list beneath it wholesale — so setting `shell.blocked_externals` to add one program would otherwise wipe the shipped block-list that keeps `vim`, `ssh`, `sudo` and friends out of the embedded shell. After every load and merge, Inkhaven unions those shipped entries back in. You can always **add** to the block-list; you cannot accidentally erase its floor. It is the one place the otherwise-permissive config refuses to let you shoot yourself in the foot.

## Editing it safely

There are two honest ways to change the file, and Inkhaven supports both.

### From inside the editor — **Ctrl+B 0**

The chord **Ctrl+B 0** opens `inkhaven.hjson` in a full-screen modal editor, without ever leaving the app. It is a real text editor over the raw file — HJSON syntax highlighting, and the same movement and editing chords the manuscript editor uses (arrows, **Home** / **End**, word jumps, **Ctrl+U** undo, **Ctrl+Y** redo, selection with **Shift**, and so on). Because you are editing the file's actual text, your comments and formatting are preserved by definition — nothing is re-serialised.

● ● ● *Ctrl+B 0 — editing the config in place*

```
+ - Edit . inkhaven.hjson -----+
| 1 {                               |
| 2   language: english             |
| 3   llm: { default: gemini }      |
| 4   backup: { max_age: "7d" }     |
| 5 }                               |
|                                   |
| Ctrl+S save   Ctrl+R review   Esc close |
+-----+
```

**Ctrl+S** saves. If the saved bytes differ from what was loaded, a **Restart required** overlay appears, because — as we saw — the config is read only at launch. Your edit is on disk; it simply applies the next time you start Inkhaven.

● ● ● *The restart-required overlay after a real change*

```
+ - Restart required -----+
| Saved. Configuration is read once, at |
| launch. Restart inkhaven to apply.   |
|           [ press any key ]         |
+-----+
```

Two more keys earn their place in this modal. `Ctrl+R` fires an **LLM review** of the buffer you are looking at — the same “reviewer model, never an executor” pattern the prompts editor uses. The model reads your config and streams back notes (a suspicious value, a key that looks misspelled, a block that contradicts another); the response lands in the AI pane, visible once you close the modal. It advises, it never writes. And `Esc` closes the editor, warning you on the status line if you have unsaved edits.

**HOW FEATURES WRITE CONFIG FOR YOU**

A few features edit the config on your behalf — the goals editor, and CLI helpers like `show-dont-tell bootstrap --update`. They do not re-serialise the whole file (that would flatten your comments). Instead they perform a **surgical splice**: each targeted leaf is patched in place if it already exists, or appended inside its parent block if it does not, leaving every comment and every untouched key exactly where it was. Before the write, the pre-patch file is copied to `<project>/ .config-backups/`, and the new file is written atomically — so an interrupted patch can never truncate your config, and a roll-back is one `cp` away.

**By hand, in any editor**

Because it is plain text, you can also open `inkhaven.hjson` in whatever editor you like — including Inkhaven’s own embedded shell, or the manuscript editor if you add the file to the project. Two safety rails are worth knowing. You can **validate** a config without launching the TUI by running any read-only command against the project and discarding its output; if the file is malformed the CLI prints a **config error: ...** and exits non-zero. And if you ever want to start over, rename the file and run `inkhaven init --force` to lay down a fresh default template, then re-merge your customisations from the old copy. Inkhaven never edits the file in place except through the surgical splice above, so what you write is what stays.

## A tour of the blocks you will touch

The full file has dozens of blocks; most you will never open. What follows is the short list of the ones that reward a visit, each with the one or two knobs that matter and a pointer to where the feature itself is documented. For the complete field list, types, and defaults of any block, see **Appendix C**.

### language and genre — who the book is for

Two top-level fields set the project’s identity. `language` (default `english`) is the primary writing language; it drives the Snowball stemmers behind the Places and Characters highlight overlays, the default grammar-check prompt, and the language of every style word-list. It accepts the Snowball set — `english`, `russian`, `french`, `german`, `spanish`, and more. `genre` (default unset, i.e. genre-blind) names the book’s genre, e.g. `literary_realism` or `fantasy`; the Inner Editor and other readers use it to tune their prompts. Change `language` and restart, and the whole tool speaks your language. See the chapters on the editor overlays and the inner readers; full list in Appendix C.

### llm — which AI answers

The `llm` block lists your AI providers and picks a default. Out of the box it ships five ready entries — `gemini` (the default), `claude`, `openai`, `deepseek`, and `grok` — each naming a `model` and the environment variable that holds its API key (`api_key_env`). Switching the default AI for the whole project is a one-word edit: set `default: claude` and restart. Local providers like Ollama need no key — omit `api_key_env` entirely. If a provider’s key variable is unset at runtime, Inkhaven declines cleanly rather than crashing, and by default (`auto_fallback: true`) it will reach for any other provider whose key is present. The AI chapters in Part III cover scopes, chat, and cost; the per-field reference is in Appendix C.

### editor — how the writing surface behaves

The `editor` block governs the manuscript pane’s behaviour (its `colours` live in `theme`). The knobs you are most likely to touch: `wrap` (default `true`, soft word-wrap), `autosave_seconds` (default `5` — idle seconds before a dirty paragraph saves; `0` disables idle autosave, though save-on-switch and save-on-quit still fire), `tab_width` (default `2`), and `auto_close_pairs` (default `true`). Nested under it is `style_warnings` — the inline overlays that underline weak prose. The whole `style_warnings`.

family is **off by default** (`enabled: false`); turn it on and the sub-detectors (filter words, repeated phrases, show-don’t-tell, anachronism) light up, each indi-



vidually tunable and each keyed to your project `language`. The Writing chapters in Part II cover the editor and these overlays end to end; Appendix C has every sub-knob.

### theme — every colour on screen

The `theme` block sets every colour the TUI paints — pane backgrounds and foregrounds, the five border states, modal windows, the syntax highlighter, the lexicon overlays. The shipped defaults are a Catppuccin Mocha dark palette, tested across many terminals. Values are hex strings (`"#RRGGBB"` or the short `"#RGB"`); an **empty string or an unparseable value** falls back to the baked-in default, which is exactly what makes a partial global theme override work — you name only the handful of colours you want to change. This is the block most people put in a user-global file so one palette follows them across every book. The panes chapter shows what each colour paints; the full colour table is Appendix C.

### backup — the safety net

The `backup` block drives both the `inkhaven backup` command and the auto-backup that fires when you quit. `out_dir` (default empty, meaning a project-sibling `inkhaven-backups/<project-name>/` directory next to your project) is where `.zip` snapshots land; `max_age` (default `"7d"`, humantime-parsed, so `"24h"` or `"12h"` also work) is how stale the last backup may get before the exit hook makes a fresh one; `keep_last` (default `0`, meaning keep all) prunes older archives; and `auto_backup_on_exit` (default `true`) is the clean on/off switch. The Backup chapter earlier in this part walks the whole cycle; the fields are in Appendix C.

### goals — writing targets and pace

The `goals` block fuels the status-bar progress widget and the progress modal. Every field is optional and **zero or empty disables that particular goal while still recording history**, so the modal always has something to show. The common ones: `daily_words` (default `0`), `active_minutes_daily`, `streak_grace_per_week` (rest days forgiven per rolling week), and per-book `target_words` and `deadline` under `books.<slug>`. `auto_promote_on_target` (default `true`) nudges a paragraph's status up a rung when it crosses its word target. The writing-goals material covers the workflow; Appendix C lists the fields.

### ai — the AI pane's behaviour

Distinct from `llm` (which picks the **provider**), the `ai` block tunes how the AI pane **behaves**. The one most writers care about is `diff_review_on_apply` (default

`true`): every AI edit to your prose is routed through a side-by-side diff you must accept before a single byte changes — the tool’s standing promise that the AI never writes to your manuscript unasked. `per_paragraph_memory` (default `false`) opts a paragraph into remembering its own chat turns. Part III covers the AI pane; the block’s fields are in Appendix C.

### embeddings — the semantic index

The `embeddings` block controls how your prose is turned into vectors for semantic search. `model` (default `MultilingualE5Small`) picks the fastembed model — keep a multilingual one if you write in any non-English language. The rest (`chunk_size 800`, `chunk_overlap 0.15`, `pool_size 4`) rarely need a hand. Changing the model triggers a one-time download and a re-index of your prose; run `inkhaven reindex` after. The search chapter in Part II covers the index; Appendix C has the model list.

### keys — rebinding the chords

A short list of global chords is rebindable through `keys`: `save` (Ctrl+s), `search` (Ctrl+/,), `ai_prompt` (Ctrl+i), pane cycling, and — most usefully — the three **prefix** chords `meta_prefix` (Ctrl+b), `bund_prefix` (Ctrl+z), and `view_prefix` (Ctrl+v). If your terminal multiplexer eats Ctrl+B — `tmux` claims it by default — set `meta_prefix: Ctrl+g` and every Meta chord follows. Deeper sub-chord remapping lives in the `keys.bindings` overlay list. The Key-binding chapter and Appendix A carry the full chord map; the config fields are in Appendix C.

## The principle behind all of it

One idea runs under every block above, and it is worth naming because it explains the defaults. Inkhaven is a **permissive** tool: it informs, it does not block. Almost everything is off, or set to its safe default, until you opt in — the style overlays, per-paragraph memory, the health monitor, the timeline, the embedded shell’s riskier corners. Cost caps and budget settings **warn** you and colour a chip; they never refuse to run the thing you asked for. The one place the tool overrides you is the security floor on the shell block-list, and even there it only **adds** protection, never removes a choice. The config file is a set of dials you may turn, not a form you must fill in — which is exactly why you can write a whole book without opening it, and bend the tool to your hand the day you decide to.

## WHAT YOU LEARNED

- Every project keeps its settings in `inkhaven.hjson` at its root — written complete by `inkhaven init`, in the comment-friendly **HJSON** dialect, and read once at launch.
- Config is **layered**: compiled-in defaults, then the project file, then partial user-global overrides that win — so missing fields never break, and a broken global file is skipped while a broken project file is fatal.
- Edit it safely two ways: `Ctrl+B 0` opens a full-screen editor (`Ctrl+S` save, `Ctrl+R` LLM review, a **restart-required** overlay), or hand-edit any way you like; feature-driven edits use a comment-preserving surgical splice with a pre-patch backup.
- The blocks you will actually touch are `language / genre`, `llm`, `editor` (with `style_warnings`), `theme`, `backup`, `goals`, `ai`, `embeddings`, and `keys` — each covered in depth by its own chapter, with every field in **Appendix C**.
- The governing principle: **cost caps inform, never block**, and almost everything is off or default-safe until you opt in.

## APPENDIX A



## The Keybinding Map

---

Inkhaven is driven from the keyboard. Almost every action in the editor has a chord, and the ones that do not have a chord have a menu you reach with one. This appendix is the complete, printable map of that keyboard — every chord the TUI honours, organised by the pane or overlay that owns it. It is the reference you keep open beside the tutorials until the muscle memory sets; it is also, on a bad terminal day, the place you come to find out which key your multiplexer just ate.

The map mirrors the shipping `Documentation/KEYBINDING.md`, cross-checked against the binding tables in `src/tui/keybind.rs` — where the two ever disagree, the code wins, and this appendix follows the code. A short note at the very end lists the disagreements worth knowing about.

## The two prefixes, and how to read a chord

Most single keystrokes belong to whatever pane has focus: a letter typed in the editor is a letter, the same letter in the tree is a command. To reach the hundreds of commands that are *not* pane-native, Inkhaven uses **prefix chords** — you press a prefix, release it, and the *next* key selects the action.

There are three prefixes, all rebindable:

|               |                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+B</b> | The <b>meta</b> prefix — hierarchy, panels, the reading intelligences, translation. The second key's meaning depends on which pane has focus. |
| <b>Ctrl+V</b> | The <b>view</b> prefix — exporters, pickers, links, bookmarks, timeline, and the prose/craft tools.                                           |
| <b>Ctrl+Z</b> | The <b>Bund</b> prefix — the embedded scripting language and the OS shell pane.                                                               |

A chord is written the way you type it: **Ctrl+B Shift+I** means *hold Ctrl and press B, release both, then press Shift+I*. A plain **Shift+I** (no prefix in front) is a single keystroke. Case is not significant to the parser — **Ctrl+s**, **Ctrl+S** and **Ctrl+Shift+S** all match the same binding, because terminals disagree about how they report case with modifiers held.

When a prefix is pending, the status bar shows a yellow **META** / **VIEW** / **BUND** chip and lists the actions available for the focused pane. **Esc** cancels a pending prefix without running anything; an unrecognised second key cancels with a hint naming which pane's table it consulted.

● ● ● *The meta prefix is pane-specific*

Ctrl+B ... (press Ctrl+B, release, then one more key)

|              |                                                                                     |
|--------------|-------------------------------------------------------------------------------------|
| Tree focus   | c chapter · s subchapter · p paragraph<br>d delete · m morph-type · Up/Down reorder |
| Editor focus | s read-aloud · n snapshot · t retitle<br>q translate · c character · p place        |
| AI focus     | c clear chat                                                                        |
| Any focus    | h help · w world · j inner readers · Esc cancel                                     |

## REBINDING

Every prefix, and every sub-chord under it, is configurable. The three prefixes live in the `keys` block of `inkhaven.hjson`; the sub-chords are listed as overrides in `keys.bindings` or bound at runtime through the `ink.key.*` Bund words. The full mechanism is in Appendix C and the configuration chapter — this appendix documents the shipping defaults.

## Global chords

These work from any focus except where a narrower table overrides them. The *italic name in parentheses* is the configurable `keys.*` slot, where one exists.

|                          |                                                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Tab</b>               | Cycle focus Tree → Editor → the active right pane → Tree ( <i>next_pane</i> ).                                             |
| <b>Shift+Tab</b>         | Cycle focus in reverse ( <i>prev_pane</i> ).                                                                               |
| <b>Ctrl+/<br/>Ctrl+I</b> | Focus the top Search bar ( <i>search</i> ).                                                                                |
| <b>Ctrl+S</b>            | Focus the bottom AI prompt bar ( <i>ai_prompt</i> ).                                                                       |
| <b>Ctrl+S</b>            | Save the open paragraph and re-embed it ( <i>save</i> ); no-op with nothing open.                                          |
| <b>Ctrl+Q</b>            | Hard quit. Autosaves the open paragraph first; refuses to quit if that save fails, so the error stays on screen.           |
| <b>Ctrl+1</b>            | Focus the Editor pane.                                                                                                     |
| <b>Ctrl+2</b>            | Open the full-screen Outline pane. Use <code>Ctrl+B Shift+0</code> if your terminal re-encodes <code>Ctrl+2</code> as NUL. |
| <b>Ctrl+T</b>            | Focus the side Tree pane.                                                                                                  |
| <b>Ctrl+3</b>            | Focus the AI pane.                                                                                                         |
| <b>Ctrl+4</b>            | Focus the Search bar.                                                                                                      |
| <b>Ctrl+5</b>            | Focus the AI prompt bar.                                                                                                   |

|                     |                                                                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+V Space</b> | Open the command palette — fuzzy-find any command by name, chord, or description and run it ( <code>global.command_palette</code> ).               |
| <b>Ctrl+Shift+M</b> | Toggle mouse capture on/off. Off lets the terminal do native drag-select and clipboard; on restores click-to-focus and wheel-scroll. Session-only. |
| <b>Alt+←</b>        | Step backward through visited paragraphs.                                                                                                          |
| <b>Alt+→</b>        | Step forward again (after stepping back).                                                                                                          |

The function keys are global too, though a few only do their work with a paragraph open in the editor:

|                |                                                                                                                                                                                         |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>F1</b>      | Open the Help-manual query pane — a grounded RAG answer over the Help book streams into the AI pane. One-shot; not added to chat history.                                               |
| <b>F8</b>      | Open the Typst diagnostics list for the most-recent paragraph; <b>Enter</b> jumps the cursor to a diagnostic. Works from any pane.                                                      |
| <b>F9</b>      | Cycle the AI scope: None → Selection → Paragraph → Subchapter → Chapter → Book → Facts → Socrates → Editor → Graph → None. Sticky scopes persist; the rest auto-reset after one prompt. |
| <b>F10</b>     | Toggle inference mode Local ↔ Full. Local confines the model to the supplied context; Full lets it draw on general knowledge. Help stays pinned to Local.                               |
| <b>Ctrl+F6</b> | Open the project-wide snapshot browser — every snapshot across all paragraphs, newest first. Works from any pane.                                                                       |

## The Tree pane

The tree is focused on launch. It shows the hierarchy with kind glyphs (📖 book, ▶ chapter, ▶ subchapter, ¶ paragraph) and a dim word count on each paragraph. The row of the paragraph currently open in the editor is marked with a green bold ▶ regardless of focus.

### Navigation

↑ / ↓      Move the cursor one row up / down.

|                           |                                                                                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| →                         | Expand the cursor's branch, revealing its children. No-op on a paragraph or an already-open branch.                                                            |
| ←                         | Collapse the cursor's branch; if already collapsed (or on a paragraph), move to the parent.                                                                    |
| <b>Home / End</b>         | Jump to the first / last row.                                                                                                                                  |
| <b>PageUp / Page-Down</b> | Move the cursor 10 rows up / down ( <code>page_up</code> / <code>page_down</code> ).                                                                           |
| <b>Enter</b>              | Open the cursor's node. Paragraphs load into the editor and shift focus there, autosaving any previously-open dirty paragraph; branches print a hint and stay. |
| <b>Shift+Enter</b>        | Pin the focused paragraph into the split-view secondary pane instead of opening it primary.                                                                    |
| <b>F2</b>                 | Open the Rename modal for the cursor node (changes the displayed title only; slug and file stay).                                                              |
| <b>F3</b>                 | Open the file picker: <code>Enter</code> on a file imports it as a new paragraph; <code>Enter</code> on a directory recursively imports the tree.              |
| <b>q / Q</b>              | Quit (autosaves the open paragraph first if dirty).                                                                                                            |
| <b>Esc</b>                | Cycle focus to the Search bar.                                                                                                                                 |

### Modifier-free shortcuts

These plain-key commands work only when the Tree pane has focus. They exist alongside the `Ctrl+B` meta chords because terminals and multiplexers often intercept the prefix. Uppercase (implicit Shift) is accepted; a real `Ctrl` / `Alt` modifier is rejected, so `Ctrl+A` will *not* fire add-subchapter.

|              |                                                                          |
|--------------|--------------------------------------------------------------------------|
| <b>B / b</b> | Add a new book at the root, inserted above the system-book block.        |
| <b>C / c</b> | Append a chapter at the end of the book's children.                      |
| <b>V / v</b> | Insert a chapter immediately after the cursor's enclosing chapter.       |
| <b>A / a</b> | Append a subchapter at the end of the chapter's children.                |
| <b>S / s</b> | Insert a subchapter immediately after the cursor's enclosing subchapter. |



|              |                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| <b>+</b>     | Append a paragraph at the end of the parent’s children.                                                              |
| <b>P / p</b> | Insert a paragraph immediately after the cursor’s enclosing paragraph.                                               |
| <b>D / d</b> | Delete the cursor’s node — only if it is a branch (book/chapter/subchapter).                                         |
| <b>-</b>     | Delete the cursor’s node — only if it is a paragraph.                                                                |
| <b>U / u</b> | Move the node up — swap with the previous sibling.                                                                   |
| <b>J / j</b> | Move the node down — swap with the next sibling.                                                                     |
| <b>Z / z</b> | Collapse the cursor’s enclosing subchapter.                                                                          |
| <b>X / x</b> | Collapse every expanded branch in the tree.                                                                          |
| <b>Space</b> | Mark / unmark the cursor row for multi-select; <b>ESC</b> clears all marks.                                          |
| <b>T / t</b> | Cycle the leaf’s type: paragraph(typst) → paragraph(hjson) → paragraph(jinja) → script. Walks the marked set if any. |
| <b>E / e</b> | New Jinja template paragraph (rendered to Typst at assembly).                                                        |
| <b>I / i</b> | New structural paragraph — pick a subtype (code / admonition / math / procedure / table) with matching boilerplate.  |
| <b>O / o</b> | Cycle status one rung up the ladder (napkin → first → ... → ready → napkin). Walks the marked set if any.            |
| <b>G / g</b> | Tag the marked set (or the cursor row) — opens the tag picker scoped to every target.                                |
| <b>y</b>     | Copy the cursor paragraph onto the cross-pane clipboard (shared with the Outline pane).                              |
| <b>m</b>     | Move (cut) the cursor paragraph onto the clipboard.                                                                  |
| <b>f</b>     | Affix the clipboard paragraph as the last child of the cursor’s effective parent.                                    |
| <b>?</b>     | Open the pane-aware Quick reference overlay (Tree pane only).                                                        |

Why kind-specific delete? Safety: **-** will not nuke a whole chapter if your cursor slipped onto it, and **D** will not kill a paragraph you meant to keep. If you want a delete that ignores kind, use the global **Ctrl+B D**.

## The Editor pane

The backing widget is `tui-textarea` driven without its built-in shortcuts, so Inkhaven can intercept the modern conventions itself. The border colour carries the dirty state at a glance while the pane is focused: green when saved, yellow with unsaved edits, white when unfocused (dirtiness then moves to the title's `[modified]` suffix and the red  chip in the status bar).

### Cursor movement

|                           |                                  |
|---------------------------|----------------------------------|
| <b>← / →</b>              | One character left / right.      |
| <b>↑ / ↓</b>              | One line up / down.              |
| <b>Home / End</b>         | Start / end of the current line. |
| <b>PageUp / Page-Down</b> | One viewport up / down.          |
| <b>Ctrl+←</b>             | Previous word boundary.          |
| <b>Ctrl+→</b>             | Next word boundary.              |
| <b>Ctrl+Home</b>          | Top of the document.             |
| <b>Ctrl+End</b>           | Bottom of the document.          |

### Editing

|                       |                                                                     |
|-----------------------|---------------------------------------------------------------------|
| <b>any character</b>  | Insert at the cursor; replaces the selection if one exists.         |
| <b>Enter</b>          | Insert a newline.                                                   |
| <b>Backspace</b>      | Delete the character before the cursor (or the whole selection).    |
| <b>Delete</b>         | Delete the character at the cursor.                                 |
| <b>Ctrl+Backspace</b> | Delete the previous word.                                           |
| <b>Ctrl+S</b>         | Save to disk and re-embed; reloads the tree so word counts refresh. |

## Selection, clipboard, undo

The editor uses non-standard cut/paste keys because the conventional ones were claimed by other functions (`Ctrl+X` is search-repeat; `Ctrl+Z` is the Bund prefix). If `arboard` fails to initialise, copy/cut/paste fall back silently to an internal yank buffer that stays within the session.

**Shift+←** / **Shift+→** Extend the linear selection by one character.

**Shift+↑** / **Shift+↓** Extend the selection by one line.

**Ctrl+A** Select the entire document.

**Ctrl+C** Copy the selection to the system clipboard.

**Ctrl+K** Cut the selection to the clipboard; marks the document dirty.

**Ctrl+P** Paste from the clipboard at the cursor (or over the selection).

**Ctrl+U** Undo.

**Ctrl+Y** Redo.

## Rectangular block selection

A second, separate selection model, always rectangular, drawn reversed on top of the syntax highlighting. Rectangular *cut* and *paste* are deferred in this release; copy covers the common cases (a column of numbers, a list of names, a verse stanza).

**Alt+↑ ↓ ← →** Enter block-select mode if needed, then move the cursor one cell without touching the linear selection.

**Alt+C** Copy the rectangle to the system clipboard as a multi-line string; clears the anchor.

**Esc** Cancel block-select; keep the document open.

## Line-targeted deletes

Each of these saves and restores the yank buffer around the operation, so a following `Ctrl+P` paste still produces the last copy.

**Ctrl+D** Delete the current line and its trailing newline.

**Ctrl+E** Delete from the cursor to the end of the line.

**Ctrl+W** Delete from the cursor back to the start of the line. (Some terminals eat **Ctrl+W** as delete-word; rebind if yours does.)

### Find and replace (regex)

In-buffer regex search over Rust's **regex** syntax. Matches highlight red; the current match brightens.

**Ctrl+F** Open the Find modal — type a regex, **Enter** to run; the cursor jumps to the first match.

**Ctrl+X** Repeat — in search mode jump to the next match; in replace mode replace the current match and advance. Active only while a search is in progress.

**Ctrl+R** First press opens Find & Replace and applies the first replacement; a second press in replace mode replaces every remaining match.

**Ctrl+B** (inside the modal) Toggle scope between this paragraph and the whole book; book scope opens a per-match review modal.

**Esc** Clear the active search — drops the highlights and exits replace mode.

### Split-edit, snapshots, files

**F3** Open the file picker; **Enter** on a file replaces the open paragraph's buffer. Directories are rejected here.

**F4** Toggle split-edit: a read-only snapshot of the buffer, captured on enter, in the lower pane.

**Ctrl+F4** Accept the split snapshot — replace the live buffer with the captured copy, exit split, mark dirty.

**Ctrl+H** (split only) Scroll the lower snapshot pane up one line.

**Ctrl+J** (split only) Scroll the lower snapshot pane down one line.

**Shift+F4** Toggle the full-screen two-paragraph split-view; the right pane is the secondary slot filled by any picker's **Shift+Enter**.

**F5** Save a versioned snapshot of the open paragraph (opens an annotation prompt).

**F6** Open the snapshot picker: **Enter** loads (safety-snapshots first), **V** diffs against current, **D** removes, **/** filters by annotation.

|                 |                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------|
| <b>F2</b>       | Rename the open paragraph.                                                                          |
| <b>F7</b>       | Grammar-check the open paragraph; the review streams into the AI pane.                              |
| <b>F12</b>      | AI critique of the open paragraph — mode-aware (critique-edit, or evaluate-changes in split mode).  |
| <b>Ctrl+F12</b> | Send the Typst diagnostic at the cursor to the AI pane with an explain-or-fix prompt.               |
| <b>Esc</b>      | Defocus to the Tree without closing the document (clears a block selection first if one is active). |

#### AUTOSAVE

You never have to save to move. Whenever focus leaves the editor — via **Tab**, **Ctrl+1..5**, **Ctrl+T**, **Esc**, opening another paragraph — a dirty paragraph is saved automatically. An idle timer (`editor.autosave_seconds`, default 5) and quit both save too. Characters typed since the last save render bold until the next save clears the marker.

## The AI pane

Focus lands here when you bounce off the AI prompt with **Esc**. The action keys fire only once an inference is **Done** and non-empty; while streaming or on error only **q** and **Esc** respond.

|              |                                                                                            |
|--------------|--------------------------------------------------------------------------------------------|
| <b>Esc</b>   | Bounce focus back to the AI prompt bar.                                                    |
| <b>r / R</b> | Replace the editor selection (or the whole document) with the AI text; refocus the editor. |
| <b>i / I</b> | Insert the AI text at the cursor.                                                          |
| <b>t / T</b> | Prepend the AI text to the top of the paragraph.                                           |
| <b>b / B</b> | Append the AI text to the bottom of the paragraph.                                         |
| <b>c / C</b> | Copy the AI text to the system clipboard only — no editor change.                          |

**g / G** Grammar-apply — lift only the corrected paragraph from the response and overwrite the buffer; the changed characters stay highlighted across saves.

**q / Q** Quit.

Where a response comes from a generator (a submission draft, a structure analysis), an extra **L** files it as a paragraph into the matching system book.

### The Output / Thoughts pane

The right region cycles between three panes; content auto-surfaces the relevant one unless you are actively reading there.

**Ctrl+B Tab** Cycle the right region Output → AI → Thoughts.

**Ctrl+B Shift+Tab** Cycle the right region in reverse.

**Ctrl+Z f** Fullscreen the current right pane (Output / Thoughts; the AI pane uses **Ctrl+B K**).

**↑↓ (or k j)** Select the previous / next message.

**g / G** First / last message.

**o / Space** Expand / collapse the selected message's structured detail.

**a** Ask the AI about the selected message (carries its full detail).

**d** Dismiss the selected message.

**p** Pin / unpin (pinned messages sort to the top).

**Enter** Primary, kind-specific action — open target, accept proposal, jump to event.

**r / e** (translations) remember / edit-and-remember.

**i / m / x** (Socratic) record-as-intent / make-note / mark-addressed.

Output filtering — the title shows **shown/total · filter**, persisted in **.session.json**:

**f** Cycle the source filter: off → fact-check → socrates → timeline → world → translation → lexicon → variety → ai → bund → other → off.

|          |                                                                                 |
|----------|---------------------------------------------------------------------------------|
| <b>S</b> | Cycle the minimum severity: off → Info → Warning → Contradiction → off.         |
| <b>t</b> | Toggle this-paragraph-only — show just the messages tied to the open paragraph. |
| <b>c</b> | Clear all filters.                                                              |

## The AI prompt bar

Activated by **Ctrl+I**. A leading **/** opens the Prompt picker; a leading **Help!** routes the line through the F1 Help-RAG flow; anything else is sent verbatim as a streaming inference.

|                                   |                                                                                                                        |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>printable char</b>             | Insert at the cursor. A leading <b>/</b> opens the Prompt picker; otherwise it closes.                                 |
| <b>Backspace</b><br><b>Delete</b> | <b>/</b> Edit the buffer; refreshes the picker if visible.                                                             |
| <b>← / → / Home / End</b>         | Cursor movement within the buffer.                                                                                     |
| <b>↑ / ↓</b>                      | (picker open) Move the selection.                                                                                      |
| <b>Tab</b>                        | (picker open) Expand the selected template into the buffer with <b>{{selection}}</b> / <b>{{context}}</b> substituted. |
| <b>Enter</b>                      | Picker open: expand the template. Otherwise: spawn a streaming inference; focus stays on the prompt bar.               |
| <b>Esc</b>                        | Picker open: close it. Otherwise: bounce focus to the AI pane to read the answer.                                      |

## The Search bar

Activated by **Ctrl+/.** Semantic search over the whole project.

|                       |                                                                       |
|-----------------------|-----------------------------------------------------------------------|
| <b>printable char</b> | Insert at the cursor; closes the results overlay (the query changed). |
|-----------------------|-----------------------------------------------------------------------|

|                     |   |                                                                                        |
|---------------------|---|----------------------------------------------------------------------------------------|
| <b>Backspace</b>    | / | Edit the query; closes the results overlay.                                            |
| <b>Delete</b>       |   |                                                                                        |
| <b>← / → / Home</b> | / | Cursor movement within the query.                                                      |
| <b>End</b>          |   |                                                                                        |
| <b>↑ / ↓</b>        |   | (overlay open) Move the result cursor.                                                 |
| <b>Enter</b>        |   | Overlay open: open the highlighted result. Otherwise: run the search and show results. |
| <b>Esc</b>          |   | Overlay open: close it. Otherwise: cycle focus to the Editor.                          |

## The meta prefix — Ctrl+B

The largest table in the app, and pane-specific: **Ctrl+B** then a key means different things in the Tree, Editor and AI panes. The sub-tables below are in that order, followed by the chords that fire from *any* pane. **Esc** cancels a pending meta prefix.

### Ctrl+B in the Tree pane

|                 |                                                                                                                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+B C</b> | Add a chapter.                                                                                                                                                                                      |
| <b>Ctrl+B S</b> | Add a subchapter.                                                                                                                                                                                   |
| <b>Ctrl+B P</b> | Add a paragraph.                                                                                                                                                                                    |
| <b>Ctrl+B D</b> | Delete the cursor's node (with confirmation) — kind-agnostic, unlike the plain <b>D</b> / <b>-</b> .                                                                                                |
| <b>Ctrl+B M</b> | Morph the leaf's type: paragraph(typst) → paragraph(hjson) → paragraph(jinja) → script.                                                                                                             |
| <b>Ctrl+B ↑</b> | Reorder — swap with the previous sibling (also <b>Ctrl+B u</b> ).                                                                                                                                   |
| <b>Ctrl+B ↓</b> | Reorder — swap with the next sibling (also <b>Ctrl+B j</b> ).                                                                                                                                       |
| <b>Ctrl+B Q</b> | Imposition preview (Q for quire) — the production-layout plan for the selected book, over its built PDF. <b>Enter</b> imposes; tree-scoped so it never collides with the editor's <b>Ctrl+B Q</b> . |

Book-adding has no meta chord — use the plain-letter **B** in the tree (see the note at the end of this appendix).



**Ctrl+B in the Editor pane**

|                       |                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+B S</b>       | Read the open paragraph aloud via the OS TTS engine. Gated by <code>editor.tts.enabled</code> .                                |
| <b>Ctrl+B N</b>       | New snapshot of the current buffer (same as F5).                                                                               |
| <b>Ctrl+B R</b>       | Cycle the paragraph's status: None → Napkin → First → Second → Third → Final → Ready.                                          |
| <b>Ctrl+B F</b>       | Open the Typst function picker — type to filter, <code>Enter</code> inserts <code>#name(...)</code> .                          |
| <b>Ctrl+B T</b>       | Retitle the paragraph from its first sentence.                                                                                 |
| <b>Ctrl+B M</b>       | Morph the paragraph's content type (as in the tree).                                                                           |
| <b>Ctrl+B P</b>       | Inside <code>#image("...")</code> , pick a sibling image; otherwise run a Places RAG over the selection.                       |
| <b>Ctrl+B C</b>       | Character RAG — query the selection against the Characters book; the answer streams into the AI pane.                          |
| <b>Ctrl+B G</b>       | Notes RAG — query the selection against the Notes book.                                                                        |
| <b>Ctrl+B Y</b>       | Artefacts RAG — query the selection against the Artefacts book.                                                                |
| <b>Ctrl+B Q</b>       | Translate the paragraph INTO an invented language defined under the Language system book; the result streams into the AI pane. |
| <b>Ctrl+B Shift+Q</b> | Translate the paragraph FROM an invented language — the reverse direction, for round-trip testing.                             |
| <b>Ctrl+B D</b>       | Deterministic (rule-based) translation of the paragraph into an invented language; result and trace land in the Output pane.   |

The editor scope also owns a large family of prose-craft and analysis chords — mostly `Ctrl+B Shift+...`. They are collected in the next table.

**Ctrl+B — prose, craft, and analysis**

Most are editor-scoped, but the toggles and overlays — `Ctrl+B Shift+F`, `Ctrl+B Shift+L`, `Ctrl+B Shift+P`, `Ctrl+B Shift+N`, `Ctrl+B Shift+G`, and `Ctrl+B Shift+Y` — fire from any pane; the rest are editor-scoped. Several open a modal or stream into the AI pane.

|                       |                                                                                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+B Shift+F</b> | Toggle the inline style-warning overlay (filter words, hedges) — a session-local override of <code>editor.style_warnings</code> .                      |
| <b>Ctrl+B Shift+K</b> | Toggle the live echo overlay — underline words echoing across nearby paragraphs of the chapter.                                                        |
| <b>Ctrl+B Shift+H</b> | Open the sentence-rhythm gauge — sentence lengths, mean/stddev/CV, and a Monotone / Steady / Varied / Choppy verdict.                                  |
| <b>Ctrl+B Shift+M</b> | AI sentence-rhythm rewrite — mix short and long sentences; auto-opens a diff modal (accept snapshots first).                                           |
| <b>Ctrl+B Shift+T</b> | AI show-don't-tell scan — flags telling passages and proposes rewrites into the AI pane.                                                               |
| <b>Ctrl+B Shift+L</b> | Open the project-wide concordance — every lexical stem with counts and KWIC samples; <code>Enter</code> jumps to a sample.                             |
| <b>Ctrl+B Shift+P</b> | Toggle the status-bar POV / character chip.                                                                                                            |
| <b>Ctrl+B Shift+N</b> | Toggle prompt-language mode: None → book-defined → paragraph-detected → None (session-local).                                                          |
| <b>Ctrl+B Shift+E</b> | Reader-pace preview — a teleprompter that advances word-by-word at <code>editor.reading_wpm</code> .                                                   |
| <b>Ctrl+B Shift+Y</b> | Next stanza — create a sibling verse paragraph of the same <code>para:verse-*</code> type, opened for editing (structure only, never a line of verse). |
| <b>Ctrl+B Shift+X</b> | AI fact-check against the Facts book — flags claims that contradict established world facts.                                                           |
| <b>Ctrl+B Shift+S</b> | Facts semantic-search modal — find facts, mark several, send them to a targeted Facts chat.                                                            |
| <b>Ctrl+B Shift+J</b> | Jump to the next fact finding from the last <code>Ctrl+B Shift+X</code> check.                                                                         |
| <b>Ctrl+B Shift+D</b> | Verify the open paragraph's <code>verify</code> -marked code listings through their runners; results to the Output pane.                               |
| <b>Ctrl+B Shift+R</b> | Save the open paragraph as an audio file via macOS <code>say -o</code> (opens a path picker). macOS only.                                              |

**Ctrl+B Shift+G** Writing-streak heatmap — the last 91 days of word deltas as a 13×7 grid, with streak and totals.

### Ctrl+B in the AI pane

**Ctrl+B C** Clear the chat history and the currently-displayed inference (F9 drives the scope cycle instead).

### Ctrl+B from any pane

**Ctrl+B H** Open the pane-aware Quick reference overlay — the live keymap plus a static cheatsheet.

**Ctrl+B V** Open the version / author / credits pane.

**Ctrl+B I** Open the current book's info panel — paths, stats, PDF status.

**Ctrl+B L** Switch the active LLM provider; the choice is persisted to `inkhaven.hjson`.

**Ctrl+B E** Toggle typewriter sound effects (Enter / focus-out clicks); persisted.

**Ctrl+B A** Assemble the book — emit a Typst-compilable tree under the artefacts dir.

**Ctrl+B B** Build the book — assemble then run `typst compile` (PDF to the artefacts dir).

**Ctrl+B O** Take the book — build, then copy the PDF (and configured extras) into the launch directory.

**Ctrl+B Shift+B** Run a project backup now, ignoring the exit-hook recency cooldown.

**Ctrl+B U** Undo the most-recent paragraph delete (restores the front of the kill-ring). From every pane EXCEPT the Tree, where **Ctrl+B U** reorders the node up (see the Tree section).

**Ctrl+B K** Toggle full-screen AI mode — the AI pane fills the window with the chat history and prompt.

**Ctrl+B W** Open the World overview — the `world.hjson` definition and every compiled layer (astronomy, geology, climate, hydrology, demographics), each marked once materialized. **C** compiles the world.

**Ctrl+B Shift+W** Toggle distraction-free / focus mode — hide every other pane and give the editor the full window.

|                       |                                                                                                                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+B J</b>       | Open the Inner Socrates overview — the entrance to the whole inner-reader family (see the next section). From every pane EXCEPT the Tree, where <b>Ctrl+B J</b> reorders the node down (see the Tree section). |
| <b>Ctrl+B X</b>       | Open the ConLang hub — a read-only overview of every constructed language: phoneme inventory, template/constraint counts, prosody, romanization, lexicon size, speakers.                                       |
| <b>Ctrl+B z</b>       | Open the knowledge-graph hub — <b>n</b> the paragraph's one-hop edges, <b>i</b> the edge inbox (P promotes, d rejects), <b>w</b> walks the graph to answer the AI-prompt question.                             |
| <b>Ctrl+B Shift+O</b> | Open the full-screen Outline pane (the reliable backup for <b>Ctrl+2</b> ).                                                                                                                                    |
| <b>Ctrl+B Shift+C</b> | Unified review pass — every fast deterministic checker at once (fact-check + Inner Socrates over the paragraph, timeline critique over the project) into the Output pane.                                      |
| <b>Ctrl+B Shift+I</b> | SENTINEL continuity ledger — the ranked deterministic continuity findings grouped by kind; <b>Enter</b> jumps, <b>k</b> runs the LLM coherence pass.                                                           |
| <b>Ctrl+B Shift+A</b> | LECTOR read-through dashboard — the measured intensity curve, per-chapter scene/sequel beat, and reader findings; <b>k</b> runs the synthetic first-read.                                                      |
| <b>Ctrl+B Shift+U</b> | CHRONICLE draft-history dashboard — the trend since your last milestone and the cleared-vs-introduced split; <b>m</b> marks this draft.                                                                        |
| <b>Ctrl+B Shift+Z</b> | KEN knowledge dashboard — epistemic continuity (premature_knowledge / leaked_secret / dropped_reveal); <b>Enter</b> jumps to the offending paragraph.                                                          |
| <b>Ctrl+B \$</b>      | AI cost dashboard — today's LLM call tallies per capped subsystem against their daily caps.                                                                                                                    |
| <b>Ctrl+B ]</b>       | Tag the open paragraph — the tag picker scoped to the buffer.                                                                                                                                                  |
| <b>Ctrl+B }</b>       | Search by tag — the tag picker in read-only mode; <b>Enter</b> lists every paragraph carrying a tag.                                                                                                           |
| <b>Ctrl+B 0</b>       | Edit project HJSON — a full-screen editor for <code>inkhaven.hjson</code> ; <b>Ctrl+S</b> saves, <b>Ctrl+R</b> fires an LLM review.                                                                            |

|                       |                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+B Shift+O</b> | Project doctor panel — the <b>doctor --scan</b> findings as a modal; <b>r</b> repairs the highlighted one, <b>R</b> repairs all.         |
| <b>Ctrl+B Shift+V</b> | Piper TTS voice picker — the catalog plus downloaded voices; <b>Enter</b> downloads/selects, <b>d</b> removes.                           |
| <b>Ctrl+B 1..7</b>    | Status filter — show only paragraphs of a given status under the cursor: 1 Ready, 2 Final, 3 Third, 4 Second, 5 First, 6 Napkin, 7 None. |
| <b>Ctrl+B &lt;</b>    | (editor) Jump the cursor to the previous scene-break line.                                                                               |
| <b>Ctrl+B &gt;</b>    | (editor) Jump the cursor to the next scene-break line.                                                                                   |

## The inner-family submenu — Ctrl+B J

**Ctrl+B J** opens the Inner Socrates overview — the doorway to Inkhaven's family of *inner readers*, each a different critical sensibility. From the overview a second key branches into the family. All of them observe and question; none rewrite your prose.

|                        |                                                                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+B J then S</b> | Select the active Reader Persona.                                                                                                                                                           |
| <b>Ctrl+B J then L</b> | View the intent ledger — the deliberate authorial choices the interrogator respects.                                                                                                        |
| <b>Ctrl+B J then F</b> | Fast-check the open paragraph (deterministic, free) → Output.                                                                                                                               |
| <b>Ctrl+B J then E</b> | Engage the slow LLM Socratic track over the paragraph.                                                                                                                                      |
| <b>Ctrl+B J then A</b> | Toggle ambient Socratic checks.                                                                                                                                                             |
| <b>Ctrl+B J then C</b> | Open a Socratic conversation about the paragraph.                                                                                                                                           |
| <b>Ctrl+B J then T</b> | Inner Theologian — a tradition-neutral reader poses moral questions through eleven lenses → the Thoughts / Output pane. It asks, never judges.                                              |
| <b>Ctrl+B J then P</b> | Inner Poet — scan the open verse paragraph's metre and rhyme against its declared <b>poem:</b> form. Sub-keys: F scan · E engage · D declare a form · T two-column translation · A ambient. |
| <b>Ctrl+B J then Y</b> | Inner Stylist (CHORUS) — synthesise voice-at-scale (distinctiveness, POV/head-hop, tense, register). Sub-keys: F pillars → Output · E engage the coach · R the voice-report dashboard.      |

- Ctrl+B J then R** Inner Rigor — run the reasoning-rigor reader over the open paragraph.
- Ctrl+B J then N** New-persona wizard — create a new Reader Persona.

#### THE INNER EDITOR LIVES ELSEWHERE

The second inner reader, the **Inner Editor**, is reached not through Ctrl+B J but through Ctrl+V 0 (O for Observe) — because Ctrl+B E was already the sound toggle. It observes literary craft as Praise / Note / Concern. Its sub-keys: E engage · C converse · A ambient · F jump to findings.

## The view prefix — Ctrl+V

The **Ctrl+V** family routes to in-process exporters, pickers, the writing-progress tools, paragraph links and bookmarks, the timeline, and the prose/craft checks. All of it is rebindable through `keys.bindings.view_sub`.

### Export, progress, targets

- Ctrl+V 1** (editor / AI) Export the open paragraph's buffer as markdown. (tree) Export the cursor node and all descendants.
- Ctrl+V 2** (editor / AI) Export the containing subchapter's subtree as markdown.
- Ctrl+V S** Toggle similar-paragraph mode — vector search opens a second editor side by side.
- Ctrl+V G** Open the writing-progress modal (today / streak / per-book pace / sparkline). **e** opens the goals editor.
- Ctrl+V T** Set / clear the per-paragraph word-count target (empty or 0 clears).
- Ctrl+V Shift+G** Project word-count goal modal — projects the finish date from the last-30-day delta.

### Links and bookmarks

- Ctrl+V A** Add an outgoing paragraph link — the tree enters select-to-link mode.

|                       |                                                                                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+V I</b>       | Add an incoming paragraph link — the reverse of <b>A</b> .                                                                                   |
| <b>Ctrl+V L</b>       | List outgoing links (floating picker; <b>D</b> removes one).                                                                                 |
| <b>Ctrl+V K</b>       | List backlinks — paragraphs that link to the open one.                                                                                       |
| <b>Ctrl+V B</b>       | Toggle a bookmark on the open paragraph.                                                                                                     |
| <b>Ctrl+V M</b>       | Open the bookmark picker.                                                                                                                    |
| <b>Ctrl+V Shift+B</b> | Sibling-book lookup — find the same-slug paragraph under a different book and pin it to the split-view secondary (the translation workflow). |

### Pickers and navigation

|                       |                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+V P</b>       | Fuzzy paragraph picker over every user-book paragraph.                                                                |
| <b>Ctrl+V Shift+P</b> | Recent-paragraph picker — the same list sorted by most-recently modified.                                             |
| <b>Ctrl+V Shift+U</b> | Kill-ring picker — choose any of the up-to-10 buffered deleted paragraphs to restore.                                 |
| <b>Ctrl+V R</b>       | Render the open paragraph — compile it in-process and float a PNG preview; <b>S</b> saves a page, <b>A</b> saves all. |
| <b>Ctrl+V N</b>       | Jump the cursor to the next Typst diagnostic in the buffer.                                                           |
| <b>Ctrl+V W</b>       | Paragraph mini story-view — a radial graph of the paragraph's links and mentions.                                     |
| <b>Ctrl+V Shift+W</b> | Book story-view — the radial hierarchy graph of the whole book, rasterised and floated.                               |
| <b>Ctrl+V H</b>       | Hidden-character report on the open paragraph (tabs, trailing white-space, CRs) — status bar only.                    |
| <b>Ctrl+V Shift+S</b> | Show the cursor's breadcrumb path on the status bar (Book ▶ Chapter ▶ Subchapter ▶ Paragraph).                        |

### Editing inserts

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| <b>Ctrl+V @</b> | Cite picker — fuzzy-find a citation defined in the Sources book and insert <b>@key</b> . |
|-----------------|------------------------------------------------------------------------------------------|

|                       |                                                                                                                                    |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+V #</b>       | Package import picker — insert an <code>#import "@preview/..."</code> line for a Typst Universe package.                           |
| <b>Ctrl+V &amp;</b>   | Cross-reference picker — insert <code>@label</code> for a label defined in the manuscript.                                         |
| <b>Ctrl+V x</b>       | Snippet <code>#include</code> — pick a reusable snippet from the Snippets book and insert (or replace) its include.                |
| <b>Ctrl+V Shift+X</b> | Snippets overview — every snippet with its reference count; <code>Enter</code> jumps to the source.                                |
| <b>Ctrl+V d</b>       | (editor) AI continuation drafting — continue the paragraph in your voice; the AI pane's <code>I</code> lifts only the draft block. |
| <b>Ctrl+V f</b>       | (editor) Insert an inline footnote — <code>#footnote[...]</code> (Typst) or a markdown <code>[^id]</code> .                        |
| <b>Ctrl+V j</b>       | (editor) Reflow the paragraph at the cursor to the editor's text width (one undoable edit).                                        |
| <b>Ctrl+V y</b>       | (editor) Style-transfer rewrite — pick a reference paragraph; the AI rewrites the open one in that voice.                          |
| <b>Ctrl+V Shift+Y</b> | (editor) WordNet thesaurus — replace the word under the cursor with a chosen synonym / antonym / hypernym / hyponym.               |

## Research and citation checks

|                       |                                                                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+V ?</b>       | Confront the open paragraph against the research corpus (Facts + ingested Sources) — graded against / supporting findings to the Output pane. |
| <b>Ctrl+V c</b>       | Lint the open paragraph's <code>@key[locus]</code> citations against their sources' reference schemes (deterministic).                        |
| <b>Ctrl+V Shift+C</b> | Sourcing pass — flag sentences that make a checkable factual claim but carry no <code>@key</code> citation.                                   |

## Structure, revision, the bible

|                       |                                                                                                                                                                          |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+V Shift+K</b> | Structure outline — the <code>plan check</code> report as a per-beat position bar with drift and a tension overlay; interactive (map beats, link threads, cycle status). |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



|                       |                                                                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+V Shift+R</b> | The Editorial Pass — one ranked worklist unifying every reader; <b>f</b> acts on a finding (rewrite / decision / brief), <b>F</b> batch-fixes. Every prose change is snapshotted first. |
| <b>Ctrl+V Shift+L</b> | Story bible — every Character with its tracked attributes, plus Places, Artefacts, Facts, and the Glossary; <b>Enter</b> jumps to a source.                                             |
| <b>Ctrl+V Shift+F</b> | Deep AI world refresh in the background — facts check + scan + drift + continuity extract; advisory, writes sidecars only.                                                              |
| <b>Ctrl+V Shift+J</b> | Manuscript intelligence dashboard — every metric (words, structure, pacing, threads, comments) in one view; <b>e</b> exports.                                                           |
| <b>Ctrl+V O</b>       | Inner Editor overview (see the inner-family callout above).                                                                                                                             |

### Threads, prose voice, dialogue, character, myth

|                       |                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+V Shift+H</b> | Threads picker over the Threads book; <b>Enter</b> opens a swim-lane weave view.                            |
| <b>Ctrl+V Shift+A</b> | AI thread audit — send every thread's beats and waypoints to the LLM for blind spots.                       |
| <b>Ctrl+V Shift+D</b> | Thread doctor modal — the same blind-spot checks deterministically (zero links / payoff unfired / dormant). |
| <b>Ctrl+V V</b>       | Prose voice check (NARR-1) in the background — chapter metrics that drift past threshold emit to Output.    |
| <b>Ctrl+V Shift+V</b> | Toggle ambient prose checks — re-run after an editing pause, gated by a cooldown.                           |
| <b>Ctrl+V Shift+Q</b> | Dialogue fingerprint — the per-character voice signature for the nearest speaker.                           |
| <b>Ctrl+V Shift+N</b> | Character arc — the declared arc, chapter-by-chapter state chain, agency scores, and coverage gaps.         |
| <b>Ctrl+V Shift+M</b> | Mythology heatmap — symbol density, motif presence, and archetype coverage per chapter → the Thoughts pane. |
| <b>Ctrl+V z</b>       | Toggle the terminology overlay — red-underline banned synonyms of canonical Glossary terms.                 |
| <b>Ctrl+V Shift+Z</b> | (editor) Declare the banned synonym under the cursor a deliberate variant (stop flagging it).               |

## Submissions and the timeline

|                       |                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+V u</b>       | Submission tracker — the <code>.inkhaven/submissions.json</code> log; <code>Space</code> / <code>s</code> cycles status, <code>d</code> removes. |
| <b>Ctrl+V q</b>       | Submission-package generator — pick a query letter / synopsis / comps / logline; streams into the AI pane.                                       |
| <b>Ctrl+V e</b>       | Chronological event picker over the story timeline.                                                                                              |
| <b>Ctrl+V Shift+E</b> | New event at the cursor — opens the timeline and the new-event prompt.                                                                           |
| <b>Ctrl+V Shift+I</b> | Edit the timing metadata of the open event paragraph.                                                                                            |
| <b>Ctrl+V Shift+T</b> | Swim-lane timeline view (lowercase <code>t</code> stays the word-count target).                                                                  |

## The Bund prefix — Ctrl+Z

`Ctrl+Z` fronts the embedded Bund scripting language and the OS shell pane.

|                 |                                                                                                                             |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+Z R</b> | Run the current buffer as a Bund script.                                                                                    |
| <b>Ctrl+Z N</b> | New Bund script.                                                                                                            |
| <b>Ctrl+Z E</b> | Open the eval modal.                                                                                                        |
| <b>Ctrl+Z ?</b> | Open the script picker — scripts in the cursor's branch; <code>A</code> toggles to the Scripts system book.                 |
| <b>Ctrl+Z o</b> | Open / close the embedded nushell pane (state and history preserved across close).                                          |
| <b>Ctrl+Z O</b> | Drop the cached shell engine and turn buffer and open a fresh shell.                                                        |
| <b>Ctrl+Z h</b> | (inside the shell) Toggle history-selection mode — <code>C</code> copies a turn, <code>i</code> inserts it into the editor. |
| <b>Ctrl+Z p</b> | Emit a haiku to the Output pane, in the book's language.                                                                    |
| <b>Ctrl+Z f</b> | Fullscreen the current right pane (Output / Thoughts).                                                                      |

- Ctrl+Z c** (editor) Add an inline comment on the selection — a sidecar `.comments.json` beside the paragraph. Moved here from `Ctrl+V c`, which the LOCI citation check now owns.
- Ctrl+Z Shift+C** Open the project-wide comments panel over every `.comments.json` sidecar. Moved here from `Ctrl+V Shift+C`, now the sourcing check.

## Modals and overlays

Overlays absorb input while they are up; the underlying pane keeps its visual focus but sees no keys until the overlay closes. The conventions are consistent: `Enter` confirms, `Esc` cancels, `Ctrl+Q` always hard-quits through a modal.

### Add modal

**printable char** Insert into the title buffer.

**Backspace** / Edit the title.

**Delete**

**← / → / Home** / Cursor navigation within the title.

**End**

**Enter** Commit — derive the slug, create the file and record, reload the tree, move the cursor to the new node.

**Esc** Cancel without creating anything.

### Delete-confirm modal

**y / Y / Enter** Confirm — remove the filesystem subtree and the records; the cursor lands on the parent.

**n / N / Esc** Cancel.

### File picker (F3)

**↑ / ↓** Move the cursor one entry.

**PageUp / Page-Down** Jump by 10.

**Home / End** First / last entry.

|              |                                                                           |
|--------------|---------------------------------------------------------------------------|
| →            | Expand a directory inline.                                                |
| ←            | Collapse an expanded directory, else move to the parent.                  |
| <b>Enter</b> | Commit — replace the buffer (editor) or import the file/directory (tree). |
| <b>Esc</b>   | Cancel.                                                                   |

### Snapshot picker (F6)

|                |                                                                                 |
|----------------|---------------------------------------------------------------------------------|
| ↑ / ↓          | Navigate the snapshots (newest first).                                          |
| <b>Enter</b>   | Load the selected snapshot (takes a pre-restore safety snapshot first).         |
| <b>V</b>       | Side-by-side diff of the snapshot vs current; <b>ESC</b> returns to the picker. |
| <b>D / Del</b> | Remove the snapshot.                                                            |
| <b>/</b>       | Enter filter mode — narrow by annotation substring.                             |
| <b>Esc</b>     | Cancel.                                                                         |

### Prompt picker

|                           |                                               |
|---------------------------|-----------------------------------------------|
| ↑ / ↓                     | Move the selection.                           |
| <b>Enter / Tab</b>        | Expand the selected template into the buffer. |
| <b>Backspace / Delete</b> | Edit the filter; the picker re-filters live.  |
| <b>Esc</b>                | Close without expanding.                      |

### Search-results overlay

|               |                                                   |
|---------------|---------------------------------------------------|
| ↑ / ↓         | Move the result cursor.                           |
| <b>Enter</b>  | Open the highlighted result.                      |
| <b>Esc</b>    | Close the overlay (the Search bar stays focused). |
| <b>typing</b> | Close the overlay and continue editing the query. |

### AI diff-review modal

Pops when an AI rewrite lands and `ai.diff_review_on_apply` is on.

|                          |                                                                     |
|--------------------------|---------------------------------------------------------------------|
| <b>a / A / Enter / e</b> | Accept — apply the rewrite (snapshot first) and refocus the editor. |
|--------------------------|---------------------------------------------------------------------|

|                      |                                                                 |
|----------------------|-----------------------------------------------------------------|
| <b>r / R</b>         | Reject — leave the buffer unchanged.                            |
| <b>↑ ↓ PgUp PgDn</b> | Scroll the diff; <b>Home</b> / <b>End</b> jump to top / bottom. |
| <b>Esc</b>           | Same as reject.                                                 |

## The mouse

Inkhaven captures the mouse on startup (toggle with **Ctrl+Shift+M**). Left-click moves focus to the clicked pane and positions the cursor there; gutter clicks in the editor are ignored.

|                       |                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------|
| <b>Left-click</b>     | Focus the clicked pane; position the tree row or editor character cursor.             |
| <b>Wheel (Tree)</b>   | Move the tree cursor 3 rows per tick.                                                 |
| <b>Wheel (Editor)</b> | Scroll the viewport 3 lines per tick.                                                 |
| <b>Wheel (AI)</b>     | Scroll the chat history (older messages on wheel up).                                 |
| <b>Wheel (modals)</b> | Scroll the shell turn buffer, HJSON editor, or picker cursor; other modals ignore it. |

To select and copy text through the terminal's own clipboard while capture is on, hold **Shift** (or **Option**, per your terminal) as you drag — that bypasses capture and uses the terminal's native selection.

### WHEN A CHORD NEVER ARRIVES

If **Ctrl+S**, **Ctrl+Q**, or the **Ctrl+B** prefix seem dead, a layer above Inkhaven is eating them. **Ctrl+S** / **Ctrl+Q** are terminal flow control — run `stty -ixon`. **Ctrl+B** is `tmux`'s default prefix — rebind `tmux`'s prefix or remap Inkhaven's `meta_prefix`. When **Ctrl+Shift+arrow** is not transmitted, use the tree's plain-letter shortcuts (**B**, **C**, **A**, **+**, **D**, **-**) instead.

## A closing note, and two honest caveats

That is the whole keyboard. The companion TUIs — `inkhaven research`, `inkhaven worldbuilder`, and `inkhaven prompts-editor` — run outside the main editor and carry their own, smaller keymaps, documented with each tool.

Two places where this map follows the code against older prose, worth knowing because the printed defaults and your habits may disagree:

**Add a book**            There is no `Ctrl+B B` for it — that chord builds the book. Add a book with the plain-letter `B` in the Tree pane.

**Ctrl+B (editor)**      `S` reads aloud, `R` cycles status, `F` is the function picker, `L` swaps the LLM — not the older `save / history / split / load` meanings some tables still show.

When in doubt, the live keymap is one keystroke away: `Ctrl+B H` from any pane prints the Quick reference for the pane you are in, generated from the same binding table this appendix was compiled against.

## APPENDIX B

# B

## The Command Line

---

Everything Inkhaven does in the terminal is one binary: `inkhaven`. Run it bare and it opens the full-screen editor on the project in the current directory. Give it a subcommand and it runs **headless** — it does one job, prints to standard output, exits with a status code, and never draws a screen. That headless surface is what this appendix catalogues: every subcommand, grouped by the work it does, with its important flags and a one-line statement of purpose.

The headless commands exist so that everything the editor can do, a script, a `Makefile`, or a continuous-integration gate can do too. Most of the reading intelligences — continuity, dialogue, sources, terms, knowledge — exit non-zero when they find a problem, which is exactly what a pre-submission check wants.

## Invocation and global options

The shape is always the same: the program name, optional global flags, a subcommand, and that subcommand's own arguments.

• • • *The general form*

```
inkhaven [--project <dir>] <subcommand> [args...]
inkhaven                               # no subcommand → open the TUI
inkhaven --version                      # print the version and exit
inkhaven <subcommand> --help          # the subcommand's own help
```

Only one option is global — it applies to every subcommand:

- project <dir>**      The project root to act on. Aliases: **-p** and the long **--project-directory**. For **init** this is the directory to create; for every other subcommand it defaults to the current working directory.
- help / -h**          Print help. Given after a subcommand, prints that subcommand's arguments and flags — the authoritative, always-current reference for a single verb.
- version**            Print the binary version (the same string **doctor** reports) and exit.

### CONVENTION IN THIS APPENDIX

Flags shown in `<angle brackets>` take a value; flags shown bare are switches. A command that “exits non-zero on findings” is meant for a CI gate — pair it with `--json` and `jq` to fail a build. Because `-p` is reserved for `--project`, no subcommand offers `-p` for anything else; provider overrides are always the long `--provider`.

## Project and setup

The lifecycle of a project: create it, grow its tree, keep it healthy, and recover it when something goes wrong.



● ● ● *Creating and scaffolding*

```

inkhaven init <path> [--template <t>] [--force]
    Create a new project. Templates: empty (default), novel,
    nonfiction, rpg-sourcebook, technical, nanowrimo.
inkhaven template list
    List every project template with a one-line description.
inkhaven add <kind> <title> [--parent <p>] [--slug <s>]
    [--after <sibling>]
    Add a node: book | chapter | subchapter | paragraph |
    script. --after inserts beside an existing sibling.
inkhaven list
    Print the whole hierarchy as an indented tree.
inkhaven outline [--filter <text>]
    Print the manuscript outline (the Ctrl+2 pane's twin).

```

● ● ● *Reshaping the tree*

```

inkhaven delete <slug-path> [--yes]
    Delete a node and its descendants. Dry-runs without --yes.
inkhaven mv <slug-path> <up|down>
    Reorder a node among its siblings.
inkhaven paragraph copy <src> <dest>
    Duplicate a paragraph under another parent (fresh uuid).
inkhaven paragraph move <src> <dest>
    Relocate a paragraph under another parent.
inkhaven reindex [--prune] [--adopt]
    Re-index every .typ file from disk into the store.
    --prune drops rows whose file vanished; --adopt registers
    orphan files under the matching hierarchy branch.

```

● ● ● *Health, backup, recovery*

```

inkhaven doctor [--scan] [--json] [--class <c>]
                [--autofix [--yes]] [--voices]
                [--tts-test <text>] [--filter-words-snippet]
Health report: binary, project, actionable notes. --scan
walks for zero-byte files, orphan rows, missing files,
corrupt comment sidecars; --autofix repairs them (logged).
inkhaven stats [--book-name <b>]
Per-paragraph table: status, word count, target %, mtime.
inkhaven backup [--out <dir>]
Zip the project into a dated archive.
inkhaven restore <archive.zip> --to <fresh-dir>
Restore a backup into a new directory.
inkhaven recover <crash-report.hjson> [--yes] [--keep]
Walk a crash report's rescued buffers and apply them back
to disk (a .pre-recover backup is written first).

```

The two standalone TUI editors for a project's own settings live here as well — they open their own screens rather than running headless:

**config** Open the schema-aware TUI editor for `inkhaven.hjson` (the `Ctrl+B 0` raw editor stays the power-user fallback).

**prompts-editor** Open the four-pane TUI workbench for `prompts.hjson` — the project's prompt overrides.

**comments** Headless view of the per-paragraph `.comments.json` sidecars: `list`, `resolve <id>`, `reopen <id>`, `delete <id>`, `export`. Mirrors the `Ctrl+V Shift+C` panel.

## Writing and search

The everyday text operations — find things, change things, watch the prose, track the daily count.

● ● ● *Finding and changing prose*

```

inkhaven search <query> [--limit <n>]
    Semantic search across the whole project (default 10 hits).
inkhaven book-rag retrieve <query> [--book-name <b>]
    [--top-k <n>] [--context]
    Show the passages Book-scope chat would ground on – the
    exact retrieval, with no LLM call. --context prints the
    composed block the model would receive.
inkhaven replace <pattern> <replacement>
    [--regex] [--substring] [--ignore-case]
    [--book <b>] [--include-system]
    [--dry-run | --yes]
    Project-wide find & replace. Literal whole-word by default;
    snapshots each touched paragraph. --dry-run previews; --yes
    applies. System books excluded unless --include-system.

```

● ● ● *Watching the prose*

```

inkhaven style [--book-name <b>] [--language <l>] [--json]
    Run the deterministic style detectors (filter-words,
    repeated phrase, show-don't-tell, anachronism). CI/CLI
    parity for the Ctrl+V w overlay.
inkhaven goals
    Writing-goals report: word totals, today vs the daily goal,
    the streak, per-book pace and deadline. Read-only.
inkhaven snippets list [--json]
inkhaven snippets check [--book <b>] [--json]
    List reusable snippets + reference counts, or validate
    every #include against the Snippets book (exit 1 on a miss).
inkhaven wordnet fetch <lang...> | import <lang> <file>
    | lookup <word> [--lang <l>] | list
    The multilingual thesaurus: fetch/import WordNet data, then
    look a word up for senses, synonyms, antonyms, hypernyms.

```

## The intelligences

These are the readers that watch a whole manuscript for what a long book gets wrong. Each has an in-editor chord; each also runs headless here. The deterministic ones are free and exit non-zero on a finding; the ones that call a model are gated

behind an explicit `--deep` / `--slow` / `--coherence` flag with an informative token cap.

### ● ● ● *The unified passes*

```
inkhaven check [--paragraph <id>] [--book-name <b>]
               [--no-fact] [--no-socrates] [--no-timeline]
               [--no-continuity] [--no-lector]

The unified review pass: every fast deterministic checker
over a scope, one consolidated summary. (Ctrl+B Shift+C.)
inkhaven edit [--json] [--only <cats>] [--book-name <b>]
               [--show-deferred] [--deep] [--provider <p>]]

The Editorial Pass: one ranked revision worklist unifying
every detector. --deep refreshes the AI sidecars first.
inkhaven revise [--book-name <b>] [--json]

REDLINE – the editorial letter: every reader's findings
synthesised into one prioritized developmental letter.
Advises; never rewrites. (Ctrl+V Shift+R.)
```

### ● ● ● *Continuity and knowledge*

```
inkhaven continuity check [--only <d>] [--skip <d>] [--json]
                         [--coherence] [--max-cost <n>]
                         [--force]]

SENTINEL – every continuity detector (co-location, timeline,
numeric, char-facts, referenced-before-introduced), deduped
and ranked. Exit non-zero on a Contradiction. (Ctrl+B Shift+I.)
inkhaven continuity extract [--provider <p>]
inkhaven continuity list

Build / dump the AI continuity bible.
inkhaven knowledge [--book-name <b>] [--json]
                  [--deep] [--max-cost <n>]]

KEN – who knows what, when: premature_knowledge,
leaked_secret, dropped_reveal. Deterministic; --deep adds
the LLM implied-irony pass. (Ctrl+B Shift+Z.)
```

● ● ● *Reading, voice, and dialogue*

```

inkhaven readthrough [--deep [--max-cost <n>] [--force]]
                        [--json]
    LECTOR – the book read forward once as a first reader: the
    intensity curve, scene/sequel beats, ranked findings
    (confusion, info-dump, dip, put-down). (Ctrl+B Shift+A.)
inkhaven chorus voices  [--book <b>] [--character <c>] [--json]
inkhaven chorus scan    [--book <b>] [--json]
inkhaven chorus report  [--book <b>] [--json]
inkhaven chorus stylist [--book <b>] [--coach]
                        [--suppress <k>] [--unsuppress <k>]
    CHORUS – voice at book scale: character fingerprints, POV /
    head-hop discipline, the book-scale report, and the Inner
    Stylist's synthesis. (Ctrl+B J → Y.)
inkhaven dialogue scan  [--book <b>] [--findings <f>] [--json]
inkhaven dialogue profile [--book <b>] [--character <c>] [--json]
inkhaven dialogue refresh [--book <b>] [--chapter <n>]
inkhaven dialogue suggest [--book <b>] [--chapter <n>]
    Dialogue quality: zero-attribution / said-bookism /
    talking-head findings + per-character fingerprints. scan
    exits non-zero on a bare span. (Ctrl+V Shift+Q.)

```

● ● ● *Prose, drift, terms, chronicle, character*

```

inkhaven prose profile [--book <b>] [--deep] [--json]
                        [--language <l>]
inkhaven prose refresh | drift | suggest ...
    NARR-1 – deterministic narrative-voice metrics per chapter
    (rhythm, diversity, hedging, interiority). (Ctrl+V V.)
inkhaven drift list [--entity <e>] [--json]
inkhaven drift scan [--provider <p>] [--json]
    WORLD-2 – descriptions of one entity that diverge across
    the book (list = deterministic; scan = the AI judgment).
inkhaven chronicle mark <label> [--ref <r>] [--book-name <b>]
inkhaven chronicle list | trend | diff <from> <to> [--json]
    CHRONICLE – draft-history: stamp the readers' metrics as a
    milestone, then trend/diff them. (Ctrl+B Shift+U.)
inkhaven character arc <name> | check | refresh | plan
    CHAR-1 – the arc report, completeness checks, agency
    re-scoring, and Planning-Board coverage gaps.

```

## The inner family

The Inner readers question rather than correct: they ask, they never edit your prose. Each keeps its findings and an **intent ledger** — the deliberate choices it should stop flagging.

### ● ● ● *Socrates and the Editor*

```

inkhaven inner-socrates check [--text <t> | --paragraph <id>
                             | --path <p>] [--slow
                             [--max-cost <n>] [--force]]
    The Dialectician's fast track over prose (--slow adds the
    LLM Socratic pass). (Ctrl+B J.)
inkhaven inner-socrates timeline | findings | ledger
                             | persona ... | suggestions ... | bundle ...
    The timeline pass, persisted findings, the intent ledger,
    reader personas, promotion candidates, and .isl bundles.
inkhaven inner-editor engage [--text <t> | --paragraph <id>
                             [--force]]
    One Inner Editor pass over a paragraph. (Ctrl+V 0.)
inkhaven inner-editor findings | intent <cat> | suggestions ...
                             | config show | usage
    Findings, declaring a category deliberate, and cost.

```

### ● ● ● *Theologian, Rigor, Poet, cockpit*

```

inkhaven theologian scan    [--book <b>] [--signal <s>] [--json]
inkhaven theologian session [--book <b>] [--chapter <n>]
                             [--category <1-6>] [--lens <code>]
inkhaven theologian suppress --para <p> --reason <r>
    The moral/theological reader: the fast ethical-signal
    detector + the slow LLM session. It asks, never judges.
inkhaven rigor scan [--book <b>] [--signal <s>] [--strict] [--json]
    The reasoning-rigor reader: false dichotomy, question-
    begging, straw man, overgeneralization, non-sequitur. Zero-AI.
inkhaven poetry forms | syllabify | metre | rhyme | scan
                             | status | trilemma | phonemes ...
    The Inner Poet: measure verse (metre, rhyme, syllables,
    forms) — never generate it. (Ctrl+B J → P.)
inkhaven companions
    The examined-authorship cockpit: findings across the whole
    inner family, the intent ledger, today's cost per companion.

```

## The world and the graph

The layer beneath the prose: the simulated world, the knowledge graph, the timeline, and the fact-checker that holds the prose to all three.

### • • • *The world simulation*

```

inkhaven realworld new <name> [--force]
inkhaven realworld validate | show [--json]
inkhaven realworld compile [--layer <l>] [--json] [--materialize]
    Scaffold / validate / compile world.hjson. Layers: all,
    astronomy, geology, climate, hydrology, demographics.
inkhaven realworld propose | propose-myth | propose-rulers
    | propose-language | proposals <list|accept|...>
    The world proposes Places, myths, rulers, languages; you
    work the queue – nothing commits until you accept.
inkhaven realworld travel --from <a> --to <b> --days <d>
    [--mode foot|horse|cart|ship]
inkhaven realworld scene | weather | gazetteer | history
    | calendar | trade | name | culture | ecology
    Derived world reference: is a journey plausible, the local
    season, a scene brief, the gazetteer, the chronology.
inkhaven realworld critique [--write-notes] [--lints-only]
inkhaven realworld map [--spec-only] [--no-ingest]
    The AI world critique and the plakart map render.

```

The `worldbuilder` command opens the interactive TUI front-end to that pipeline; `world` is its read-only snapshot:

● ● ● *Worldbuilder, snapshot, graph, timeline*

```

inkhaven worldbuilder [--session <s>] [--interview] [--from-map]
    The interactive worldbuilder TUI – a front-end to realworld.
inkhaven world [--json] [--deep] [--provider <p>]] [--entity <e>]
    A world-consistency snapshot: facts, contradictions, drift,
    continuity coverage, anachronisms. Sub-checks: utopia-check,
    utopia-model, utopia-suppress, utopia-refresh (coherence).
inkhaven graph stats | rebuild | neighbors <n> | paths <a> <b>
    | contradicting <n> | loci <n> | pending
    | promote <e> | dismiss <e> | link <n> | ask <q>
    | lexical
    SEMNET – the typed-edge knowledge graph: inspect it, walk
    it, triage advisory edges, or ask it a question. (Ctrl+B z.)
inkhaven event add <title> --start <t> [--end <t>] [--track <k>]
inkhaven event list | show <path> | critique [--track <k>]
inkhaven event link-character <path> <name>
inkhaven event link-place <path> <name>
    Story-timeline events (needs timeline.enabled). link-character
    / link-place attach an explicit participant (feeds KEN presence).
inkhaven fact-check [--text <t> | --paragraph <id>] [--slow]
    [--timeline-aware auto|on|off] [--timeline-only]
    Check prose against the simulated world; respects the
    magic: ledger. (Ctrl+B Shift+X.)

```

The AI world scans that populate the `edit/world` sidecars also live as their own verbs: `facts scan|check|extract|import|init|list`, `tension scan|list`, and `drift scan|list` (above).

## Language

The constructed-language suite is Inkhaven’s single largest command family – over a hundred verbs under `language`, all headless twins of the interactive `linguistic` companion. The full treatment is in the LINGUISTIC companion book; the groups below are the map.



● ● ● *The linguistic companion and its verbs*

```

inkhaven linguistic [--language <name>] [--session <s>]
    Launch the interactive linguistics TUI (dev/verify/
    analyze/research over the Language system book). Ctrl+B X.

inkhaven language <verb> ...    – the headless twin, by group:
    Build      init, add-word, remove-word, define-rule,
                generate-word, generate-lexicon, scaffold
    Phonology  syllabify, ipa, stress, tone, harmony, romanize,
                transliterate, distribution, suggest-phonemes
    Grammar    parse, tree, gloss, igt, paradigm, sentence,
                agree, relative, coordinate, complement, movement
    Analyse    audit, stats, metrics, universals, naturalness,
                frequency, concordance, collocations, texts
    Translate  translate, reverse, cross, corpus, eval, remember
    Diachrony  reconstruct, cognates, sound-change, family-tree,
                derive-lexicon, borrow, areal, propose-loans
    Socio      lect, dialects, varieties, speakers, idiolect,
                ecology, propose-dialect
    Produce    export, dictionary, grammar-book, tutorial, igt
    Script     font-build, font-compose, glyph-draft, glyph-lint
    Link       link-place, link-character, scan-manuscript, query
    Check      doctor, check, check-clause, check-agreement,
                grammar-check, realism-check, areal-check

```

The multilingual-coverage helpers for the **manuscript's** natural language (not a conlang) are separate:

- |                                              |                                                                                                                                           |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>lang status</b>                           | The coverage matrix — stemming, detector word-lists, prompts, embeddings — for the project (or <code>--language</code> ) language.        |
| <b>lang bootstrap &lt;lang&gt;</b>           | Generate the full detector vocabulary for any language via one LLM pass; <code>--yes</code> patches <code>inkhaven.hjson</code> in place. |
| <b>prompts bootstrap &lt;lang&gt;</b>        | Generate per-language variants of the seven embedded prompts; <code>--update</code> merges them into <code>prompts.hjson</code> .         |
| <b>show-dont-tell bootstrap &lt;lang&gt;</b> | Generate the four show-don't-tell word lists for a language; <code>--update</code> merges them in place.                                  |

## Research and scholarship

The Research Assistant is its own TUI (`research` with no research flag), but almost all of its acquisition and analysis runs headless through flags on the same command — one non-interactive step per flag.

● ● ● *The Research Assistant, headless*

```

inkhaven research [--thread <name>]
    Open the Research Assistant TUI (its own screen).
inkhaven research --import <path> | --sync <folder>
    Ingest a document (md/txt/pdf/.bib) or register a folder
    for re-import-on-change.
inkhaven research --agentic <topic> [--out <f>]
    Autonomous deep research: decompose, gather, emit Facts
    (each with model provenance) for review.
inkhaven research --batch <file> [--auto-confirm]
    [--confidence <0..1>] [--out <f>]
    Research a question list headlessly into a report.
inkhaven research --gutenberg <q> | --archive <q>
    | --wikisource <q> | --bible <ref>
    | --quran <s> | --bookofmormon <ref>
    Ingest a public-domain source by search or reference.
inkhaven research --snowball <seed> | --contradict | --converge
    | --socrates <topic> | --report | --bibliography
    Cross-check the corpus: citation neighbourhoods, source
    contradictions, triangulated convergence, the Dialectician's
    questions, the clustered report, a BibTeX export.
  
```

● ● ● *Sources, terms, the scholarly apparatus*

```

inkhaven sources check    [--book-name <b>] [--json]
inkhaven sources coverage [--book-name <b>] [--ai] [--json]
inkhaven sources list | import <file.bib> | export [--format <f>]
    Validate every @key against the Sources book; flag uncited
    factual claims (coverage); import/export BibTeX / CSL-JSON.
inkhaven terms check     [--book <b>] [--json]
inkhaven terms suggest   [--book <b>] [--auto-create]
    Flag banned synonyms of canonical Glossary terms; cluster
    drifting terminology into proposed entries.
inkhaven index-locorum   [--book-name <b>] [--format <f>] [--strict]
    Index Locorum – every @key[locus] cited, grouped by source.
inkhaven index-verborum  [--book-name <b>] [--format <f>]
    Index Verborum – every scholarly-lexicon term, by sense.
inkhaven lexicon list    [--book <b>] [--watched] [--json]
    The scholarly lexicon: original-language forms + senses.
inkhaven argue           [--book-name <b>] [--provider <p>] [--json]
    Extract each chapter's central claims and their support;
    flag unsupported claims and orphan citations (exit non-zero).
inkhaven index           [--book-name <b>] [--format md|typst|json] [--out <f>]
    Generate a back-of-book index from the Glossary terms.

```

## Producing the book

The finishing line: turn the tree into a file someone can read — a PDF, an EPUB, a Word manuscript, a web page, an audiobook — or operate on the PDF itself.

● ● ● *Assembly and export*

```

inkhaven build [--book-name <b>] [--compile]
    Assemble a book into the artefacts tree; --compile runs
    typst on the root .typ. Headless twin of Ctrl+B B.
inkhaven export <format> [--output <o>] [--book-name <b>]
    [--status <floor>] [--tag <t>] [--profile <d=v>]
    [--blind] [--bundle <path>]
    Export to typst | pdf | markdown | tex | epub | html.
    --status/--tag filter paragraphs; --blind strips identity;
    --bundle writes a self-contained arXiv LaTeX submission.
inkhaven epub [--book-name <b>] [--output <o>] [--title <t>]
    [--author <a>]
    Export a book to EPUB 3.
inkhaven import-epub <file.epub> [--book-name <b>] [--dry-run]
    Import an EPUB as a user book (the inverse of epub).
inkhaven manuscript [--book-name <b>] [--output <o>] ...
inkhaven docx [--book-name <b>] [--font times|courier] ...
    Shunn standard manuscript format: typst or Word (.docx).

```

● ● ● *Timeline, concordance, audio, docs*

```

inkhaven export-timeline [--book-name <b>] <typst|svg|png>
    --output <o> [--track <k>]
    Render a book's timeline as a listing or a swim-lane.
inkhaven export-concordance <csv|json> --output <o>
    [--min-count <n>]
    Export the project-wide concordance (stems + KWIC).
inkhaven audiobook [--book-name <b>] [--output <o>] ...
    Synthesise a book to a chaptered .m4b (needs ffmpeg + TTS).
inkhaven tts engine | test <phrase> [--voice <v>]
    | binary <status|download>
    | voice <list|download <n>|remove <n>>
    | catalog refresh
    Headless management of the Piper TTS stack (Ctrl+B Shift+V).
inkhaven docs verify [--yes | --dry-run] | links [--external]
    | review [--floor <f>] [--since <ref>]
    TDOC – verify marked code blocks, check links, review
    currency. (Ctrl+B Shift+D.)

```

Most of the `pdf` family operates on an existing PDF — typically Inkhaven's own `build` output. Those mutating ops write a `<stem>-<op>.pdf` sibling unless `--out` is given, and writes are atomic. The exceptions are `cover`, `barcode`, and `merge`: they take no single input PDF to derive a stem from, so `--out` is required.

## ● ● ● PDF page and print operations

```

inkhaven pdf info <in>
    Page count, size, metadata, outline size.
inkhaven pdf extract | delete --pages <2-4,7> [--out <o>]
inkhaven pdf rotate --pages <p> --degrees <90|180|270>
inkhaven pdf reorder --mapping <3,1,2>
inkhaven pdf split --every <n> | --at <p,p>
inkhaven pdf merge <in...> --out <o>
    Page-level surgery: keep, drop, rotate, reorder, split, join.
inkhaven pdf metadata <in> [--title ...] [--strip]
inkhaven pdf outline <in> [--set <toc>]
    Read / set document metadata and bookmarks.
inkhaven pdf impose | booklet -- print-ready signatures.
inkhaven pdf cover --pages <n> --out <o> [--isbn <i>][--image <p>]
inkhaven pdf barcode <isbn> --out <o>
    A full cover-and-spine PDF, or a standalone EAN-13 barcode.
inkhaven pdf preflight | grayscale | optimize | watermark | sample
    Print-readiness check, colour, size, stamping, proofs.

```

The submission apparatus tracks where a finished manuscript went and drafts its package:

**submissions add / list / status / add-note / remove**    The submission tracker — record a market, a status, a response, a note.

**submission digest / query / synopsis / comps / logline**    Build the book digest, then draft the query letter, synopsis, comp titles, and logline from it (AI).

## Scripting

One command reaches the embedded Bund language directly:

**bund <code>**    Evaluate a Bund expression against the Adam VM and print the top of the workbench — e.g. `inkhaven bund "40 2 + ."`. A phase-0 smoke command that does not open the project store; use it to verify the scripting layer works and to experiment with syntax. The full `ink.*` surface is documented in the Bund guide.

**output show** / The CLI surface over the Output message channel (the pane itself is  
**emit / dismiss** / a TUI feature) — useful when scripting, or when ssh'd into a project  
**clear** with no TUI.

## Planning and threads

Two families sit between structure and prose — the Planning Board and the plot-thread ledger:

● ● ● *The Planning Board and plot threads*

```
inkhaven plan init [--framework <f>]
    Scaffold a framework's beats: three_act (default),
    save_the_cat, story_circle, hero_journey, seven_point.
inkhaven plan check [--book-name <b>] [--drift <pct>] [--json]
inkhaven plan analyze [--book-name <b>] [--provider <p>]
    Diagnose beat coverage / pacing (deterministic), then map
    beats to chapters and name structural problems (AI).
inkhaven plan map <beat> <chapter> | unmap <beat>
inkhaven plan scaffold [--premise <p>] [--chapters]
inkhaven plan scene ... | sequel ... | tension rate
    Map beats, expand a premise, manage scene / sequel cards,
    and rate chapter intensity.
inkhaven thread add <name> [--status <s>] [--weight <w>]
inkhaven thread list [--status <s>] [--weight <w>]
inkhaven thread doctor [--json] | export [--format <f>] [--out <o>]
    Plot-thread paragraphs under the Threads book, with a
    health report. (Ctrl+V Shift+D.)
inkhaven myth scan | check | profile | refresh | suppress
    Mythological & symbolic pattern library. (Ctrl+V Shift+M.)
```

## Miscellaneous and imports

The remaining verbs: a one-shot model call, the Help-book importers, the Scrivener bridge, and the cost dashboard.

● ● ● *Odds and ends*

```

inkhaven ai <prompt> [--provider <p>]
    One-shot AI inference from the command line.
inkhaven cost
    Today's LLM call tallies per capped subsystem vs their caps.
inkhaven import-scrivener <path.scriv> [--draft-as-book <t>]
    [--skip-research] [--dry-run]
    Import a Scrivener project (RTF → Typst, single-binary).
inkhaven import-help --documents-directory <dir>
    Import a directory tree into the Help system book.
inkhaven import-typst-help
    Refresh the bundled Typst reference in the Help book (F1).

```

**HIDDEN COMMANDS**

A handful of verbs are hidden from `--help` because they exist only for the benchmark and endurance harnesses — `gen-fixture`, `_bench-load`, `_bench-render`, `_bench-embed`, `_bench-graph`, `_bench-report`, and `_soak`. They build synthetic projects and time the store, render, embed, and graph paths. An end-user project never needs them; they are listed here only so an unexpected entry in a process list is not a mystery.

**THE SINGLE SOURCE OF TRUTH**

This appendix is a curated map, and Inkhaven's command surface grows with every release. The authoritative, always-current reference for any single verb is the binary itself: `inkhaven <subcommand> --help` prints that subcommand's exact arguments, flags, defaults, and value sets as the installed version defines them. When this page and `--help` disagree, `--help` is right.

## APPENDIX C

# C

## The Configuration Reference

---

Every Inkhaven project carries one configuration file, `inkhaven.hjson`, written verbatim by `inkhaven init` and read once at startup. It is HJSON — strict JSON with comments, unquoted keys, optional commas, and multiline strings — so every example below pastes straight in. This appendix is the field-by-field reference Chapter 29 points to: one block per subsection, each field with its type, its compiled-in default, and a one-line meaning. Omit any block and you get exactly these defaults, because every field is `#[serde(default)]`.



**HOW THE LAYERED LOAD WORKS**

Precedence, lowest to highest: (1) compiled-in defaults, (2) the project's `inkhaven.hjson`, (3) `~/.config/inkhaven/config.hjson`, (4) every `~/.config/inkhaven/conf/*.hjson` in sorted filename order. A global file **wins over the project** and may be **partial** — put in only the keys you want to change. A malformed global file is skipped with a warning; only a malformed project file is fatal. Unknown fields are ignored. See Chapter 29 for the narrative treatment.

**DEFAULTS ARE FROM THE CODE**

Every default here is the value in `src/config.rs impl Default`, which is authoritative where it and the prose docs disagree. Four such corrections are flagged in place: `sync_interval_seconds` (600, not 60), `backup.out_dir` (empty, not `backups`), `book_rag.exclude_system_books` (eight entries), and `utopia.stage2_max_pairs` (present in code only).

## Top-level fields

The keys that sit at the root of the file, beside the block tables.

|                                       |                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>language</b>                       | string · <code>english</code> — primary writing language; drives Snowball stemmers + the F7 grammar prompt. Empty falls back to <code>editor.stemming.languages</code>                                                                                                                                                                                                               |
| <b>genre</b>                          | string · <code>(none)</code> — declared genre ( <code>literary_realism</code> , <code>fantasy</code> , <code>scifi</code> , <code>mystery</code> , <code>memoir</code> , <code>nonfiction</code> , <code>technical</code> , <code>academic</code> , <code>philosophy</code> , <code>theology</code> , <code>utopian</code> , <code>poetry</code> , ...); feeds genre-aware prompting |
| <b>prompts_file</b>                   | path · <code>prompts.hjson</code> — the prompt library, resolved against the project root                                                                                                                                                                                                                                                                                            |
| <b>inner_socrates_default_persona</b> | string · <code>utopia</code> — the project's default reader persona until one is explicitly activated                                                                                                                                                                                                                                                                                |
| <b>artefacts_directory</b>            | string · <code>empty</code> — where per-book build output lands; empty resolves to the OS cache dir, keeping artefacts out of the project tree                                                                                                                                                                                                                                       |

**sync\_interval\_seconds** `64 · 600` — seconds between background `Store::sync()` flushes of the HNSW index + DuckDB checkpoint; 0 disables the timer. (Code says 600; older prose said 60.)

## Everyday blocks

### embeddings

How paragraph bodies become vectors for semantic search (`fastembed`).

|                      |                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------|
| <b>model</b>         | <code>string · MultilingualE5Small</code> — <code>fastembed</code> model; pick an E5 for any non-English writing |
| <b>chunk_size</b>    | <code>int · 800</code> — approx characters per embedded chunk                                                    |
| <b>chunk_overlap</b> | <code>float · 0.15</code> — overlap fraction between adjacent chunks                                             |
| <b>pool_size</b>     | <code>int · 4</code> — r2d2 pool size per backing DuckDB file; clamped to a floor of 2 at open                   |

Model names: `MultilingualE5Small` (default, 384-dim), `MultilingualE5Base`, `MultilingualE5Large`, `BGEM3`, and the English-only `BGESmallENV15` / `BGEBASEENV15` / `BGELargeENV15`. Switching models triggers a one-time download; run `inkhaven reindex` afterward.

### llm

AI providers and the default. The shipped defaults define five providers; `default` picks one, `auto_fallback` lets a failed call retry a sibling.

● ● ● *llm — shipped defaults*

```
llm: {
  default: gemini
  auto_fallback: true
  providers: {
    gemini: { model: gemini-2.5-pro,   api_key_env: GEMINI_API_KEY }
    claude: { model: claude-sonnet-4-5, api_key_env: ANTHROPIC_API_KEY }
    openai: { model: gpt-4o,           api_key_env: OPENAI_API_KEY }
    deepseek: { model: deepseek-chat,  api_key_env: DEEPSEEK_API_KEY }
    grok:    { model: grok-2-latest,    api_key_env: XAI_API_KEY }
  }
}
```

|                                           |                                                                                          |
|-------------------------------------------|------------------------------------------------------------------------------------------|
| <b>default</b>                            | string · gemini — which provider is used with no <code>--provider</code> flag            |
| <b>auto_fallback</b>                      | bool · true — retry a failed call on another configured provider                         |
| <b>providers.&lt;name&gt;.model</b>       | string · varies — model id passed to genai (it picks the adapter)                        |
| <b>providers.&lt;name&gt;.api_key_env</b> | string · varies — env var holding the key; omit entirely for local providers like Ollama |

If an `api_key_env` is set but unset at runtime, Inkhaven refuses the call with a clean status message rather than crashing.

### editor — core

Editor-pane behaviour. The visual look lives in `theme`.

|                                  |                                                                                      |
|----------------------------------|--------------------------------------------------------------------------------------|
| <b>theme</b>                     | string · default — reserved; the visual theme is the top-level theme block           |
| <b>tab_width</b>                 | int · 2 — informational; tui-textarea inserts a literal tab                          |
| <b>wrap</b>                      | bool · true — soft word-wrap; false gives horizontal scroll                          |
| <b>autosave_seconds</b>          | int · 5 — idle seconds before a dirty paragraph auto-saves; 0 disables idle autosave |
| <b>startup_splash</b>            | bool · true — 7-second launch splash with today's stats                              |
| <b>mouse_captured</b>            | bool · true — initial mouse-capture state; Ctrl+Shift+M toggles                      |
| <b>confirm_quit</b>              | bool · false — confirm modal on Ctrl+Q                                               |
| <b>pov_chip_enabled</b>          | bool · true — status-bar POV/character chip                                          |
| <b>show_glossary_chip</b>        | bool · true — the C·P·A worldbuilding-density chip                                   |
| <b>show_facts_chip</b>           | bool · false — the [?]N Facts-count chip                                             |
| <b>startup_haiku</b>             | bool · true — emit a baked-in haiku at startup / new paragraph / Ctrl+Z p            |
| <b>continuation_anchor_count</b> | int · 1 — previous paragraphs sent as voice anchors for Ctrl+V d                     |
| <b>footnote_style</b>            | string · typst — typst (#footnote[...]) or markdown ([^id]) for Ctrl+V f             |
| <b>stemming.languages</b>        | list · [english, russian] — legacy; superseded by top-level language when non-empty  |

**editor — prompt language**

Which language each AI prompt resolves against.

**prompt\_language\_mode** `enum · book_defined` — `book_defined` uses the language field; `paragraph_detected` runs whatlang on the live paragraph

**prompt\_language\_detection\_min\_chars** `int · 10` — minimum characters before whatlang detection is attempted

**editor — durability**

Crash-safety and behaviour knobs (1.3.37).

**crash\_mirror\_seconds** `int · 2` — cadence of crash-rescue buffer mirrors

**deleted\_paragraph\_history** `int · 10` — kill-ring depth for Ctrl+V Shift+U undelete

**external\_change\_auto\_reload** `bool · true` — false warns instead of silently reloading a clean buffer changed on disk

**fact\_check\_idle\_seconds** `int · 5` — idle delay before the auto fact-check fires

**visited\_history\_cap** `int · 0` — 0 is unbounded; caps the back/forward visit list

**disk\_warn\_mb** `u64 · 100` — low-disk startup warning threshold (MiB); 0 disables

**warn\_uncommitted\_on\_exit** `bool · true` — confirm on quit if the git repo has uncommitted changes

**editor — reading & echo**

Reading-pace chips (1.2.18) and the echo-repetition scan (1.2.19+).

**reading\_time\_chip** `bool · false` — the 📖 remaining/total reading-length chip

**reading\_wpm** `u32 · 200` — words-per-minute for reading-time / pace / audiobook timing

**paragraph\_long\_secs** `u32 · 180` — read-time threshold flagging a paragraph-too-long finding

**echo\_window** `usize · 5` — consecutive-paragraph window for the echo scan

**echo\_min\_repeats** `usize · 3` — occurrences within the window required to flag

**echo\_max\_global** `usize · 40` — distinctiveness ceiling; words used more often are common vocabulary

**echo\_overlay** `bool · false` — default state of the live echo overlay (Ctrl+B Shift+K)

**editor.tts — read-aloud**

Text-to-speech (1.2.9 System backend + 1.2.17 Piper). Master switch off.

|                             |                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------|
| <b>enabled</b>              | bool · false — master switch for Ctrl+B S read-aloud                                |
| <b>voice</b>                | string · Milena — case-insensitive voice-name fragment (prefers Enhanced/Premium)   |
| <b>speed</b>                | f32 · 1.0 — rate multiplier over the engine's normal rate                           |
| <b>greeting</b>             | string · empty — spoken at startup; empty skips                                     |
| <b>goodbye</b>              | string · empty — spoken at shutdown (blocks up to 5s); empty skips                  |
| <b>engine</b>               | string · auto — auto   piper   system backend selector                              |
| <b>voices_dir</b>           | string · .inkhaven/voices — Piper voice cache (sandboxed to project root)           |
| <b>auto_download</b>        | bool · true — stream a missing Piper voice on first use                             |
| <b>catalog_url</b>          | string · piper-voices voices.json — Piper voice catalog URL                         |
| <b>catalog_ttl_hours</b>    | u32 · 24 — catalog cache freshness window                                           |
| <b>binary_path</b>          | string? · null — explicit piper binary; null auto-resolves via PATH then cache      |
| <b>auto_download_binary</b> | bool · true — the tts binary download CLI fetches Piper; the TUI never auto-fetches |
| <b>cache_max_voices</b>     | usize · 5 — LRU cap on the voices dir (models are 25–100 MB)                        |
| <b>play_command</b>         | string? · null — playback command override; {path} is the WAV                       |
| <b>sample_rate_hz</b>       | u32 · 22050 — Piper synthesis sample rate                                           |
| <b>auto_gitignore</b>       | bool · true — append the voices dir to .gitignore on first download                 |

**editor.style\_warnings — inline overlays**

The amber inline overlays flagging weak prose (1.2.9+). Master switch off; Ctrl+B Shift+F toggles per session.

|                             |                                                                      |
|-----------------------------|----------------------------------------------------------------------|
| <b>enabled</b>              | bool · false — master switch for all inline style overlays           |
| <b>filter_words.enabled</b> | bool · true (if master on) — intensifier/hedge/sensory-verb detector |

**filter\_words.use\_stemming** `bool` — match via Snowball stems so list entries are lemmas

**filter\_words.<lang>** `array · []` — per-language list; empty uses the built-in, non-empty REPLACES it

**filter\_words.extra\_words** `· []` — words added on top of the language default

**repeated\_phrases.enabled** `bool` — true (if master on) — flags n-grams repeating within a paragraph

**repeated\_phrases.n** `int · 4` — window size (consecutive non-stop words)

**repeated\_phrases.threshold** `int` — minimum occurrences before flagging

**repeated\_phrases.use\_stemming** `bool` — stem before n-gram comparison

**repeated\_phrases.<lang>.stop\_words** `array · []` — class words excluded from comparison; empty uses the built-in

**show\_dont\_tell.enabled** `bool · true` (if master on) — flags telling patterns (was angry, angrily, realised)

**show\_dont\_tell.use\_stemming** `bool` — stem entries so inflections collapse

**show\_dont\_tell.<lang>.linking\_verbs** `array · []` — category lists; empty uses the built-in, non-empty REPLACES  
**\_emotion\_adjectives**  
**\_manner\_adverbs** /  
**\_cognition\_verbs**

**anachronism.year** `int · (unset)` — the setting year; the detector is OFF until this is set

**anachronism.terms** `array · []` — additive term/earliest pairs on top of the ~35-term built-in lexicon

Any language can be curated through per-language **maps** under `filter_words.languages.<lang>`, `show_dont_tell.languages.<lang>`, and `repeated_phrases.languages.<lang>` (1.3.13); an uncurated language gets empty built-ins, never the English lists. `inkhaven lang bootstrap <lang>` populates them via the LLM.

## theme

Every colour Inkhaven paints, as `#RRGGBB` (or `#RGB`) strings; empty or unparseable falls back to the baked-in default. The shipped palette is Catppuccin Mocha.

● ● ● *theme — pane, border, modal, lexicon*

```

theme: {
  pane_bg: "#1e1e2e", pane_fg: "#cdd6f4"
  line_number_fg: "#6c7086", current_line_bg: "#313244"
  border_focused: "#cba6f7", border_unfocused: "#45475a"
  border_dirty: "#f9e2af", border_saved: "#a6e3a1"
  border_readonly: "#94e2d5"
  modal_bg: "#181825", modal_fg: "#cdd6f4", modal_border: "#cba6f7"
  places_fg: "#89dceb", characters_fg: "#f9e2af"
  language_word_fg: "#b4a8e1" // italic, per foreign-inclusion
  convention
  search_match_bg: "#f38ba8", search_current_bg: "#f5c2e7"
}

```

● ● ● *theme — tree, chips, overlays, syntax*

```

theme: {
  tree_open_marker: "#a6e3a1", tree_book_fg: "#f5c2e7"
  tree_chapter_fg: "#89b4fa", tree_subchapter_fg: "#94e2d5"
  tree_paragraph_fg: "#cdd6f4", editor_position_fg: "#89dceb"
  ai_scope_fg: "#fab387", ai_infer_fg: "#94e2d5"
  grammar_change_fg: "#f38ba8"
  style_warning_anachronism_fg: "#eba672"
  style_warning_echo_fg: "#b48ead"
  style_warning_banned_synonym_fg: "#e05a5a"
  pov_chip_bg: "#8b1d88", pov_chip_fg: "ffffff"
  syntax_heading: "#cba6f7", syntax_bold: "#f9e2af"
  syntax_italic: "#94e2d5", syntax_string: "#a6e3a1"
  syntax_number: "#fab387", syntax_comment: "#6c7086"
}

```

The per-detector `style_warning*_modifier` keys (`filter_word`, `repeated_phrase`, `show_dont_tell`, `echo`) default to the empty string, which maps to `underline`; accept `bold`, `dim`, `reversed`, `italic`, `none`, or `+`-combined forms like `underline+bold` for terminals where the default reads faint. The thirteen `syntax_*` keys drive the tree-sitter Typst highlighter.

## keys

The configurable global chords; everything else is hard-coded.

**save**                      chord · Ctrl+s — save the current paragraph

|                    |                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------|
| <b>search</b>      | chord · Ctrl+/ — focus the top Search bar                                                          |
| <b>ai_prompt</b>   | chord · Ctrl+i — focus the AI prompt bar                                                           |
| <b>next_pane</b>   | chord · Tab — cycle focus forward                                                                  |
| <b>prev_pane</b>   | chord · Shift+Tab — cycle focus back                                                               |
| <b>page_up</b>     | chord · PageUp — page up                                                                           |
| <b>page_down</b>   | chord · PageDown — page down                                                                       |
| <b>meta_prefix</b> | chord · Ctrl+b — the Meta prefix (set Ctrl+g if tmux eats Ctrl+B)                                  |
| <b>bund_prefix</b> | chord · Ctrl+z — the Bund prefix                                                                   |
| <b>view_prefix</b> | chord · Ctrl+v — the View prefix                                                                   |
| <b>bindings</b>    | array · [] — user overlay rebinding sub-chords (layer: meta_sub   bund_sub   view_sub   top_level) |

## backup

The `inkhaven backup` CLI and the auto-backup-on-exit hook.

|                                  |                                                                                                                                                                                                           |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>out_dir</b>                   | string · empty — snapshot destination; empty resolves to the OS data dir (not the literal backups/ the prose once implied). Empty-string never means disabled here — use <code>auto_backup_on_exit</code> |
| <b>max_age</b>                   | humantime · 7d — max age of the last backup before the exit hook makes a fresh one; 0s disables the auto hook                                                                                             |
| <b>wait_for_key_after_backup</b> | bool · true — hold the splash with a Press any key prompt                                                                                                                                                 |
| <b>amber_threshold</b>           | f32 · 0.5 — fraction of max_age at which the health chip turns amber; 0 disables the amber tier                                                                                                           |
| <b>auto_backup_on_exit</b>       | bool · true — the clear toggle for the exit-hook backup                                                                                                                                                   |
| <b>keep_last</b>                 | usize · 0 — 0 keeps all; else prune the oldest zips beyond this count                                                                                                                                     |

## goals & project

Writing-progress goals feed the status bar and the Ctrl+V G / Ctrl+V Shift+G modals. All optional; zero/empty disables a goal but still records events.

**goals.day\_boundary** enum · utc — utc | local; when the writing day rolls over for streaks + caps



**goals.daily\_words** int · 0 — project-wide daily word target

**goals.active\_minutes\_daily** int · 0 — daily active-time target (save-to-save gaps capped at 5 min)

**goals.streak\_grace\_per\_week** int · 0 — missed days forgiven per rolling 7-day window

**goals.books.<slug>.target\_words** int · 0 — book total; 0 hides the pace line

**goals.books.<slug>.date** string · empty — YYYY-MM-DD target date  
line

**goals.status\_ladder.<status>** string · {} — trailing-7-day promotion targets keyed by lowercased status  
tus>

**goals.auto\_promote\_book** bool · false — advance a paragraph's status when a save crosses its  
target\_words

**project.word\_count\_goal** int · 0 — total across counted\_books; 0 disables the goal display

**project.target\_date** string · empty — ISO date for the projection verdict

**project.counted\_books** array · [] — books that count (by title); empty means every user book

## ai

AI-pane behaviour not tied to a provider (1.2.6+).

**per\_paragraph\_memory** bool · false — record Paragraph-scope turns onto the paragraph's  
ai\_memory

**per\_paragraph\_memory\_max\_turns** int · 0 — turns kept per paragraph; 0 disables

**diff\_review\_on\_apply** bool · true — route r/g rewrites through a side-by-side diff modal first

**reseed\_prompt\_examples** bool · false — re-seed the Prompts book with the embedded .example  
prompts (idempotent)

## Retrieval & the semantic net

### book\_rag

The AI pane's always-on Book-scope retrieval (BOOK\_RAG-1). Tunes what a Book-scope prompt grounds on.

**top\_k** int · 5 — direct semantic hits retrieved per query

**context\_expansion** int · 1 — sibling paragraphs kept on each side of a hit

**max\_context\_token** int · 8000 — budget for the composed context (chars ÷ 4)

**include\_system\_books** list · notes, research, places, characters, artefacts, world, language — author-content books joined to the manuscript pool

**exclude\_system\_books** list · scripts, prompts, typst, help, intent, sources, glossary, snippets — meta books never searched (code default; the prose docs list only the first five)

## graph

GRAPHMIND — the Graph AI scope + **graph\_ask** walk. Bounds the cost of a walk; retrieval width is shared with **book\_rag**.

**ask\_max\_steps** int · 8 — max LLM turns a graph walk takes before a forced synthesis

**ask\_search\_width** int · 6 — seed nodes each search step returns

## continuity

SENTINEL — the unified, zero-AI continuity ledger. On by default and free.

**enabled** bool · true — master switch for the review-pass ledger

**ambient** bool · false — re-check the edit's scope on every save

**ambient\_cooldown\_secs** int · 30 — throttle floor between ambient re-checks

**co\_location** bool · true — a character in two places at overlapping times

**timeline** bool · true — orphaned events / fuzzy overlaps (standalone check)

**numeric** bool · true — direction reversal / conflicting durations (EN/FR/ES)

**char\_facts** bool · true — an established fact changed across chapters

**introduce** bool · true — an entity referenced before it is introduced

**introduce\_tolerance** int · 0 — earlier-chapter references tolerated before introduce flags

## lector

LECTOR — the read-through. Deterministic core on by default.

**enabled** bool · true — master switch for the read-through review line

**framework** string? · null — structure framework the shape is read against (three\_act, save\_the\_cat, story\_circle, hero\_journey, seven\_point, kishotenketsu); null suggests one from genre

## prose

NARR-1 — deterministic, zero-AI narrative-voice profiling per chapter.

**deep\_metrics** bool · false — include Tier-2 (sensory balance + active/passive ratio)

**matrr\_window** int · 100 — MATTR sliding-window size (tokens)

**baseline\_chapter** int · 1 — drift is measured against this chapter

**language** string? · null — override (en/ru/de/fr/es); null uses project language then EN

**ambient** bool · false — Ctrl+V Shift+V default; re-run after an editing pause

**ambient\_cooldown\_secs** 90 — floor between ambient runs

**extra\_modal\_tokens** array · [] — appended to the language's modal/hedging list

**extra\_interiority\_phrases** array · [] — appended to the language's FID phrase list

Its **thresholds** sub-block sets the per-metric crossing that raises a finding: **sent\_len\_cv** 0.15, **burstiness\_b** 0.15, **matrr** 0.05, **modal\_density** 0.020, **interiority\_ratio** 0.10, **de\_erlebte\_rede\_particle\_density** 0.05 (DE only), **sensory\_channel\_max** 0.15, **active\_passive\_ratio** 1.5.

## chorus & stylist

CHORUS — voice and style at book scale, plus the Inner Stylist coach. All advisory.

**chorus.distinct\_threshold** 0.5 — RMS z-distance below which two voices are flagged indistinguishable

**chorus.distinct\_ignore\_pairs** list — pairs never flagged, as Name|Name (order/case-insensitive)

**chorus.register\_drift\_threshold** register-metric change vs chapter 1 that flags a drift

**stylist.enabled** bool · true — master switch for the Inner Stylist

**stylist.session\_budget** float · 0.15 — informative daily USD budget for the coaching track

**stylist.language** string? · null — coaching-prompt language override

**dialogue**

DIALOG-1 — deterministic dialogue quality & attribution.

**attribution\_window** int · 60 — token window for the attribution name search

**unattributed\_run\_threshold** ~~int~~ — unattributed turns tolerated before the zero-attribution finding

**talking\_head\_threshold** 6 — dialogue-only paragraphs before the talking-head finding

**beat\_min\_words** int · 8 — minimum words for a non-speech sentence to count as an action beat

**said\_bookism\_threshold** 0.15 — said-bookism density delta above baseline that triggers a finding

**fingerprint\_min\_utterances** attributed utterances before a character fingerprint shows

**language** string? · null — override (en/ru/de/fr/es); null uses project then en

**extra\_neutral\_verbs** array · [] — verbs added to the neutral list (SF/fantasy speech verbs)

**extra\_said\_bookism** array · [] — verbs added to the said-bookism list

**drift**

Semantic-drift retrieval (1.3.10) — divergent descriptions of one entity.

**top\_k** usize · 24 — vector hits pulled per entity before name-filtering

**max\_snippets** usize · 8 — descriptions kept per entity (bounds the judge prompt)

**pronouns.<lang>.{character,place,artefact}** map · {} — per-language coref pronoun sets (1.3.13); written by lang bootstrap

**facts**

Series-shared canon (1.3.8). One field.

**shared\_path** string? · null — a directory of plain-text fact files layered under each book's Facts book (local wins on a clash)

## The Inner family & readers

### inner\_editor

INNER\_EDITOR-1 — the literary/stylistic companion. LLM-only, paragraph scope; caps inform, never block.

● ● ● *inner\_editor — the shape*

```
inner_editor: {
  enabled: true
  engagement: { idle_threshold_seconds: 60, cooldown_seconds: 120,
                max_findings_per_paragraph: 3 }
  context:    { preceding_paragraphs: 3, following_paragraphs: 0 }
  persona:    { tone: balanced, verbosity: concise,
                praise_frequency: moderate, genre_aware: true,
                belief_stance_enabled: true }
  output:     { severity_threshold: note, group_by_paragraph: true,
                always_show_persona_label: true }
}
```

**enabled** bool · true — false fully disables (the manual chord then just informs)

**engagement.idle\_threshold\_seconds** paragraph-pause wait before auto-engaging

**engagement.cooldown\_seconds** same-paragraph cooldown; an edit re-arms it

**engagement.max\_findings\_per\_paragraph** per paragraph

**context.preceding\_paragraphs** interpretation context sent before the paragraph

**context.following\_paragraphs** context sent after

**persona.tone** enum · balanced — critical | balanced | encouraging

**persona.verbosity** enum · concise — concise | standard | detailed

**persona.praise\_frequency** moderate — rare | moderate | frequent

**persona.genre\_aware** bool · true — fold the project genre into the prompt

**persona.belief\_stance\_enabled** — allow the does-the-prose-believe-itself category

**persona.categories.\*** bool · true — the eight categories, each toggleable (literary\_richness, tautology, style\_observation, style\_instability, dictionary\_richness, belief\_stance, craft\_praise, editorial\_suggestions)

**output.severity\_threshold** note — praise | note | concern visible floor; at note, Praise is persisted but not pushed to Output

**output.group\_by\_paragraph** bool · true — group findings by paragraph

**output.always\_show\_persona\_label** bool · true — always show persona label every finding with the persona

**llm.editor\_engagement.max\_calls\_per\_session** int · 60 — max calls per session, never blocks)

**llm.editor\_engagement.confirm\_above\_calls** int · 40 — confirm above calls session exceeds this

**llm.editor\_engagement.max\_calls\_per\_day** int · 200 — max calls per day in inkhaven cost

**llm.editor\_engagement.max\_calls\_per\_month** int · 4000 — max calls per month

**llm.conversation.max\_calls\_per\_session** int · 80 — max calls per session in conversation-mode caps \_per\_day

**llm.backoff\_max\_retries** int · 3 — retry count on transient failure

**llm.backoff\_initial\_seconds** int · 1 — initial backoff delay

## theologian

INNER-THEOLOGIAN-1 — the tradition-neutral moral/theological reader. Eleven lenses, never a verdict.

**enabled** bool · true — master switch; false gates everything including the fast-track

**on\_paragraph\_idle** bool · true — fire a Category-1 question on paragraph idle

**idle\_threshold\_seconds** int · 8 — idle wait; null uses Inner Socrates' Slow threshold

**session\_budget** f32 · 0.15 — slow-track USD sub-budget; informs, never blocks

**fast\_track** bool · true — run the deterministic fast-track in the review pass

**moral\_invisibility\_window** int · 3 — paragraphs after a harm event checked for acknowledgment

**consequence\_gap\_window** int · 5 — paragraphs after lethal violence checked for consequence

**sacred\_levity\_signal** bool · true — emit the sacred-vocabulary-in-levity signal

**disabled\_lenses** array · [] — lens codes to exclude from slow-track hints

**language** string? · null — question/marker language override; null uses project then en

### rigor

RIGOR — the deterministic, zero-AI reasoning-rigor reader.

**enabled** bool · true — master switch

**fast\_track** bool · true — run in the review pass / deep-refresh

**language** string? · null — marker-language override; null uses project then English

**false\_dichotomy** bool · true — flag forced-binary framings

**question\_begging** bool · true — flag unargued assertions (obviously, of course)

**straw\_man** bool · true — flag dismissive characterizations

**overgeneralization** bool · true — flag strong absolutes (always, never)

**non\_sequitur** bool · true — flag a conclusion connective with no warrant

**equivocation** bool · true — flag a Glossary watch\_equivocation term used without pinning a sense

### myth

MYTH-1 — the symbolic pattern library over the declared Mythology book.

**enabled** bool · true — master switch for the review-pass findings + heatmap chord

**heatmap\_buckets** usize · 8 — chapter buckets the heatmap collapses the book into

**consistency\_min\_chapters** usize · 5 — chapters a symbol must span before the LLM consistency check

**motif\_min\_occurrences** usize · 3 — occurrences a motif needs before the LLM completeness check

**final\_act\_pct** u32 · 25 — the final act is the last this-percent of chapters

**check\_cost\_warn** f32 · 0.08 — warn when an myth check LLM run exceeds this USD

## utopia

WORLD-6 — utopian/dystopian coherence over declared premises.

**stage2\_cost\_warn** f32 · 0.10 — Stage-2 cost-warning threshold (USD); informs, never blocks

**stage2\_max\_pairs** int · 200 — cap on premise pairs compared in Stage 2 (code-only; absent from the prose docs)

**stage3\_batch\_size** int · 5 — chapters per Stage-3 background pass

**stage3\_min\_chapter\_words** int · 200 — minimum chapter words for the entailment scan

**group\_gap\_threshold** int · 1 — non-claim paragraphs that break a premise group; 0 makes all one group

## char

CHAR-1 — character-arc tracking.

**stall\_threshold** int · 4 — unchanged chapters after the baseline before a stall fires

**active\_window\_before** int · 5 — tokens before a verb a name may sit and still be the actor

**active\_window\_after** int · 8 — tokens after a verb a name may sit and be the patient

**min\_chapters\_for\_check** int · 3 — chapters of state before the LLM arc checks run

**enrich\_from\_dialog** bool · true — enrich the state chain with DIALOG-1 signals

**enrich\_from\_voice** bool · true — enrich with NARR-1 chapter interiority

**language** string? · null — arc language override; null uses project then English

**extra\_action\_verbs** array · [] — genre verbs the agency scorer treats as action

**extraction\_cost\_warn** f32 · 0.20 — state-extraction cost-warning USD; informs, never blocks

## world

WORLD-12 — the AI world-critique pass (**realworld critique**).

**critique\_enabled** bool · true — false runs the deterministic lints only, skipping the LLM

**critique\_max\_tokens** size · 24000 — per-call soft token cap; 0 disables it

**critique\_cost\_warn** f32 · 0.10 — warn when a run exceeds this USD



## Timeline

### timeline

Story timeline (1.2.6+). Off by default; set `enabled: true` plus a calendar to turn on event tracking.

**enabled** bool · false — master switch; every chord/CLI/Bund hint no-ops when off

**default\_track** string · main — track label when an event has none

**calendar.preset** string · custom — gregorian | sols | custom (custom honours every field)

**display.show\_orphan** bool · true — synthetic orphan row at the bottom of the swim lane

**display.swim\_lane\_max\_rows** int · 10 — truncate beyond this with a +N more row

**display.default\_zoom** float · 1.0 — initial ticks-per-cell

**display.grid\_every\_days** int · 7 — faint vertical bar every N days; 0 disables (code-only field)

The `calendar` sub-block (preset `custom`) carries `base_unit`, a base-first `units` stack (each with `per_parent` + optional `names`), `seasons`, `epoch_label` / `epoch_before_label`, a `display_format` with `{year}`, `{month}`, `{month-name}`, `{day}`, `{hour}` tokens, and `parse_aliases`. `sols` and `gregorian` expand to preset stacks.

### timeline.critique

The refactored timeline critique — orphan + fuzzy-overlap + optional LLM elaboration. Gated by `timeline.enabled`.

**enabled** bool · true — master switch for the critique

**orphan.enabled** bool · true — the orphaned-event check (code-only sub-block)

**orphan.min\_orphan\_age\_days** int · 0 — suppress orphan findings younger than this; 0 emits immediately

**orphan.min\_significance** string · low — lowest significance surfaced (low | moderate | high)

**fuzzy\_overlap.enabled** bool · true — the fuzzy-precision overlap check

**fuzzy\_overlap.min\_suspicion** moderate — lowest suspicion surfaced (low | moderate | high)

**fuzzy\_overlap.cluster\_min\_size** minimum events for a cluster (vs pairwise) finding

**elaboration.enabled** bool · true — LLM elaboration when a provider exists; else pattern-only text

**elaboration.max\_calls\_per\_run** hard cap on elaboration calls per run

**elaboration.confirm\_above\_calls** confirm once a run would exceed this

**legacy\_flag\_deprecation.warn\_on\_use** on the retired event critique --legacy path

## scrivener

Scrivener-importer behaviour (1.2.8+). Gated by **timeline.enabled**.

**date\_fields** array · [Date, Story Date, Event Date] — CustomMeta field names interpreted as event dates

## Research

### research — the assistant

RESRCH-1 — the separate **inkhaven research** TUI. Omit the block for these defaults.

**default\_thread** string? · null — thread to open (null = picker / default)

**rag\_top\_n** usize · 5 — max Facts paragraphs prepended per query as RAG context

**session\_budget\_warning** f64 · 0.50 — per-session cost-cap warning (USD); informs, never blocks

**max\_pinned\_nodes** usize · 3 — max pinned Facts nodes (Ctrl+P)

**show\_keybind\_hints** bool · true — show the keybind hints bar by default

**min\_width** u16 · 80 — minimum terminal width before a resize message shows

**split\_ratio** u32 · 4 — Facts-tree columns out of 10 (4 = 40% tree)

**diff\_top\_n** usize · 3 — /diff: similar facts shown

**verify\_min\_sentence\_words** — /verify: minimum sentence words for claim extraction

|                           |                                                                                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>dedup_warn_score</b>   | f64 · 0.92 — /fact: near-duplicate similarity that warns; informs, never blocks                                      |
| <b>triangulate_gate</b>   | bool · false — /fact from model/web/document is triangulated across sources before commit (code-only; network-heavy) |
| <b>refute_gate</b>        | bool · false — /fact from model/document gets one adversarial refutation pass before commit (code-only)              |
| <b>import_chunk_chars</b> | size · 1500 — /import: max characters per embedded chunk                                                             |

The `research.agentic` block gates the autonomous deep-research loop: `enabled` (bool, `true`), `max_subquestions` (usize, `6` — the total Facts a run may emit), `max_rounds` (usize, `3` — gap-driven iterate rounds; `1` is a single pass). The `research.web` block gates `/web: enabled` (`false`), `provider` (`none` — `tavily` | `searxng` | `none`), `api_key` (empty), `endpoint` (empty), `max_results` (`5`), `fetch` (`true`), `pipeline` (`chat` — `chat` | `ingest`).

### research source blocks

Each keyless source under `research.max_chars` bounds the embedded portion; `auto_cite` mints a SOURCES-1 entry on `/fact`.

|                   |                                                                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>geonames</b>   | <code>enabled true · endpoint http://api.geonames.org · username empty</code><br>(empty keeps /geonames unavailable)                        |
| <b>gutenberg</b>  | <code>enabled true · endpoint https://gutendex.com · max_chars 300000 · auto_cite true</code>                                               |
| <b>archive</b>    | <code>enabled true · endpoint https://archive.org · max_chars 300000 · auto_cite true</code> (Internet Archive; absent from the prose docs) |
| <b>wikidata</b>   | <code>enabled true · endpoint https://www.wikidata.org · max_statements 24</code> (top of the trust ladder; skips the fact-check gate)      |
| <b>wikisource</b> | <code>enabled true · default_lang en · max_chars 300000 · auto_cite true</code>                                                             |
| <b>scholarly</b>  | <code>enabled true · mailto empty · auto_cite true</code> (OpenAlex + arXiv)                                                                |

The `research.scripture` block (`/bible`, `/quran`, `/bookofmormon`) defaults: `enabled true`, `bible_endpoint` `https://bolls.life`, `quran_endpoint` `https://api.alquran.cloud/v1`, `bom_url` the bcbooks 1830 Book of Mormon JSON, `bible_translation` `null` (`en=WEB`, `ru=SYNOD`, `fr=FRLSG`, `de=LUT`, `es=RV1960`), `quran_translation` `null`, `max_chars` `200000`, `auto_cite true`.

## Content authoring

### sources

SOURCES-1 — the bibliography engine over the Sources book.

**all** bool · true — collect every Sources entry into every book; false scopes per book

**bibliography\_style** string · ieee — CSL style passed to Typst's #bibliography (apa, chicago-author-date, mla, ...)

**auto\_bibliography** bool · true — append the #bibliography(...) line during assembly

### snippets

Trigger-keyed editor text expansions (1.2.14+), under `editor.snippets`. A trigger followed by Space expands. Placeholders: `{today}` (optionally with a format, `{today:%Y-%m-%d}`), `{now}`, `{paragraph_slug}`, `{paragraph_title}`, `{selection}`, `{author}`, `{cursor}`, the picker forms `{char_lookup}` / `{place_lookup}` / `{artefact_lookup}`, and a `bund:` body prefix. The trigger map lives under `.triggers`.

• • • *editor.snippets — a few bindings*

```
editor: {
  snippets: {
    triggers: {
      "\\dt": "{today:%Y-%m-%d}"
      "\\au": "- {author}"
      "\\todo": "TODO ({today}): {cursor}"
    }
  }
}
```

### jinja

STRUCT-1 — Jinja template paragraphs. Self-gating; one knob.

**continue\_on\_error** bool · false — false aborts assembly on a render error; true writes a visible error block and continues

### images

Image-node handling in the tree and editor.

|                           |                                                                             |
|---------------------------|-----------------------------------------------------------------------------|
| <b>preview_enabled</b>    | bool · true — render image previews                                         |
| <b>allowed_extensions</b> | list · png, jpg, jpeg, gif, webp, svg — extensions accepted for image nodes |
| <b>max_size_bytes</b>     | u64 · 33554432 (32 MiB) — a larger file is rejected with a status message   |

## Typst output & export

These blocks feed the synthesised `settings.typ` / `globals.typ` prepended to every assembled book. Empty/zero generally falls through to Typst's own default.

### typst\_compile

The compile engine and diagnostics (`Ctrl+B B` / `Ctrl+B 0`).

|                    |                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------|
| <b>engine</b>      | string · external — external (shells out to typst on PATH)   inprocess (typst as a library) |
| <b>diagnostics</b> | bool · true — run typst-syntax on save/idle; parse errors on the status bar                 |

**diagnostics\_idle\_seconds** — idle debounce before the recheck; 0 runs every tick

**semantic\_diagnostics** bool · false — full typst::compile on the open paragraph (inprocess only; false positives expected)

**bundle\_fonts** bool · true — ship Computer Modern + Linux Libertine in the binary (inprocess)

**use\_system\_fonts** bool · true — also search host system fonts (inprocess)

**packages\_enabled** bool · true — fetch @preview/<pkg> from packages.typst.org (inprocess)

**wait\_for\_key\_after\_compile** bool · true — hold the splash after compile finishes

**error\_system\_prompt** string · empty — override the AI compile-error prompt

### typst\_page, typst\_fonts, typst\_layout

Page geometry, face, and paragraph layout.

**typst\_page.paper** string · us-letter — anything Typst's paper: accepts

**typst\_page.margin\_top** string · 2.5cm / 2.5cm — top and bottom margins  
**\_bottom**

**typst\_page.margin\_inside** string · 3cm / 2cm — binding-edge and outside margins  
**\_outside**

**typst\_page.page\_numbering** string — page-number format; empty = none

**typst\_page.columns** u32 · 1 — column count; 0/1 fall through to single-column

**typst\_fonts.body** string · Linux Libertine — body typeface (emitted with a bundled fallback)

**typst\_fonts.body\_size** string · 11pt — body size

**typst\_fonts.monospace** string · DejaVu Sans Mono — raw/code typeface

**typst\_fonts.language** string · en — two-letter tag for #set text(lang:)

**typst\_layout.justify** bool · true — justify body paragraphs

**typst\_layout.leading** string · 0.7em — inter-line leading

**typst\_layout.paragraph\_indent** string — first-line indent; empty = none

**typst\_layout.heading\_numbering** string — #set heading(numbering:) arg; empty = unnumbered

## typst\_templates

The eight `wrap_*` functions written into each book's `globals.typ`. Each is a string; an empty string falls back to the shipped default for that one function (so you override just the wrapper you care about). Fields: `wrap_book`, `wrap_chapter`, `wrap_subchapter`, `wrap_paragraph`, `wrap_image_book`, `wrap_image_chapter`, `wrap_image_subchapter`, `wrap_image_inline` — all default empty. The shipped bodies live in `default_wrap_*` and are regenerated on every assembly, so edits belong in the config, not the generated file.

## typst\_universe

Source for the `Ctrl+V` # package-import picker (1.6.15+).

**url** string · `orangex4 typst-universe-with-stars packages.json` — the community package manifest

**ttl\_hours** u32 · 24 — cache-freshness window in hours

**frontmatter**

Journal-article front matter (1.6.15+). All-empty renders nothing.

|                          |                                                                                              |
|--------------------------|----------------------------------------------------------------------------------------------|
| <b>abstract</b>          | string · empty — a single plain-prose paragraph (Rust field <code>abstract_text</code> )     |
| <b>keywords</b>          | list · [] — keyword list                                                                     |
| <b>authors</b>           | list · [] — Author records: name, affiliation, orcid, email, corresponding (all empty/false) |
| <b>funding</b>           | string · empty — funding statement; identifying, dropped under --blind                       |
| <b>data_availability</b> | string · empty — kept under --blind                                                          |
| <b>code_availability</b> | string · empty — kept under --blind                                                          |

**output**

Multi-format export for `Ctrl+B 0`.

|                            |                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------|
| <b>extra_formats</b>       | list · [] — formats built alongside the PDF (markdown, tex, epub, docx, imposed_pdf, cover_pdf) |
| <b>extras_step_pause_m</b> | 64 · 400 — ms the extras splash holds each format; 0 disables the pause (code-only)             |
| <b>extras_wait_for_key</b> | bool · false — hold the final splash frame for a keypress (code-only)                           |
| <b>imposed_pdf_config</b>  | string · default — the imposition profile used when imposed_pdf is listed                       |

**docs**

TDOC — the technical-documentation blocks. Only `docs.html` appears in the prose docs; `docs.verify`, `docs.variables`, and `docs.index` are code-only.

|                               |                                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>verify.enabled</b>         | bool · false — master switch for verified code blocks; nothing runs unless true                                                               |
| <b>verify.timeout_seconds</b> | 14 · 30 — per-block wall-clock cap                                                                                                            |
| <b>verify.runners</b>         | map · {} — language → shell command ( <code>{file}/{dir}</code> substituted, run via <code>sh -c</code> )                                     |
| <b>verify.extensions</b>      | map · seeded — language → temp-file extension ( <code>rust→rs</code> , <code>python→py</code> , ...); unknown falls back to <code>.txt</code> |

**variables** map · {} — TDOC-3 {{key}} substitutions applied at assembly across every export

**index.from\_glossary** bool · true — INDEX-1: include every Glossary term (synonyms as see-refs)

**index.terms** list · [] — extra index terms beyond the Glossary

The `docs.html` static-site export (`inkhaven export html`):

**html.site\_title** string? · null — null uses the exported book's title

**html.theme** string · default — bundled theme name (only default today)

**html.template\_dir** string · html — project override root for functional/ and theme/ files

**html.variables\_file** string · html.hjson — HJSON exposed to templates as site

**html.search** bool · true — build the client-side search index (accepted, wired later)

**html.citation\_style** string · author-year — author-year | numeric (accepted, wired later)

**html.include.sources** bool · true — fold Sources / Glossary into the site sary

**html.include.characters** bool · false — places the remaining companion books, off by default  
`guage / .world / .mythology / .notes / .index`

## **tex\_export**

LaTeX export document class + preamble (1.6.15+). All-empty leaves `tylax's article` output untouched.

**document\_class** string · empty — journal class (IEEEtran, elsarticle, article, ...)

**class\_options** string · empty — class options (conference, twocolumn, 11pt,a4paper)

**extra\_packages** list · [] — extra usepackage lines (bare names or full lines)

**preamble** list · [] — raw preamble lines before `begin{document}`



## PDF production (1.3.0)

Three blocks feed the `inkhaven pdf ...` subsystem. `imposition.profiles` names folding-signature recipes for `pdf impose` (six are built in — `default`, `chapbook`, `us_perfect`, `us_chapbook`, `thick`, `a5_book`); a profile carries `style` (`perfect_bound` | `saddle_stitch` | `side_stab`), `sheets_per_signature`, `target_sheet_size`, `orientation`, `margins`, a `creep` sub-block, `marks` toggles, and `blank_page_policy`. `cover` sets house cover/spine defaults — `front_width_mm` 152.0, `front_height_mm` 229.0, `bleed_mm` 3.0, `interior_stock` `uncoated_80gsm`, `cover_stock` `cover_250gsm`, `spine_font_size_pt` 11.0, `image_fit` `cover`. `preflight` sets DPI targets — `default_profile` `hand_binding`, `hand_binding_dpi` 300, `print_shop_dpi` 300.

## Environment, safety & audio

### health

Background health monitor (1.2.15+). Off by default.

**enabled**                    `bool · false` — spawn the monitor task; the status chip stays hidden when off

**auto\_repair.rescue\_duplicates** `bool · false` — auto-delete \*.inkhaven-rescue files older than 30 days

### project\_lock

Single-instance advisory lock (1.3.36) — informs, never hard-blocks by default.

**enabled**                    `bool · true` — acquire the lock; false disables guarding entirely

**on\_conflict**                `string · prompt` — `prompt` (interactive y/N) | `warn` (always proceed) | `refuse` (never open a second session)

### scripting

The Bund sandbox policy. Destructive categories deny by default; writers opt in.

**trust\_decision**            `string · ask` — `ask` (needs a .inkhaven/trust marker) | `trust` (run unconditionally) | `deny` (never run)

**fs\_unsandboxed**           `bool · false` — false confines `ink.fs.*` to the project root via `path_safety`

Category gates (`fs_write`, `net`, `shell`, `code_eval`, ...) default to deny; list the categories or individual words a project may use. See `SECURITY_WARNING.md`.

## shell

The embedded nushell pane (1.2.8+, `Ctrl+Z o`).

**enabled** bool · true — open the shell pane; false makes the chord a no-op

**max\_buffered\_turns** int · 50 — in-memory (command, output) pairs retained

**max\_output\_lines** int · 1000 — per-turn stdout/stderr lines kept; excess is truncated with a marker

**blocked externals** list · ~40 basenames — full-screen TUI apps refused before spawn (vim, less, top, tmux, ssh, ...); [] disables the guard

**external\_timeout\_secs** int · 30 — wall-clock budget per command before a watchdog interrupt

**insert\_template** string · `#raw(block: true, lang: shell, `{output}`)` — Typst wrapping a `Ctrl+Z h→i` insert

## hierarchy

**unbounded\_subchapters** bool · false — false keeps Book→Chapter→Subchapter→Paragraph; true nests subchapters arbitrarily

## sound

TUI sound effects (opt-in; `Ctrl+B E` toggles).

**enabled** bool · false — master switch, off so launch is silent

**volume** f32 · 0.6 — master volume 0.0–1.0, clamped at load

## oracle

ORACLE (1.7.9+) — the conlang phonotactic guardian on save.

**enabled** bool · true — master switch; false gates the on-save scan

**on\_save** bool · true — run the guardian when a paragraph is saved

## cost

The AI-cost dashboard (`Ctrl+B $ / inkhaven cost`). Caps **inform**, never gate.

**world\_daily\_call\_cap** int · 200 — world fact-check slow-track ceiling

**inner\_socrates\_daily\_call\_cap** int · 50 — Inner Socrates slow-track ceiling

**usage\_retention\_days** · 30 — days of per-category tallies kept before pruning

**default\_input\_per\_1m** · 3.0 / 3.0 — fallback USD-per-1M-token prices

**default\_output\_per\_1m**

The **pricing** map ships list prices per model-name substring (gemini-2.5-pro 1.25/10.0, claude-sonnet 3.0/15.0, claude-opus 15.0/75.0, gpt-4o 2.50/10.0, deepseek 0.27/1.10, and more); override an entry as prices move.

#### THE KNOWLEDGE / KEN BLOCK

The 2.6.0 KEN feature (epistemic continuity) has an RFC but no config block yet — there is no **knowledge:** key in **Config**. When it ships it will follow the same **#[serde(default)]** contract as every block above, so an older config keeps working unchanged.

Beyond these blocks, **inkhaven.hjson** is forward-compatible by construction: old configs run on new releases (missing fields take the default), obsolete fields stay parseable (so downgrading is safe too), and Inkhaven never rewrites your file in place. To reset to shipping defaults, rename the file and run **inkhaven init --force**, then re-merge your customisations from the annotated template at **assets/default\_project.hjson** — the same file **init** writes.

## APPENDIX D

## D

## The Feature Index

---

This is the single map of the whole product: every feature Inkhaven ships, wired to the four ways you reach it. Read a row left-to-right — the feature name, then its terminal verb (`inkhaven <verb>`), its in-editor chord, the **Manual chapter** that teaches it, and the `ink.*` Bund namespace that scripts it. Use it two ways: forward, when you know a feature and want its command or chord; and backward, when you meet a `Ctrl+...` chord or an `ink.*` symbol and want to know what it belongs to. It mirrors `Documentation/FEATURE_INDEX.md` and adds the chapter column so the map points back into this book. Chords are authoritative in Appendix A; verbs in Appendix B; every config block in Appendix C.

## HOW TO READ A CELL

Each right-hand cell reads `CLI · chord · ch N · ink.*`, in that order, and drops whatever does not apply. A missing verb means the feature is chord-only; a missing chord means it is terminal-only; **ch** — means no single chapter owns it (reach it by CLI or its topic guide); a missing namespace means it is not scriptable from Bund. Where two chapters share a feature both are named.

## Manuscript & editor

**The editor & tree**    chord (pane keys) · **ch** 4, 5 · ink.tree ink.node  
ink.paragraph

**Search (semantic  
+ full-text)**    CLI search · chord Ctrl+V S · **ch** 7 · ink.search

**AI writing assis-  
tant**    chord F9 (scope cycle) · **ch** 9 · ink.ai

**Chat with your  
book (RAG)**    CLI book-rag · chord F9 → Book · **ch** 10 · ink.book\_rag

**Snapshots &  
split-edit**    chord F6 / Ctrl+F6 · **ch** 6 · ink.snapshot

**Outline pane**    CLI outline · chord Ctrl+2 / Ctrl+B Shift+0 · **ch** 5 ·  
ink.outline

**Structural para-  
graphs (`para:\*`)**    chord i (Tree) · **ch** 5

**Jinja template  
paragraphs**    chord e (Tree) · **ch** 5

**Reusable snip-  
pets**    CLI snippets · chord Ctrl+V x · **ch** 5 · ink.snippets

**Style & filter-  
word overlays**    chord Ctrl+B Shift+F · **ch** 8

**Output pane**    CLI output · chord Ctrl+B Tab · **ch** 3 · ink.io ink.pane

**Writing goals & streaks** CLI `goals` · chord `Ctrl+V g` / `Ctrl+B Shift+G` · **ch** —

## The world & knowledge layer

**World simulation** CLI `realworld` · chord `Ctrl+B W` · **ch 14** · `ink.world`

**Interactive worldbuilder** CLI `worldbuilder` (own TUI) · **ch 14** · `ink.world`

**Utopia / dystopia coherence** CLI `world utopia-check` · **ch 14** · `ink.utopia`

**Facts & fact-check** CLI `fact-check` · chord `Ctrl+B Shift+X` · **ch 14**

**Semantic net (graph)** CLI `graph` · chord `Ctrl+B z` · **ch 15** · `ink.graph`

**Timeline & events** CLI `event` (export `export-timeline`) · **ch 16** · `ink.event`

**Places / Characters** chord `Ctrl+B P` / `Ctrl+B C` · **ch 13** · `ink.char`

**Character arcs** CLI `character` · chord `Ctrl+V Shift+N` · **ch 13** · `ink.char`

**Plot threads** CLI `thread` · chord `Ctrl+V Shift+D` (doctor) · **ch** — · `ink.thread`

**Mythological patterns** CLI `myth` · chord `Ctrl+V Shift+M` · **ch** — · `ink.myth`

## The intelligences — the book watches & reads itself

**SENTINEL — continuity** CLI `continuity check` · chord `Ctrl+B Shift+I` · **ch 17** · `ink.continuity`

**KEN — who knows what, when** CLI `knowledge` · chord `Ctrl+B Shift+Z` · **ch 17** · `ink.knowledge`

|                                           |                                                                                                                                 |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>LECTOR — read-through</b>              | CLI <code>readthrough</code> · chord <code>Ctrl+B Shift+A</code> · <b>ch 18</b> · <code>ink.readthrough</code>                  |
| <b>CHORUS — voice at book scale</b>       | CLI <code>chorus</code> · chord <code>Ctrl+B J → Y</code> · <b>ch 18</b> · <code>ink.chorus</code><br><code>ink.stylist</code>  |
| <b>NARR — narrator voice</b>              | CLI <code>prose</code> · chord <code>Ctrl+V V</code> · <b>ch 18</b> · <code>ink.prose</code>                                    |
| <b>Dialogue quality &amp; attribution</b> | CLI <code>dialogue</code> · chord <code>Ctrl+V Shift+Q</code> · <b>ch 18</b> · <code>ink.dialogue</code>                        |
| <b>REDLINE — revision partner</b>         | CLI <code>revise</code> · chord <code>Ctrl+V Shift+R</code> · <b>ch 19</b> · <code>ink.revise</code><br><code>ink.review</code> |
| <b>CHRONICLE — draft history</b>          | CLI <code>chronicle</code> · chord <code>Ctrl+B Shift+U</code> · <b>ch 19</b> · <code>ink.chronicle</code>                      |
| <b>Unified review pass</b>                | CLI <code>check</code> · chord <code>Ctrl+B Shift+C</code> · <b>ch 19</b>                                                       |

## The inner family — readers who question

|                                      |                                                                                                                                              |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inner Socrates (Dialectician)</b> | CLI <code>inner-socrates</code> · chord <code>Ctrl+B J</code> · <b>ch 20</b> · <code>ink.inner_socrates</code>                               |
| <b>Inner Editor</b>                  | CLI <code>inner-editor</code> · chord <code>Ctrl+V 0</code> · <b>ch 20</b> · <code>ink.inner_editor</code>                                   |
| <b>Inner Theologian</b>              | CLI <code>theologian</code> · chord <code>Ctrl+B J → T</code> · <b>ch 20</b> · <code>ink.theologian</code> <code>ink.inner_theologian</code> |
| <b>Inner Poet</b>                    | CLI <code>poetry</code> · chord <code>Ctrl+B J → P</code> · <b>ch 20</b> · <code>ink.poem</code>                                             |
| <b>Reasoning rigor</b>               | CLI <code>rigor</code> · <b>ch 20</b>                                                                                                        |
| <b>Companions cockpit</b>            | CLI <code>companions</code> · <b>ch 20</b>                                                                                                   |

## Language & poetry

**ConLang suite** CLI `language` · chord `Ctrl+B X` · ch 21 · `ink.lang`

**Linguistic analysis** CLI `language scan` · ch 21 · `ink.lang`

**WordNet thesaurus** CLI `wordnet` · chord `Ctrl+V Shift+Y` · ch 21

**Poetry (measure verse)** CLI `poetry` · chord `Ctrl+B J → P` · ch 22 · `ink.poem`

## Research & scholarship

**Research assistant** CLI `research` (own TUI) · ch 23

**Sources / bibliography** CLI `sources` · chord `Ctrl+V @` · ch 23 · `ink.sources`

**Terminology / glossary** CLI `terms` · chord `Ctrl+V z` · ch 23 · `ink.terms`

**Index Locorum / Verborum** CLI `index-locorum` · `index-verborum` · ch 23

**Scholarly lexicon** CLI `lexicon` · ch 23

**Argument outline** CLI `argue` · ch 23

**Back-of-book index** CLI `index` · ch 23

## Export & production

**Typst / PDF assembly** CLI `export` / `build` · chord `Ctrl+V R` · ch 24 · `ink.export`  
`ink.typst`



|                                    |                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>EPUB (import + export)</b>      | CLI <code>import-epub / export</code> · <b>ch 25</b> · <code>ink.export</code>                            |
| <b>HTML website</b>                | CLI <code>export html</code> · <b>ch 25</b>                                                               |
| <b>Technical-doc checks (TDOC)</b> | CLI <code>docs verify / links / review</code> · chord <code>Ctrl+B Shift+D</code> · <b>ch 26</b>          |
| <b>arXiv / scientific bundle</b>   | CLI <code>export --bundle &lt;path&gt;</code> · <b>ch 25</b>                                              |
| <b>Audiobook / TTS</b>             | CLI <code>tts</code> · chord <code>Ctrl+B S / Ctrl+B Shift+V</code> · <b>ch 25</b> · <code>ink.tts</code> |
| <b>Scrivener import</b>            | CLI <code>import-scrivener</code> · chord <code>F3</code> · <b>ch 25</b>                                  |

## System & scripting

|                                   |                                                                                            |
|-----------------------------------|--------------------------------------------------------------------------------------------|
| <b>Bund scripting</b>             | CLI <code>bund</code> · Script nodes · <b>ch 27</b> · the whole <code>ink.*</code> surface |
| <b>AI cost dashboard</b>          | CLI <code>cost</code> · chord <code>Ctrl+B \$</code> · <b>ch 12</b>                        |
| <b>Project doctor</b>             | CLI <code>doctor</code> · chord <code>Ctrl+B Shift+0</code> · <b>ch 28</b>                 |
| <b>Config editor</b>              | chord <code>Ctrl+B 0</code> · <b>ch 29</b> · <code>ink.db</code>                           |
| <b>Keybindings</b>                | <b>ch 29</b> (+ Appendix A) · <code>ink.key</code>                                         |
| <b>Backup / restore / reindex</b> | CLI <code>reindex</code> · <b>ch 28</b>                                                    |
| <b>Prompt plates</b>              | CLI <code>prompts</code> · <b>ch 11</b>                                                    |

### KEEPING IT CURRENT

When a feature is added, it earns a row here, in Documentation/FEATURE\_INDEX.md, and in Appendices A–C — a feature with no doc and no chapter is not finished. If a chord in this map disagrees with your terminal, Appendix A and Documentation/KEYBINDING.md are the arbiters; a cell left blank there is deliberate, not an omission.



AFTERWORD

# About the author

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*Vladimir Ulogov.*

**V**ladimir Ulogov has spent decades building infrastructure for distributed systems — the kind of software that watches other software. Early in his career he worked on monitoring and telemetry platforms; later years took him into federated observability, telemetry buses, and the architecture of systems that have to make sense of millions of data points without losing the thread.

Observability, in the end, is the art of **holding a system too large for one head** — of watching it without losing the thread, and never reporting a thing it cannot back up. It is not an accident that a tool he built for writers carries the same instinct to the desk: a book, past a certain size, is exactly such a system, and it too

deserves an instrument that holds it for you.

What makes him slightly unusual in his corner of the industry is a tendency to write his own tools — not in the sense of small utilities, but in the sense of programming languages. The Bund language (its compiler, its VM, its document store, its parser) lives in a long series of Rust crates on crates.io. `rust_dynamic`, `rust_multistack`, `rust_multistackvm`, `bundcore`, `zbus_universal_data_gateway` — each is a building block that exists because the off-the-shelf options didn’t fit the shape of the work.

## Why Inkhaven exists

Inkhaven is Vladimir’s personal reflection on how a single tool can hold the **whole** of writing a book. The frustration it answers is one every long-form writer knows: that the work is scattered across a dozen apps that don’t talk to each other — a browser for research, one program for the prose, another for the characters, a spreadsheet for the word count, a design tool for the final typesetting — and every seam between them is a place the book leaks.

Inkhaven’s wager is that all of it belongs in one place, next to the keyboard: the prose and the world it is set in, the facts it must not contradict, the assistant you steer, and the finished PDF. Not because integration is elegant, but because a book past a certain size outgrows the mind writing it, and a tool that holds your words should also be able to **hold the book** — its facts, its continuity, its shape — so the writer is free to do the one thing no tool can, which is write.

## A work of love

Inkhaven is open source. The licence is permissive — you can read it, fork it, study it, modify it, and pass it on. Strictly speaking, the licence also lets you sell it. The author would, gently but firmly, disagree with you doing that. Inkhaven was not designed as a *for sale* project. It is a work of love made for the authors who can least afford to pay for software, and turning it into a commercial product would betray the reason it exists. So please — don’t.

It carries no analytics, no telemetry, no upsell. The binary will never phone home; the project will never have an “Enterprise” tier you have to escape from.

This was a deliberate choice. There are excellent commercial tools for writing. They cost money — which is fine for many writers, and a barrier for many more. Inkhaven exists for the second group: for the graduate student writing a dissertation on a battered laptop, for the novelist who shouldn't have to pick between rent and software, for the engineer drafting in the same terminal where they already write code, for anyone who would benefit from a tool that respects their work without asking for a credit-card number.

## A note on cooperation

Vladimir believes firmly in the human capacity for mutual help — that we make better work, and live better lives, when we share what we know and what we build. Open source is one of the most concrete expressions of cooperation our era has produced: code read, improved, and passed forward without payment, without permission, by people who will never meet.

If Inkhaven helps you finish your book — that is enough. If it gives you a chord pattern you adapt into your own tool — that's a gift back to the larger project of making software writers can love. As a society, we achieve the greatest things when we help each other rather than compete with each other.

## Where to find more

|                    |                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>GitHub</b>      | @vulogov — the source for Inkhaven, Bund, and the dozen-plus Rust crates that carry the infrastructure. Issues and PRs welcome. |
| <b>LinkedIn</b>    | /in/vladimirulogov — posts on observability, the occasional long-form essay.                                                    |
| <b>YouTube</b>     | @vulogov — talks and walkthroughs from the conference trail.                                                                    |
| <b>X / Twitter</b> | @vladimir_ulogov                                                                                                                |

*If a fact you grounded ever surprises you, or a command trips you up and you can't find the answer in this book — open an issue on GitHub. The author reads them.*

